ARCHITECTURAL AND URBAN REFLECTIONS AFTER DELEUZE AND GUATTARI

EDITED BY

CONSTANTIN V. BOUNDAS
AND VANA TENTOKALI

GLOBAL AESTHETIC RESEARCH

Architectural and Urban Reflections after Deleuze and Guattari

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For his indelible trace in digital regionalism

We dedicate this volume
to Professor Dimitris Papalexopoulos
Ph.D Paris I (Sorbonne), D.P.L.G.,
Architect, National Technical
University of Athens

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Introduction

Constantin V. Boundas and Vana Tentokali

Most of the chapters of this book had an earlier life as contributions to the International Conference, *Gilles Deleuze and Félix Guattari: Refrains of Freedom*, held in Athens, Greece, between the 24th and the 26th of April 2015. They have been revised and are now surrounded by chapters written for the sake of the present publication. The editors of this volume (an architect and a philosopher) chose to include chapters with three thoughts in mind: (1) to highlight reflections of architects, landscape architects, urban architects and planners that have been, in the past twenty years, working and writing under the influence of Deleuze and Guattari; (2) to make the collection useful to those who are curious to find out what a philosopher/creator of concepts (Deleuze) and a schizoanalyst/political activist (Guattari) have to do with designers (of architecture, landscape architecture and urbanism); and (3) to offer it as a road sign at the intersection of philosophy, architecture and ethics that would be an introduction to the discourse and practice of experts in the fields, without being boring or trite.

The inscription of philosophical thought in the theory of architecture is not something new. There has been a long tradition of interconnection between the two, since architecture, as it builds up its own theoretical discourse, never fails to seek awareness with the preoccupations of the philosopher, the questions he or she raises and the tentative answers that he or she gives. Unlike those humanities which, according to Michel Foucault, try to be self-founded, architecture has resisted the temptation to proclaim its autonomy. Its resistance may have been helped by the fact that it comes with two hypostases, floating as it is between science as a toolkit, and art, as a fertile source of affects and qualities. Naturally, the kind of interconnection between architecture, science and art, being always open to new questions, evaluation and revisions, depends on parameters determined by the varying historical

conjunctures and contexts (Sykes 2010, 18). Ever since the 1960s, for example, architecture seems to have shifted closer to science (particularly to the 'new sciences') importing and adapting concepts and techniques from the theories of chaos, complexity and the non-linear or topological geometries, as much as exploring and making use of them as methodological tools in its own everyday practices. But, on the other hand, in being the art of building form, architecture, during the same time period, did not fail to be mindful of the theoretical frames wrought by philosophers with problems and solutions of their own, whose image of thought could prove congenial to its own experimentation and work.

Ever since the 1960s, two philosophical trends – the deconstructive philosophy of Jacques Derrida and the difference and repetition philosophy of Gilles Deleuze and Félix Guattari – have been promising to provide the philosophical support that was needed. These two bodies of discourse replaced the idea of static structure with the search for modulation, transformation and 'difference' and undertook to explain the emergence of structure and form from the formless and the chaotic. The emphasis shifted from an architecture and urbanism of objects to an architecture and urbanism of process. Our volume focuses on the dialogue that has been established between architecture, urbanism and the Deleuzoguattarian image of thought – a dialogue that, for the moment, at least, has brought close together the philosopher's creation of concepts, the production of affects of the artist and the delineation of functions of the 'new scientist'. We do not believe that this dialogue is based on causal connections between the concepts invented, the affects experienced and the functions combined. It is rather a task fulfilled by the practitioners in each field (philosophy, art and science) who,in the pursuit of problems proper to their discipline, have become aware of the fact that practitioners in neighbouring disciplines confront questions and problems of a similar kind.

Dismantling and reconnecting 'otherwise' have been *à l' aire du temps*. Philosophers were ready to put an end to the soliloquy of the Cartesian subject and to chip away at the foundations of the ghostly transcendental ego the same time that architects and urbanists were looking for ways to shake up the foundations of *venustas*, *firmitas* and *utilitas* (beauty, strength and utility) for the sake of what Simone Brott sees as a project for a different, free subjectivity, beyond the Kantian binary of interiority and exteriority (Brott 2013, 152). This was the time of the philosophies of difference when the old privilege of the straight line was given up for the flexibility of Gothic and serpentine lines on the designing pads of the inventors of concepts and of the framers and builders of space alike. Active and reactive forces were celebrated by those friends of wisdom who were in the process of overturning Platonism the very moment that architects were beginning to talk of matter as a force field capable of engendering, transforming and dissolving forms. The concern

for the disruptive questions of genesis or emergence proved fertile for the imaginary of the architect and the urbanist.

As philosophers were unleashing another language underneath the language, allowing it to speak the death of the eternal verities of good and common sense, architects, designers and urbanists unleashed technology, embraced the digital and allowed the virtual to be the *sentiendum* and the *cogitandum* of their art. The applications of digital technology began to serve, not only as a powerful tool for designing, modelling and manufacturing architectural forms but also as a powerful and efficient means for thinking about the objects resulting from its use and the subjects that choose to legitimise it as manifestations of a certain way of (re)conceiving, (re)thinking, contemplating and experimenting with architecture and urbanism.

Philosophers in the process of rediscovering the Kantian sublime did not hesitate to express their impatience with the symmetrical and the well-formed object of aesthetic appreciation at the same time as architects and urbanists were experimenting with 'destabilisation', 'deformation', 'decomposition', 'deconstruction' and 'different/ciation'. The violence of sensation tormenting painting canvases, sculpted artworks and built space trades off representation for the exploration of a world never seen before but yet strangely familiar and near. It is worth remembering here the words of Deleuze about the painter: 'The painter's problem is not how to enter the canvas, since she is already there . . . but rather how to get out of it' (Deleuze 2005, 62). Painting, therefore, he used to say, understood as *hysteresis* is not the filling of an empty surface with the representation of an object. It is imagination aggressively reasserting its rights. In this context, sensation is no longer the response to a form. It is intimately related to forces. It is the capturing of forces. We believe that the new architect and the new urbanist discover themselves at this point in agreement with the painter. Moreover, they welcome the void, no longer as a site of absence and lack but rather as a structural element intrinsically bound with the solid, not only as an isolated or exceptional residual event but also as an operative system associated with the channeling of flows. Tentokali and Constantin Boundas' chapter (chapter 4), included in this collection, shows that as a channeling of flows, the system of void unifies the space of the city, invalidating the boundaries between the urban tissue of the city and the landscape of the suburban sprawl. As a result, a series of new epistemological hybrids emerge covering the in-between spaces among the traditional fields of space, such as landscape urbanism and ecological urbanism. A new condition of blurring epistemological boundaries among the fields of space is established.

Things proved to be more complicated in the relationship between the Deleuzoguattarian theory and the architectural and urbanist thought, in the context of the political. A scanning of the relevant literature shows that, after an earlier helpful exchange between the inventors of concepts and the

builders of frames, architecture and urbanism shed the politically emancipatory skin that had been theirs in their dialogue with the political ontology of Deleuze and Guattari. In 2011, Simone Brott was already sounding the alarm: 'The 1990s re-inscription of Deleuze into Architectural thought, after the initial phase in the 1970s and 1980s) [is] one that largely evacuated the political aspect from Deleuze and Guattari's work' (Brott 2011, 7). Several critics today do not even hesitate to express their suspicion that the eclipse of the political dimension is itself politically motivated, marking the strengthening of the grip of the neo-liberal ideology upon the theory and the practice of architecture and urbanism. A few of them claim to have found elective affinities between neo-liberal ideology and the Deleuzoguattarian minoritarian discourse. Andrew Spencer, for example, wrote:

> Both parties [Deleuzian architecture and neoliberal thought] profess their hatred of hierarchical planning and their enthusiasm for spontaneous ordering and self-organization. . . . What they share, most of all, is a conception of the nature of the human subject, of its relations with the world around it, and of how it should be governed. This shared understanding of subjectivity has furnished architecture with the opportunity to design and build for the continued expansion of neoliberalism into the world of work, education, culture and consumption. (Spencer 2016, 1–2)

It seems that Spencer's claim that the neo-liberal ideology has created the ground for the elimination of the emancipatory line of escape from architecture and urbanism is not far from the truth. But we refuse to attribute the responsibility to Deleuze and Guattari who have repeatedly stated that 'before being there is politics' (Deleuze and Guattari 1987, 249). We find it difficult to accept the depoliticisation of Deleuze and Guattari whose nomadic deconstruction was undertaken for the sake of *la prise de la parole* by those who have no voice and as the anticipation of, and the call for, the coming of a new people and a new earth. We believe, along with Paul Patton, that their jointly authored works, *A Thousand Plateaus* and *Anti-Oedipus*, constitute 'a work, if not of political philosophy that provides tools for the justification or critique of political institutions and processes, but at least . . . a political ontology that provides tools to describe transformative, creative or deterritorialising forces and movements' (Patton 2000, 9).

We acknowledge that this is a very complicated and controversial subject that has not yet been sufficiently discussed. To be sure, attempts have been made to attribute the depoliticisation of architecture and urbanism to the digital technologies and the ubiquity of computer programmes. For example, Michael Hays links 'the new paradigm' of architecture to the digital technologies and computer programmes that coordinate and synthesise multiple parameters and different sorts of data into smooth, frictionless flows (Hays

2010, 473) (cited in Jobst 2013, 64). In a similar vein, Krista Sykes laments the 'lack of a single theoretical discourse during the current period which is marked by the absence of an "overarching concept", since critical theory had entered a period of being "in transition, if not in crisis" ' (Sykes 2010, 12–16). She maintains that this transition is precipitated by the new pragmatism that experimentation and experience sustain (ibid.). However, we are not convinced. Holding the new technology responsible for the eclipse of the political concerns and for the absence of theory explaining the reasons for the eclipse and proposing ways of rekindling the civic persona of the architect does not seem to provide the explanation we need. It is not clear why the advantages of the new technologies could not be appreciated and put to work side by side with the pursuit of justice and emancipation. This is the reason why we think that the responsibility lies with the avant-gardes of architecture which, since the beginning of this century, devote their fancy work to dress up the new cities of the world trade or to give a facelift to the old ones under the demands of the market – with the avant-gardes that stopped doing theory.

Perhaps Peter Eisenman was right when, in a 2008 public dialogue with Greg Lynn and Kurt Forster in New York City, he drew a distinction between the compositional and the performative aspects of architecture, claiming that 'architecture, when "equated" with music can be distinguished into the compositional and the performative parts. The computational paradigm . . . unilaterally matched . . . the performative part and not . . . the compositional' (Eisenman 2008). This was his response to the question of the moderator concerning a possible epistemological shift as a consequence of the computational paradigm. And Eisenman (2008) concluded: 'As far as the computational is not introduced to the compositional but only to the performative, it is impossible for a shift to take place'. This may very well be true, but it does not prevent the rise of a new question: Why does performance eclipse theory from the composition? And the answer to this question seems to us to require the recognition of the relationship between architecture, urbanism and the market as an essential one.

The building of frames is not only an art; the marketability of the built frames affects them to the core. And to the extent that their marketability is determined nowadays by the axioms of a neo-liberal market, the neo-liberal ideology affects their very being. Frichot and Loo raise an important point when they write that 'the schema laid out in *What Is Philosophy?*, which distinguishes the plane of art from those of science and philosophy, and places architecture clearly on the side of art, yet doesn't seem to account for the aspects of the discipline that surpass art. . . . [T]he case of architecture is clearly complicated by its dual role as art and industry' (Frichot and Loo 2013, 73). In the same spirit, writing on the disastrous effects that the 'iconic architecture' of the 2004 Athens Olympics has had on this country's

economy, Simone Brott has this to say: 'The iconic architecture industry was enabled precisely by the historical convergence of the advanced technologies of transnational finance and the advanced technologies of architecture and city making. . . . The material goals are the same, capital accumulation' (Simone Brott 2017, 26).

Since 2008, a number of studies dealing on design as a unifying process in architecture and the visual arts have suggested that the design process cannot be divided into parts – it is simply computation. Few other studies maintain that computation can best be equated with *analysis*, since it is both creation as perspective and planned action (Charidis 2017). In other words, we are once again confronted by two identities: creation and planned action, and a new term covering them both or being substituted for them – *analysis*. Remembering though that, for Deleuze, what matters is the process that makes the difference, and not the molar identities, it is, we think, high time for architects and urban theorists to work again for the restoration of the dialectical relationship between creation and planned action, composition and performance, and for the establishment of their tensive balance.

PART 1. ARCHITECTURE AND URBANISM: ARTS OF THE BUILT SPACE [BALLANTYNE, JOBST, CHATZISAVVA, TENTOKALI WITH BOUNDAS, AND SMITH]

The definition of architecture as the art of the built space does not of course limit 'built space' to the city; built space in the countryside may well be the affair of the architect as much as the construction of edifices in the city is. But the characterisation of urbanism as the art of the city design has traditionally promoted this identification – urbanist or urban planner is the Greek *poleodomos*, the city builder, the one who structures the city. This ambiguity prompted us to begin our collection with Andrew Ballantyne's chapter (chapter 1), 'Schizoanalytic City' because, if the city, as Ballantyne argues, is an engine of connection-making that generates multiplicities of becomings, and if technology today makes connections at a distance possible, the hackneyed binary, city/countryside, may have to be reconfigured, and reconfiguring it will be a helpful introduction to an alternative architectural and urbanist thought and deed that we would like to call 'nomadic'.

Ballantyne's eloquent survey of the city is not another example of the reports of the old flâneur. His survey of the assemblage of assemblages that a city represents is more Deleuzean and less Baudelairean – and this, for the following reason: Victor Fournel in 1867, drawing a distinction between *flâneur* and gawker (*badaut*), noted that unlike the impersonal being that the gawker becomes, absorbed as he is by the outside world and the encompassing

crowd, the *flâneur* is in full possession of his individuality (Fournel 1867). However, Ballantyne's chapter shows that the binary, subjectivity/lack of subjectivity (*flâneur/badaut*), involved in Fournel's distinction is overcome when the report of the stroller is couched in the idiom of the material vitalism that comes with the texts of Deleuze and Guattari.

Rather than thinking of the city as a citadel, bounded by a wall, we should think of the city, Ballantyne suggests, as a cloud or a swarm with dynamic relations between its particles. His survey of the city is not conducted in the context of the traditional view of subjectivity. Rather, and, as Cameron Duff already argued, 'there is not a subject and its object, but rather events of subjectivisation, effectuated in the meeting of elements in space and time. . . . Subjectivity is the achievement of bodies acting together' (Frichot and Loo 2013, 222–23). The thing that makes a city, writes Ballantyne, is the connectedness of its citizens, which makes possible a range of specialisation that is impossible in an agricultural village. If we are rooted, it is not so much in a specific location as in specific networks. According to Andrew Ballantyne, "Identities are fluid and are activated by circumstances. It is the connections that matter, and the edges of the entity fade away: we can participate in urban networks and urban prosperity, even if we live in a place that looks like the countryside and do our networking online from there. Inside and outside no longer have the physical meaning that they used to have when we could only communicate effectively by being physically close to other people" (Ballantyne, in the present volume, p. 29).

With Marko Jobst's chapter (chapter 2), 'Deleuze, Space and the Architectural Fragment', it is space that becomes our focus. In architectural discourse, space, having gone through long discussions of its use, misuse or lack thereof as a concept, remains today, Jobst deplores, an 'unthought problem'. Theorists like Doreen Massey attribute its devaluation to Bergson's linking of space with representation, and despite Elizabeth Grosz's and Gregory Flaxman's attempts to offer alternative ways of thinking about space – ways that draw from the relevant interpretations by Deleuze of Bergson (in the case of Grosz) and Kant (in the case of Flaxman) – the embarrassment remains: It is as if, writes Jobst, 'that, in architecture in particular, it is no longer particularly useful to mention space'. Jobst, however, remains convinced (following Flaxman in this respect) that novel definitions of space are both necessary and possible, but not, he emphasises, 'without philosophy's closer attention to speak of space'. He issues therefore a warning to the effect that, as long as space remains architecture's unthought problem and philosophy oscillates between a Bergsonian subsumption of it under extensity and a Kantian formalism, space will be under-theorised, and Deleuze's appropriation by architects – an appropriation that Jobst cautiously welcomes – will be uneasy. He then concludes that the blind spot may very

well be an opportunity: 'The problematics of space in architecture and philosophy make us think that we must investigate the role of architecture in philosophy's spatial discourse'.

The middle section of Jobst's chapter suggests ways for the reinscription of space in architectural discourses, consonant with the materiality of architecture and therefore attractive to a Deleuze-inspired vital materialism: these ways are heterogeneity, close-up/any-space-whatever and fragment. 'In the architectural close up', writes Brott that Jobst endorses, 'what takes place is an architectural faceification, where the architectural visage, the surface of encounter, is subjectivized and incorporates the individual subject. Here, close-up means the architectural subjectivity itself isolated and laid bare as the feature of a face' (Brott 2011, 58). Any-space-whatever describes the 'emptied spaces' of personal subjectivity. Jobst remarks that the close-up and the any-space-whatever have been explored by Deleuze and his commentators exclusively with reference to cinema, and that the architectural close-up, with its own facialisation and architectural subjectivity has been neglected (with the exception of Simone Brott). As for fragment, it is not the part that precedes totality because fragment is said outside truth. 'It has to do with an intermittent, discontinuous discourse, an affirmation of philosophical becoming. Becoming is no longer the fluidity of an infinite duration (as in Bergsonism) but rather the dismemberment of Dionysos' (Manola Antonioli, 'Nietzsche et Blanchot. Parole de fragment', http://books.openedition.org/pupo/1109?lang=en).

This redirection of attention to space in architecture leads Jobst to the question of affect. He urges his readers to address the question of the affect more thoroughly. Philosophy must acknowledge that all spatial questions – including those on the affect – are inextricably linked with architecture. The ground that architectural phenomenology has claimed as its own, for half a century or so, should be remapped through the Deleuzoguattarian approach to affect. Phenomenologists, in architecture, tend to be concerned with spatial and tactile experience, with the 'atmospheres' of architecture, with all the senses (as opposed to vision as the dominant and problematic major lens). But theirs is, ultimately, a very conservative notion of what humanity is, how it relates to the environment and so on; it does not allow for more complicated ways in which the human 'subject' and the architectural, that is, man-made environment ('the object') interact, come into contact, encounter and influence each other. He maintains that Simone Brott's take on our encounters with architecture and architecture's effects – even if she does not write primarily on the question of the affect as such – is the best, most sensitive, way of trying to grapple with a very fine-grain of how we interact with architecture, and how it influences us, and vice versa. (These last lines come from a private exchange with Jobst.)

The last part of this chapter presents the reader with a brief review of two recent writings: Douglas Spencer's *The Architecture of Neoliberalism* and Nadir Lahiji's (2016) *Adventures with the Theory of baroque and French Philosophy*. Spencer's effort to show that Deleuze's entry in the discourse on architecture strengthened the grip of neo-liberal ideology and Lahiji's exploration of the Deleuzean baroque are based more on the arguments of post-Deleuzean critics (Badiou, Žižek) and less on the original texts, and that they show as a result a considerable misuse of Deleuze's own discourse on the affect and the baroque. Jobst calls for more complex conceptual tools if the swing between the critical and the post-critical is to be escaped.

We share Jobst's complaint that the philosophical discourse on space fails to be sufficiently informed by the architectural experience. But we cannot assume that our readers are very familiar with the philosophical (especially the Deleuzean) discourse on space. We reckon therefore that a brief reference to its salient features will allow them to appreciate Jobst's point more adequately. Nomadic architecture, given its rejection of the Cartesian container-like space, has been after alternative ways of thinking space. Its starting point is that a theory of difference that wants to be consistent cannot afford to leave the manifold of time and the manifold of space uncoordinated. If space were foreign to time, the differential theory of time would be powerless to show that difference is logically and ontologically prior to identity. To show that time and space are indeed coordinated in favour of difference, we should begin with space, not with extension or extended things in space. Space – unlike time – gives the appearance of a homogeneous manifold, in other words, of a manifold without qualities or differences among qualities. Given its alleged indifference to quality, as well as to movement and duration, space gives the appearance of being infinitely divisible.

This notion of space – the space of Galileo and Newton – refuses to be dislodged from our subconscious, despite the arguments that the theory of relativity piles up against its coherence. This space is foreign to duration and, as a result, no change or motion can ever take place in it. Change and motion through the continuous succession of instantaneous spaces, with each one of them representing the state of the world at a given instant, is as incoherent as the reconstitution of time on the basis of instantaneous points. Deleuze, therefore, following Bergson, argues that the space of Galileo and Newton is an ideal, never-realised limit of the Einsteinian dynamic time-space. Since the concepts of absolute simultaneity and of instantaneous space are basically identical, there is no space separable from time, and there is no such thing as a purely spatial distance. To conclude from this that absolute motions simply do not exist, since absolute frames of reference do not exist, was but one small methodological step that Einstein took. It amounts to the daring claim that there is no such thing as an 'everywhere-now'. There is no such thing

as simultaneity of distant events; consequently, there is also no such thing as immediate action at a distance in the sense of Newtonian mechanics.

The extreme theoretical limit of the process of the dilation of duration would be a complete suspension of time and its complete transformation into a homogeneous and static space. As long as this absolute, external limit is not reached, different degrees of spatiality correspond to different degrees of durational tension, or even better, different degrees of spatiality are different aspects of one and the same thing. Matter tends towards a complete juxtaposition of its parts, without ever attaining it. But if space, extension and extended things in space cannot be properly understood without the logical and ontological primacy of duration, then real movement would be better explained as the transference of a state, instead of being conceptualised as the transposition of a thing. The Aristotelian opposition to the atomistic kinetic tradition, which interprets the phenomenon of spatial displacement as an outward, sensory manifestation of a more pervasive qualitative transformation, seems vindicated. The philosophy of difference can safely base its theory of the natural world upon the string of events between the emission of photons and their absorption or 'annihilation' rather than upon particles persisting through time (Deleuze 1988b; Čapek 1971, 232–33).

In the sequence, our volume transitions from the concept of space to the concept of place. Dimitra Chatzisavva's chapter (chapter 3), 'Architectural Translations of Deleuze and Guattari's Thought on the Concept of Place', begins with the old idea of space as an enveloping mass waiting to be built or left undeveloped, and then moves onto the phenomenological notion of the (lived) place that displaced it and to the interest that Heidegger's writings on this subject generated among their readers in the 1960s. Heidegger's thought, Chatzisavva argues, becomes the main impetus towards an ontological dwelling, an authentic inhabitation placed against the alienated, homeless mode of modern technological man. By contrast, in the case of Deleuze and Guattari, place is perceived as a permanent intermediate condition between other concepts rather than as a constant ground of the convergence of sense. As John Rajchman also puts it, Deleuze and Guattari 'depart from the traditional phenomenological view of corporeally grounded lived space, in which the Earth does not move and in which the body is thought to be "situated" or "thrown into the world" and hence from the senses of grounding that Heidegger would associate with historical peoples' (Rajchman 1998, 85). Place, Chatzisavva proposes, should be perceived as a potential of difference, as an open field performing and constructing its own sense through its relations to wider processes, material and conceptual. Places are not locations occupying a certain extent in space, they have 'intensity'. A place-as-assemblage is always a coherent 'multiplicity' of parts, a hotchpotch with no pre-existing whole (Dovey, 2010, 26–27). The singularity of a specific place, therefore, should be investigated in relational terms. This is the topological

comprehension of place, as landscape of differences, and at the same time the acknowledgement of the spatial and material contribution of the concept of place to the production of sense. In the sequence, Chatzisavva chooses to illustrate her point through a reference to Enric Miralles and of his associates Carme Pinós and Benedetta Tagliabue, who show no interest in the forms of things but rather in the forms between things, and in the forces that produce and overflow forms. Miralles's architecture attempts to capture and maintain the vital vibration and the temporal dynamics of a place. He is excavating, Chatzisavva states, the places of human and non-human life where radical otherness is embraced.

Next, Vana Tentokali with Constantin V. Boundas's' chapter (chapter 4) 'The Skin of the Public Space' draws the attention of the reader to the important, albeit often undertheorised, role that the empty space, the non-built or the void, plays in the determination of the filled and the built. With the public space of the Greek city as her field of exploration, with Walter Benjamin's *flâneur* as her guide, and Lewis Carroll's Alice as her conceptual persona, Tentokali formulates few initial observations. Public space in the Greek city looks today, she says, as an undefined, unstructured, unformed farrago, floundering between huge quantities of built (full) space with distinct boundaries and filings of unbuilt (void) space with indeterminate ones. It seems to want to say something, but it does not quite articulate it – to hide something stubbornly, without revealing it. It hides a secret that our writer would like to explore.

Walter Benjamin, and his *flâneur* lend her the notion of the labyrinth of the metropolis; Carroll's Alice shows her how to dive into the void (the rabbit's hole), the reversibility and simultaneity of opposites, and the centrality of the surface. It looks as if Tentokali and Boundas mobilise Derrida's techniques for the sake of Deleuzean conclusions. The reversal of the binaries, built/unbuilt, filled/void and depth/surface, leads to the inclusive disjunctions of the statements 'depth is no longer the complement' and 'the deepest is the skin'. Void and built public spaces mutually determine each other allowing the architect to trace the mystery of the labyrinth of the Greek city through the channel of the void at the surface of the public space. Starting with the surface of the void, the excavation tracing the mystery can now proceed with a dual target. From the point of view of the urbanist in pursuit of her or his everyday tasks, it is the void that permits the depth to rise to the surface and to express the depth's sense. Unless our urbanist, in agreement with Deleuze, decides to read sense as the construct of events, in which case the depth rising to the surface will also be the depth of events, and the excavation that makes the revelation of the secret possible will have to be genealogical. The call therefore by Colin Rowe and Fred Koetter for another architecture, 'which will start from the historical tradition of the past and not from the utopia of the future' (Schumacher 1971), may be accepted without modifications by the

architect, urbanist and the archaeologist in pursuit of their everyday tasks. But it has to be modified by those wishing to capture the sense of the Deleuzean sense. Nomadic architecture, in this case, will begin with the genealogical investigation of the forces, the counterforces and their accidental encounters engaged in their own *gigantomachia* inside the durée. For, after all, the events of the virtual are not states of affairs, nor are they becoming history.

Our search for the kind of architecture and urbanism that would escape the traditional traps continues with Chris L. Smith's chapter (chapter 5), 'Bodies without Organs and Cities without Architecture'. Smith reminds his readers of Peter Eisenman's rejection of the traditional reception of urbanism working with areas filled with objects, and his preference for urban landscapes constituted by events. He explores the possibilities that the notion 'body without organs' holds for a city without architecture. With reference to Antonin Artaud's 'Letter to Hitler' and Deleuze and Guattari's chapter on 'body without organs' (BwO) in *A Thousand Plateaus*, he notes that Artaud had fused the body and the street and paved the way towards his suggestions concerning the making of a body without organs. Using this notion as a metaphor, Smith brings it to bear on the manner by which a city might be lived without recourse to its habitual modes of organisation – as a veritable city without architecture. He develops his argument with reference to the Berlin *Stolpersteine*, small cobblestone blocks inserted into the ground, each one of which is covered with a bronze plaque bearing the name of a Jew, victim of the Holocaust. The *Stolpersteine* project proceeds in an incisive and subversive manner, neither by traditional architecture, nor by roadblocks, but by cobblestones placed carefully, one at a time, and mindful of the warning that the making of a BwO should preserve enough of the body's organisation to allow it to function. The pebbles do not constitute a wholesale attack on the city, but rather a careful redirection of attention, away from the traditional organs of the city and their habitual functions – onto the material that come to constitute the very streets upon which intensities circulate. As such, Smith contends, *Stolpersteine* suggests new ways for making the city, a new city of mobile intensities that belie the habitual demarcations of architecture – a city without architecture that might operate with the logic of the body without organs.

PART 2. ARCHITECTURAL AND URBANIST TOOLKITS [ZOULIAS, HARRIS, KOUSOULAS, PROÏMOS, MORAÏTIS, VERYKIOU]

The second section of our volume focuses on theoretical frameworks and basic concepts, which provide nomadic architecture with the theoretical and conceptual armature that permit it to displace the paradigm of arborescent

architecture and urbanism. Deleuze and Guattari hold theories and the concepts that populate them to be toolkits, useful in the construction of planes of immanence but always ready to be dispensed or substituted by other theories and different concepts, better suited for the task at hand. This section introduces the reader to chaos theory (Zoulias), critical urban theory and assemblage urbanism (Harris), assemblage thinking (Kousoulas), fractal geometry (Zoulias), fold (Proïmos and Moraïtis) and topology (Moraïtis and Verykiou).

Stathis-Alexandros Zoulias's chapter (chapter 6) 'Gilles Deleuze and Chaos Theory' provides the reader with a helpful discussion of the main tenets of chaos theory and with an elucidation of the key concepts that populate it, including immanence, orgiastic representation, smooth/striated space, line of escape, deterritorialisation, strange attractor and dark precursor. The centrality of fractal geometry for this theory is being underscored. So is the notion of the fold, which, ever since Leibniz, is taken to be the smallest element of existence, replacing the point. In thinking of chaos theory, Paul Rabinowicz's witticism is worth keeping in mind: 'In the architectural composition the geometric order, as well as chaos are the basic components. . . . Elimination of chaos from the architectural composition causes "spatial boredom". Elimination of geometric order causes the illegibility of compositions' (Rabinowicz 2000, 200).

Zoulias, in the closing pages of his chapter, introduces few remarks on the impact that chaos theory has had on architecture: For him, the fundamental consequence of a chaotic dynamic in the morphology of the architectural object is the refutation of a belief that holds matter to be something formed by the application of external forces to it, and the manipulation of matter, as something that reveals an immanent to it form. He suggests that the use of the abstract diagram and the algorithm that forms spatial structures could be interpreted, in the context of architecture, with the help of De Landa's 'abstract building', which provides only a few topological traits but is open to innumerable final adjustments depending on its surroundings, as well as on the nature of its inner traits. Finally, Zoulias adds, chaotic dynamics in architecture makes the holistic parameterisation of the architectural object possible. The elements that compose the object will not be separated and idle, each fulfilling its purpose, but they will enter a field condition of a continuous flux of energy and information between them. He does not think it superfluous to remind us that the chaos that non-linear systems display has nothing to do with the idea of disorder and that it is a purely deterministic phenomenon.

In the sequence, Keith Harris's chapter (chapter 7) 'A Thousand Models of Realisation: Towards a Deleuzoguattarian Critical Urban Theory' stages a debate between critical urban theory and assemblage urbanism. Returning to Neil Brennen's championship of the former and Colin McFarlane's support of the latter, Harris is highlighting similarities and differences between

a Frankfurt school–inspired urban theory and a Deleuze/Guattari-inspired discussion which rejects abstract, macrostructural tendencies in view of place-based narrations. He accepts McFarlane's concession that assemblage thinking can contribute to critical urban theory a robust descriptive orientation that would account for the historical urban inequalities as well as for potential directions that alternative urbanisms might take. He also accepts that assemblage thinking can contribute a reconceptualisation of agency with its focus on sociomaterial relations and that, given the affinity between assemblage and the idea of cosmopolitanism, it can provide the imaginary for thinking about potential urban relationships where difference is central. Harris accepts for the most part the criticism that the critical urban theory brings against the urban assemblage thinking, that is, that the latter seeks to displace urban political economy as the fundamental analytical orientation of the way urbanisation occurs. He goes on, though, to show that assemblage urbanism can still be defended and maintained if grounded in the political and economic dynamism that Deleuze and Guattari lay out in the chapters 'War Machine' and 'Apparatus of Capture' of *A Thousand Plateaus*. Using the South Lake Union project as a case study, Harris concludes that, although Deleuze and Guattari's political philosophy was not developed in order to theorise urbanisation, it can serve as a basis to that end, but it nevertheless requires a degree of transformation to address current conditions. Although we agree with Frichot and Loo's claim that 'assemblage avoids reduction to essentialism and reduction to text, [and that it] has a capacity to move architecture away from a focus on fixed form towards process and transformation' (Frichot and Loo 2013, 131), we believe that the reader of Harris's essay will do well to keep in mind Ian Buchanan's skepticism concerning the legitimacy of the derivation of the assemblage theory from the writings of Deleuze and Guattari (Buchanan 2015, 382–92).

More of an insight into assemblage thinking may be gained from Stavros Kousoulas's chapter (chapter 8), 'Non-Correlational Athens', which confronts correlationism to Deleuze and Guattari's 'flat ontology' and assemblage thinking. A correlationist approach to urban entities, according to the author, considers them as a seamless whole, a totality whose parts formulate relations of intensity leading later to its emergent properties. This process, warns Kousoulas, generates eventually the inability of conceiving architectural objects and urban entities as emergent products, enforcing a constant resemblance to archetypical taxonomies. In the sequence, the chapter confronts correlationism with the flat ontologies of assemblages. In the words of Kim Dovey, 'assemblage avoids reducing to essentialism and reducing to text' (Dovey 2010, 16). Flat ontologies consist of self-organising systems or assemblages, where the dynamic properties of matter produce a multiplicity of complex relations and singularities that sometimes lead to the creation

of new, unique events and entities, while more often to relatively repetitive orders and practices. The rest of the chapter focuses on the superiority of the non-correlationist approach to Athens, the Greek capital, in a case study for the analysis of its urban unit, the *polykatoikia*.

A correlational approach, Kousoulas argues, would advance the archetypical resemblance of the emergent urban unit with the dominant modernist principles of the time, privileging the agency of a meta-subject, the architect, who being in control and detached from the complexities of the urban assemblage, performs as a solution-making machine. On the contrary, from a non-correlational point of view, the *polykatoikia* has to be seen not only as a solution but also as a problem-making machine. The emergence of form is neither the input nor the output of a linear process but the complexity of micro-macro couplings which occur in the mesoscale. The chapter concludes by focusing on the problems generated by seeing forms as solutions to problems. The 'question-problem complex' requires us to ask, not only what is a form, but rather who, which one, how many, how, where, when, in which case and from which point of view. In these minor questions lies the potential of tracing a formal continuous variation. The *polykatoikia*, the unit of the Athenian urban reproduction, has to be placed in an environment of practices: a relational field of gravitational forces and attractors, which eventually give spatial and temporal shape – as spaces and rhythms of inhabitance – to the everyday life.

Constantinos V. Proïmos chapter (chapter 9), 'Architecture at the Age of Its Digital Production: The Force, Differentiation and Humanity of the Fold as an Architectural Principle', stresses that the fold is a cultural force carrying the traits of many eras of cultural development but originating in the baroque. For, in the baroque, the fold leads to infinite inclusion, representing constant movement and continuous reference to other folds. Proïmos sides with Matthew Krissel, who argues that the fold is a differentiating principle in a constantly evolving architecture that has learnt to defy the classicist premise, *firmitas* (strength or longevity), and to replace it with an architecture in constant progress. The reader is helped by the many examples of such a constantly evolving architecture: Stephen Perella's pixel architecture which investigates formal plasticity with no regard to function or other determining factors of architecture; Marcos Novak's liquid architectures; and Greg Lynn's exact, anexact and inexact architectural making and formless design, made possible through computational processes. The chapter is sensitive to the charge that the architecture exemplified in these cases by no longer seeking stable forms but only continuous variation and movement through computational processes may strike some of us as particularly inhuman. Nevertheless, Proïmos contends, the fold gives the opportunity to raise the question of humanity while at the same time conceptualise and theoretically ground developments in design issuing from digital technology that is now vastly

employed in architectural practice and thought. The stake then, for him, is an immanent ethics of sustainable design issuing from the fold.

The reader will find more on the concept of the fold in Konstantinos Moraïtis chapter (chapter 10), 'Design of Earth Movement: Objects, Buildings and Environment conceived as Landscape Formations', which focuses in particular the relation of the Deleuzoguattarian theory of the fold to the contemporary conception of landscape, conceiving earth surface as a morphogenetic environment under constant transformation. Moraïtis text is sensitive to the fact that the fold finds itself at the centre of the contemporary interest in topological geometry and moreover to the possibility of expressing a contemporary topological 'anxiety' in design and constructional practices. Topology deals with continuity of transformation. The architect's raw material is no longer form but deformation. Form emerges from the process. Originally initiated as a seventeenth-century Leibnizian effort, topology was not largely developed until the end of nineteenth century and the beginning of twentieth century, while its application in urban and landscape design occurred only recently, as a consequence of the advancement of contemporary computational simulation techniques. Moraïtis estimates that Deleuze's book *The Fold: Leibniz and the baroque* represents the first pole of the growth of topological thinking: that of the seventeenth century. The second pole, the contemporary one, is given in Deleuze's relation to Bernard Cache and through him to the maturation of the computational design practices. What is missing, Moraïtis concludes, from the Deleuzean fold project is a detailed reference to the geometrical *anamorphosis* brought to bear on the baroque garden design. Through anamorphic treatment, a hidden undulation of earth surface was being created, clearly related to landscape-folding formations, as conceived by contemporary architectural, urban and landscape design. One will do well to wonder, at this point, whether Deleuze's lack of attention to *anamorphosis* to which Moraïtis points out is due to his determination to stay away from resemblance and representation altogether because, as Brian Massumi writes, the 'more distorted (anamorphic) or unanchored practices of simulation play on resemblance, but in needing it to play on, hold fast to it' (Massumi 1998, 18).

The last chapter of part 2, Anthi Verykiou's chapter (chapter 11), 'Spatial Transcriptions of the Concept of the Fold in Architecture as a Landscape-Sensitive Approach' suggests that the parallel-in-time elaboration of the post-structural assemblages as differential and differentiating, by Ron Thom and Gilles Deleuze in the field of mathematics and the field of philosophy, respectively, yields the departure from a static perception of space towards a dynamic performative engagement with it in a specifically extended topological approach. This topological mathematical processing relates to spatial assemblages of complex elementary objects which result in a view of *topos* as a coherent theoretical field. Hence the abstract and multidimensional

topological spaces relate to spatial articulation of theoretical approaches, and through the architectural design, to procedures assisted by contemporary digital constructions of imageries performing expressive approaches to the complex cultural identity of a place and its variations. These imageries, she maintains, can be described as multiple alternative realisations of the singular. All this, claims Verykiou, refers to a landscape-sensitive approach. According to the extended topological view, *landscape* refers to the coherent condition of theoretical reflection as the deeper meaning of the notion of *topos*, but by emphasising the singularities and the intensities of a particular real place, it provides a landscape and mental topological continuity.

PART 3. VITAL MATERIALITY [CACHE, MOUSTAKA, HALE, KOSMA, KOUZOUPI]

Alternative architecture and urbanism, in their critique of the traditional hylomorphic image of thought, opt for the model of vital materiality that Deleuze inherited from Bergson. Part 3 of our volume includes five chapters that illustrate the transcendence of the old by the new in the fields of designing (Cache), diagramming (Kosma), mapping (Kouzoupi), and, finally, in the rehabilitation of matter and sensation and the overcoming of the subject/object binary relationship (Moustaka and Hale).

In *Difference and Repetition*, Deleuze laid to rest the traditional fears at what differs, along with the epistemological axiom that only identity, resemblance and analogy grant knowability; he did it by pointing out that, thanks to the differential calculus, differentials are determinable and therefore knowable. But when his 'flat ontology' came to be populated with an infinity of 'folds within folds', (instead of points, lines and solids), the ability of serpentine lines to escape, fold, unfold and refold raised a new fear – the fear of formlessness and the question of the generation of forms. This is why we have placed at the beginning of Part 3 an excerpt from a larger work of Bernard Cache, student initially and then associate of Deleuze whose work Deleuze acknowledges in his *Difference and Repetition*. In his chapter (chapter 12), 'Laocoon and the Snakes of History', through a marriage of concepts coming from past and current practices, Cache focuses on the intriguing serpentine curves of the marble sculpture of the Trojan seer Laocoon and on the work of the German artist Albert Dürer, in order to unfold a dynamic logic for defining formlessness and to offer an intriguing possibility for understanding the generation of forms. Aided by Dürer's work, particularly the *Underweysung der Messung*, and centred on the concepts of multiplicity, changeability and complexity, Cache analyses Dürer's numerical model in terms of a creative computational machine. In so doing, he opts to work with the Deleuzean concept of becoming – a highly constructive procedure of natural systems where

distinct entities coexist and closely communicate with each other. With this as a background, Cache intends to define form by means of sequential instances that derive from the evaluation and recombination of distinct parameters, as he goes on to propose a few new possibilities.

In the sequence, Athena Moustaka's chapter (chapter 13), 'Reterritorialising Concrete as an Actor of Comfort in Architecture', attempts to refute the traditional hylomorphic view of matter as a passive and inert substrate lending itself to the imposition of form from the outside. As Frichot and Loo put it, in nomadic architecture 'hylomorphism is replaced by vital materialism. . . . No longer can we remain at a distance from the materials . . . instead we are required to work with the materials, experiment with what they can do for us, become receptible to particularities they emit' (Frichot and Loo 2013, 112–13).

Using the Park Hill estate in Sheffield as a case study, stripped to its concrete building material, and with the help of an elaborate discussion of the mutating conditions for comfort and feeling comfortable, Moustaka suggests that properties emerge in assemblages with open associations and through the continuously changing relationships of their entities. The emergent properties do not pre-exist in the individuated entities but are only revealed through their interaction. What is perceived as a comfortable condition has been mutating constantly to include varying concepts and definitions. In this respect, the work of Deleuze and Guattari offers a helpful terminology to describe comfort as the emergent property of an intricate assemblage of components, some material, some immaterial, albeit all interacting together. It is the ability of the material to serve multiple purposes and morph into different roles, Moustaka concludes, that influences the behaviour of the building towards comfort or non-comfort.

However, there are still pertinent questions that must be raised: In his chapter, 'Sensing the Virtual, Building the Insensible', to which we already referred, Brian Massumi, for example, in an effort to clarify the relationship between virtual and actual, emerging and emerged, raises one of these pertinent questions that we wish to reproduce here in full: 'But does [the topological turn] live up to the project of drawing on the virtual to draw out the new? The question remains: how could it, if its end product is recognisably still standing-form? By virtual definition, the built form does not resemble its conditions of emergence. . . . The virtual gives form, but itself has none (being the unform of transition). The virtual is imperceptible . . . A building is anything but that' (Massumi 1998, 11–12). The built and the conditions of its emergence do not resemble each other. Where then, in the final product lies the virtual? And then Massumi adds:

> There is, perhaps, a way out of the impasse. But only if there is willingness to reentertain questions about perception, experience and even consciousness that

> have been anathema for some time now to many in architecture, as well as in other domains of cultural theory and production. (Ibid., 13)

It is Mike Hale's chapter (chapter 14), 'Radicalising Architecture by Redefining the Monument', that takes it upon itself to reexamine these questions. It centres on monument and monumentality in order to illustrate Deleuze and Guattari's original theory of materiality (their material vitalism) at work. Material vitalism helps redefine the monument in terms of sensations, enduring beyond the materials by means of which an artwork has been created. The monument is a being of sensation and nothing else, Hale asserts – a claim that presupposes the distinction between affect and affection, percept and perception. Compounds of materials give way to compounds of sensation they create, which take on a trajectory of their own without further reliance on the structure that supported their emergence. As Kim Dovey again puts it, sensation is 'a raw experience of perception prior to cognition, language of meaning . . . a kind of animal condition strongly linked to desire' (Dovey, 1998, 15). Sensation is preserved in duration, so once it is created through the architect's use of steel and stone or the painter's use of colour and brushstroke (in other words, their materiality), the unleashing of sensation is present in duration and filtered into the virtual reality ready to eternally return and affirm its ongoing productive affectivity – 'recalls of emergence, reminiscences of newness', in the words again of Massumi. Hale then concludes that architecture's task is to make perceptible the imperceptible forces that populate the world. Architecture must strive to allow us to feel these forces, and this is something different from rendering them visible or habitable.

Now, vital materiality, being both producer and product of non-standard architecture and urbanism, is facilitated in its creations by the availability of new techniques and tools or by the adjustment of old ones to novel usages. The two chapters that follow – Kosma's and Kouzoupi's – are about the tools of diagramming and mapping, albeit about the kind of diagramming and mapping that stay as far away from representation and resemblance as possible.

Anthia Kosma's chapter (chapter 15), 'Diagrammatic Narratives: Graphic Fields of Rupture and Catastrophe', is the one that chooses to discuss the Deleuzean concept of the diagram, not as a drawing or as an ideal picture, but rather as a tool for the presentation of the way something works, instead of the way it looks. 'The concept of *Diagram*, writes Kosma, is being displaced away from an explanatory tool-operation-drawing towards a dynamic image of action. Being 'a snapshot of a multiplicity in a constant state of flux', as Zdebik remarks, 'the diagram provides information about something incorporeal' (Zdebik 2012, 1, 12). Although the interest in this function of the diagram is strong today, it has often been ignored or misunderstood by the traditional attitudes and perceptions of an ocularcentric culture, thanks to its

own clichés and rules of representation. Borrowing from Allen Stan, Kosma argues that the diagram is not a thing in itself but a description of potential relationships among elements; not an abstract model of the way things behave in the world but a map of possible worlds. The centrality of the importance of the diagram in architecture is aptly characterised by Gregory Cowan in his thesis *Nomadology in Architecture: Ephemerality, Movement and Collaboration*. Cowan designates the diagram as a form of visual thinking – a 'thought-image' (2002, 17), and Kosma seems to agree with him. Returning to the function of the diagram to offer information about something incorporeal, Kosma chooses to refer to a mural of a black background with white random marks and spots, which, until recently, was displayed on the face of a building associated with the National Technical University of Athens, a symbol of the students' uprisal against the Greek Junta in 1973. The defacement of the historic building, the catastrophe of the mighty symbol, Kosma explains, is the diagrammatic erasure of old clichés and sedimented meanings, for the sake of the emergence of the new.

Part 3 closes with Aspassia Kouzoupi's chapter (chapter 16), 'The Concept of Map in the Homeric *Odyssey*'. This chapter is an interesting contribution to the burgeoning literature surrounding the concept and the function of the map in architecture and urban thought. Without reference to him or to his work, Kouzoupi reaches some of the most intriguing conclusions of James Corner's research in *The Agency of Mapping*. In this 1999 text, Corner wrote: 'The function of mapping is less to mirror reality than to engender the re-shaping of the worlds in which people live. . . . Mapping acts may emancipate potentials, enrich experiences and diversify worlds' (225). 'Maps mediate between the real and the virtual, between past and future, between history and design' (Frichot and Loo 2013, 137). With reference to the Homeric *Odyssey* and to Ulysses's *nostos* through the lenses of Deleuzoguattarian concepts, Kouzoupi observes that Ulysses's navigation covers unknown and known terrains, smooth in the case of the unknown (striated only by the rhythmic reiteration of refrains) and striated in the case of known territories, with the help of seemingly available 'encapsulated maps'. The experimentation [which according to Dovey] distinguishes mapping from tracing . . . 'infuses the map with a desire to understand how the place may be navigated or changed . . . making it a mediator between the real and the virtual' (Dovey 2010, 29). The creation of new vistas that replace the representation of what is already there (Zdebik 2012, 10–11) and the uncovering of hidden forces that underlie the workings of a given place (Corner 1999, 124) are attributes of the Homeric narrated maps. It is remarkable to read in Kouzoupi's Deleuzean reception of Homer's maps that the emancipation of the map from its appropriative shackles did not have to wait for our post-colonial times to emerge. We are thinking here of Corner again: 'As we are freed from the old limits of frame

and boundary, *preconditions for the survey and "colonization" of wilderness areas* [emphasis ours], the role of mapping will become less one of tracing and re-tracing already known worlds, and more one of inaugurating new worlds out of old. Instead of mapping as a means of appropriation, we might begin to see it as a means of emancipation and enablement, liberating phenomena and potential from the encasements of convention and habit' (Corner 1999, 252). Homer's narrated maps, trusted to Ulysses and his companions – intruders-turned-explorers – show that the emancipation of the map from the appropriative fury of the one-eyed Cyclops was accomplished long time ago. The emancipation that Corner is talking about was already an attribute of maps in Homer's times (Corner 1999, 213–52).

PART 4. THE CLINICAL [FRICHOT]

Nomadic architecture and urbanism can no longer sustain the image of the neutral builder and designer pursuing their art undisturbed by the clamour of the social and political forces and counterforces that surround them. Matthew David Allen correctly identifies the predicament of us all when he writes that

> the orientation to a center, carried out by a set of stratifying machines . . . does, in fact create us. We are embedded in a structure of signs and symbols attached to things in places and time that literally direct our movements (both mental and physical). . . . We are always turning up against the built environment that pre-exists our actions. But we are not merely instruments of the State, the semi-permanent built environment is a social construct and therefore requires human complicity. (Allen 2005, 52)

Of course, nomadic architecture does not promote the idea of the politically responsible architect and urbanist planting themselves 'slightly ahead' or 'slightly to the side' 'so [they] may speak the silent truth of each and all' (Deleuze 2004, 207). It advocates instead that we resist our own power regimes, immanent to our theories and practices, to the extent that we ourselves are both instruments and objects.

The last section of our volume (part 4) has one only chapter – Hélène Frichot's (chapter 17). Writing in 1996, Francesca Hughes welcomed the 'moment when the architectural profession . . . beginning to shift from its traditionally male domination, [witnesses] the introduction of women to the main body of architecture [that] might bring about a reconstruction of the orders that pervade architectural production and consumption' (Hughes 1996). It is precisely the reconstruction of these orders that Hélène Frichot, a well-known Australian architect, has undertaken to illustrate in her contribution to this volume. Her chapter, 'From the Exhaustion of the Dogmatic

Image of Thought That Circumscribes Architecture to Feminist Practices of Joy' intends to stake out the meeting place between feminist architectural ecologies of practice and the enduring legacy of Gilles Deleuze and Félix Guattari's conceptual creations. Critical of architects that have turned a blind eye to the politics underpinning Deleuze's work, Frichot draws attention to the concept of *superfold*, introduced by Deleuze in his essay on 'control societies'. She uses it as a key for her exploration of the indebted women-form going through the dynamic passage of becoming-woman and her practices of joy in architecture – an architecture, whose task is to conserve life and to shelter (human) existence from external forces. If, as Frichot writes, 'the fold (the operation proper to man) allows Deleuze to think creatively about the production of subjectivity,' and, ultimately the possibilities for, and production of, 'non-human' forms of 'subjectivity', and, on the other hand, the superfold is synonymous with the superman – understood as that which 'frees life' from within man, Frichot's discussion speaks in an anticipatory way at the possibilities that feminist architectural practices hold for an 'unlimited finity' where 'a finite number of components produce an infinite number of combinations'. The 'digital turn' in architecture with which the fold came to be associated, the advances in technology and the vital materialist 'palpations' of sensation and the affect render the opening of the human to non-human forces more plausible than ever before, without of course eliminating the ambivalence of these openings and the capacity of these assemblages for both good and evil. Frichot argues that a sensitivity towards 'following the material' of a fold might also be fostered. Condemning 'the lack of engagement with socio-political phenomena . . . such as neo-liberal forces of privatisation, consumption, competition, and commodification' (Frichot and Loo 2013, 203), in other words, condemning the 'neo-avant-garde' practice in architecture that focuses on the autonomy of exquisite architectural form, Frichot repeats Jennifer Bloomer's call for the decentring of architecture's hidden anthropomorphism, as well as the inherent phallocentrism of its aesthetic paradigms; welcomes the feminist practices of joy that stay alert to the sufferings and enjoyments of an ecological encounter; and attempts to 'reappropriate the commons'. Perhaps, Frichot's feminist practices of joy begin to intersect, in their own way, with Deleuze's vision of architecture as a distribution of light. '[Architecture's] basic medium is light', writes Deluze, 'It uses concrete and stone, metal and glass, to sculpt light in ways that either direct the fixation of attention steadfastly from their confounded conditions of emergence, or on the contrary enable it sporadically to fold back into them'. One can only hope that nomadic architecture – the first of the arts – will stay capable of sculpting the light and focusing it on the task of translating the Deleuzean gerund towards its actualisation: from the *sentiendum* to the sensible – from that which ought to be sensed to what is actually sensed.

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Part I

ARCHITECTURE AND URBANISM: ARTS OF THE BUILT SPACE

Chapter 1

Schizoanalytic City

Andrew Ballantyne

Abstract

A city can sometimes seem to be a clearly defined entity, but only when we do not look at it too closely. The thing that makes a city is the connectedness of its citizens, which makes possible a range of specialisation that is impossible in an agricultural village. The people associated with a city come and go like molecules in a body. If we are rooted it is not so much in a specific location as in specific networks. Identities are fluid and are activated by circumstances. The city organism does not have consciousness in the same way as us and does not give an account of its state of mind, but the non-conscious processes that make it work are sometimes like and sometimes the same as the unconscious processes that produce our conscious states of mind – desiring machines. It is the connections that matter, and the edges of the entity fade away: we can participate in urban networks and urban prosperity, even if we live in a place that looks like the countryside and do our networking online from there. Inside and outside no longer have the physical meaning that they used to have when we could only communicate effectively by being physically close to other people. Rather than thinking of the city as a citadel, bounded by a wall, we should think of the city as a cloud or a swarm with dynamic relations between its personal particles. Its collective will is not directly articulated but can sometimes be inferred. The city as the original engine of communication-making has generated multiplicities of becomings that we do not see in advance because it is not our conscious will that is making them but something that happens between us when we assemble unknowingly in an ill-defined entity much greater, more complex and diffuse than we can grasp.

After dark, approached from the air, the city looks serene: an accumulation of points of light. A barely indicated line of dots here in the darkness marks

an outlying road. It gets brighter along the arterial highways where red and white corpuscles flow in opposite directions. A sprinkling of lights over the suburbs, and then more spilled light in the places where there are more people than cars in the streets – a central square, a mall. We know what it is like to be in the city – that is where most people now live – but even when we are not actually in it, it is in us, and we can travel across the distances between cities without ever ceasing to be urban. Even in private in my car when I can see rolling fields on either side of the motorway, I am in an environment made in a factory, made comfortable so I do not feel the weather outside, and I can learn from my radio about the new developments in government, or listen to music that I have brought along with me, which might have originated at the court of Versailles, from imperial Vienna, or from a garage in Indianapolis. I am not, here in the car, going to hear the singing of skylarks – the road noise masks any real sounds from the locality – and even if I turn off the road and put myself quietly in a field so that I can listen, I hear the skylarks as an urbanite with a sense of the pastoral. If I am going to escape the city I need to do better than this.

The city pulses with life. Its glow masks the stars, and it has its own weather – the smoggy haze that shows up in beautiful sunsets and stealthily stifles its inhabitants. It is the city's connections that are important, more important than the things connected, but we do not always notice that because we have names for the things, so we can focus on them more easily. Things can be delineated and be in identifiable places, but the connections arise between things and can disappear with a shift of the things or even a shift of attention. When we meet, our eyes make contact: we have things to say to one another, something will happen between us. You have a name, I have a name. I can talk about your eyes and your voice, but it is more difficult to say what it is that we have between us. A space, yes, but it is a space with a certain charge, and it is easier to demonstrate it than to talk about it.[1] The best way, as in a novel or a play, is to act it out and feel what it is that happens. At the scale of the city some of the connections become more concrete because they leave tracks. 'The town is the correlate of the road';[2] it arises because of the connections between things – connections with other towns, and in principle with the farmland in between them. The connections themselves are not human, even when they are connections between humans. The eye contact between us is not in itself more human than the road that joins my house to yours. The connections are what matters: the roads bring the traffic, the commerce, the new ideas and people, the food. The town needs these flows coming into it and leaving it, just as a body needs food and produces waste, and if it is cut off from them then it expires – undernourished or overwhelmed by its own waste products. There is an old idea of a town with a fortified wall around it, making a clear zone inside and outside, but even in such a case – where the

boundary is announced and solidified into a thing – there is a need for traffic in and out. The surest way to bring such a town to its knees was to lay siege to it, to stop the flows, and then it becomes very clear that the boundary is normally a control rather than a barrier. The barrier was needed when the place came under attack, not when things were working normally, and the reason it might come under attack would be because towns were places where valuable things accumulated and could be plundered. Nowadays that sort of thing is more likely to be done electronically, but in the times when walled towns were built they were refuges where the citizens could barricade themselves and their livestock until the immediate threat had passed. It is those moments of emergency that gave the towns their defined militarised form, but what made them work as towns – as generators of ideas and prosperity – was quite different. The boundary gives the impression that the town might be a contained entity, but it is not. The city always necessarily reaches out beyond its limits, drawing in the supplies that it needs.

The great thing that the town offers is the possibility of specialising. When people hunted and gathered to sustain themselves, they did not accumulate wealth and power. The history of human settlement stretches back only to a limited time – roughly 10,000 years – and before that the human population was relatively small and limited in the places it could inhabit. The earliest towns we know about were in Mesopotamia, where agriculture seems to have begun, and where people started making food to come to them rather than going out looking for it. In those early days everyone would have been in touch with food production in one way or another, growing crops, husbanding animals or hunting. As towns grew, it became possible for people to develop more specialised skills such as making ceramics that demanded practice and expertise. The larger the population of a town, the more specialised its inhabitants can become. Instead of producing my own food, I might pay someone else to supply me, while I concentrate my energies on something else, like making clothes that people will pay for. Towns have to be places for commerce in a way that the countryside does not, as it would be possible to subsist there on a smallholding without the same need to connect with the outside world. In a town there is always a deal, and the food chain is longer. Even today most food is grown in the countryside, but usually it is nowhere near the place where I buy it. The green beans in the supermarket come from Kenya. The asparagus comes from Peru. I have no idea how the food arrived here, or through how many hands it has passed, or by what transport it was carried. There are specialists who do those things – people who work in docks and warehouses, truck drivers and sales staff. Each has their own set of skills, their own world that overlaps with other people's without being identical. In a small town there will be some commerce, dealing with the essentials for daily life. In a large city there will be a much wider range of professional

services – lawyers' offices, a hospital, and maybe things that only the prosperous can afford to buy – fashionable furniture, luxurious clothes – as well as the necessary things that the smaller town would have. There is something about the proximity of other people and services that makes this diversity and specialism possible and practicable. There is some mechanism at work, which we need not fully understand, that certainly makes this happen.

In recent centuries cities have changed. The state used to be a city, but now the state is usually a nation, and national boundaries are the places where we find the defensive barriers, and they are not walls for military defence but places that mark the edge of jurisdiction and control the movement of people. As late as the early nineteenth century, cities were still walkable, but they started to spread as suburban railways extended the distances across which viable daily connections could be made. Then there were the effects of the automobile and electronic communications. The city now can be quite diffuse – I can live in the countryside, and might work from home on a networked computer, physically travelling to a workplace for some meetings, or to different places to meet colleagues at convenient places – a hotel lobby, a coffee bar, a club. We travel more than ever before and maintain links with people across long distances, but we still do not feel that we know someone we have not actually met in person.

In the city we keep meeting people we have not met before. That may be the defining characteristic of urban experience.[3] If I live in a village, then the people I meet are very likely to be people I have met before, and I will acknowledge them when we pass each other in the street – maybe stop for a chat. In the city that is much less likely to happen, and we usually avoid personal interactions with the people in the street. In the city I meet some familiar faces – friends, colleagues, shopkeepers – but usually when we meet it is by some sort of arrangement. If I have a club or a favourite cafe and fairly regular habits, I might meet up with friends in a semi-random way, but the key thing with the city is that I have ways of disconnecting from other people as well as meeting them. If I find someone lying in the street in my village, then I will certainly try to do something to help. In a big city it is by no means certain that that would happen, especially if we have seen other people walking past. We connect in the city in different ways, and it is not to be taken for granted that we feel a strong-enough bond of humanity with everyone else there if we feel that there are others around who might be more closely bonded, or more properly responsible. Diogenes lived (like a dog) on the streets of Athens and was the first person to call himself cosmopolitan – a citizen not of this particular state or that, but of the cosmos – but although he has an imperishable reputation, it is all too easy to imagine his presence as difficult for respectable citizens to accept, and if they went about their business in the city, would they stop and listen to his ideas?[4] The city brings us

closer together in some ways, and has been an engine for the opening up of human possibilities, but at a personal level it changes us and makes certain sorts of disconnection not only possible but also normal.

The city is a plateau in which certain sorts of interactions occur, and it does not have clear boundaries – they are not the point. Once we have learnt to behave in certain ways, we do not immediately abandon them with a change in location.

John Ruskin explains a game of molecules in his *Ethics of the Dust*. The 'dust' in the title was his attempt to coin an Anglo-Saxon substitute for the Greek-derived 'atoms' – the unsplittables that found their way from Democritus and Epicurus into modern physics.[5] In the game he has his class of schoolgirls move about in different ways – running about at high speed in different directions, to enact being a gas, standing in a more regular configuration to represent a crystal and so on. It is a problematic text for a modern audience, because it tries to explain womanly virtues along the way, but the game of molecules is an invitation to empathise with minerals, which is a challenging thought experiment for anyone. It is an attempt to illustrate the configuration and mutability, not an ascription of feelings to the particles. I am making the same move but in the opposite direction, thinking about the connections between people as if they were chemical bonds at a molecular level. The analogy is not perfect: the relations between people are much more complex and are affected by more factors, but the numbers involved are much smaller. The move from the molecular to the molar (which Deleuze and Guattari often make in a figural way) involves Avogadro's number (6.023×10^{23}) which is many times the population of the earth (7.4×10^{9}) so the scope for complexity and emergence is vastly different, and we must not ask too much of the material. Michel Serres invokes the same model when commenting on Livy's descriptions of crowds behaving in different ways in connection with the founding of Rome.[6] They go through phase transitions – a loose assembly like a gas that turns into a turbulent liquid and then an organised solid – just as in Ruskin's game, as a larger crowd – maybe thousands-strong, but nowhere near the numbers of molecules in a breath of air. The crowd seems to take on different characteristics as the links between people are intensified or dispersed. The social fabric takes on different properties, just as a physical substance (water, e.g.) takes on different properties (depending on whether we are dealing with it as solid ice, boiling water or condensing vapour).

Elias Canetti in *Crowds and Power* made a sustained study of the psychology of crowds, having noticed that when we are in crowds we behave differently from when we are alone. The demagoguery he witnessed during the 1930s induced crowds to behave in ways that the individuals in the crowds would not have countenanced as individuals, but they found themselves being

carried along and implicated. Canetti's analysis defines the physical properties of crowds as impersonal tendencies that seem to be independent of the people who constitute any particular crowd – such as the tendency of crowds to grow: a small crowd attracts people to it; a larger crowd attracts people to it. Crowds can 'crystallise' around 'crowd crystals'.

The crowd crystal is constant; it never changes its size. Its members are trained in both action and faith. They may be allotted different parts, as in an orchestra, but they must appear as a unit, and the first feeling of anyone seeing or experiencing them should be that this is a unit which will never fall apart. Their life outside the crystal does not count.[7]

The thing that makes the crowd cohere has nothing to do with the capabilities and capacities of these people as individuals but the elemental interpersonal forces. Canetti makes it clear that crowds have many different kinds of behaviour, but there are characteristics that come into play, so we can begin to understand that we are dealing with one recognisable type of crowd rather than another. The connections mean that the assembly takes on a life of its own, with its own collective will that is not the same as the life or the will of the individuals.[8] That may remain unconscious or unarticulated, but it might be voiced or signalled by the crowd crystal, a demagogue or as a dispersed murmur rippling through the crowd, and can then catalyse the crowd's action. The crystal structures the crowd and gives it a particular set of potentials.

The extraordinary difference between the capacity of an isolated organism and the crowd is demonstrated very powerfully with slime mould, which has been coaxed into doing astonishing feats of computing. The 'individuals' are tiny and simple cells that can respond in very limited ways only to a limited range of stimuli. We notice them only when they cohere in huge masses, when they look like a slow-moving blob with no discernible constraints on its form. This blob has the capacity to disappear, which happens when the individuals disperse, because as individuals they are tiny and pass without being noticed – they become a cloud of dust. The cells are of such simplicity that their behaviour can be modelled mathematically, and their crowd behaviour can then be predicted, so it is clear that there is no need for there to be more complex 'leader cells' in the group (as biologists used to think there must be). Despite the individual cells being incapable of anything we would want to call 'thought', once there is a crowd of them they are capable of doing things that look very much like thought – finding their way through a maze, or modelling the German motorway system.[9] The model of thought here has a correspondence with the analysis made by Marvin Minsky in *The Society of Mind*, in which on page after page there are diagrams of processes that might be too elementary to be called 'thought', but which, when they are combined, can produce results that look very much like the products of thinking.[10] The brain itself is a mass of connections, and the connections are what make it possible

to do the things it does. Without them there would be little elements switching on and off and it would not amount to anything but working together – usually unconsciously – and in a distributed way through the nervous system, not all in the brain but spread around the body and in the environment around the body – we build up the ways of responding to things and to each other that we recognise as thoughtful and human, maybe as rational or empathetic.

We need to hardly translate this to the scale of the city: we are already in ourselves as complex as cities and as inhabited by multiplicities. These things run in parallel, not as analogies, but because they have the same abstract machine powering them – albeit at different degrees of complexity.[11] There is a whole field of natural computing, including studies of swarm intelligence, in which we see alternatives to the digital binaries operating in the computations of (for example) slime mould that can collectively respond to interdependent variables in ways that they seem to take in their stride, when digital systems are thrown into a spin.[12] When we see human activity speeded up (as in the film *Koyaanisqatsi*),[13] we can distance ourselves from feelings of empathy, and the people who move in swarms through the streets and transport interchanges can be divested of their humane attributes and turned into flows of particles. If I am navigating my way across a station concourse then I arrive, look around for signs that will tell me where I need to go and then I will make way in that general direction, maybe with some distractions along the way, I might browse in a shop. I might buy a coffee. I might take avoiding action because someone looks as if they are going to walk across my path. If I am late then I might move more quickly and in a determined way. If I am early I will actively look for distractions. But from a distance, speeded up, I am a particle and I predictably do what the other particles do: pass across the concourse and on to a train. My anxieties and apprehensions, my satisfaction at finding what I need – these things are drained from the picture as I become part of the particulate mass. It is for this reason that slime mould can make very adequate models of certain sorts of human behaviour, where the humane complexities are taken away and we are left with patterns of movement, patterns that emerge from the complex interactions of very simple things, where it turns out the connections are more important than the things being connected, in the sense that it is the connections that generate the patterns and give the crowd its character, or the matter its grain. It is the connections that constitute the abstract machine, and being distinct from the things they have characteristics that recur when the type of connections recur, even when the things being connected are quite different. There has to be a certain level of complexity before there is the onset of a new character, or new individuations, and a system with thousands of billions of connections can be nuanced and novel in ways that are impossible for a system with only a few tens of thousands, but in these elementary systems we can begin to see the processes at work.

It is from the tissue of interconnections that the city emerges. It has a machinic unconscious of its own, distinct from that of its inhabitants.[14] There are potentials in it that we do not know until they emerge from it – at which point we can see that they were latent all along. The specialisation that the city permits makes it possible to live in ways that experiment and discover what the possibilities are, given the technologies, ideas and cultural mores that are in play. A rural settlement that is dominated by agricultural production for subsistence will probably be found to have very traditional ways of life continuing in it, even if there is a place in it for factory-made vehicles and television sets. There can be exceptions, but it is what we expect. By contrast the disconnections that are possible in the urban milieu make possible ways of life that experiment and depart from tradition in ways that sometimes scandalise when they come to wider attention. Simondon's term for this emergence is 'individuation', and the city is a plateau of intensities in which individuations occur and cascade.[15] It is like a charged chamber, across which a bolt of electricity might spark, or a saturated solution from which crystals might form. The milieu gives the conditions for individuation and becoming, and the particular milieu of the city involves a certain level of intensity of contacts with other people, ideas and stimuli. We can retreat to a closed room, or to the countryside, to make sense of things – to put them in order and into words, but the quality that makes a city something we want to call a city is this urban milieu – the fact that we encounter strangers, meet people with ideas that are new to us – that is the city's great cultural value. 'The city is a tool', said Le Corbusier, and in a manner of speaking so it is, but a tool for what?[16] That is up to us. Above all, though, it is a tool for individuation – a cloud chamber. Of course we might have been drawn to it for other reasons, and if we look back 10,000 years to the first would-be permanent settlements then they did not have the quality of urbanism as we now experience it. Their milieu was different, and they emerged in it and from it to make a new zone that at first must have been part of nomadic traditions, but perhaps the need to maintain a watch to irrigate or to guard the crops from predators would have made some people stay with the crops while others did their traditional thing and returned for the harvest. Through most of the time that humans have existed, we have managed without permanent settlements, but their potential for individuation and transformation has been extraordinary. Nine thousand years ago, perhaps the largest settlement in the world, Çatalhöyük, had maybe 3,000 inhabitants, and the great city of Babylon – a byword for sophistication and (in a different tradition) decadence – had a population of approximately 150,000 when it was the capital of the Persian Empire and slaughtered the Spartans at Thermopylae.[17] The first city to have a million inhabitants was probably Rome, about 2,000 years ago. There are now over 300 cities of that size, and 35 megacities of over ten million. Once people find

out about settlements, there seems to be something that they find compelling. They are growing all the time, and growing with increasing rapidity. Maybe it is connected with the tendency of crowds to grow, or maybe there is an engine of prosperity or the lure of individuation – of finding a potential that would otherwise go undeveloped, even if we do not know quite what it might be. We have moved from a position where people handled tools and developed expertise in using them, to a position where we have built machines that wield tools for us;[18] but in the city we have assembled a diffuse tool that handles us – a vast machine, or multiplicities of machines, acting on us and through us, producing the people and the identities that are becoming. We notice others who can be of use to us in actualising our desires or our dreams, and we link up with them, so between us and through us there are new possibilities, new senses of what we can become. A group identity may be fleeting – the people who had an intense life-changing conversation together and then never met again – or it might be contracted as solemnly and as irrevocably as a marriage, like the 'crowd crystal'. It is never the whole of who we are, but it can unleash a new behaviour – alliances usually give a boost of confidence, as we realize that our hitherto private thoughts resonate with others'.

The conditions that produce 'urban' milieus always support a multiplicity of individuations, and they are fluid: you are part of my milieu, I am part of yours. Furthermore the milieu is not only external to our bodies but also is in contact with them and flows through them – we have crowds within us, and once we have learnt to live as a citizen, there are enriching encounters to be had by introducing the affects and experiences to one another. Think of the hermit Saint Anthony in isolation on top of his mountain near Thebes.[19] In Flaubert's description of his condition, he has escaped from the city but not from the crowds: he is visited by a swarm of demons, visions, mythical and literary creatures and personages. The clamour of voices continues, and his aim is to find harmony and peace by becoming mineral, like the earth, participating in the rocks on which he sits. In the mind the connections around the body, through the nervous system and the peculiar specialism of consciousness in the brain, give rise to the desires and our decisions about how to deal with them. The particles have properties that do not need feelings in our account of them, but remember Spinoza! Writing to a friend in 1764, Spinoza asked him to imagine

> that a stone, while continuing in motion, should be capable of thinking and knowing, that it is endeavouring, as far as it can, to continue to move. Such a stone, being conscious merely of its own endeavour and not at all indifferent, would believe itself to be completely free and would think that it continued in motion solely because of its own wish. This is that human freedom, which all boast that they possess, and which consists solely in the fact, that men are conscious of their own desire, but are ignorant of the causes whereby that desire has been determined.[20]

We feel that we have free will, and can give an account of our actions in which we take responsibility for our decisions and the things we decide to do. Spinoza's observation is that we are in the grip of our desires, and we do not know where they come from – we can make only the most limited inferences about the desiring machines that produce them. In fact it has been a perennial concern of philosophers and sages down the ages to learn how to escape being at the mercy of one's desires. The disciple of asceticism, which has been advocated in many different philosophical and religious contexts, is a discipline of self-control that makes the conscious will work against everyday impulsive appetites, until those appetites are conquered and no longer ask to be appeased. We might have no idea what causes our desires, but there have been many attempts to devise routines that help us to manage them, so we are not altogether under their control. Against this in more recent times the skills of advertisers have been increasingly developed to prompt desires, so unselfconsciously we act on the desire by buying a product. The control of desire is still very much an issue for us as individuals and societies. The point I want to reach here is that the reason we discuss 'will' at all is because we have consciousness and are consciously aware that we have feelings – which maybe controllable, or may seem irresistible – but if we are free to do what they lead us to want to do, then we feel we have all the freedom we need, and if our way is blocked then we feel thwarted. On the other hand, where there is no consciousness we do not think of 'will' as having a role to play. Spinoza's rock just has physical properties, which can be predicted and described. Similarly if we look at the game of molecules as a model of a gas, we have no interest in what the feelings were of the people playing the game. Were they having fun? Maybe, and that might be a reason to play the game, but it is not the point of it as a model of a gas. The feelings of the particles are not an issue: it is the relations between the molecules that are being modelled – the looseness of the fit, their mobility and fluidity.

Elisée Reclus, a nineteenth-century anarchist and geographer, said that 'mankind is nature becoming conscious of itself'.[21] It is a description that places people firmly in nature as a not-separate part of it. Within the body, consciousness is the body-becoming-conscious – it is not a separate free-floating spirit that somehow inhabits the body and that can come and go. Consciousness is a part of the body and a part of nature. It is not in everything, but there are things going on in non-conscious bodies (like the rock hurtling through space) that are not so very different from non-conscious things going on in our bodies that somewhere further down the line generate, for example, a conscious craving for cheese.[22] At the scale of the planet James Lovelock has been speculating about the global intelligence that he has personified as Gaia, suggesting the presence of a will to make corrections that stabilise the earth's ecosystem that might mean making it inhospitable to humans.[23]

What I am wanting to do here is to argue that there is a real slime-mould-like machinic intelligence operating at the scale of the city. There is a problem in inferring its will, because consciousness is in the organic human bodies, not in the superhuman city that is composed of the links between humans. Thomas Hobbes discussed this problem at the scale of a nation in his *Leviathan* (1651) in which he imagined the monarch as the mouthpiece through which the collective will of a nation would find expression.[24] His frontispiece illustrated a giant crowned figure, holding a sword and a crozier, looming over hills and a town. On closer inspection, the giant's body is made up of crowds of ordinary people. It is a depiction of the 'commonwealth', an artificial man, and the book is a study of its construction and working. 'Why may we not say', he asks, 'that all *Automata* have an artificial life?'[25] The problem we have in dealing with these superhuman entities is that they have no consciousness of their own, but must co-opt humans' consciousness to give voice to their thinking. At least that would be a human perspective on the matter. Cities and commonwealths do what they do whether they are conscious of it or not. The human hope would be that if humans can understand what is happening and give voice to it and discuss it, then the city or commonwealth might be guided to do things that are in the interests of humans, but of course there may be other imperatives that the humans do not know about, or do not act upon. The city seems to want to say 'feed me!' (like the monstrous plant in *The Little Shop of Horrors*) and does not have an upper limit on its appetite. For example, some of the early agricultural settlements were abandoned because their soils were depleted. The novel activity of growing crops in the same place year after year without knowing how to replace the soil's nutrients turned out to be unsustainable, so a settlement that ran on that basis turns out not to be a permanent settlement after all. The know-how about agriculture built up gradually over many generations, and now it is to be hoped that it is part of the conscious thought that informs farming communities. The soil does not have a voice of its own, but people have learnt about its needs.

If we look at things at molecular scale then all the entities we deal with as humans turn out to be rather permeable and to have cloudy boundaries. If I am dealing with things in everyday life then that is no problem at all: I have ways of perceiving things and of describing them that help me to do what needs to be done. I pick up the papers, pour coffee, turn on the lights, without worrying about what is happening at a molecular level. I can eat and digest food without noticing how the molecules migrate. I feel like eating cheese. A calcium molecule in the cheese might follow its own inclinations and find itself for a while in my gut and then later be lodged in a bone, which is useful to me but was not the conscious reason that I felt I was reaching for the cheese (though if I were calcium-deficient perhaps I would find myself craving it). Moving abruptly up to a different scale, we, the dust, circulate

like molecules in much larger entities – cities, corporations, societies – that have a fabric made up of people and the things that connect them. The bonds are sometimes strong, sometimes weak. Sometimes there is a clear functional connection where we look for cause and effect, but sometimes the relation is less clear – a random conjunction, vague attraction, that settles into place and produces effects that were not anticipated but which might nonetheless be valued. These larger entities do not have consciousness the way people we do, but they have ways of operating that demand certain things from us and are indifferent to our state of mind. We are caught up in them, and we can be dependent on them for our sustenance and our sense of identity. Once co-opted there is no easy escape. We are free to come and go as we like, but if we participate in these networks we find ourselves connected with a supply of things that we desire, including food, shelter, sociability and status, usually these days mediated through a supply of money.

A city can sometimes seem to be a clearly defined entity, but only when we do not look at it too closely. Even in the old days when cities had walls with gates that were locked at night, there were counter examples, like Sparta – which was the most powerful of the ancient Greek city states, but its built fabric made it look like a collection of villages.[26] The thing that makes a city is the connectedness of its citizens, which makes possible a range of specialisation that is impossible in an agricultural village, where everyone participates in growing food. The diversity and multiplicity is what makes everything else possible: and without the urban milieu it remains virtual – unformed and unconceptualised. In a city most of the food comes from outside, and there are people who do things like keeping shops, making ceramics, programming computers or running banks. Anyone can tell a joke, but it takes an extended network of connections to be able to make a living as a professional entertainer – a network that used to be possible only in a city, but now we have electronic connections that make the physical city less necessary as a real thing, even as our urban populations grow ever faster. The bigger the city, the more specialised the citizens can become, and the further removed they can be from the fundamentals of agriculture that ultimately support the food supply, even if we buy our food in a highly processed form. An individual can be born in one city and move to another, or migrate from the countryside to the town, and the city sustains itself through the participation of these individuals who are relatively free to come and go, but they will find themselves dependent on their connections if they are to thrive. It is a major decision in a life to move from one place to another, to uproot and make new connections and find new means of sustenance. Once we have found a place that sustains us, most of us stay with it and gravitate back to it – call it ‘home’. What would make us move on? The feeling that we could do better elsewhere, or disruption of the place. We could feel free to move, or might feel tied to the place,

by family or debt – and might feel secure, or trapped. The individual finds a way to deal with the situation, or fails to deal with it, and we have human accounts of what that feels like. Maybe this description sounds too rooted in place for a contemporary sensibility. If we are rooted, it is not so much in a specific location as in specific networks. There might well be a place we call home, but we keep travelling in between going back to it, keeping in touch with the people we keep in touch with, meeting new people, losing touch with others. To be properly nomadic, one would move on, finding new networks as well as new places, not going home: feeling at home while shedding familiar ideas and trying out new ones, new identities, new possibilities. Where do we learn the habits of nomadic thinking? If it is defined not as travelling from one place to another but as adopting new ideas, mobile ideas, then surely the city is the place to come. 'Nomadic theory', says Rosi Braidotti, 'practises philosophy as the art of connection making'.[27] When I am in my place of work, I am from elsewhere – the town where I grew up. If I travel further afield, I am from the north. If I go to France then I am English. If I go to Asia then I am European. Within the city I meet different people and I tell myself who I am when I think I belong with one group, or feel distant from another. Do I attract attention? Do I look smart, or queer or alien in this place? Do I look like a threat or a victim? And am I really any of these things in relation to these people? To what class do I belong? In one group the important thing about me is my architectural training, in another it's that I'm an academic. The colour of my skin might be insignificant or might be crucial to how I am treated here. None of those identities matters at all when I am parking my car, crossing the road as a pedestrian or at home peeling vegetables. The identities are fluid and are activated by circumstances. In the city there is a greater range of possibilities open to me – just look around. There are people around who have made decisions about how to present themselves that I would never make for myself: for me they are part of the scenery. There are people who have found ways to sustain themselves in specialised environments like street markets, building sites or night clubs that seem to have a whole culture of their own. The people involved know how to behave there, where others might be bewildered, but again it is only a part of a life, and there are other roles to play at other times of the day and the week.

The city organism does not have consciousness in the same way as us and does not give an account of its state of mind, though we might hope for a mayor who can articulate something like a collective will. Whatever is said or not said, some cities grow and prosper, while others do not. Rome grew, became immensely powerful, and then fell, became depopulated, and then was reoccupied by newer civilisations in the fifteenth and then the nineteenth centuries. Detroit's flourishing, fall and rebirth has been more recent. Even when a city continues without being abandoned at any stage, it can – it

must – transform itself over the centuries. Medieval Paris is profoundly different from nineteenth- or twenty-first-century Paris, even though some of the monuments have been preserved. Venice as a thirteenth-century imperial power had a very different ethos from the city that built the Rococo ballrooms along the Grand Canal, and different again from the provincial tourist spectacle it has more recently become. The city works now in a different way. The connections between people are different, and people's sources of sustenance are less apparent. We move about faster and further, and connect electronically as well as by meeting other people face-to-face, but physical cities seem to grow in size and importance, even as we seem to be physically less firmly attached to them. It is still the connections that matter, even if the edges of the entity are less apparent, now that we can participate in urban networks and the city's prosperity, even if we live in a place that looks like the countryside and do our networking online from there, with only occasional visits to urban buildings. Inside and outside no longer have the physical meaning that they used to have when we could only communicate effectively by being physically close to other people. Rather than thinking of the city as a citadel, bounded by a wall, we should think of the city as a cloud of dust, or a swarm of slime mould, with some physical attributes and regularities, with dynamic relations between its personal particles. Its collective will can manifest itself in gradual growth or abandonment, attraction or dispersal, tensions or sereni-ties, just as our own will, produced by non-conscious mechanisms, can take us by surprise. We come to the city in the hope of finding – what? – our fortune? stimulating company? an alternative to the stifling identity of our parents' home? But having engaged with the city, we are changed – we become more urbane, more guarded, more confident in some ways, more inhibited in others – maybe richer, maybe addicted and destroyed. The city as the original engine of connection-making has generated multiplicities of becomings that we do not see in advance because it is not our conscious will that is making them but something that happens between us when we assemble unknowingly in an ill-defined entity much greater, more complex and diffuse than we can grasp.

NOTES

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Chapter 2

Deleuze, Space and the Architectural Fragment

Marko Jobst

Abstract

This chapter provides an overview of the role of space in the architectural discourse surrounding the philosophy of Gilles Deleuze and his collaborations with Félix Guattari, in order to forward an argument about the importance of considering the propositions made with regard to spatial experience, architectural encounters and the subsequent notion of architectural fragmentation. In order to do so, the text distinguishes between the treatment of space in architecture and philosophy, and borrows from Deleuze's cinematic concepts and discussions to indicate that a certain level of architectural experience, deemed of crucial importance in our encounters with it, often tends to elude architectural discourses both within the discipline and in philosophy.

Furthermore, the chapter considers the recent evaluations and reception of Deleuze's work and its (mis)application in architectural theory, in order to suggest that the current architectural resurgence of the discourse of criticality simultaneously misses its architectural target by overlooking the nature of architectural experience and fails to utilise the full potential of Deleuzian thinking.

We live today in the age of partial objects, bricks that have been shattered to bits, and leftovers. We no longer believe in the myth of the existence of fragments that, like pieces of an antique statue, are merely waiting for the last one to be turned up, so that they may all be glued back together to create a unity that is precisely the same as the original unity. We no longer believe in a primordial totality that once existed, or in a final totality that awaits us at some future date. We no longer believe in the dull grey outlines of a dreary, colorless dialectic of evolution, aimed at forming a harmonious whole out of heterogeneous bits by rounding off their rough edges. We believe only

in totalities that are peripheral. And if we discover such a totality alongside various separate parts, it is a whole of these particular parts but does not totalize them; it is a unity of all of these particular parts but does not unify them; rather, it is added to them as a new part fabricated separately (Deleuze and Guattari 2012: 57).

In this chapter, I will outline a trajectory of interest when discussing architecture in the context of Gilles Deleuze's philosophy and his collaborations with Félix Guattari. The aim is to show the ways in which certain aspects of engagement with Deleuze's work appear in architectural discourse, especially in relation to the question of space. At the same time, when discussed within philosophy, spatial discourse comes with its own set of agendas that aim to surpass those discussed in architecture, yet remain largely blind to architecture. While engaging the same concept, the writings on space in the discourses of architecture and philosophy pursue somewhat different trajectories, resulting in a formulation of concerns that don't necessarily intersect. While space, which is still taken for granted in architectural theory as one of the key disciplinary terms, remains a contentious discursive reference point within the discipline – making it ultimately possible to question its usefulness, in philosophy, the discussions surrounding space are yet to engage architecture in all its specificity. Ultimately, it is not so much the use of the concept of space in architecture that should be interrogated from either of these two disciplinary fields but the use of architecture in philosophy's spatial discourses, part of which is the question of architectural experience.

In *Words and Buildings: A Vocabulary of Modern Architecture*, in the chapter on space, Adrian Forty delineates the history of the term in relation to architecture (2000). Forty insists that it is only towards the end of the nineteenth century that space finally entered the discipline as one of the key defining concepts. By the end of the 1960s it would have become dominant in architecture's disciplinary definitions, causing Robert Venturi and Denise Scott-Brown to famously proclaim it to be the most oppressive term in modern architecture. Coupled with Henri Lefebvre's severe critique of architecture's abuses of (and misconstrued claims to) the term, the concept, at least during the period of 'postmodern' re-evaluations of the role of history and architecture's regimes of signification, lost its dominant position (Forty 2000). But with the reactions to the excesses of postmodern architecture and a renewed interest in spatial concerns, albeit formulated differently by the 1990s, and with the introduction of issues particular to new technologies more recently, the term regained currency both in its social/cultural and material/technological senses of the word.

What follows is a brief take on some of the key discursive issues surrounding space since the turn of the millennium. Significantly, they are

inseparable from the influence of Gilles Deleuze and Félix Guattari's writing on architectural and philosophical discourses and, as such, are indicative of the uneasy albeit productive relationship architectural theory (a contentious term itself) forges with philosophy. The focus is on the following questions: space in relation to time (primarily the outcome of Deleuze's engagement with Bergson), the notion of spatial heterogeneity (revived in recent years with parametric design in mind) and the question of space implicit in the notion of the fragment.

Further to this, and more broadly, this chapter reflects on the more recent evaluations of the uses of Deleuze's philosophy in the context of architectural theory and practice. In this respect, Deleuze's influence remains a troubled one in the context of architectural discourse, suspended between uncritical appropriations and critiques unwilling to utilise his project to its full potential.

SPATIAL DISCOURSE(S)

In *For Space*, Doreen Massey writes about what she identifies as the negative association between space and representation. Massey identifies Henri Bergson as one of the key influences and notes that his focus on time and duration had 'devastating consequences' on the way space would have been subsequently conceptualised. Echoing Edward Soja, she points out that this form of prioritisation of time over space in Bergson was 'one of the most forceful instigators of a more general devaluation and subordination of space relative to time which took place during the second half of the nineteenth century' (Massey 2005: 21). Bergson's understanding of space in relation to duration, by now well rehearsed, is inseparable from the question of representation, which is associated with space, devaluating it in the process. Regarding Bergson's argument concerning the way duration is irreducible to 'spatialised' sections of time, she writes:

> Does not the argument . . . imply that the 'space' which comes to be defined, via a connotational connection with representation, must likewise be impossible? Does it not rather mean that space itself (the dimension of a discrete multiplicity) can precisely not be a static slice through time? With that kind of space it would indeed be impossible to have history as becoming. In other words, not only can time not be sliced up (transforming it from a continuous to a discrete multiplicity) but even the argument that this is not possible should not refer to the result as space. (Massey 2005: 23)

In an earlier work, Elizabeth Grosz had moderated this take on Bergson by showing that within his own conceptualisations of duration and space there

already existed a seed of possible 'rescue' from the clutches of representation. Grosz wrote that:

> In a certain sense Bergson acknowledges the becoming one of the other, the relation of direct inversion between them, when he conceptualizes space as the contraction of time, and time as the expansion or dilation of space' (Grosz 2001: 115). It is through the 'habit of thought', she writes, which 'inverts the relations between space and objects, space and extension' that space was made to appear to come before objects, precede them, 'when in fact space itself is produced through matter, extension, and movement'. (Grosz 2001: 115)

This is an aim akin to Massey's, even if Bergon's role is framed somewhat differently. In both cases, what is being asked is how space can be conceptualised outside the negative connotations with representation; and in both cases, Deleuze's and Deleuze and Guattari's presence hovers over the project as the original source of interest in Bergson, as well as the potential avenue of discursive rescue.

Grosz concludes with the following:

> The very configuration of space may be heterogeneous, just as the movements or configurations of duration vary. Perhaps, in other words, there is a materiality to space itself, rather than materiality residing with only its contents. (Grosz 2001: 128)

The question of materiality is, no doubt, where architecture still isn't sufficiently written into philosophical discourse, despite its *material* ubiquity. But I would like to focus here on the former, the issue of heterogeneity. Observing it from an architectural perspective, it is instructive to note that the architectural reader on space edited by Hensel, Hight and Menges in 2009 took the notion of heterogeneity as the key conceptual tool for rescuing the concept of space from the clutches of discursive negativity.

Hensel, Hight and Menges claim that 'complex, multiplicitous or heterogeneous' discussion of space has been missing from architectural discourse, unlike the discussions of complexity in relation to form and programme (Hensel et al. 2009: 9). This is manifest in the way other disciplines, such as social geography, would have been discussing the term as early as the late 1980s, and the authors conclude that space remains an 'unthought' problem, reduced once again to the questions of form or programme (Hensel et al. 2009: 11).

This question of 'unthought' problems is, I would argue, where the discussion around space should be turned on its head, since it is precisely through space that other disciplines – like philosophy – touch upon architecture, yet don't often confront it head-on. The fact that social geography dominated the

discourse in the late 1980s means that not only the discussions concerning space often took place in that arena and were only then filtered into architecture but also they tended towards a very specific scale of phenomena, that is, those of the urban realm.

The three authors' solution is to reinscribe space in architectural discourse through the concept of heterogeneity. Here we have an echo of what both Massey, a social geographer, and Grosz, a philosopher writing in architecture, would have already stated, without it being particularly clear how the discussion has become specific to the discipline of architecture. While the authors' apparent aim is to point out that architecture needs to keep up, if it is still to lay any claim to the multi/cross/trans-disciplinary discourses of space, the discourse itself comes from a tellingly reduced number of philosophical sources reliant heavily on Deleuze and Guattari and authors like Von Uexkull who have been reintroduced into the various disciplinary discourses due to their influence. Most importantly, as Noam Andrews pointed out, Hensel, Hight and Menges's discourse on heterogeneity was a theoretical platform for legitimising parametric design (Andrews 2010). As such, it was not only reductive in aim but also oriented towards the so-called new technologies, implicitly relating all novelty to that of technology while managing to lose sight of the broader picture in the process and, almost by default, erase the political from the enquiry.

Coming from philosophy, on the other hand, Gregory Flaxman writes that space is 'one of the most perplexing and elusive concepts in all of Gilles Deleuze's philosophy' and that this is because it doesn't correspond to 'any kind of traditional definition' (Flaxman 2005: 176). He discusses Delezue's engagement with space not in the context of Bergson, as would have been the case with the previous examples (and from where a substantial portion of writing on Deleuze in architectural theory comes) but by going back to the relationship between Deleuze's thought and Kan't philosophy. Flaxman writes that Deleuze reverted to Kant's 'sense' of space but rewrote it by 'dissolving' the separation between inside and outside. The result is that 'we have neither a determinate space to inhabit nor determinate principles of space to inhabit us' (Flaxman 2005: 177). This operation makes 'the very concept of space no longer sustainable', he claims, and precisely at that junction Deleuze's most challenging philosophical proposition ushered in, that of finding 'new determinations with which to characterise the immanence of perception and thought' (Flaxman 2005: 177). In a key turn of the argument he notes that:

> The impoverishment of space by representation returns us to the fate of sensation, for if we are ever to go beyond or beneath extensity we are bound to slough off the transcendental conditions of possibility in favour of the singularity of

> sensation – as Deleuze would say, of this percept or that affect. (Flaxman 2005: 182–83)

It is because of this, he writes, that we see Deleuze returning repeatedly to the arts 'and especially what could be called "the arts of depth", in order to provide the basis for a philosophy of sense' (Flaxman 2005: 183). Deleuze, Flaxman claims, starts with the work of art as 'experimentation with sensibility', with the aim to define 'the operative transcendental conditions. The artwork thus consists in an experiment in experience, a form of "experientiation"' (Flaxman 2005: 183) – a notion that is, I would argue, of critical relevance to architecture, yet missing from the majority of discussions around architecture, except in the corners concerned with phenomenology, of which I have written elsewhere (Jobst 2013).

Flaxman concludes by questioning whether it is even possible to speak of space in such a context. As he writes, we might, but:

> Perhaps only as a kind of conceptual reminiscence. In the contortions and distortions of modern art and modern cinema, the metaphysical ground of space is foreclosed, and this is why Deleuze calls for 'new determinations of space and time'. We cannot reconcile aesthetics without also revising our sense of space, and – even more to the point – we might say that the very reconciliation of aesthetics must begin from the very point of revising space. (Flaxman 2005: 185–6)

This carries a couple of key implications for architecture. First, however timely Hight, Hensel and Menges's attempt to reinvigorate spatial discourse in architecture might have been – and there is certainly still something to be said about space from within the discipline *for* the discipline, even if we overlook the stress on technological paradigms and aims evacuated of politics – it is, nevertheless, precisely for the reasons that Flaxman points out (i.e., the 'foreclosure of the metaphysical ground'), in many ways an exhausted project, and not only in the move from Kant to Deleuze but more broadly perhaps. It might well be that, in architecture in particular, it is no longer particularly *useful* to speak of space.

Second, and more importantly, I would argue that from the position of philosophy, novel definitions of space – which Flaxman implies were pursued by Deleuze and deserve to be developed further – simply cannot be addressed without a closer look at architecture, however broadly, or indeed narrowly, we define the discipline. Within writing in philosophy, architecture is still very much overlooked, especially as in the instance of 'experientiation'.

The question of space within philosophy still rarely considers architecture beyond the general realm of the urban and/or the social (i.e., the territory claimed by social geography). Architecture, as the designed, built and inhabited material environment, organised by man and experienced at the scale of

perception our bodies stubbornly retain despite all the supposed technological paradigm shifts; architecture, as the increasingly inescapable object as well as subject of living; architecture, as precisely the form of 'experientiation' that Flaxman writes about, is largely missing. In the context of Deleuze's philosophy in particular, where architecture is declared to be 'the first of the arts', this should be understood as a failure to think immanently (Deleuze and Guattari 1994: 186).

CLOSE-UP, ANY-SPACE-WHATEVER AND FRAGMENT

There are a number of directions that can be pursued from this point on. The one I would like to sketch out marks a trajectory that runs across philosophy, moving images and architecture, while remaining relevant to the discussions surrounding the status and understanding of space. Although this trajectory still needs to be contextualised more widely in architectural theory by expanding the question of perception and its relation to an immanent understanding of images, I will indicate here one very specific conjunction: that of Deleuze's cinematic close-up – as closely related to the concept of *any-space-whatever* – and the notion of the fragment. In this, Simone Brott's take on 'effects-image' (in keeping with the conceptual vocabulary she developed around the 'impersonal effects' of architecture) helps elucidate some key points.

Writing about a scene from Bergman's *Through a Glass Darkly*, Brott points out a difference between Deleuze's understanding of the close-up in film and the notion of an architectural close-up. The latter, she writes, operates by 'isolating the event of contact between a character and an architectural series', which, in turn, 'makes manifest a blurring of subjectivity' (Brott 2011: 57). What the character in question discovers is that: 'her subjectivity is not an individuation, an occupation of space, but emerges from the very materiality of the effects-image' (Brott 2011: 57). Deleuze equates the cinematic close-up with the affection-image, and this is significant, since 'what it renders up close is the effective merging of subject and object' (Brott 2011: 57). Unlike the cinematic close-up, though, which is inextricably linked to the human face, the architectural close-up does something different, Brott claims:

> In the architectural close-up what takes place, rather, is an architectural faceification, where the architectural visage, the surface of encounter, is subjectivized and incorporates the individual subject. Here, close-up means the architectural subjectivity itself isolated and laid bare as the feature of a face. (Brott 2011: 58)

The consequence of the possibility of having 'architectural subjectivity isolated and laid bare' is a rare opportunity to discuss architecture against such

a fine constitutive, as well as experiential, grain. Furthermore, if we think this in the context of space, it is significant that this should be understood to happen precisely at the 'up close' scale of the fragmented architectural object, where the loss of expected spatial connections (the extraction from spatio-temporal coordinates, as Deleuze writes in *Cinema 1*) is as subtle as it is ubiquitous. The architectural close-up, Brott writes, 'envelops the entire psychokinesthetic situation between the subject and the material encounter' (Brott 2011: 58), bringing us directly to the concept of any-space-whatever.

> Deleuze favored Pascal Auge's term espace quelconque ('any-space-whatever') to describe the 'emptied spaces' of an extinct personal subjectivity. But it could be argued that in any-space-whatever, the personal subjectivity is only extinct insofar as she is absorbed into the field of subjectivization. For Deleuze, absorption in espace quelconque means literally 'an effacement', yet we might say that in the effects-image absorption is production. (Brott 2011: 60)

This notion of 'absorption as production' should be understood to be an integral, all-pervasive aspect of architecture, making it impossible for philosophy not to take 'encounters' with architecture into account more systematically. If we are to speak of processes that cut across apparent divisions such as subject/object, then Brott's inquiry is a crucial move towards fleshing out how we might begin to think in more specific terms what takes place in our encounters with the realm of architecture. Furthermore, 'in the architectural space of pure withdrawal we could say the effects-image engulfs all other effects in order to produce a situation of what I call withdrawn affect. Any-space-whatever produces affect precisely by its mechanism of withdrawal' (Brott 2011: 60).

While obviously stemming from the discussion of a film scene – and this is, incidentally, something Brott repeatedly does, confirming to what extent we are yet to confront Deleuze's writing on cinema with architecture in a more systematic way – the key relevance of this series of propositions lies in the way it offers a model of conceptualising space in relation to a very particular spatial 'register' of architecture. Any-space-whatever should, perhaps, be understood to be critical for the 'architectural encounter', and it is not just the merging of subject and object, Brott's primary concern, that makes it so, but the very emergence of *thought* that it is wed to. 'Space' may have become a blind spot in architectural discourse, and reviving it might well be a lost cause, as I have already noted; but architecture itself remains a blind spot when it comes to the conceptualisations of space in philosophy. In other words, while the concept itself might be of limited use to architecture (and often abused in architectural discourse), it still needs rethinking in the philosophical discourse, with architecture piggybacking it, as it were.

My suggestion therefore would be that one way of starting this would be through the reworking of the status of the *fragment* in architecture and, with it, a scale of scrutiny our designed and built environments are not often subjected to. Close-up, any-space-whatever and the fragment indicate that we are asking what gives rise to thought in our encounters with architecture, a political question par excellence.

This is a question that repeatedly presents itself in Deleuze's oeuvre, in various guises. Fragment is discussed, for example, in his book on Proust, where Deleuze aims to relate fragmentation to time and, more pertinent for architectural discussions, its relationship to the question of what constitutes a whole. This relationship between the parts and wholes is also implicit in his reading of Leibniz. It is also present in *Anti-Oedipus*, where it is framed within the context of psychoanalysis and linked to the question of schizo-analysis. As Brott points out, in Deleuze, fragment could be seen to have a double lineage: as part-object derived from psychoanaysis (via Melanie Klein and Jacques Lacan) and in the aesthetically framed considerations contained in *Proust and Signs*, which have their roots in the German Romantic movement, to begin with (Brott 2011: 86). Brott already employs an approach to discussing the architectural fragment in her engagement with Guattari's own writing on Japanese architecture in the 1980s. This is, however, still conceived with an acknowledgement of the building as an identifiable whole, that is, to a different sense of scale of fragmentation, and therefore potential unity, from the one I pursue here.

In *Anti-Oedipus* we find addressed both the questions of heterogeneity and fragmentation. As Deleuze and Guattari write:

> Schizzes have to do with heterogeneous chains, and as their basic unit use detachable segments or mobile stocks resembling building blocks or flying bricks. We must conceive of each brick as having been launched from a distance and as being composed of heterogeneous elements: containing within it not only an inscription with signs from different alphabets, but also various figures, plus one or several straws, and perhaps a corpse. Cutting into the flows . . . involves detachment of something from a chain; and the partial objects of production presuppose stocks of material or recording bricks within the coexistence and interaction of all the syntheses. (Deleuze and Guattari 2012: 54)

The metaphor of the brick is a tellingly architectural one. On a traditional reading, bricks presuppose already constituted, standardised elements of architecture. As such, they are fragments only insofar as they are, quite literally, the pre-existing 'building blocks' used in the construction of larger wholes. Nonetheless, if we follow the suggestion that each of the bricks mentioned here is itself 'composed of heterogeneous elements', it becomes clear that the fragmentation implied is not one that relies on standardisation of

basic units so much as on a continual process of 'cutting into flows', 'detachment of something from a chain' and the production of 'syntheses'.

> Breaks or interruptions are not the result of an analysis; rather, in and of themselves, they are syntheses. Syntheses produce divisions' (Deleuze and Guattari 2012: 56). Or, differently put, '[t]o withdraw a part from the whole, to detach, to "have something left over," is to produce, and to carry out real operations of desire in the material world'. (Deleuze and Guattari 2012: 56)

Finally, I include here briefly Deleuze's writing on Beckett's TV plays in 'The Exhausted', as it performs a potentially interesting dual role: it not only describes space in a way which should be understood to be highly important for architecture (via the notion of any-space-whatever) but also manages to outline a very particular way of seeing, and therefore *thinking*, architecture. The fact that Deleuze's examples are 'framed' by moving images should be seen as a broader suggestion regarding what we might extract from the arts, note through them and then take back into the immediate experiences of architecture as a mode of thinking new connections into existence. Deleuze writes:

> Ghost Trio is made up of both voice and music. It is still concerned with space, with exhausting its potentialities, but it does so in a completely different manner than does Quad. One might first think it is an extended space qualified by the elements that occupy it: the floor, the walls, the door, the window, the pallet. But these elements are defunctionalized, and the voice names each of them successively while the camera shows them in close-up – homogenous, gray, rectangular part homologous with a single space distinguished solely by nuances of gray: in the order of succession, a sample of the floor, a sample of the wall, a door without a knob, an opaque window, a pallet seen from above. These objects in space are strictly identical to the parts of space. It is therefore an any-space-whatever, in the previously defined sense: it is completely determined, but it is determined locally – and not globally, as in Quad – by a succession of even gray bands. It is an any-space-whatever in fragmentation, in close-ups, whose filmic vocation was indicated by Robert Bresson: fragmentation 'is indispensable if one does not want to fall into representation. . . . Isolate the parts. Make them independent as a way of giving them a new dependence'. Disconnect them to allow for a new connection. Fragmentation is the first step in a *depotentialization of space, through local paths* [my emphasis]. (Deleuze 1997: 164–5)

So, let's imagine what it might mean to 'exhaust' space, not only as a discursive referent, which I discuss in the first segment of this chapter, but also as a concrete, daily spatial experience of architecture, inextricable from time, movement and matter, and impossible to consider immanently without taking the experience of architecture into account. The myriad 'impersonal effects'

of architecture that Brott writes of are omnipresent. They ceaselessly contribute to our experience of the world. How do we exhaust such a condition if we haven't fully *closed in* on it yet?

In discursive terms, there are always two 'spaces' at play in philosophy and architecture, despite the seeming inter/cross/trans-disciplinarity of broader contemporary discourse. Despite their best intentions, writers in architecture tend to tweak the discussion to fit their needs (see Menges et al., but also Forty's general overview of the history of the concept in architecture), while writers in philosophy tend to overlook architecture, dismissing its ubiquitous presence. In terms of the *experience* of architecture, on the other hand, and its potential to fragment and 'schiz' rather than simply 'materialise', we will have to look much closer. Escapes are always already performed in architectural encounters, which render architecture inevitably, ceaselessly fragmented.

CODA: ZOOM-OUT

It is interesting to note in this context two recent publications that reflect on Deleuze's work in the context of architectural history and theory. The first one is Douglas Spencer's *The Architecture of Neoliberalism: How Contemporary Architecture Became an Instrument of Control and Compliance*; the second is Nadir Lahiji's *Adventures with the Theory of baroque and French Philosophy*, both of which came out in 2016.

In the former, the author addresses the role architecture plays in the context of contemporary economic paradigms and puts forward an approach to architectural criticism. Taking to one side the important role his work plays in the broader landscape of writing in and of architecture, especially in relation to its practice (which often goes unchecked), I would like to focus here on what happens in Spencer's work to the discourse surrounding Deleuze's philosophy.

The author writes about the instances of what he refers to as 'architectural Deleuzism'. Similar arguments, against the same range of culprits, have been made before, by Harry Francis Mallgrave, Simone Brott and Hélène Frichot, among others. In short, Spencer's argument is that, despite claims to the opposite, the evacuation of overt and clear critique both from certain forms of twenty-first century architecture and the theoretical writings propping it up, turned them into instruments of neo-liberalism. One of the key themes he discusses in this context is precisely the discourse surrounding *affect*, which was imported from Deleuze via a number of philosophical mediators, and adjusted to their own, particular purposes by a generation of practicing architects and academics (and often both in the same person), primarily in the context of the US academia.

The very title of Spencer's book posits that the architectures in question have become an instrument of control and compliance, which reveals the crucial problem in his argument. Like the architects he critiques, Spencer takes it for granted that the actual buildings under scrutiny – such as Foreign Office Architects', Yokohama Ferry Terminal or Ravensbourne College – are, indeed, effective in what they aimed to accomplish. In the same way that Farshid Moussavi, one of the architects in question, claims to have 'designed' affect, or to put it more generously, designed the formal conditions that allow particular forms of affect to emerge and be experienced, the implication in Spencer's argument is that these buildings are – precisely in those terms and through the means utilised by the designer – effective instruments for channelling neo-liberal values, deliberately or not. This becomes particularly problematic if we look at Spencer's discussion of Ravensbourne, which is an *ineffective* building, formative of frustrated architectural 'users', rather than potential neo-liberal 'entrepreneurs'. Simply put, the building doesn't fulfil its primary purpose, repelling the people who inhabit it through its deliberately ambiguously designed spaces – let alone lend itself to implementing the values of neo-liberalism.[1]

The problem, I would argue, is that these buildings can't be effective as instruments because they do not retain such a perfect hold over the architectural subjects they do affect, as implied. Architecture, while being a constant participant in the production of affect and, as I traced earlier, critical in the production of spatial experience, precisely cannot be said to produce outcomes framed in such absolute terms, be they the supposedly liberating ones of Moussavi's claim or the neo-liberal enslavements of Spencer's. The fragmentations and absorptions of architectural form take place both within and beyond such design gestures, and therefore construct a much more complex field of architectural encounters.

Differently put, both of these positions place too much faith in a particular set of design features. What Spencer criticises, ultimately, is simply the discourse that avoids addressing the effects of neo-liberalism (and leads to the rise of the so-called post-critical architectural theory); and what Moussavi instrumentalises – not just in her defence of design approaches and 'philosophies' but also in her definition of what architecture is, or can be – is, once again, the discourse itself. It is here that Mark Wigley's blunt commentary on the 'post-critical' generation of architectural theorists is, perhaps, telling:

> For the advocates of post-criticality, it's hard to imagine a more embarrassing argument to have made. There are always new forms of idiocy being invented, but that's a pretty hard one to beat. All of them ended up running schools of architecture, and I don't think they are interested in any kind of memory project that would remind them of what they were saying earlier since now they are devoted to being awake. (Wigley 2015: 266)

Broad, if amusing, tone aside (however appropriate in reflecting the industrialisation of the production of academic thought and its institutional context), in both cases – Moussavi's (mis)use of Deleuze and Spencer's critique of such uses through the reintroduction of the project of critique – it is the actual complexity of Deleuze and Guattari's thinking around affect that suffers, and consequently its usefulness for raising questions regarding the spatial experience of architecture.

Furthermore, in terms of Spencer's book, the underlying issue is the implicit dismissal of anything that might have come out of the context of Deleuze and Guattari's thought. It is not just 'Deleuzism' that he critiques by getting only as close as Brian Massumi and Simon O'Sullivan to the actual work of Deleuze and Guattari on the question of affect; by eschewing a more thorough engagement with original texts, as well as the extensive body of commentary that exists on it by now, Spencer simply refuses to tackle the question of affect. I suspect that it simply doesn't agree with the very form of critique his project depends on.

Nadir Lahiji's book represents an interesting 'generational' companion to Spencer's. What Spencer discusses as 'Deleuzism', Lahiji labels as 'hyper-Deleuzian' theory. His book centres on the baroque and engages much more committedly with Deleuze's own work – even if his key guides are clearly not particularly receptive to the aims of Deleuze's philosophical project (Badiou, Žižek, Hallward). Setting aside the discussion that could be had about Žižek's evaluations or interpretations – precisely yet another manifestation of neoliberal mechanisms within philosophy (a charge against Deleuze that Žižek makes and Lahiji repeats) – Lahiji engages *The Fold* head-on, touching on that perennial source of Deleuzian ideas transplanted in architecture, often misread to serve dubious purposes.

The question of the fold is simultaneously a spatial question (due to the literal, formal appropriations of 'folding' in architectural theory since the turn of the millennium, if nothing else) and one that, in Lahiji's clearly masterful (and frequently humorous) treatment, remains oddly divorced from more substantial interrogations of the actual *experience* of architecture. In the section on Frank Gehry, this becomes clear: while the dissemination of architectural images that bypass all immediate spatial experience of buildings, they are still buildings that should be evaluated and critiqued through just such encounters. Which is why, if we speak of experience, as well as the relationship between space and affect, the actual architectures simply slip through in terms of what effects they do produce – as much as the ones they are said to produce, but don't.

If, as Lahiji writes, 'euphoric "hyper-Deleuzeans" have, inadvertently or not, come to occupy the center of the philosophical ontology opened up by the conspicuous absence of the critical project of the Left' (Lahiji 2016: 130), and he does, unlike Spencer, make the effort to show that Deleuze's was a political and critical project (however 'critique' itself might be defined

in relation to 'thought'), the question, for those of us who feel more closely affiliated to Deleuze than Badiou or Žižek, is how to show that neither the reduced-Deleuzian nor anti-Deleuzian approaches really get to grips with the complexities of architecture: the ways it is experienced, as well as what might be at stake if we move beyond the critiques of current architectural production, utterly necessary as they are.

The *encounter* with architecture, in other words, remains under-theorised. I maintain that this would mean addressing the question of affect more thoroughly, not reverting to the notions of critique that don't offer sufficiently nuanced models of conceptualising architectural experience – let alone its political reach. To my mind, Brott's lens when discussing architectural encounters remains the most sophisticated conceptual framework on offer. But Brott's (2015) own abandonment of Deleuze in recent years in favour of, say, Adorno is equally telling: in some contexts, too much subtlety can blunt the edge of what needs to be said.

So, this is by no means a defence of the various forms of 'Deleuzism' that open Deleuze's philosophy to the charge of neo-liberal evacuation of politics from architecture and its current pursuits. But it is precisely by asking for more complex conceptual tools that we will ever be able to go beyond such crude swings between the 'critical' and the 'post-critical', as seems to be the current tendency these two recent tomes of architectural theory signal.

As for philosophy beyond the discourse of architectural theory: until it acknowledges that all spatial questions, as well as the question of affect, are inextricable from the realm of architecture, the sooner we will have a way of approaching the myriad ways in which our man-made environments influence and shape us.

NOTE

1. In the several years that I have been interviewing applicants into Year 1 BA Architecture course at the University of Greenwich, there have been a remarkable number of students undertaking the so-called foundation, that is, preparatory courses at Ravensbourne, who have commented on the difficulties of working inside the building – open plan, noise levels and so on. Such comments, subjective as they are, come from a generation that could be seen as the prime target for indoctrination into the dogmas of neo-liberalism that this building, in Spencer's analysis, performs. But his critique aims at a hypothetical target, not the actual building.

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Chapter 3

Architectural Translations of Deleuze and Guattari's Thought on the Concept of Place

Dimitra Chatzisavva

Abstract

The concept of place in architectural discourse has been limited to interpretations that relate it to a phenomenological reading, derived mainly from the romantic tradition, as a concrete existential framework, in permanent dialectical opposition in relation to the abstract, dynamic space. This chapter examines the origins (late nineteenth century) of this opposition and proposes the concept of *place as a potential of difference*, following the thought of Deleuze and Guattari for a conceptualisation of architectural place beyond its dialectical relationship with space. The notion of *place* according to this view is not understood as a negation or reversal of features associated with space but rather with other places and times that its porous, precarious boundaries permit. Crucial for my chapter is Deleuze's thought on *difference* beyond negativity and identity. In this view, the place does not work anymore as a constant ground that guarantees the identity of some sense or other but offers an opportunity for its repetitive differentiation. Within this framework, the chapter investigates similarities and differences between Merleau-Ponty's and Deleuze's thoughts on perception and spatial experience. Finally, the influences of the thoughts of French thinkers, Deleuze and Guattari, and the successive transcriptions of it in contemporary architecture will be examined. Granting that digital design has become the privileged field of this transfer, the chapter chooses to explore other possibilities of relating the work of Deleuze and Guattari to architecture, centring on the work of Enric Miralles. In this context, the chapter attempts to investigate whether the spatial *ethos* that Deleuze and Miralles suggest offers support to the invention of new existential territories and new possibilities of life.

INTRODUCTION

> If human physical and philosophical pursuits demand that man fathoms the Cosmos, art reverses such a demand. The terrifying demand posed by art is that man be fathomed by the Cosmos. (Heimonas, 1984: 122)

The cosmological logos of the author Giorgos Heimonas is exposed to the exhaustion of sense in order to extract sensation. The rhythmic disruption of sense runs as a repetitive refrain in the punctuation of the author's thought. In his work, the quality of man becoming minor signifies a constant exchange of associations with becoming inorganic and animal; as he puts it, it is a 'stingray thought which suddenly shudders' (Heimonas 1987: 67), a common Logos that 'incorporates human and non-human molecules, the rational and the *alekton* (the unspeakable) . . . consciousness and mountain' (Heimonas 2007: 36). His writing exposes Logos through what can, but predominantly what cannot, be spoken. He talks about the 'unspeakable hints', these 'volatile fumes of the words', for 'the mythological drought which drained off our cosmological landscape' (Heimonas 1988: 126, 129).

The author's corporeal and affective discourse seems to have many resonances with the work of the French thinkers Deleuze and Guattari. It is not a narrative of affects but rather one of their actualisations. Affects, percepts and sensation neither depend nor are placed within the perceptual horizon of a subject in its relation to objects; they belong to an immanent, expanded pre-individual field, to a zone of indeterminacy. Deleuze talks about non-visual, inhuman and even molecular perceptions. He examines how the human eye has come to be considered the subjective starting point of perception – the way subjects and objects are formed out of specific perceptions. With respect to spatial perception, Deleuze and Guattari critically focus on associating the idea of space with a long tradition that brings together vision and geometry, spatiality and a humanistic point of view.

A genealogical investigation into the conceptual origins of spatiality reveals vision as the leading factor for the foundation and the conceptualisation of space in architecture.

SPACE, PLACE AND DIALECTIC

Architectural discourse, from its first organised form in the Renaissance to the late nineteenth century, dealt mainly with the aesthetic forms defining space rather than with architectural space per se. Space as a category proper to architecture appeared for the first time in art theories of the late nineteenth

century (Riegl, Schmarsow). It was then that the idea of space gained a new perspective in the aesthetic theories, giving rise to the immaterial understanding of architecture which opposed the already existing stylistic revivalism.[1] Amorphous, abstract space would provide an escape for the materialistic interpretation of changes in architectural style (Semper) and a resolution of the conflicts on the concept of style (eclecticism) (Van de Ven 1993: 357–60). It's the era when the perceptual subject had to abandon its projective identification with the morphological qualities of objects (empathy), so that the theories of 'pure visibility' could provide the necessary distance for the emergence of the concept of architectural space.

Aesthetic theories of pure visibility and the Vienna School would be proven decisive for the emergence of architectural space as an autonomous aesthetic category along with the idea that architecture is a *spatial action* (Sola-Morales Rubio 2003). Specifically, considering space as the dominant element of architectural formatting, Alois Riegl seeks to perceive historical architectural analysis in spatial terms, replacing the abstract idea of style with a more flexible concept adaptive to the historical change processes.

Riegl (1985 [1901]) would associate the formation of architectural space with specific modes of vision distinguishing thereby three phases in the evolution of art. The first phase, exemplified by Egyptian architecture in projects such as the pyramids, is dominated by the gaze on a two-dimensional surface, turning into a haptic (tactile) vision acting like the sense of touch (tangibility) and denying the depth of a three-dimensional space. In the second phase, that of Greek art, vision is integrated into tactile-optical, featuring some depth gradients. However, space in its tactile impermeability is not yet present. In the third phase, that of the late Roman era – with the Pantheon as a starting point – vision is optical; space is now acknowledged as an enveloping mass. In the Pantheon, space is the protagonist with all its internal forms ranging in depth and defining space. Progressively, Basilica acquires an even more dynamic internal space. Space gradually emancipates itself into an amorphous volume, as its non-delimitation receives maximum emphasis. It is in this way that Riegl describes the gradual transition from object to subject, from haptic to optical vision, from surface to space.

Such an optical, immaterial, abstract dimension of architectural space would be widely used in modern architecture as a normalisation tool, as a neutral field for applying rational decisions to the planning and regulation of a new reality. This interpretation of space, in conjunction with the new abstract geometry, would then serve as the arsenal of modernism for the rejection of previous forms and local restrictions. The purist view of space would work just as a hygienic interventional mechanism for the resolution of contradictions concerning its origin.

In fact, several theorists would trace other latent routes behind the dominant and seemingly constant interpretation of functionalist space. According to Vidler (2000: 3), remnants of the empathic projection and the will for expressive form remain in a state of constant dialogue with the dominant stage of modernism, while for Colquhoun (1997: 13–23), rationalism has been only one aspect of the Modern Movement; two parallel movements, romanticism and enlightenment, *Kultur and Zivilisation* (*Culture and Civilisation*), have dialectically coexisted ever since the eighteenth century.

Thus, among the different modes of space conceptualisation, the operating space of New Objectivity eventually dominated, only to let its dialectically opposite characteristics be attributed to a concept that after the war would affect architecture: the concept of *place*. Indeed, in the early 1960s, the concept of space-time is replaced by the more existentialist notion of 'place'. The thought of Heidegger is the main impetus towards an ontological dwelling, an authentic inhabitation placed against the alienated, homeless mode of modern technological man.

For post-war architectural theories, akin to this philosophical meditation, the sense of place is prior to any objectification and interpretation process within the world horizon. It is a sense withdrawn into a 'timeless time' in which case the architectural gesture corresponds to a 'hearing' and thus becomes the allusive movement that allows the genius loci to emerge. The architect, just like the poet, attempts to locate and retrieve such notion against the distance imposed by space and time. Architectural space has to facilitate this original experience, working as a stable ground to guarantee the identity of sense.

In this approach, space is not a norm of collective behaviour but rather a hermeneutic place revealing our unique and inherent relationship with things. The notion of place in this phenomenological view is perceived as a negation or a reversal of features that are associated with the space in modernism (the haptic, tectonic, closed, concrete, etc., are now promoted). Such reversal of hierarchy in terms of the binary opposition of the space/place or rationalism/romanticism tries once more to resolve the contrasts and the affinities that these notions entail. As a result, the conceptualisation of architectural place beyond its dialectical relationship to space is still a challenge for architecture.

PLACE AS A POTENTIAL OF DIFFERENCE – G. DELEUZE

Gilles Deleuze refers to Alois Riegl regarding a new kind of Egyptianism, that of haptic rather than optic space, where perception is no longer driven by a single transcendent, external eye – the one developed by classical

perspective – but rather by a close vision, much like touch (tangibility), which allows singular, unexpected and simultaneous events to emerge. In this perspective, the eye and spectacle share the same plane, which leads to a non-narrative spatialisation, a diagrammatic operation of space and its deliverance from any sense of an overall configuration (Deleuze 1981). The terms of dialectical opposition between tangibility and vision interchange in unexpected encounters. Here the notion of place is no more perceived as a negation or a reversal of features associated with space. It unfolds with other places and times permitted by its porous boundaries.

Deleuze and Guattari refer to a virtual, molecular, undefined space whose fragmented components can be assembled in multiple combinations, both conceptual and material. The virtual appears as a transcendental space-time, immanent to the real and open to the possibilities of the real that are not yet actualised. To Deleuze (2001: 25) a transcendental field is 'a qualitative duration of consciousness without a self'. Experience is 'sensations that no longer rely on a perceiving subject or a constituted world' (Colebrook 2006: 95).

The assumption that the perceptual order comprises something expressed and non-expressed shares resonances with Maurice Merleau-Ponty's viewpoint (1964) in which something invisible – the invisible of this world – is associated with the visible – as a gestalt of the perceived world. Merleau-Ponty invokes the encounter, the *chiasm* between antithetical conditions (the *flesh*), the place of potential, where the intertwining of Man and World takes place in the original conjunction of the sensing and the sensed, the perceptual and the perceived. According to A. Mouriki the idea of *flesh*

> defined by reversibility removes the subject-object dualism, not through a dialectic reconciling the opposites, but rather through a hyper dialectic allowing the merging of the conditions governing the relation, their constant reference to one another without ever overlapping; permitting the maintenance of the multiple, the existence of a 'creative imbalance' within the Self. (Mouriki 1990: 87)

Gilles Deleuze and Merleau-Ponty, therefore, avoid the binary relations, and opt for the direct association between mind and sensation, gaze and touch. While both affect a revitalisation of structure by throwing it into motion, Merleau-Ponty places limits and attaches values to this motion by situating it in perception, thereby giving it a 'corporeal intention' (Taylor 1992). In Deleuze's thought, things exist as they appear, not necessarily in a man's consciousness but as a force which manifests itself in opposition to another force. The phenomenological-existential body to Merleay-Ponty remains trapped in an original world of lived experience, while Deleuze's transcendental empiricism refers to an experience and a corporeality which is more radical than that of the Man-World relation.

Merleau-Ponty attempts, in a rather enigmatic way though, to subdue the differences by means of convergence, unlike Deleuze and Guattari who acknowledge that there is not a unifying system of viewing the world but just a dissemination of fragmentary perspectives. In Deleuze's case, *difference* is not based on identity, it is the vital distance between two terms, and there is no way it can be bridged; it does not lead to convergence and asserts itself through repetition. The existential territory for Deleuze and Guattari is nomad and relational; far from being a revealing clearing, it is rather a creative, immanent potential.

Actually, the association of spatiality with the experience of an embodied subject, the spatial grounding is to Merleau-Ponty the warranty of temporal uncertainty (Taylor 1992: 157), while, in Deleuze's case, temporality consistently avoids itself, ensuring thus the perpetual flux of eternal return as a difference. Nothing restricts such lack (*manque*), neither truth nor identity nor the body. Such repetition of difference, the perpetual flux of sense, the *being* as *becoming* are referred to in his work as a temporal uncertainty assuring spatial uncertainty. The *place* does not work anymore as a stable ground to guarantee the identity of sense but is an opportunity for its repetitive differentiation. Therefore, Merleau-Ponty's notion of flesh constitutes a plane of immanence, which is already there and of which we are all part, while in Deleuze the plane of immanence is an *exteriority* to be activated and in constant recreation.

In their book *What Is Philosophy?*, Deleuze and Guattari comment on the relation between *flesh* and *dwelling*:

> The flesh . . . is no longer the inhabitant of the place, of the house, but of the universe that supports the house (becoming). It is like a passage from the finite to the infinite, but also from territory to deterritorialization. (Deleuze and Guattari 1991: 180)

And they continue:

> Flesh is only the thermometer of a becoming. (Deleuze and Guattari 1991: 179)

> Art begins not with flesh but with the house. That is why architecture is the first of the arts. (Deleuze and Guattari 1991: 186)

In the same book, they consider architecture as a construction on a plane of consistency, as an *agencement* (assemblage) of frames, a joining of levels in an interlocking of differently oriented frames; they actually comment on the consistency of the frames and their joints holding the compounds of sensations rather than representational and functional properties. Architecture releases expressive material, frames our territory, marks the ground, but is

also opened to deframing, the virtuality of a place, the perpetual passage from small to large refrains, from the finite to the infinite.

Thus, contrary to Heidegger and Merleau-Ponty's ideas of a dwelling rooted in a place, Deleuze and Guattari's deterritorialising thought focuses on singular ways of life, not rigid but experimental, moving in an unstable place which, just like a map, precedes and remaps its boundaries. It is the virtual ungrounded depth of an intensive space into an extensive space. In the French thinkers' perspective, *place* is perceived as a permanent intermediate condition between other concepts rather than as a constant ground of the convergence of sense. In this context, place is perceived as a potential of difference, as an open field performing and constructing its own sense through its relations to wider processes, material and conceptual. The singularity of a specific place, therefore, is investigated in relational terms. This is the topological comprehension of place, as landscape of differences, and at the same time, the acknowledgement of the spatial and material contribution of the concept of place to the production of sense.

SPATIAL AND LANDSCAPE ASSEMBLAGES – E. MIRALLES

In architectural approaches sharing a similar attitude towards place as a landscape, a potential of difference, interest shifts from the closed spatial object to its actions and affects. In this perspective, landscape is seen as the connective tissue organising not only objects and spaces but also the dynamic processes and the events among them. In such topological perception of place, architects work with rhythms and textures instead of forms, and they compose unstable spatial structures without any representational references but rather processing arrangements of material and ground.

Such architectural projects base their organisational logic on connections and interactions, putting the emphasis on relations instead of the terms of multiplicity. Proximity and distance relations in such architectural approaches alter perpetually enabling intermediate intervals, frames within other frames, or passages to continuous thresholds. These thresholds are not the boundaries of new digital form generating process, but perpetual transition processes from conceptual to sensory situations. Their work attempts to elaborate through design all place qualitative states and experiences.

Digital design constitutes the privileged field of association between the thought of Deleuze and architecture. However, propositions best ascribing the notion of *place as potential of differences* are those which appeal to the transition from motion (and the productive effectiveness of information) to emotion, affect and material increase of meaning.

The architecture of Enric Miralles and his successive associates Carme Pinós and Benedetta Tagliabue moves within this very frame. The understanding of place within their work is bound to strengthen those relations which produce a singular idea for the place, an embrace of the virtual. While considering the place as a frame of coexistence of unusual correlations, they oppose the static notion of *differing* in favour of the dynamic idea of *becoming*. The compositional refrain in Miralles's work is set to question the stereotypical integration of vision and spatiality, of spatial perception and human experience. His representational lenses do not seek scientific affirmations; they assume responsibility to magnify, to view, as if myopic, and to listen for what they localise as significant in one place and that is often imperceptible. In his architecture there is no difference between touch and vision, sensation and thought.

The Catalan architect constructs his own compositional tools, assembles his conceptual machines making them operate as the engines of his architectural wanderings. Miralles is interested not in the forms of things but rather in the forms between things; he works with abstract tools such as rhythm, changes in scale and continuous variation. It is a diagrammatic process that 'does not function to represent a real, but to construct, to bring it into existence' (Deleuze and Guattari 1987: 142).

Miralles's projects avoid closed recognisable contours, autonomous forms and symbols, as they do not represent forms but rather expose the forces that produce and overflow them. His compositional tools are abstract diagrams exposing the possible lines of development of an architectural project, the map of its motions and conversions. Such expressive maps guide thinking and sensation towards the direction introduced by that project. What we call *a map* or *a diagram*, states Deleuze (1995: 33) 'is a set of various interacting lines . . . lines are the basic component of things and events. So everything has its geography, its cartography, its diagram'.

Miralles's works display the nerve lines they consist of; Miralles (2000: 309) comments 'A project consists of knowing how to tie up multiple lines, multiple ramifications that open up in different directions'. Lines are for Miralles the scaffolds where he hangs the complex, rhizomatic web of his thought and are constantly in a mutual transformation. He accumulates and multiplies these lines always adding new ones, never erasing anything.

The diagrammatic maps of Miralles are mainly geological; they expose the processes, the construction materials of a concept, a place; they make visible the successive layers of their significance, their coincidental configuration, the forces and elements of which a form is made up. He understands architectural place as a site-specific excavation; he starts with the elements he encounters; he is interested in everything a place is made of.

Earthworks, excavations are to Miralles the confirmation of the project in place, its registration in time. However, it is not a nostalgic, phenomenological revelation of a pre-existing sense but rather a discontinuous excavation carried out in sections and routes without any regular succession, a rupture of the time layers of a place towards multiple directions, a geology of urban memory investigating the imperceptible – less defined spatial qualities rearranged into new relations. Exaggerated details, a paradox in Miralles's works, reverse their internal consistency and open a line of flight, evoking associations, accelerating inscriptive experience in the new space.

The landscape method enables the architect to move in-between, to introduce distances and differences in the midst of things, an attitude that brings to mind the words of Deleuze and Guattari (1987: 380): 'Habiter ce monde signifie: developper les arts des intervalles' (To inhabit this world means: to develop the arts of intervals). Miralles (1997: 71) in his turn states, 'Work is done between the things. The idea foremost in my mind is to work in their relationships. The project has to reveal them, and at the same time build them'. In this approach, each individual component of his composition is in many respects connected with all other parts; it is treated less as an autonomous entity and rather as a fragment of broader systems. Thus, the relationship between space and landscape is defined as the adjustment of distance between things, a singular topology of heterogeneous connections between matter and mind.

Miralles constantly works on intermediate degrees of perception which do not necessarily refer to corporeal experience. Shorthand sketches, characteristic photographic views in his presentations demonstrate the stitching and juncture of different shots and time intervals involved in every decision, colourised as if everything is seen from a distance that joins them together by a 'distracted gaze' which overturns any temporal sequence and imaging sharpness.

In a way reminiscent of Deleuze's *time-image*, Miralles's compositional apparatuses do not represent but rather present time. His architecture attempts to capture and maintain the vital vibration, the temporal dynamics of a place. 'Inhabiting a place is not other than moving across a place's time' (2000: 34, 36), he claims, 'To repeat, to make it again. The project must not insist in a particular moment in time, but in inhabiting it'. Miralles is concerned with the intensive, rather than the extensive, dimension of temporality.

Working with time as a medium, Miralles lets different stories to be revealed and intimate times to unfold, along with actions of objects and situations that create a place. He dares to expose intimate details, experiences embedded in each situation in a way that makes them almost natural requirements, necessary for the place. This shift from singular discourse to the collective experience, from the subjective to the cosmic, is a constant in his work.

Such peculiar attitude does not however come under a subjective, intuitive and arbitrary hermeneutics. Miralles does not downgrade the sensation to an intimate experience or the memory to a contemplation of an original time, familiar to phenomenological interpretations. His architecture is interested in the shadows, the breaths produced by buildings, the traces left in their context, the new percepts and affects caused by correlations. The production of place continues to be possible in his architecture – not as revelation of permanent ground but as the production of an event, as the territory of the consistency of a particular intensity. His building assemblages do not simply seek to be integrated into place. They rather establish their own rhythm in the landscape, interrelating becomings with it.

Miralles's work, just like that of Deleuze and Guattari, investigates a pluralist sense-experience that exceeds the specific occasion of existence, becoming a signal for an opening to some world. These creators make room for a geometric, and at the same time, ethical disposition towards heterogeneous existential encounters. They are excavating the places of human and non-human life where the other, the radical otherness, is embraced. Their landscape, topological research on the concept of place allows new associations through motions erasing the border between thought and sensation, between place and what a place can do.

NOTE

1. Throughout the nineteenth century, most architectural theorists struggled with the significant question, 'What is style?'. The dispute around the concept of 'style' was aggravated by the seemingly endless conflict between the appearance and reality of material form in architecture. The identification of architecture with space in the early 1890s promoted architecture as the Ars Magna, because space is, by definition, the most immaterial of all means of artistic expression. This coincided with the visible emergence of modern architecture, in particular, with the Art Nouveau movement. It can be argued that the Art Nouveau represents the first real modern movement in architecture because it broke definitely with the eclectic tendencies of the nineteenth century. In Art Nouveau both ornament and construction were fused in a new unity, but above all, it was the first movement that visualised the new consciousness of spatial abstraction (Van de Ven 1993).

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Chapter 4

The Skin of the Public Space[1]

Vana Tentokali with Constantin V. Boundas

We begin with what the urban space of the Greek city conceals, with the one that is not disclosed and we choose to ignore. In a way this introductory phrase constitutes the inquiry of this chapter which possesses a triple character. It consists of a finding, an observation and a question.

Finding: Public space in the Greek city seems that it wants to say something, but it does not articulate it, that it hides something in a stubborn way, but it does not reveal it. Observation: Public space in the Greek city looks today like an undefined, unstructured, unformed farrago, since it flounders between huge quantities of built (full) with distinct boundaries and fillings of unbuilt (void) with ambiguous boundaries. It balances out irrationally between the chippings of the natural (overshadowed by the constructed) and the constructed (which ignores the natural). Question: What does public space conceal? And why? Is the initial finding reasonable? What message is being conveyed? We could hypothesise that 'the public space of the Greek city conceals something'. In that case, a process of tracing and decoding should be initiated with the intention to retrieve and understand what can be disclosed by the public space in the Greek city.

Our chapter focuses on a single parameter of the public space, the spatial, leaving untouched the public one. Although it would be naïve to accept the idea that the parameter of the public could be entirely excluded from an interrogation like the one that we are proposing, we are adapting here a reading, indeed a 'rewriting' procedure of space that allows traces of the writer's questions to be read. One of them is bound to reach the parameter of the public. At any rate, space is taken here to be 'a product of subjective projection and introjection, as opposed to a stable container of objects and bodies' (Vidler 2001, 1). Deleuze in *Difference and Repetition* also claims that 'Space is not a neutral container without qualities, but the totality of qualified ideas/

structures which contracts in a "place" and presents itself to experience' (Langer 2002, 30).

We start pondering over the secret of the public space by focusing on space itself. The pondering is following a course that could lead us to it. It seems that our course is almost imperceptible, being more of a sensuous experience and less of a conceptual search for its meaning. However, a conceptual guide for the detection of its secret in this case exists, and it belongs, according to David Frisby, to Walter Benjamin, who has already tried to trace and decipher the secrets hidden in the architecture of the city. Architecture for him is the most significant indication of the latent 'mythology' of modernity reconstructing and crossing its 'dream world'. Referring to the flâneur that wanders in the topography of Paris, in the most hidden side of the new metropolis of the period, Benjamin traces the historic object which constitutes the realisation of the architectural dream of the ancients: the labyrinth. 'No image is more strongly associated with Benjamin's analysis of the topography of Paris than that of the labyrinth' (Frisby 2009, 830). The labyrinth of the metropolis.

We follow then the magnetised 'sight' of Benjamin who is captivated by the well-hidden architectural dream of the ancients at the centre of the metropolis. But our look is not directed to the secret of the public space the way that Walter Benjamin proposes. It is turned towards the proposed course that leads to the labyrinth of the metropolis. In our case, both literally and metaphorically, this course leads to the labyrinth of the Greek city. It concerns a labyrinth of the twenty-first century. This is where its secret is located, this is also where the decoding must take place.

The labyrinth of the Greek city, like every other labyrinth, consists of full and void regions, being unevenly distributed. The full parts overwhelmingly predominate over the void ones, while the void parts tend towards their extinction. It is widely known though, at least in Greece, that the majority of the urban space (especially the public one) is built or better over-built. There is only a small percentage left unbuilt, so small that it acquires an invaluable and multilateral symbolic significance. Our observation focuses on this small percentage of the public space, namely the unbuilt, and we savour the many names that are ascribed to it – names with subtle differences: unbuilt, not built, void, landscape, natural environment, sprawl, and so on.

WHY THE VOID

Flirting with the concept of landscape – the epistemological object of landscape architecture – architecture discovered many of its properties and embedded them within its own discourse. 'Landscape' is not synonymous with environment, it is rather 'the environment perceived' (Bourassa 1988).

'Perceiving' the environment (built and unbuilt, solid and void, edificiary and empty), architecture focused on the 'void' and on all its scaling formations (going from the urban tissue to the suburban tissue and to the 'sprawl') and placed them all to its own curriculum.

Architecture has now recognised that the concept of void, in any scale from the urban tissue to the suburban sprawl, is a factor of a new unifying condition of urbanity between urban and suburban space. Aaron Betsky arguing for the unifying character of the void in architecture maintains that 'Architecture must be an unfolding of the landscape' (Betsky 2000). Bridging the gap between architecture and its related epistemological fields, Rem Koolhaas introduces the idea that 'sprawl' must be 'architectonised'. Albert Pope, in his book *Ladders* broadens the concept of urbanity, examines the role of the void as a unifying factor between urban and suburban space, and states: 'the voids of the contemporary city, the gaps and the lacunae that have come to characterize contemporary urban space, are not without meaning' (Pope 1996, 17).

WHY THE SURFACE

The question that rises at this point is this: Why focus on the unbuilt or the void? We prefer to think of the question, not in the context of the traditional query 'what is the void?', but rather, along with Vidler, in the context of 'what the void does'. From among all the many meanings of the void, Vidler explores its usefulness. He is interested in 'what the void can hold' or in everything 'that may be produced in a vacuum, or by the vacuum' (Vidler 2000, 14). We hypothesise that the void would also be useful in our attempt to resolve the mystery held by the labyrinth of the public space.

Now, the void of the urban tissue, as any other void, has a structure – a structure with three dimensions (if we deliberately want to forget the fourth one): two horizontal and one perpendicular. It has its surface (length, breadth) and its depth. Which of the dimensions offered by the void is going to be more useful in the unveiling of the secret? Or better: which is more appropriate for the detection of the course that leads to the secret? The surface or the depth? The answer seems obvious: How could it be possible for depth to be fathomed when surface is not penetrated? The surface is obviously our starting point in the unveiling of the secret of the labyrinth of the urban space.

But of course Deleuze does not ground the importance of the surface on the fact that it is the obvious starting point of our investigations and discoveries. It was not the surface as a starting point that made him confess that 'he needed many years in order to understand and realize that between the . . . two notions [surface, depth] the surface is the most privileged' and that 'Depth is no longer a complement' (Deleuze, 1990,9). We will begin to understand the

significance of the surface only after we focus on what the surface does in his work – what its usefulness is and how it is deployed.

His 1969 book, *Logic of Sense*, contains the answer to this question in the way that it relates surface and 'sense'. 'Sense', he writes, 'is what is formed and deployed at the surface' (Deleuze 1990, 125). At the beginning of the *Logic of Sense*, Deleuze pays particular attention to the part of Alice's adventures, when she, following her instinctual inclination, seeks the secrets of the events and of her becoming. With the help of Lewis Carroll's literary texts *Alice in Wonderland* and '*Through the Looking Glass*', Deleuze follows the course of Alice and watches her 'falling' into the Wonderland.

WHY THE EVENT

In his readings of Stoic philosophy, Deleuze focuses on the notion of the event that he holds to be 'coextensive with becoming and . . . becoming . . . itself, coextensive with language' (Deleuze 1990, 8). It is language that fixes limits (the moment, e.g., at which the excess begins), and language again that transcends the limits and 'restores them to the infinite equivalence of an unlimited becoming'. It is Carroll's language that inspires Deleuze and allows him to play with limits and going beyond them: 'In Carroll's work, everything that takes place occurs in and by means of language' (Deleuze 1990, 22). Language, for Deleuze, has three figures: the metaphysical or transcendental surface, the incorporeal abstract line and the decentred point. These figures correspond to the surface effects of events; the line of sense immanent to the event, at the surface; and the point of nonsense, surface nonsense, on the line, being co-present with sense (Deleuze 1990, 183).

Now, the event is not what occurs (an accident), it is rather inside what occurs, it is the result of actions and passions. The event is not a particular state of affairs or a happening but rather something made actual in the state of affairs. Events exist only in states of affairs, however, they differ from them, being not identical with them and never exhausted in their becoming actual (Deleuze 1990, 9). Events are neither agents nor patients. 'They are impassive entities-impassive results. They signal and await us' (Deleuze 1990, 149). In Deleuze's ontology, the virtual and the actual are two mutually exclusive, yet jointly sufficient, characterisations of the real. Actual/real are states of affairs, that is, bodies and their mixtures or individuals existing in the present.

Virtual/real are incorporeal events and singularities in a plane of consistency. The kind of process that we find in Deleuze's ontology is not properly captured in the scheme, actual/real actual/real; its correct schematisation is rather this: virtual/real actual/real virtual real, in other words, becoming, instead of being a linear process from one actual to another, should rather be

conceived as the movement from an actual state of affairs, through a dynamic field of virtual/real tendencies to the actualisation of the field in to a new state of affairs. Events subsist in language by taking the form of a verb: 'There are verbs carrying off with them becoming and its train of reversible events and infinitely dividing their present into past and future' (Deleuze [1969] 1993, 23). Lewis Carroll's metaphors intrigue him. In one of these, Deleuze finds a reference to events as crystals: 'Events are like crystals, they become and grow only out of the edges, or on the edge'. Elsewhere, Deleuze refers to them as being in between humour and irony: 'Paradox appears as a dismissal of depth, a display of events at the surface, and a deployment of language along the limit. Humor is the art of the surface, which is opposed to the old irony, the art of the depths and heights' (Deleuze 1990, 9).

Architects have found Deleuze's ontology of the event helpful for their own purposes. Hélène Frichot, for instance, stresses the role that the event plays in the contemporary architectural discussion: 'The architect', she writes, 'is interested in how the surface effects produced by the circulation of events might be created in material forms of expression' (Frichot 2005, 66). Manuel de Landa also pays attention to the different engagements between scientists and philosophers over the notion of the event and has this to say:

> Scientists, tracking the actualization of the virtual, examine the actual beings, as well as intensive processes. They are focused on functions which have states of affairs or mixtures for reference through which science continually actualizes event in a state of affairs, thing or body that can be referred to. Philosophers, tracking the opposite motion, that which reconstitutes virtual multiplicities (as ideal events) out of actual entities and gives them consistency as a space, support that philosophical concepts have events for consistency, through which philosophy extracts a consistent event from the states of affairs. (de Landa 2005, 86)

Tamsin Lorraine, reading *The Logic of Sense*, points out that Deleuze distinguishes events of sense from specific propositions and states of affairs and concludes: 'Concepts are events of philosophical thinking that hover over concrete states of affairs' (Lorraine 2005, 175).

WHY BECOMING

Incorporeal, virtual events form series, and these series subsist by inhering in the series of states of affairs, made up of actual bodies and their mixtures. Following Carroll and Alice in her adventures, Deleuze introduces the idea of reversibility of the events in the series. There is nothing static about these series. Their being is becoming, modulation and transformation. They are simultaneously characterised by the temporality of the 'not yet' and the

'already past', inclusively disjoint. They eschew the present. They are to be thought, with the help of Bergson's *durée*, as processes of virtual (and, therefore, real) memories, 'hovering over' the actual, and therefore not at all subjective—series of events, pure becoming. This is how Deleuze puts it:

> Every event can be said to have a double structure. On the one hand, there is necessarily the present moment of its actualization: the event 'happens' and gets embodied in a state of affairs and in an individual. . . . Here the time of the event, its past and future, are evaluated from the perspective of this definitive present and actual embodiment.
>
> On the other hand, the event continues 'to live on', enjoying its own past and future, haunting each present. (Deleuze 1990, 151)

According to Deleuze, a process or becoming is an ongoing stratification. The stratified formations do not remain identical to themselves and unchanging when new strata are being deposited 'on top of them'. They are in a constant state of motion and transformation. The objective memory we alluded to earlier is stratified; language itself attempting to express the sense of this memory is stratified.

Since they are not limited by the present, and they are simultaneously past and future, the reversibility of events is constitutive of their becoming. Deleuze, following Carroll, observes Alice's course as she digs a hole, a shaft that reaches the earth's depth and finds a mixture of bodies which coexist mingling with each other (the actual states of affairs). Gradually, as the story unravels, Alice becomes less capable of pursuing her course towards the depth. She releases her double, immaterial subsistence, and instead of digging, she goes on sliding from right to left and from left to right (in the realm of virtual events). Alice has graduated from being to becoming lived through a sequence of reversions: Her body size oscillates between bigger and smaller; temperature intensities that her senses experience revert from cold to warm and vice versa; gender differences with reference to her behaviour and role are also moving between the active and the passive, surface and depth, cause and effect. Past and future turn into inclusive disjunctions (Deleuze 1990, 9). Becoming is the process of actualising the virtual or better, the process by means of which the virtual is being actualised by being differenciated. Even 'thinking is becoming' (Colebrook 2003, 125–26). 'Depth is no longer a complement' states Deleuze (Deleuze 1990, 9) and then concludes, 'Indeed, this is the first secret. . . . If there is nothing to see behind the curtain, it is because everything is visible'. Derrida seems to agree on this point, when he writes, 'If you want to hide something, you can place it in the most obvious place' – on the surface, we add.

We now ask: does the surface that the void provides constitute the solution of this mystery? As we already mentioned, the surface of the void in

the Greek city is severely restricted by the built parts. In the context of the labyrinth of the urban tissue, the surface of the void consists in a system of (void) regions. Our system comprises a linear continuum of voids (the streets network) that crosses both the densely built urban tissue and the almost unbuilt suburban space. It comprises also the partial local voids (squares and residual voids). Thus, the system of voids we are being faced with limits seriously the choice of the course we may take in whatever direction. Nevertheless, like the rabbit's hole in Alice's course, the surface of the voids of the labyrinthine Greek city is offered for our adventure in becoming. There is no other possibility either for Alice or for us. In our adventure, our motive duplicates that of Alice's: the unveiling of the secret. We ignore what can be found. We just know that we are going to find something. In any case, if we have learnt anything from Alice's experience, it is that we can move in many ways: perpendicularly as far as it concerns our diving and horizontally from left to right or from right to left.

We start our diving perpendicularly. Starting at the surface of the void, the unbuilt, we follow the channel on the surface that leads to the depth of the lower hidden layers. These are also built. The fact is that layers or planes are the discrete strata which, as they settle into place offer continually something new to see or speak, or better, something not new but very old but hidden. The Greek city is stratified, often reminding of a palimpsest. Lower layers correspond to a historic period of the city and bear witness to the built cultural assets of the past. The only way to reach the lower layers of the history of the city is to start at the surface of the unbuilt or void of the actual highest layer. This is a perpendicular movement cutting through all the layers of the history of the city. One can then encounter, detect, read, show and bring to the surface cultural assets for preservation and management.

Next, our movement becomes horizontal from left to right and from right to left. It follows a process of becoming which obeys the system of the unbuilt labyrinth of the urban tissue and runs on the surface of the Greek city as a horizontal channel, connecting the centre with the periphery. This horizontal movement finds a new function for the unbuilt or void, namely, the integration of the urban with the suburban void, the landscape, the sprawl. It proposes a new perspective for design that cannot strictly distinguish the architectural from the urban and the landscape.

What motivated the adventures – the presumed secret – turns out now to be common knowledge. The perpendicular dive to the palimpsest and the horizontal exploration of the surface bequeaths us history. This is not something new of course. It places the discussion introduced by Colin Rowe and Fred Koetter in 1988 with the terms of contextualism in a contemporary framework. Rowe and Koetter alluded then to 'an architecture that starts from the tradition of the past instead of a utopia of the future. . . . History and tradition

is for architecture what supposition is for science' (Rowe and Koetter 1988). Tradition, in this sense, and history give us the states of affairs of architecture. The virtual events have not yet been named. The past strata brought to the surface turn actual. From one actuality to another. But we know now that history belongs to the actual and to the virtual, becoming – or better: we know that the actual has the becoming that the virtual, in the process of its own differenciation, grants it.

NOTE

1. This chapter is based on Vana Tentokali's essays, "What does the Public Space hide?" and "Digging the Surface."

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Chapter 5

Bodies without Organs and Cities without Architecture

Chris L. Smith

Abstract

In a chapter titled 'November 28, 1947: How Do You Make Yourself a Body without Organs', Gilles Deleuze and his accomplice Félix Guattari cite Antonin Artaud's 'Letter to Hitler'. On its surface, the letter is a note to the führer reminding him of a discussion he'd had with Artaud 'in 1932 in the Ider Café in Berlin'. The café discussion, it seems, had concerned itself with access to Paris; one assumes military access to the city. However, as is the case with much of Artaud's intense and manic writing, the surface is particularly deep. Artaud refers to 'a map which was not just a map of geography' and to 'roadblocks' that are not just blockages of streets. In the letter, Artaud endows Paris with the qualities of troubled bodies. Barricaded streets start to sound more like constipated digestive tracts. Artaud's 'Letter to Hitler' is a confusion and fusion of geographies and bodies. What we get in this short letter is a sense of the interchangeability of organs, architectures, bodies and cities that are under threat. This chapter will turn to another such confusion and fusion of geographies and bodies: to the intermingling of the geo-historic and bodily that might constitute a city. I will turn to Berlin to explore the outcome and affect of the *Stolpersteine* project. The *Stolpersteine* project was conceived by the German artist, Gunter Demnig, and involves an incursion into the cobblestones of a city in order to commemorate the lives of those lost to the Nazi regime. The project is a challenge to traditional forms of memorial and to the habitual means by which we constitute a city itself. The *Stolpersteine* project suggests a manner by which a city might be lived without recourse to its habitual modes of organisation (architecture) and as such suggests a procedure by which we might make the city a body without organs.

In a chapter titled 'November 28, 1947: How Do You Make Yourself a Body without Organs?' in *A Thousand Plateaus* (1980), Gilles Deleuze and his accomplice, the psychoanalyst Félix Guattari, cite Antonin Artaud's 'Letter to Hitler':

> Dear Sir, on 1932 in the Ider Café in Berlin on one of the evenings when I made your acquaintance and shortly before you took power, I showed you roadblocks on *a map that was not just a map of geography*, roadblocks against me, an act of force aimed in a certain number of directions you indicated to me. Today Hitler I lift the roadblocks I set down! The Parisians need gas. Yours, A.A. – P.S. Be it understood, dear sir, that this is hardly an invitation, it is above all a warning.[1]

The letter was written by Artaud in 1939 while he was a patient at the Ville-Évrard psychiatric hospital in Saint-Denis, Paris.[2] On its surface, the letter is a note to the then *führer* reminding him of a discussion they had shared 'in 1932 in the Ider Café in Berlin'. The year 1932 had been a productive one for Artaud. He had published 'The Mise en Scène and Metaphysics' and sketched out a manifesto for the 'Theatre of Cruelty'. He had also spent time in Berlin that year performing in a thriller directed by Serge Poligny, *Coup De Feu à l'Aube* and had played the role of Le Trembleur, the leader of a gang of jewel thieves. It is indeed possible that Artaud met Hitler in a café in Berlin that year. The essayist Anaïs Nin wrote early in 1933 that Artaud was a 'lean, ghostly figure who haunts the cafés'.[3] Hitler was also known to frequent cafés. While it is possible that Artaud met Hitler in the Ider café, it is unlikely. There is no evidence for the existence of an 'Ider Café' in Berlin, and Nin would go on to suggest that at cafés Artaud, 'is never seen at the counter, drinking or sitting among people. He is the drugged, contracted being who walks always alone'.[4] Artaud's preoccupation with the alleged meeting would, however, persist. Four years later, in 1943, he dedicated a copy of *The New Revelations of Being* (1937) to Hitler: 'in memory of the *Romanisches café* in Berlin one afternoon in May 1932'.[5] While the Romanisches Café did in fact exist, there is no evidence that Hitler had ever visited it. (Though known to frequent cafés, Hitler was not known for indulging in anything particularly Romanisches.) By the time of this dedication, Artaud had been transferred from Ville-Évrard to the psychiatric hospital at Rodez. His psychiatrist at Rodez, Gaston Ferdière, would use the reference to Hitler and the Romanisches Café meeting as 'a specific example of Artaud's mental derangement. . . . One recognizes here the faulty memory (so frequent with Artaud along with mistaken identity), mystical ideas, glossolalia, etc'.[6] It is possible that the 'Ider Café' Artaud had earlier referred to may be a slippage. The Ider may have been the Romanisches. Likewise, this slippage may be less a matter of mistaken identity than a slippage between a person and a place.

There are accounts of Hitler having been referred to as 'Mohammed Ider' in Cairo during the war.[7]

The geo-historical truth (or otherwise) of the event does not disrupt the generative effect of its recount. There is a feverish character to Artaud's 'Letter to Hitler'. It is short and intense, and the points are exclamatory. There is also a sense of ambiguity to the letter. Some of the ambiguity is due to the shifting subject: the text floats between references to Hitler and to Artaud himself. Some of the ambiguity is due to the shifts that occur between objects and subjects: the text slips between roadblocks and blocks that are more visceral in character. On its surface, Artaud's letter had concerned itself with access; one assumes military access. The signifiers point to geographic access to a city, Paris, or more specifically to a people, Parisiens: 'The Parisians need gas'. The letter contains many references to this geo-historic, social and military concern: references to 'power', 'roadblocks', 'maps', 'an act of force', 'warning' and 'gas'. The gas [*gaz*] may be town gas [*gaz de ville*], natural gas or petroleum gas [*gaz naturel* or *gaz du pétrole*], tear gas [*gaz lacrymogéne*], poison gas [*gaz asphyxiant*] or gas chambers [*chamber à gaz*]. However, as is the case with much of Artaud's intense and manic writing, in this letter the surface of the world coincides with Artaud's own skin. That is to say, the letter wavers between the geo-historic and the bodily. The letter includes reference to: '*not just a map of geography*', 'roadblocks against *me*', '*I* lift the roadblocks *I* set down' and 'gas'. This gas [*gaz*] may be a stepping on the gas [*metre les gaz*], methane gas [*gaz de méthane*] or any manner of indigestion and flatulence [*gaz intestinal* or *gaz d'échappement*]. One imagines all the geo-historic and bodily forms are implicated, and more beside gas leak [*fuite de gaz*]. Artaud's 'Letter to Hitler' is a confusion and fusion of geographies and bodies. Streets and intestines.

This chapter will turn to another such confusion and fusion of geographies and bodies: to the intermingling of the geo-historic and bodily that might constitute a city. I will turn to Berlin to explore the outcome and affect of the *Stolpersteine* project. The *Stolpersteine* project was conceived by the German artist, Gunter Demnig and involves an incursion into the cobblestones of a city in order to commemorate the lives of those lost to the Nazi regime. The project involves the insertion of small cobblestone blocks into the existing pavement. The blocks are covered with a simple embossed brass plaque. The plaque commemorates an individual victim of the Holocaust. The project is a challenge to traditional forms of memorial and to the habitual means by which we constitute a city itself. The *Stolpersteine* project suggests a manner by which a city might be lived without recourse to its habitual modes of organisation (architecture), and as such suggests a procedure by which we might make the city a body without organs (BwO).

ARTAUD'S BODY WITHOUT ORGANS

Deleuze and Guattari refer to Artaud's 'Letter to Hitler' in a chapter titled 'November 28, 1947: How Do You Make Yourself a Body without Organs?' The chapter is concerned with the manners by which organs and bodies might be liberated from the habits of organisation that occur under the title 'organism'. The wavering that occurs in Artaud's letter is itself a characteristic of the BwO. For Deleuze and Guattari, the BwO 'swings between two poles, the surfaces of stratification into which it is recoiled, on which it submits to the judgment and the plane of consistency in which it unfurls and opens to experimentation'.[8] The text *A Thousand Plateaus* also swings between these two poles of geo-historic stratification and conceptual experimentation. The chapters that constitute the text are all stratified by dates. The dates are noted in the titles of each chapter. They correspond to a fixed historical moment in which the experimental and conceptual content of the chapter may have exhibited its most profound expression. Brian Massumi would note in the 'Translator's Foreword' of the English edition of *A Thousand Plateaus* that '[t]he date corresponds to the point at which that particular dynamism found its purest incarnation in matter, the point at which it was freest from interference from other modes and rose to its highest degree of intensity'.[9] The date centred upon for the BwO chapter is 28 November 1947. It was the day on which Artaud completed his final work, the recording of the intense experimental radio play, *To Have Done with the Judgment of God:*

> and it was then
> that I exploded everything
> because my body
> can never be touched.[10]

Artaud had emerged from years in psychiatric institutions to write and configure the radiophonic work. The piece was commissioned by Radio Française and was subsequently banned due to its anti-American and anti-religious content. In the recording, Artaud's voice is heard in various modes: measured and strained; distressed and at breaking point; in script and incomprehensible. In addition there are the voices of the actor and director Roger Blin, the actress María Casares, Artaud's companion Paule Thévenin and percussion sounds generated by Artaud and Blin that seem to mark moments of transition in the *logue* (neither monologue nor dialogue).[11] Artaud reinvigorates elements of his 'Theatre of Cruelty', and the recording concerns itself with the torment of the body: 'must it be reduced to this stinking gas?'[12] Artaud also engages in a glossolalic language which deterritorialises words into sound in order to affect: 'caca'. In *The Theatre and Its Double* (1938), Artaud had

referred to such expressions as 'a kind of language somewhere between gesture and thought'.[13] *To Have Done with the Judgment of God* is difficult to listen to, but ease and comfort are hardly the point. The phrase under which Deleuze and Guattari summate the conceptual concern of their chapter 'How Do You Make Yourself a Body without Organs?' is voiced by Artaud in the final strained stanzas of the performance:

> For you can tie me up if you wish,
> but there is nothing more useless than an organ
> When you will have made him a body without organs,
> Then you will have delivered him from all his automatic reactions
> And restored him to his true freedom
> Then you will teach him again to dance wrong side out
> as in the frenzy of dance halls
> and this wrong side out will be his real place.[14]

According to Deleuze and Guattari, it is this radio play that marks the moment at which 'Artaud declares war on the organs'.[15] The declaration had been simmering for some time. The concerns that may have found themselves at their purest point in *To Have Done with the Judgment of God* recur across Artaud's oeuvre. Across his work there is an ongoing battle between sites and selves, geographies and bodies. One can note a particular moment of slippage between 1923 and 1925. It is the period of his correspondence with the writer Jacques Rivière. The slippage can be most clearly noted, however, not in that correspondence but in the move from the short review 'From Bilboquet' (1923), through the essay 'The Umbilicus of Limbo' (1925), to his 'From the Nerve Meter' (1925). In 'From Bilboquet', Artaud would note (in measured phrases) of Paul Klee, 'I rather like some of his nightmares, his mental syntheses conceived as architectural structures (or his architectural structures with a mental quality)'.[16] The architectural and the psychological are placed in relation. Carefully placed. One is interchangeable with the other – by the simple act of inversion. In 'The Umbilicus of Limbo', the boundaries which might traditionally have defined bodies and differentiated them from architecture (or any context) would be further challenged. Artaud would write, '[t]he belly evokes surgery and the Morgue, the construction yard, the public square, and the operating table. The body of the belly seems made of granite or marble or plaster, but a plaster that has set'.[17] Here, the body and its parts 'evoke' and 'seem' to acquire architectural, urban and sculptural features. The analogical relations are established by the transitive or linking verb. These lexical qualifications would soon slip away, and the analogical confusion of geographies and bodies would become a fusion. In Artaud's 'From the Nerve Meter' the lexicon that mediated between geographies and bodies has all but vanished. Artaud would write of 'a whole pain, a beautiful

pain, a dense and fleshy anguish, an anguish which is a mixture of objects, an effervescent grinding of forces rather than a suspended point'.[18] This slippage would continue across Artaud's work. Though Artaud would write in 1925 that '[b]etween the world and ourselves the rupture is complete',[19] this 'rupture' would, for Deleuze and Guattari, reach its 'purest incarnation' in *To Have Done with the Judgment of God*.

DELEUZE AND GUATTARI'S BWO

In *The Logic of Sense* (1969), Deleuze adopts Artaud's term 'body-without-organs' to describe a body 'without limbs, with neither vice nor sex'.[20] It is a body that operates outside the habits of thought to which a body is traditionally subjected. The body without organs referred to in *The Logic of Sense* is particularly *done* with the habits of psychoanalysis. A decade later, in *A Thousand Plateaus*, the BwO would come to 'recoil' from all manner of habit and concomitant judgements but would continue to posit itself most directly against the organism: '[t]he BwO is opposed not to the organ but to that organization of the organs called the organism'.[21] This opposition is not particularly gentle or conciliatory. Deleuze and Guattari note: '[a] perpetual and violent combat between the plane of consistency, which frees the BwO, cutting across and dismantling all of the stratum and the surfaces of stratification that block it or make it recoil'.[22] The forces against which the BwO battles are habitual systems of hierarchy and organisation (blocks) which otherwise narrow or direct the flows of desire (and particularly revolutionary forms of desire). The three blocks, or what Deleuze and Guattari refer to as the three 'great strata' which operate as impasse or point of recoil for the BwO are 'the organism, signifiance, and subjectification'.[23] The BwO is a liberation of a body from the three great strata that otherwise appropriate it, along with the systems of judgement (psychoanalytic, theological, medical, legal) associated with these strata that otherwise relegate it. The organ, as such, is the enemy only insofar as it prefigures a system of 'organism'. The organism is the enemy only insofar as it prefigures a system of judgement (including the judgement of god). Artaud was to use the word 'parasitic' in respect to the manner by which an organ functions in support of the organism.[24] Deleuze and Guattari note that '[i]t is true that Artaud wages a struggle against the organs, but at the same time what he is going after, what he has it in for, is the organism'.[25] On this point Deleuze and Guattari quote Artaud's 'Theatre of Cruelty':[26] 'The body is the body. Alone it stands. And in no need of organs. Organism it never is. Organisms are the enemies of the body'.[27]

While Artaud is evoked by Deleuze and Guattari to denote a 'high degree of intensity' in the generation of a BwO, they also draw on Artaud to caution

of the dangers involved in the making of a BwO. Despite joyous intensities, the chapter 'How Do You Make Yourself a Body without Organs?' can be read, like Artaud's 'Letter to Hitler', as 'above all a warning'. Across the chapter, the key examples of BwO processes include *the hypochondriac body*, where there is a sense that the damage is already done and the organs aren't functioning anymore; *the paranoid body*, where 'the organs are continually under attack by outside force, but are also restored by outside energies';[28] *the schizo body* 'waging its own active struggle against the organs, at the price of catatonia';[29] *the drugged body*, 'the experimental schizo';[30] and *the masochist body*: 'poorly understood in terms of pain; it is fundamentally a question of the BwO'.[31] *Lovers* also get a minor mention. Deleuze and Guattari tell us that '[d]rug users, masochists, schizophrenics, lovers – all BwOs pay homage to Spinoza'.[32]

The masochist was a body that Deleuze knew well from his earlier *critical* work on Baron von Sacher Masoch.[33] In his long essay 'Coldness and Cruelty' (1967), Deleuze would describe the operations of a masochist and the dangers involved. The masochist is suspended (in space and in time) and the genitals are divested of traditional intensities and functions (sewn up, whipped). This geo-historical block is the first part of a two-part process that allows the body to be activated afresh: First 'tie me down' and *then* 'sew the skin'.[34] The masochist transmutes the intensities habitually associated with the genitals across the entirety of the body. Desire passes over the skin, and the entire body becomes super-sexualised, pulsing like an engorged penis. There is a line, it seems, at which the suspension of the organism may threaten the body itself. The body can be strung up and its traditional or habitual modes of operation suspended – but only for so long and only to certain extents. Masochists tend to mark this line in a contract and with the utterance of a 'safe word'. The act of signifiance restores the body to organism. (Undo the ropes, put your pinstriped suit back on and leave quietly via the back door.)

The bodies that Deleuze and Guattari engage as exemplars of BwO processes tend to be those involved in particular attacks on the organs. The risk of discussing the BwO in terms of such bodies is to associate the BwO with suicidal tendencies and the negative ramifications of wholesale attacks on organs. (There are no safe words for the hypochondriac, paranoid, schizo, drugged body or, indeed, for the heartbroken lover.) Deleuze and Guattari were to note the 'paradox of those empty and dreary bodies . . . [that] *had emptied themselves of their organs* instead of looking for the point at which they could patiently and momentarily dismantle the organization of the organs we call the organism'.[35] Artaud, however, had very little patience. The BwOs may pay homage to Spinoza, but it is Artaud who made the ultimate sacrifice in charting what he calls the 'clotted nothingness' of the body.[36]

For Deleuze and Guattari '[t]he BwO is desire; it is that which one desires and by which one desires'.[37] In line with their broader philosophical position, desire is not associated with loss, nor is there anything particularly measured or inherently pleasant about desire. Desire is assembled and can operate at the level of any assemblage – any machine, any organ, any individual body or any social body. Deleuze and Guattari would suggest that 'even fascism is desire',[38] and Deleuze would note in an interview that 'people got hard-ons for Hitler, for the beautiful fascist machine'.[39] If the desires that ran a locus about Hitler demonstrate anything, it is that BwOs can be suicidal or genocidal. An army can form on the BwO just as easily as an intense gushing orgasm. A BwO can equally be full, fascist or empty. Deleuze and Guattari would come to speak of the 'three-body problem' of the BwO and trace it back to Artaud. The three body problem is an acknowledgement of the risks associated with making yourself a BwO. Deleuze and Guattari provocatively ask: '[h]ow can we fabricate a BwO for ourselves without its being the cancerous BwO of a fascist inside us, or the empty BwO of a drug addict, paranoiac, or hypochondriac? How can we tell the three bodies apart?" '[40] The philosopher and the psychoanalyst don't answer the question. They can't. Any answer would likely take us down paths that would merely reinstate another judgement system in the wake of the judgement of God. In lieu of any answer, Deleuze and Guattari invoke a cancerous, fascist, drug addict, paranoiac, hypochondriac BwO and write: 'Artaud was constantly grappling with all of that, and flowed with it'.[41] It is at this moment in *A Thousand Plateaus* that Deleuze and Guattari turn to Artaud's 'Letter to Hitler'.

In *A Thousand Plateaus*, Deleuze and Guattari cite *Cause Commune* (No. 3, October 1972) as their source for Artaud's 'Letter to Hitler' (figure 5.1).[42] There are two points to note about the reference. The first is that Deleuze and Guattari shift the italics in their quotation. In the *Cause Commune* printing of the letter, the phrase *Today Hitler I lift the roadblocks I set down!* is italicised. In *A Thousand Plateaus*, Deleuze and Guattari italicise the phrase *a map that was not just a map of geography*. This shift is telling. It speaks of the change in emphasis between Artaud and Deleuze and Guattari. Consistent across Artaud's writing is a personalisation of the body. It was *his* body with which he was concerned. The roadblocks were against *him*. It was *he* that would lift them. Susan Sontag suggests that for Artaud *the* body is '[b]oth the obstacle to and the locus of freedom'.[43] It would be even more accurate to suggest that it was *his* body that was always at stake. The self, hisself, is a constant point of fixation across the work of Artaud. He would write in 'Fragments of a Diary from Hell' (1925), '[h]e talks to me about Narcissism, I retort that it is a question of my life'.[44] The impulse of Deleuze and Guattari to make subjectification one of the three great strata against which the BwO might recoil meant that it was necessary to de-subjectify the organ and body.

It cannot be my organ or yours. It would have to be assigned the indefinite form. For Deleuze and Guattari, 'the BwO is never ours or mine, it is always *a* body. . . . "A" stomach, "an" eye, "a" mouth: the indefinite article does not lack anything, it is not indeterminate or undifferentiated but expresses the pure determination of intensity, intensive difference'.[45] The removal of the emphasis from the exclamatory sentence in Artaud's letter, which refers twice to 'I', may be understood as a shift from the language of subjectification. The intent of the (re)placement of the emphasis on the phrase *a map that was not just a map of geography* is made clear in the sentences that follow Deleuze and Guattari's reference to Artaud's letter. Deleuze and Guattari write, '[t]hat map that is not a map of geography is something like a BwO intensity map, where the roadblocks designate thresholds and the gas, waves or flows'.[46] But for Artaud, the waves and flows of gas were often personalised, reconstructing the subject at the very point of its destruction:

> And truly
> must it be reduced to this stinking gas,
> my body?
> To say that I have a body
> because I have a stinking gas that forms
> inside me?[47]

The second point to note in the move of Artaud's 'Letter to Hitler', as it occurs in *Cause Commune*, into *A Thousand Plateaus* is the point at which Deleuze and Guattari cut short Artaud's postscript. They quote only the fragment 'P.S. Be it understood, dear sir, that this is hardly an invitation, it is above all a warning'. The postscript, however, goes on:

> If you prefer, as any enlightened man who does not consider, or who does not appear to consider, take this at your leisure. I won't judge.
> Watch yourself!
> The rottenness of the enlightened French has reached its spasmodic pinnacle.
> In any case you knew this.[48]

The 'three body problem' concerns itself with the possibility of a BwO to be either full or fascist or empty. Artaud indulges in a type of critical and clinical operation: a literary, textual, play and sociomedical diagnostics that takes cities and streets as well as bodies as its target. Oddly, Artaud was less enamoured of Paris than Hitler. The architect Albert Speer quotes Hitler in June 1940 during his three-hour first (and thankfully last) trip to Paris: 'It was the dream of my life to be permitted to see Paris. I cannot say how happy I am to have that dream fulfilled today'.[49] It turns out Hitler had a hard-on for Paris, more so than he had for Parisiens. It was a particularly cancerous

BwO that Hitler unleashed across the city. An estimated 160,000 people were deported from Paris to concentration camps. There was no safe word uttered that would stop the decimation. The attack on the organs was genocidal. This horrific fascist BwO was, as with all BwOs, a move against the three 'great strata' of 'the organism, signifiance, and subjectification'.[50] The Jewish population of Paris wasn't treated as if it were composed of organisms any more. But they weren't necessarily reified as bodies either. The trains, camps, uniforms and gas chambers made anonymous of the most personalised attributes. An organ of worth was liberated from bodies. Teeth were a gold mine. Hair could be felted to line the boots of soldiers. Signification was also a site of attack. Names were erased and replaced with tattooed numerals. The population was de-subjectified, made animal in political rhetoric, and extermination was conducted *en masse*. That is to say, individuals were defined only insofar as they constituted the collective. The tattooed letter A or B before a number indicated Jews. Z for the Romanisches [*Zigeuner*]. The fascist BwO swept organisms away, but bodies too were lost. Artaud's indulgence in the BwO was indeed 'above all a warning'.

Cities without Architecture

Monuments and memorials to the Holocaust are prominent in many cities across Europe. Yet, one wonders how it is that they can be produced at all, how it is that architecture can negotiate the immensity of that horror. In German there are two words which tend to translate as 'memorial': *denkmal* and *mahnmal*. *Denk* refers to thought or thinking and *mahn* refers to a caution or warning. Perhaps the most forceful of the architecture of the memorial has valiantly attempted to embody the sum total of loss and occur as *mahnmal*. I am referring to structures such as Rachel Whiteread's cold concrete Judenplatz Holocaust Memorial (2000) in Vienna and Peter Eisenman's Memorial to the Murdered Jews of Europe (2004) in Berlin. They are concrete monoliths that bare little trace of the body. These architectures groan deeply and endeavour to deal with the immensity of the whole and come to encompass the Holocaust en masse. While powerful, these architectures also tend to reinforce the organisational strategies of the state and the city. (The state which identified bodies for deportation and the same city that subjected them to the cruellest of judgements.) The urban fabric itself constitutes a system of judgement reinforcing in concrete the three strata of organism, signification and subjectification. Architecture and urban planning have a tendency to enclose and demarcate functions: Here we shop. Here we work. Here we sleep. Here we mourn. Such demarcations both define behaviours and dictate access and exclusion. On broader scales our planning mechanisms divide our cities into

zones: retail, residential, educational, urban, suburban, rural. The architectural monument reinforces such organisational strategies. These memorials aim to suspend a population for a short intense moment or two – before one returns to the street and its work, consumerism, tourism and trivialities. While the architectural memorial operates as a territory where remembrance is fostered, it sanctions a comfortable forgetting across the remainder of a city. This organisation of the city conducted by planners and architects is surely what a full (as opposed to a fascist or empty) BwO would be *done* with.

In Berlin, not far from the site where the Romanisches Café once stood (where Artaud may have met Hitler); and not far from Eisenman's field of concrete monoliths, there are memorials of a kind that challenge the orders of architecture and the city. These memorials are called 'Stolpersteine' (stumbling stones or stumbling blocks). The name recalls an old anti-Semitic saying that Germans use when they would trip over or kick a raised cobblestone: 'a Jew could be buried here'.[51] These 100 × 100 × 100 millimetre cobblestones are set among the other cobblestones of the pavement but are covered with simple brass plaques (figure 5.2). These blocks are set in the pavement outside the places where individuals lived or worked and note the site of their deportation to concentration camps. A Stolpersteine contains a small amount of geo-historic and biographical information related to an individual: a name, a given name and a surname, sometimes a maiden name, a date of birth. They also note the date of murder or suicide and the names of the death camps implicated: 'Here lived Jüdel Laufer, Born 1890, Deported 1943, Auschwitz, Died in Oranienburg'. The four Stolpersteine dedicated to Laufer, his wife and his two daughters are set alongside each other among the pavers of the footpath. Blocks like this are spread over the city. The Stolpersteine project was initiated by the German artist Gunter Demnig. The first Stolpersteine were installed illegally in Berlin in 1996.[52] They were part of Demnig's contribution to the Artists Search for Auschwitz exhibition [*Künstler forschen nach Auschwitz*] in the Kreuzberg district of the city. The title of the exhibition would recall Theodor Adorno's sober provocation as to 'the impossibility of poetry after Auschwitz'.[53] Groups and institutions conduct research on each person to be commemorated on a Stolpersteine and then fund the laying of the stone. Michael Friedrichs-Friedländer makes and embosses the Stolpersteine, and Demnig travels across Europe to lay them. There are now over 56,000 Stolpersteine laid across Europe. It is as yet a small number, given the scale of the genocide. Adorno had cautioned that our sense of the Holocaust might degenerate into 'idle chatter', but there is something particularly poignant about the incessant chatter of these particular cobblestones.[54] These stones come to operate as both *denkmal* and *mahnmal*: an invitation to thought and a warning.

Along the streets of Berlin these stones form an odd patterning. They occur here and there. Sometimes in clusters and sometimes singly. Prominent in some areas and absent in others. There's no discernible rhythm to their placement. These memorials are not zoned or situated in a manner that allows them to function as a memorial might normally function.[55] However, every one of these cobblestones is a *block* of a kind. They occur in disrespect to the forms of organisation (plans and blueprints) to which cities are usually subject to. They pass under feet. They have multiplied rhizomatically. They come to occupy a small place in a pavement and provide a discontinuous link to the other Stolpersteine of the city but maintain a presence of their own – they are tied to one body. Sontag would note that for Artaud '[t]he vision of a total art has the same form as the vision of the redemption of the body. . . . Art will be redemptive when, like the redeemed body, it transcends itself – when it has no organs (genres), no different parts'.[56] She would qualify the statement and suggest that '[i]n the redeemed art that Artaud imagines, there are no separate works of art – only a total art environment, which is magical, paroxysmic, purgative, and, finally, opaque'.[57] A total art environment might recoil from the fixations of architecture and urban planning; fixations with property, boundaries and borders, controls and demarcations. Its defining feature would likely be circulation and the passing of intensities. Such an environment would not fixate on forming citizens but rather happily invoke the street as a point of dispersal; where even one's fixations with the self might depart. There is something particularly paroxysmal, purgative and opaque about the Stolpersteine. They signal something that is at once an invitation and a warning. This is why the Stolpersteine project suggests a procedure by which we might make of the city a full BwO.

The construction of any BwO involves two distinct phases. Deleuze and Guattari write, '[o]ne phase is for the fabrication of the BwO, the other to make something circulate on it or pass across it'.[58] The Stolpersteine are sharp geo-historic pinpoints. They are fabricated to mark a 'here' point in space and a 'then' moment in time and are, like the construction of a BwO 'measured with the craft of a surveyor'.[59] They operate much like the dates of the chapter titles of *A Thousand Plateaus* in isolating a geo-historic moment. But this moment is not the marking of a 'purest incarnation' of a concept, rather it is the marking of a moment of being set adrift – the moment of forced homelessness and removal from a city, towards a removal from life itself. The Stolpersteine do not invoke a sense of the immensity of the Holocaust as any totalising summation like a field of concrete monoliths might but rather the particularity and poignancy of a singular body taken from a singular home. The dates of *A Thousand Plateaus* mark a 'highest degree of intensity', whereas the Stolpersteine mark a desperately intense moment repeated (particularised) thousands of

times across thousands of thresholds. The Stolpersteine form an imprint, a seizure, that occurs between geo-historical fixities and the fluidities of the body. These blocks mark points where territory and territorialisation fuse in trauma.

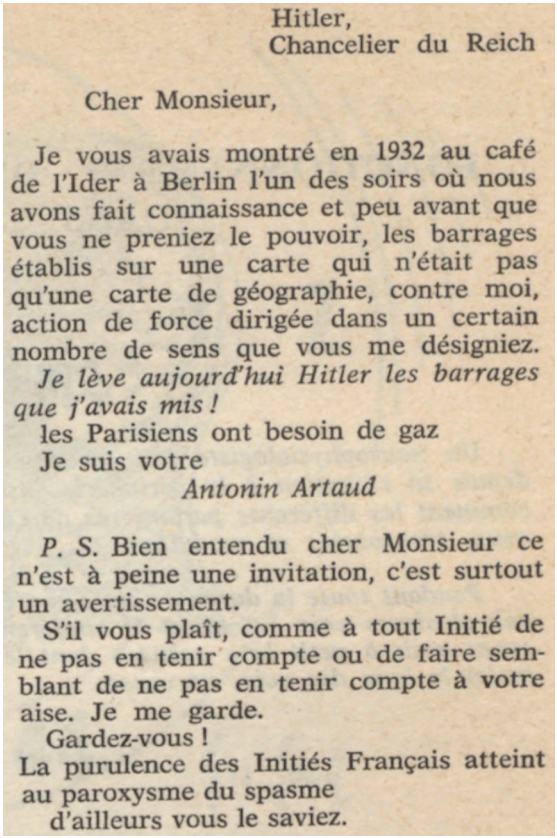
The outcome of the careful geo-historic construction of the Stolpersteine is a flow of intensities that proceed in a similar manner to the way that Deleuze and Guattari were to speak of the operations of the BwO of masochism. The Stolpersteine incite a 'conversion of forces and inversion of signs':[60] Here lived Jüdel Laufer, where I now stand. Here lived Czarna Laufer where I now live. And 'little by little all opposition is replaced by a fusion of my person with yours'.[61] This affect is not en masse. It doesn't involve generalisations or attempts to summate or totalise. The Stolpersteine tend to be engaged one at a time. The Stolpersteine neither control nor contain us as habitual architectures and traditional urban formations might. We can walk on. We do walk on. We step over the Stolpersteine quite simply – just as we might step over other paving stones. Our feet pass over these streets more simply than our thoughts, but neither are blocked. That is not to say that they aren't affecting. They are. Their intensities are however more liberated than those that we experience in traditional architectures, that demarcate particular experiences in a city and compartmentalise sensations: theatres, galleries, public squares, shrines of remembrance and memorials. Of the traditional means by which architecture marks moments, pins down dates and remembers events in stone, we can say that little to nothing comes to pass. The architects of such memorials may have fostered what Deleuze and Guattari refer to as the 'right Place, Power (*Puissance*), and Collectivity (there is always a collectivity even when you are alone)', but if nothing passes – that is, there is no passage of intensity – we suffer a 'blockage'. These architectures operate as Artaud's roadblocks did – severing the flow of intensities.

The affect of the Stolpersteine is a complex thing to describe. They might be particularly blunt objects, but the affect is the sharpest of stabs from the finest of blades. A paper cut of a kind. A thousand paper cuts across a city. These stabs prompt a type of refocusing and adjustment. A momentary realigning of the self, as one has to find oneself freshly upright and to walk with a new pace (but not to a set rhythm); a new determinacy to raise one's feet higher (but not too high). The Stolpersteine are a warning about the fascisms we internalise and extend across the planet; the fascist desires we foster when we forget the particularities of outcomes and affects when pursuing our own 'spasmodic pinnacles'. The Stolpersteine are, however, also an invitation: an invitation to rethink the city, to reorganise. There is a restrained cautiousness to the Stolpersteine project – it is less of the measured craft of a surveyor and more of the precise art of an urban acupuncturist. Pinpointing through the body precisely in order to liberate flows and allowing intensities

to pass. The Stolpersteine operate as a full BwO in 'looking for the point at which they could patiently and momentarily dismantle the organization'[62] we call the city.

CONCLUSION

Artaud's 'Letter to Hitler' was a confusion and fusion of the body and the street. For much of his adult life, Artaud had been affecting a dislocation of habitual modes of expression and formations, including the habits of the body under the organisations implied by the term 'organism'. Artaud always laid



Hitler,
Chancelier du Reich

Cher Monsieur,

Je vous avais montré en 1932 au café de l'Ider à Berlin l'un des soirs où nous avons fait connaissance et peu avant que vous ne preniez le pouvoir, les barrages établis sur une carte qui n'était pas qu'une carte de géographie, contre moi, action de force dirigée dans un certain nombre de sens que vous me désigniez.
Je lève aujourd'hui Hitler les barrages que j'avais mis !
les Parisiens ont besoin de gaz
Je suis votre
Antonin Artaud

P. S. Bien entendu cher Monsieur ce n'est à peine une invitation, c'est surtout un avertissement.
S'il vous plaît, comme à tout Initié de ne pas en tenir compte ou de faire semblant de ne pas en tenir compte à votre aise. Je me garde.
Gardez-vous !
La purulence des Initiés Français atteint au paroxysme du spasme
d'ailleurs vous le saviez.

Figure 5.1. Antonin Artaud, 'Letter to Hitler', *Cause Commune* No. 3 (October 1972), 41. *Source:* All reasonable effort has been made to obtain permissions where needed. Where it has not been possible to trace the rightsholder, we would be happy to hear from anyone with further information regarding this.

himself on the line, like one might lay a roadblock. Deleuze and Guattari's engagement with the BwO owes a particular debt to Artaud. The depersonalisation of the notion in the work of the philosopher and psychoanalyst extends the concept in a manner that challenges all the systems of judgement to which bodies are habitually subject to. Artaud was *done* with the judgement of God, and the BwO would come to be *done* with judgement altogether. Any BwO can equally be full, fascist or empty, though Deleuze and Guattari caution that one must beware the 'demented or suicidal collapse'.[63] This was the lesson of Artaud, and it is why Deleuze and Guattari suggest, 'Even if Artaud did not succeed for himself, it is certain that through him something has succeeded for us all'.[64]



Figure 5.2. Stolpersteine, Berlin. ***Source:*** **Photograph by author, 2016.**

As with the destruction of an organism, there is a necessary caution to the dismantling of the city. Deleuze and Guattari would note of the BwO that one needs to keep enough of the organism, significance and subjectification 'if only to turn them against their own systems'[65] and that '[y]ou don't do it with a sledgehammer, you use a very fine file'.[66] The same is true of our engagements with architectures and cities. The Stolpersteine project proceeds in an incisive, perhaps subversive, manner. Not by traditional architecture nor by roadblocks, but by cobblestones placed carefully, one at a time. The cautions raised in respect to the dangers of making yourself a BwO seem to have been carefully negotiated in the project. The Stolpersteine occupy locations in the existing patterning of cobblestone pavements and come to form 'continuous connections and transversal tie-ins'[67] that allow all manner of intensity to circulate. They do not constitute a wholesale attack on the city but rather a careful redirection of attention away from the traditional organs of the city (architecture) and their habitual functions – onto the material that comes to constitute the very streets upon which intensities circulate. As such, Stolpersteine suggest new ways for making the city, a new city of mobile intensities that belie the habitual demarcations of architecture: A city without architecture that might operate with the logic of a BwO.

NOTES

1. Gilles Deleuze and Félix Guattari, *A Thousand Plateaus* (Minneapolis: University of Minnesota Press, 1987), 163–64. Translation of *Mille plateaux*, volume 2 of *Capitalisme et schizophrénie* (Paris: Les Editions de Minuit, 1980), by Brian Massumi.

2. Evelyne Grossman, *Artaud Works* (Paris: Quarto Gallimard, 2004). According to Grossman's chronology, Artaud was a patient at Ville-Evrard from 22 February 1939 to 22 January 1943. Refer also to Stephen Barber, *Antonin Artaud: Bombs and Blows* (London: Faber & Faber, 1993).

3. Anaïs Nin, *The Diary of Anaïs Nin: 1931–1934* (San Diego: The Swallow Press, 1966), 187.

4. Nin, *The Diary of Anaïs Nin*, 187.

5. Antonin Artaud, 'Letters from 1943–1945 (Rodez)', in Susan Sontag (ed.), *Selected Writings* (New York: Farrar, Straus and Giroux, 1976), 644. The dedication in *Les Nouvelles Révélations de l'Être* is dated 3 December 1943.

6. Gaston Ferdière, 'I Treated Antonin Artaud', *La Tour de Feu* (December 1959). Cited in Jack Hirschman (ed.), *Antonin Artaud: Anthology* (San Francisco: City Lights Books, 1965), 105.

7. James Hamilton-Paterson, *Loving Monsters* (London: Granta Books, 2001), 260 and 263.

8. Deleuze and Guattari, *A Thousand Plateaus*, 159.

9. Brian Massumi, 'Translator's Foreword', in Deleuze and Guattari, *A Thousand Plateaus*, xiv.

10. 'To Have Done with the Judgement of God' from *Antonin Artaud: Selected Writings* by Antonin Artaud, 566. English translation copyright 1976 by Farrar, Straus & Giroux, Inc. Originally published in French as *Oeuvres Completes*, copyright 1956, 1961, 1964, 1966, 1967, 1969, 1970, 1971, 1974 by Editions Gallimard. Reprinted by permission of Georges Borchardt, Inc., for Editions Gallimard.

11. Felipe Otondo, 'Rediscovering Artaud's Sonic Order', in *JMM: The Journal of Music and Meaning* 9 (Winter 2010): 5.

12. Artaud, 'To Have Done with the Judgement of God', 566.

13. Artaud, *The Theatre and Its Double*, in *Collected Works*, vol. 4, (London: Calder and Boyars, 1974), 68. Translation of *Le Théatre et son double* by Victor Corti.

14. Artaud, 'To Have Done with the Judgement of God', 571.

15. Deleuze and Guattari, *A Thousand Plateaus*, 150.

16. Antonin Artaud, 'From Bilboquet', in Susan Sontag (ed.), *Selected Writings* (New York: Farrar, Straus and Giroux, 1976), 27.

17. Antonin Artaud, 'The Umbilicus of Limbo', in Susan Sontag (ed.), *Selected Writings* (New York: Farrar, Straus and Giroux, 1976), 67.

18. Antonin Artaud, 'From the Nerve Meter', in Susan Sontag (ed.), *Selected Writings* (New York: Farrar, Straus and Giroux, 1976), 80.

19. Antonin Artaud, 'The Activity of the Surrealist Research Bureau', in Susan Sontag (ed.), *Selected Writings* (New York: Farrar, Straus and Giroux, 1976), 106.

20. Gilles Deleuze, *The Logic of Sense* (London: Athlone Press, 1990), 129. (Refer also to 188–195.) Edited by Constantin V. Boundas. Translation of *Logique du Sens* (Paris: Les Editions de Minuit, 1969) by Mark Lester and Charles Stivale.

21. Deleuze and Guattari, *A Thousand Plateaus*, 158.

22. Deleuze and Guattari, *A Thousand Plateaus*, 159.

23. Deleuze and Guattari, *A Thousand Plateaus*, 159.

24. Antonin Artaud, 'The Theatre of Cruelty', in *Semiotext(e)*, 'Anti-Oedipus' II (3) (Cambridge MA 1977): 59. Translation by Roger McKeon.

25. Deleuze and Guattari, *A Thousand Plateaus*, 158.

26. Deleuze and Guattari, *A Thousand Plateaus*, notes 17, 532. Reference is to Artaud, *Semiotexte*, 'Anti-Oedipus', 59.

27. Deleuze and Guattari, *A Thousand Plateaus*, 158.

28. Deleuze and Guattari, *A Thousand Plateaus*, 150.

29. Deleuze and Guattari, *A Thousand Plateaus*, 150.

30. Deleuze and Guattari, *A Thousand Plateaus*, 150.

31. Deleuze and Guattari, *A Thousand Plateaus*, 150.

32. Deleuze and Guattari, *A Thousand Plateaus*, 154.

33. Gilles Deleuze, 'Coldness and Cruelty', in *Masochism* (New York: Zone Books, 1989), 7–138. Translated by Jean McNeil. This essay is an expansion of ideas first developed in 'De Sacher-Masoch au masochisme', in *Arguments* 5 (2), (January–April 1961): 40–46.

34. Wanda Sacher-Masoch. 'Contract between Wanda and Sacher-Masoch', in *Masochism* (New York: Zone Books, 1989), 279.

35. Deleuze and Guattari, *A Thousand Plateaus*, 160–61.
36. Artaud, 'The Theatre of Cruelty', 59.
37. Deleuze and Guattari, *A Thousand Plateaus*, 165.
38. Deleuze and Guattari, *A Thousand Plateaus*, 165.
39. Gilles Deleuze, 'Capitalism: A Very Special Delirium', in Félix Guattari, *Chaosophy: Texts and Interviews 1972–1977* (Los Angeles: Semiotext(e), 2009), 43–44. Edited by Sylvère Lotringer. Translated by David L. Sweet, Jared Becker and Taylor Adkins.
40. Deleuze and Guattari, *A Thousand Plateaus*, 163.
41. Deleuze and Guattari, *A Thousand Plateaus*, 163.
42. Deleuze and Guattari, *A Thousand Plateaus*, notes 22, 532. Reference is to *Cause Commune* 3 (October 1972): 41.
43. Susan Sontag, 'Artaud', in Susan Sontag (ed.), *Selected Writings* (New York: Farrar, Straus and Giroux, 1976), xlviii.
44. Antonin Artaud, 'Fragments of a Diary from Hell', in Susan Sontag (ed.), *Selected Writings* (New York: Farrar, Straus and Giroux, 1976), 93.
45. Deleuze and Guattari, *A Thousand Plateaus*, 164.
46. Deleuze and Guattari, *A Thousand Plateaus*, 164.
47. Artaud, 'To Have Done with the Judgement of God', 566.
48. Antonin Artaud, 'Letter to Hitler', in *Cause Commune* 3 (October 1972): 41. All reasonable effort has been made to obtain permissions where needed. Where it has not been possible to trace the rightsholder, we would be happy to hear from anyone with further information regarding this.
49. Albert Speer, *Inside the Third Reich* (New York: Macmillan, 1970), 172.
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58. Deleuze and Guattari, *A Thousand Plateaus*, 152.
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62. Deleuze and Guattari, *A Thousand Plateaus*, 160–61.
63. Deleuze and Guattari, *A Thousand Plateaus*, 161.
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65. Deleuze and Guattari, *A Thousand Plateaus*, 160.
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Part II

ARCHITECTURAL AND URBANIST TOOLKITS

Chapter 6

Gilles Deleuze and Chaos Theory

Stathis-Alexandros Zoulias

In Deleuze's philosophy, chaos is never granted transcendental or cosmogonic foundation – as it is, for example, in Pre-Socratic philosophy – but seems to be grounded more in the contemporary scientific perspective exemplified by chaos theory. Deleuze sets out to construct a new ontology of chaos based on the developments in the study of the behaviour of non-linear dynamical systems. Although generally proceeding by analogy, there are also direct references to certain important elements of chaos theory, such as fractals. In this study, we will attempt to decode and illuminate, up to a point, the dialogue between Gilles Deleuze's philosophy and the science of chaos, always through the prism of a particular notion of space. This study is composed of six parts divided into two sections: the first section deals with general correlations between Deleuzean philosophy and fractal geometry; the second is concerned with more specialist aspects of this theory, such as the butterfly effect, strange attractors and the morphological metamorphosis that occurs when entering a chaotic condition.

To begin with, we will attempt to correlate the spatial substance of chaotic systems,[1] which is meticulously described in fractal geometry and in some fundamental concepts of Deleuzean philosophy, such as plane of immanence, orgiastic representation and smooth space.

Fractal geometry is the geometry of chaotic systems, systems which will concern us in this study primarily in terms of their philosophical extensions. These systems are usually described by chaos theory, physio-mathematical science and philosophy that studies the non-linear dynamical systems theory that originated with Henri Poincaré (1880) but has been flourishing as a field of research since the 1960s. The geometry necessary for their description has shifted from Euclidean to fractal geometry,[2] the basic principles of which were largely formulated by mathematician Benoit Mandelbrot (1982).

The main characteristic of this geometry is that it functions with fractional rather than integral dimensions. Thus, the figures described are not strictly defined in the space of the integer dimensions where they are represented (e.g., a rectangle or a cube of specific dimensions and coordinates) but seem instead to escape through infinities. It is essentially processes of formation[3] rather than specific forms, procedures that are constantly evolving towards increasingly smaller spatial scales at which to take place. The only fixed elements that we can locate in these procedures are some measurements acting as degrees that define the way in which the process of formation will evolve, inscribing guidelines of figuration. Such measurements include non-integer dimensions, the degree of abnormality or the manner in which the principle of self-similarity[4] is constantly expressed, the reduplication and repetition of a basic geometric structure at all scales. This is not the case in Euclidean geometry, for example, when we approach a circle's area of periphery, it appears as a straight line.

The introduction to non-integer dimensions describes the intermediate and ambiguous state of geometrical forms in relation to the terms of the traditional three-dimensional Euclidean space. Specifically, it is possible that a figure be neither completely straight (one dimension) nor planar (two dimensions) but something in-between, something more than a straight line but less than a plane. Likewise, it can be more than a plane and less than a solid. The evolving figure will appear with one dimension in the integer values of the dimensions of the line, the plane or the solid – for example, dimensions such as 1.26 or 2.34. Thus a wide range of fractal values, whose dimensions have various values between the integral values of zero dimensions and three dimensions, emerges.

THE PLANE OF IMMANENCE

What Deleuze and Guattari define as immanence, or as the plane of immanence, is essentially a mediation between the undefined and unpredictable forces of chaos and the emergence, through them, of specific arrangements. It is an abstract mechanism that, by intersecting the forces of chaos in a sort of sieve, gathers together all of the directions and possible formations that accompany them, as a kind of fractal impression. It is an *ens realissimum*[5] as the condition of the possibility of objects from which established arrangements, and subsequent creations of concepts, objects and functions, emerge diagrammatically. As Deleuze and Guattari say, 'The plane of immanence is like a section of chaos and acts like a sieve. . . . By making a section of chaos, the plane of immanence requires a creation of concepts' (Deleuze and Guattari 1994, 42). If these concepts are specific arrangements, as of

a typical machine, the plane of immanence is an abstract machine that can provide a plethora of such arrangements, offering their parts as components for the simultaneous creation of other arrangements. It is a sphere that operates diagrammatically, with guidelines that can potentially lead to specific formations:

> But in reality, elements of the plane are *diagrammatic features*, whereas concepts are *intensive features*. The former are movements of the infinite, whereas the latter are intensive ordinates of these movements, like original sections or differential positions. . . . The former are *directions* that are fractal in nature, whereas the latter are absolute *dimensions*, intensively defined, always fragmentary surfaces or volumes. The former are *intuitions*, and the latter *intensions*. (Deleuze and Guattari 1994, 40)

Immanence is expressed through the infinite – against the finite nature of the concepts – eternally woven into the in-between, into that space of mediation which is not defined by lines, surfaces or volumes but by the fluidity of their interactions. Like fractals, the plane of immanence always slips between the definite figures of integral dimensions (which here represent the concepts) and provides us with non-linear movements that precede the formation and the shaping of these concepts. Movements of interference, folding and multiplication, by spreading simultaneously across the entirety of the plane, overcome any possible detection and always maintain an active element of the indeterminate, which opposes the limitations of the integral formation:

> It is this fractal nature that makes the phenomenon an infinite that is always different from any surface or volume determinable as a concept. Every movement passes through the whole of the plane by immediately turning back on and folding itself and also by folding other movements or allowing itself to be folded by them, giving rise to retroactions, connections, and proliferations in the fractalization of this infinitely folded up infinity. (Deleuze and Guattari 1994, 38–39)

THE ORGIASTIC REPRESENTATION

G. Deleuze in his book *Difference and Repetition* urges us to consider the concept of difference not as something that exists between things (i.e., as something that distinguishes itself from anything else), but rather to regard it as a relationship to itself, as something that distinguishes from its own self, or otherwise distinguishes itself from that which does not distinguish itself from it. This rather retrospective approach to difference, which Deleuze defines as 'difference in itself', is firstly placed in a correlation context between form and ground, between an object and that in terms of which it is visible and

representable. This relationship seems to be invalidated when the difference becomes a difference in itself and ground emerges to the surface of the object with the status of depth, causing what we might call a kind of formless and extensive drilling. According to Deleuze:

> Something of the ground rises to the surface, without assuming any form but, rather, insinuating itself between the forms; a formless base, an autonomous and faceless existence. This ground which is now on the surface is called depth or groundlessness. Conversely, when they are reflected in it, forms decompose, every model breaks down and all faces perish. (Deleuze 1994, 275)

The description of the penetration of the ground through the form provides us with a strong analogy regarding the process of fractional perforation that occurs in fractals. Similarly, fractal geometry permits the emergence of ground in the form and the negation of their separation. This is accomplished through the introduction of fractional or non-integer dimensions. When the shape loses the clarity and distinctness conferred on it by integral dimensions, it enters into a field of successive infinities through which the ground enters into the form as depth, as the infinite depth that is disclosed in the depiction of these infinities.

The ground, both in difference itself and in fractal geometry, appears as a projection on the surface of something deeper that could be called bottomless. Through this projection, the object enters into a direct relationship with its own depth, where the negation of limits and the activation of figure and ground become possible:

> The law of figure and ground would never hold for objects distinguished from a neutral background or a background of other objects unless the object itself entertained a relation to its own depth. The relation between figure and ground is only an extrinsic plane relation which presupposes an internal, voluminous relation between surfaces and the depth which they envelop. (Deleuze 1994, 229)

On the occasion of this new condition of correlation between figure and ground, or surface and depth, Deleuze describes the possibility of another form of representation, which he calls ‘orgiastic representation’. This, in contrast with organic representation, which has finite limits, relates in every aspect of itself to infinity, which is in a constant correlation with the ground, and the definition of any fixed limits becomes impossible. According to Deleuze, orgiastic representation has the ground as its principle and the infinite as its element, by contrast with organic representation which retains form as its principle and the finite as its element (Deleuze 1994, 43). Within the orgiastic representation, the ground is no longer useful for facilitating

representation but instead introduces infinity to the representational process. Infinity itself enters the heart of representation in infinitely small and infinitely large forms and also causes the convergence of all the finite perspectives of organic representation. In other words, there is a proliferation of perspectives and a concomitant change in their distribution, such that there is no clear orientation towards one particular point of view but rather a dispersal within the field of morphological reflections, where every perspective reveals a new world:

> Infinite representation may well multiply points of view and organize these in series; these series are no less subject to the condition of converging upon the same object, upon the same world. Infinite representation may well multiply figures and moments and organize these into circles endowed with self-movement. . . . To every perspective or point of view there must correspond an autonomous work with its own self-sufficient sense. (Deleuze 1994, 68–69)

The two characteristics of the orgiastic representation that distinguish it from the organic, that is, the introduction of the infinite against the finite and the proliferation (through reflection) of points of view and perspectives, could be depicted in fractal geometry. The fractals are completely dominated by assemblies or outbreaks of infinities around which a structure of reflective self-similarity appears. Regardless of which area we focus on, we continuously discover new similar outbreaks, new dynamical perspectives. The fractal figure incorporates infinity and is, perhaps, the only geometric approach that makes it partly representational. Thus, it can provide us with the basis for the visual depiction of orgiastic representation, a foundation of successive vanishing points towards infinity (or vice versa, a movement of infinity towards the surface). On the other hand, it can provide retrospective self-reflections following the reversal of the difference in itself.

SMOOTH SPACE

In one of the models of description for the conceptual correlation between smooth and striated, Deleuze and Guattari use the example of fractal geometry in order to express, on the one hand, the strange properties of smooth space, and on the other, its differences from the striated. The smooth and striated are, for Deleuze and Guattari, two elementary pluralities, which interchange and complement each other so that every evolving process takes place inside the striated, while each return to being takes place inside the smooth.

In order to describe the essential distinction between two forms of multiplicity, he also deploys various dualist concepts such as metric and non-metric, extensive and qualitative, centric and non-centric, arborescent and

rhizomatic, dimensional and directional, optical and haptic. From a mathematical perspective, this distinction is exemplified in fractal geometry and in particular in the relationship between integer and non-integer dimensions. The striated space is expressed by integer dimensions, through lines, surfaces or volumes. The concept of the metric distance between the specific points is especially important. The smooth space, on the other hand, is expressed by non-integer dimensions, by shapes that develop in the intermediate dimensional space of the respective Euclidean forms: lines are almost lines, from surfaces that are almost surfaces, or from volumes that are almost volumes. The key feature of these forms and their relation to space is that the space itself and that which occupies it are identified. In other words, there is no differentiation of ground and object, only a spreading process of continuous feedback and interpenetration between them. As Deleuze and Guattari say:

> A smooth, amorphous space of this kind is constituted by an accumulation of proximities, and each accumulation defines a *zone of indiscernibility* proper to 'becoming'. (Deleuze and Guattari 1987, 488)

In the geometry of smooth space, the concept of defined boundaries and contours is negated, and lines acquire their own spatial reality, reflecting through them all the aspects of the conterminous areas into infinity.

A typical example of the relationship between smooth and striated space can be found in the field of fluid dynamics and in particular in the transition from laminar to turbulent flow. One of the features of laminar flow is a structure in discrete parallel zones separated by a series of virtual line surfaces (Schwenk 2000). When such a flow is filtered through a system of barriers, an intense change in the geometrical formation of fluid is observed: the boundary surfaces between the distinct flows fold around themselves and create vortices, which then spread and multiply, occupying the entire flow and entering in this way into a new state – the turbulent flow. Turbulent flow – unlike laminar flow, with its integral dimensions (distinctive lines and discrete zones of specified scale) – is defined by the fractal geometry of intermediate dimensions and by the diffusion of elements of space at all possible scales. The element that is repeated in this fractal formation is vortices. As Richardson says, 'Big whorls have little whorls which feed on their velocity. And little whorls have lesser whorls. And so on to viscosity'. For Deleuze and Guattari, this transition to the turbulent condition is a way of escaping the space restrictions that inevitably bring about the striation process, precisely through the complete overthrow of any linearity and distinctiveness. As he says:

> At the other pole, it escapes them by the *spiral or vortex*, in other words, a figure in which all the points of space are simultaneously occupied according to

> laws of frequency or of accumulation, distribution; these laws are distinct from the so-called laminar distribution corresponding to the striation of parallels. (Deleuze and Guattari 1987, 489)

Smooth space is a space of escape; it constantly changes and opposes all forms of organisation, while fractal geometry is what opens up this possibility of escape, reversing the limitations of integer dimensions.

On another level, we will attempt to examine some of the more specific correspondences that occur in basic individual characteristics of chaotic systems and in various theories of Gilles Deleuze.

Specifically, we will examine:

(A) The status of sensitive dependence on initial conditions (small causes – big results) in chaotic systems with regard to the meaning of deterritorialisation
(B) The status of strange attractors in relation to the concept of the 'dark precursor'
(C) The morphological modification of systems when entering the chaotic condition will be correlated with the concept of the Fold.

A1. Points of Crisis and the Spread of Fluctuations – Small Causes, Large Effects

In chaotic systems small fluctuations of certain points (crisis points) can cause large changes in the evolution of a system. Through a climactic process, a small fluctuation can be magnified and spread throughout the system, thus creating a new operating regime. Within the chaotic systems there are scattered *points of instability* around which various changes in the microscopic condition climax, causing significant changes to the macroscopic condition or to the transformations of the system. This was observed by E. Lorenz while analysing the climate system through a graph that encoded the movement of warm air currents. He observed that from the same starting point, two graphs which should be the same, showed significant deviation. This occurred because the starting points, that is, the initial conditions, were not exactly the same, there was an infinitesimal difference in the fourth decimal (i.e., 0.506 instead of 0.506217).

The idea of major effects resulting from small causes was formulated by Ilya Prigogine for various chemical reaction systems. It refers more specifically to the importance of fluctuations for the transition into chaos and to the change of conditions in chemical systems. He observed that some systems in thermodynamics can escape the type of order governing equilibrium and lead

to instability, if 'certain fluctuations, instead of regressing, may be amplified and invade the entire system, compelling it to evolve toward a new regime that may be qualitatively quite different from the stationary states corresponding to minimum entropy production' (Prigogine and Stengers 1984, 200). Of great significance is the observation that in order to make this transformation of condition, a communication between the individual components, that is, their molecules, and a self-organising process are developed into a different condition, so that the system can cope with the changes of conditions and survive.[6] An awakening is triggered, of individual elements into conditions other than equilibrium, that do not exist in a state of equilibrium; according to Ilya Prigogine, molecules ignore one another and exist in a dormant state. The systems that develop away from equilibrium are generally dissipative structures[7] or dispersive structures that create new conditions in the system's matter. When, for example, a liquid changes from laminar to turbulent flow, despite the seeming disorder, a highly organised structure with multiple spatial and temporal scales is revealed, as a result of intercommunication and self-organisation of the system molecules.

A2. Deterritorialisation and Flight Lines

The significance of the aforementioned fluctuations in the formation of an individual condition for the function of chaotic systems, as well as in transitions to new functional systems, has a number of similarities with Deleuze's concept of the line of flight (*ligne de fuite*) and with the way it causes deterritorialisation – an abandonment of a striated structure and a return to the state of flux that precedes the synthesis of another. An abandonment of an operating condition is caused by an activation of some of the lines of flight inherent in a dormant state inside any system. The line of flight refers to the fluctuation in chaotic systems opening up the possibility of transition into a new state, through the deterritorialisation of an existing[8] one. However, such a fluctuation can be overcome by the pre-existing tendencies of the system without causing a change in its condition, so the line of flight, depending on the intensity with which it is charged, can lead to different types of deterritorialisation. Deleuze distinguishes three types of this process, depending on the intensity and formation of the line of flight inside a structure:

> The function of deterritorialization: D is the movement by which 'one' leaves the territory. It is the operation of the line of flight. There are very different cases. D may be overlaid by a compensatory reterritorialization obstructing the line of flight: D is then said to be *negative*. . . .
>
> Another case is when D becomes positive – in other words, when it prevails over the reterritorializations, which play only a secondary role – but

> nevertheless remains *relative* because the line of flight it draws is segmented, is divided into successive 'proceedings', sinks into black holes, or even ends up in a generalized black hole (catastrophe). . . .
>
> D is absolute when it conforms to the first case and brings about the creation of a new earth, in other words, when it connects lines of flight, raises them to the power of an abstract vital line, or draws a plane of consistency.
>
> (Deleuze and Guattari, 1987, 508–10)

All three modes of deterritorialisation can be compared to the three stages of a system's transition into chaos, through the diffusion of fluctuations. In the first case, fluctuations are overcome by the existing energies that have developed in the system and function as resistances (negative deterritorialisation). In the second case, the fluctuations are diffused in such a way that they find themselves in an intermediate, semi-chaotic state – on the threshold of chaos, where the system of operation begins to transform, while yet retaining some elements of its order (relative deterritorialisation). Finally, a case in which fluctuations spread everywhere, having gained consistency while occupying the entire system, and establish a new type of organisation of its elements (total deterritorialisation).

B1. The Hidden Structure of Chaotic Systems – Strange Attractor

The concept of the strange attractor means that in a chaotic system with an irregular data flow, there may be a hidden order structure, a shape with a defined way of developing and forming inside the phase space. This figure reflects all possible states within which the system can be found, regardless of the time sequence. It is a defined shape, since it occupies a particular area of the phase space, and yet at the same time it is not ultimately specified, since it is endlessly unfolding within it. In other words, it is a fractal structure, a continuous process of compression and folding of the phase space. Besides the fractal dimension, another dimension, which depends on the system's degrees of freedom, is also important for the definition of a strange attractor. Degrees of freedom, in the case of chaotic dynamics, are interactors which affect everything, so that each element of the system carries information on all the other elements.

Through the presence of the hidden structure, a spatial limitation in the representation of chaotic systems in the phase space is revealed, which involves the definition of a particular outline of the space area determined by each system. Within this area, however, the concept of a clear boundary is negated since, by following the structures of fractal geometry, the formation process unfolds to infinity. The positions of these systems are located in a formation that is simultaneously fixed and infinitely changing, and not located just

anywhere in space. If there is an infinite number of possible positions through which the system runs, there are also countless other positions, among and around them, where it can never be found.

The moments when the system finds itself in positions specified by a strange attractor are not defined in the same way as spatial moments. The time sequence, in which the signs that represent each condition in phase space appear, remains unpredictable and perhaps random, causing the shape of the attractor to achieve its final form through the sequential and irregular diffusion of points, rather than in a continuous way. The strange attractors, while recording the behaviour of a chaotic system, appear piece by piece, like ghosts. As Gleick points out, 'the frightful result of the strange attractor can be appreciated in another way, as the figure occurs with the passage of time, point by point, like a ghost emerging from the mist' (Gleick 1990, 201–2).

The significance of the strange attractor in a chaotic system is that it reveals a kind of stability. This stability is expressed through the existence of a hidden structure, *invisible* in the system itself but capable of revealing itself once the behaviour of the system has been recorded for a considerable period of time. This structure, without any physical presence in the system, is a sort of point-guide, providing guidelines consistent with the system's possibilities and excluding those positions in which a specific chaotic system, though unpredictable or indeterminable, cannot be found. The potentiality that develops within the system, depending on its endogenous characteristics, finds morphological expression in the strange attractors. Even though there are infinite potential conditions, the potentiality as size is nonetheless definable through a specific schematic process in the development of the system. Conversely, the absence of schematic restrictions that emerges from strange attractors would render the size of potentiality infinite and therefore inaccessible. The existence of a hidden structure in chaotic systems makes it possible to discern and describe their potentiality and differentiation from random noise.

B2. The Concept of the Dark Precursor

As Deleuze points out, we look for the 'structure' that is the form which can be filled with these descriptions and accounts (since it makes them possible) (Deleuze 1990, 280–81). Namely, the form that does not require an absolute presence in itself but functions instead as a receptor of/for partial presences. The form is constantly supplemented by new events, leading gradually to its completeness and eventually disclosing the plurality of potential states of the system. Such an invisible structure appears to respond partly to the concept of the dark precursor, as described in *Difference and Repetition*.

Gilles Deleuze arrives at this concept by seeking out a mediator to correlate the difference between differences and the difference from itself, a condition

that facilitates communication between heterogeneous systems and divergent lines, and necessitates the existence of at least minimal similarity between them. The dark precursor defines the invisible trace that synthesises all possible positions in which the association and coexistence of divergent series are facilitated and in which a connection between their differences occurs. According to Deleuze:

> Thunderbolts explode between different intensities, but they are preceded by an invisible, imperceptible dark precursor, which determines their path in advance but in reverse, as though intagliated. (Deleuze 1994, 119)

The concept of relief, as referred to in this study, resembles a mould. It has to do with an apparent absence, and at the same time, an expectation of completeness, an overlapping of possibilities:

> The path it traces is invisible and becomes visible only in reverse, to the extent that it is travelled over and covered by the phenomena it induces within the system, it has no place other than that from which it is 'missing', no identity other than that which it lacks. (Deleuze 1994, 119–20)

By hiding itself and its function, the dark precursor retrospectively maps out the meetings and connection points of the differences in intensity which develop within the system. In addition to this, the dark precursor, through constantly shifting towards itself, reveals a morphological uncertainty similar to that of the strange attractor, which tends to wind endlessly within the space it defines.

Another significant similarity between the dark precursor and the strange attractor is the characteristic of timelessness. This phenomenon may be perceived in its totality only from a point that moves in chaos, that is, where all divergent lines converge, where all possible lines are perplexed. On the other hand, from the perspective of a given present, which is connected to the representation, the lines appear to be consecutive, one before and one after (Deleuze 1994, 124). As it occurs with the strange attractors, partial observations will only give us fragmental perspectives, while it is only through a long-term monitoring from the position of infinity that we can formulate a clear picture of the entire phenomenon. The dark precursor chooses the sequences from chaos and introduces them into partial correlations with an undefined temporality that makes finite representation impossible, finding its expression only in an infinite representation.

C1. Morphological Organisation in Chaotic Conditions

The systems that prefigure a chaotic situation are formed through a process that begins with awakening, the intercommunication and self-organisation of

elementary parts into larger structures and extends to the macroscopic scale. This organisation on the microscopic scale, as indicated by Ilya Prigogine, is extremely accurate, even though the final image that the system gives macroscopically often looks desultory. The chaotic condition doesn't eliminate the morphological features of the system but transforms the condition of their creation. It imposes on the system a new organisational regime of material singularities, impacts the existing material as a symptom that affects all constituent parts and ultimately leads to a new level of inter-correlation, which is profoundly different from the one that exists in a state of stability.

This new morphological structure is based on a *regression*, a procedure that creates symmetry. Morphological information that the system 'exports' returns to it in the form of repetitions and self-reflections, having first acquired definite and established substance inside the system. Each established formation is not only the end of a process but the beginning of a new retrospective activity. In systems entering a chaotic situation, all the morphological potentialities and hidden aspects of their morphology – which prior to the transition to chaos were in a hypnotic state – emerge simultaneously. Suddenly, the whole range of details that are found inside each other or derived from one another become apparent. All of the material possibilities of formation, inherent in the system and depending on the endogenous characteristics and behaviour of the elementary parts, become visible and *available*. For example, when entering the vortex, an accumulation and synchronisation of different and conflicting frequencies occur, as opposed to a dominance of one frequency over another: 'The entire range of frequencies is present simultaneously' (Gleick 1990, 187). This accumulation of frequencies in the vortex flow is morphologically expressed by the continuous creation of folds within folds until the system finally exhausts the formation limits. This retrospective process of continuous emergence of new morphological information negates the concept of fixed and absolute form and introduces *movement* to the formation process. Through infinite unfolding, the material becomes movement, or is otherwise being created through movement.

Finally, another important element of the morphology of chaotic systems is the non-discreteness of the various areas defined within them. This occurs, for example, when some systems end up with more than one possible stable condition of equilibrium, thereby defining respective areas in the phase space. In this case, chaos occurs at the lines dividing these areas, which prove to be fractal and display infinite details. As we approach, the line stops existing and becomes a mosaic of fragments of one or the other side. Finally, in a chaotic situation, no point can be defined as part of a boundary. The boundary line, suggestive of antithetical relationships, becomes a world through unification and the topological mixing of all of the distinctive areas.

C2. The Concept of the Fold – Matter as a Perpetual Unfolding

This idea of the simultaneous emergence of all underlying morphological micro-environments during the transition of a system into a chaotic condition reveals to us an image of the continuity of matter, similar to the one that we encounter in Deleuze's concept of the Fold. Based on the mathematical theories of Leibniz, Deleuze describes a change in the perception of the structure. It shifts from the analysis of the individual elements that compose it to the logic of continuous unfolding, the logic of routes within its infinite folds. He refers to an important idea of Leibniz's philosophy, which states that the smallest element of existence is not the point but the fold. In other words, a process of reducing matter into its elemental form would not mean its dispersal into distinct points, but, rather, an unfolding through infinite folds. As he writes:

> Thus a continuous labyrinth is not a line dissolving into independent points as flowing sand might dissolve into grains, but resembles a sheet of paper divided into infinite folds or separated into bending movements, each one determined by the consistent or conspiring surroundings. . . . A fold is always folded within a fold, like a cavern in a cavern. The unit of matter, the smallest element of the labyrinth, is the fold, not the point which is never a part, but a simple extremity of the line. (Deleuze 1993, 6)

Based on this contemplation, we could infer the existence of certain 'sensitive' areas within the folds, on the surface of which a new environment is inscribed every time, thus opening up the possibility of transition to other folds. Various relationships are created in these areas, through which succeeding folds will be born. It is a way of perceiving the totality of being, that is based on an unfolding fold by fold, rather than on a detailed transition from point to point. The perception of a structure through meticulous unfolding and not through its analysis into discrete components is, as we would say, another formulation of the morphology of chaotic systems, a key feature of which is indivisibility. The impossibility of performing any separation and analysis into separate, discrete elements is common. Both in the transitional process 'from fold to fold' described by Deleuze and while going through the retrospective repetitiveness of folds in chaotic situations, what we are ultimately performing is a wander through the multiplicity of morphological possibilities.

Another key point in Deleuze's concept of the Fold is the emergence of a correlation between the Inside and Outside, according to which the Outside is inherent in the Inside and the Inside is merely an *elaboration* of the Outside. The outside is not meant here as a distant vacuum, a condition of immobility, but rather as an energy that seethes in search of a condition of inside-ness,

into which it will be channeled. The Fold becomes the bearer of the chaotic movements of the outside, the force that filters them and creates from them an Inside. The possibility of a fold precludes any complete separation between one inside and one outside, and sets in motion a continuous energy flow between them. As Foucault states:

> The outside is not a fixed limit but a moving matter animated by peristaltic movements, folds and Foldings that together make up an inside: they are not something other than the outside, but precisely the inside of the outside. (Deleuze 1988, 96–97)

This idea of the absence of boundaries between the outside and the inside, of their interpenetration or even of their meaning, simply as different expressions of the same primary forces, entails the active presence of both the outside-in and the inside-out, as well as their total topological contact:

> If the inside is constituted by the folding of the outside, between them there is a topological relation: the relation to oneself is homologous to the relation with the outside and the two are in contact, through the intermediary of the strata which are relatively external environments (and therefore relatively internal). On the limit of the strata, the whole of the inside finds itself actively present on the outside (Deleuze 1988, 119).

A similar total topological contact between 'different' areas is largely reflected in the morphology of chaotic systems, in which each area is always in contact with every other area, while simultaneously containing them.

Possible architectural extensions of the aforementioned correlations: Attempting to translate the aforementioned correlations into thinking about Space and Architecture, we could, by analogy, group the characteristics of a chaotic dynamics space into three points:

(1) Holistic Parameterisation – Diffused Variability

A first characteristic of chaotic dynamics in architecture is the holistic parameterisation of the architectural object, that is, the capability of the interaction between its fundamental elements and the diffusion of unidentified points of variability. The elements that compose the object will not be separated and idle, each fulfilling its purpose, but they will enter a field condition, of a continuous flux of energy and information between them. This will be a field condition,[9] in which the local interconnectivity between the elements will expand to all of the possible correlations, establishing, in the system of the architectural object, a condition of non-local interconnectivity. Architecture moves from a mechanistic to an organic mode. Dr. Mae-Wan Ho describes this transition using a comparison of the mechanistic and the organic universe.[10]

(2) The Process of Distribution and the Abstract Diagram as an Invisible Guide

A second characteristic, in an analogy to the strange attractors, would be a dynamic of distribution of spatial elements based on an implied diagrammatic guide that wouldn't function in a way that would limit the joints and the final correlations. The use of the abstract diagram and the algorithm to form not only spatial structures but also search structures could be interpreted, in the context of architecture, as what De Landa calls 'abstract building', which provides only a few topological traits and is open to innumerable 'final' adjustments depending on its surrounding, as well as on the nature of its inner traits (De Landa 2001). As the strange attractors function as point-guides for the chaotic system, defining neither its position nor its status at every point in time, the 'abstract building' can provide a sampling of the possible solutions, without determining a priori the one that will be chosen. The abstract building, like a chaotic system, can coil around itself 'like the motion of a coil that compresses all the information – varied and infinite – in a trajectory of consensus' (Ekeland 1995).

(3) The Process of Formation as Something Immanent in Matter

Finally, the fundamental consequence of a chaotic dynamic in the morphology of the architectural object would be the refutation of a belief that holds matter, as something that is formed by the application of external forces to it, and the manipulation of matter, as something in which form is immanent. According to Manuel De Landa, within a given area of matter, all possible formations are preserved. They depend each time on both the endogenous characteristics and the external conditions of the selected area. Far from the condition of stability but, as he says 'a whole range of persistent topological forms is revealed' (De Landa 2000, 4). The topological forms are inherent in matter as possibility, but in a chaotic condition, the entire range of these possibilities appears simultaneously, all of the different possible formation processes are displayed. All the morphological outcomes of the area of matter become possible, composing a map of possible morphological consequences of matter, complementary and intermixed. Thus, a rich material world is revealed; one that is full of possibilities.

In summary, one could conclude that the association developed in this study – between the philosophy of Gilles Deleuze and chaos theory – could finally be reflected in the architectural space through seeing it as a holistic system governed by the principle of non-divisibility, a system which cannot be analysed into individual elements (spatial elements in this case) without altering its substance. This property of non-divisibility in an architectural spatial system could be explained by two main features: on the one hand, by the existence of a widespread and collaborative interaction between the internal spatial components of the system, such that the role and intensity of each component depends directly

on all others and varies according to their variations and, on the other hand, the non-locality of the joints between the internal components of the system so that both the position and the nature of the joint can be continuously altered resulting in a multitude of possible correlations (always temporary in nature). The joints are smooth and morphologically adaptive according to neighbourhoods generated each time between the elements. These two characteristics are clearly opposed to the perception of architectural object as a linear mechanical system, where the role and the action of each element are discrete, fixed and not affected by those of other elements, and the joints between them are localised, pre-planned and consistent throughout the system evolution. So the theory of chaos, urges us to radically rethink the concept of architectural space not as a fixed arrangement but as a non-linear morphogenetic process.

NOTES

1. A system is an assemblage of entities/objects – material or abstract, which form a whole, such that each element interacts or associates with at least one other element in the set. Each object that does not associate/interact with another element of the system is not a part of the system. A subsystem is a data set which constitutes a system in itself but is at the same time part of the whole system. Chaos theory studies the behavior of non-linear dynamical systems that under certain conditions exhibit the phenomenon known as chaos. The best known is that of Devaney, which states the following: In order to characterise the behavior of a system as chaotic, the system must present the following properties:

- It must exhibit sensitive dependence on initial conditions.
- It must be in a transient state topologically.
- All of the periodic orbits must be dense.

Sensitivity to initial conditions means that two points in such a system can follow completely different courses in the phase space, even if the difference in the initial conditions is miniscule. The systems behave identically only when the initial conditions are exactly the same. Therefore, in order to predict how the system will behave in any case other than that of a limited 'time horizon', one must determine the initial conditions with infinite precision. In practice, of course, we can only determine the initial conditions with limited accuracy.

2. The theory of fractals is generally associated with non-Euclidean geometries, which reject the axiom of Euclid. These are: elliptical geometry, which is based on the work of Riemann; hyperbolic geometry, which is based on Lobachevsky's axiom; and projective geometry, which examines the projective properties of figures. It is especially associated with hyperbolic geometry.

3. The concept of process is central to fractal geometry and is ultimately what is represented in a fractal structure. It is similar to fractal forms and images that we find in nature, which are nothing but the de-crystallisation of dynamic processes in natural forms. The impressive fractal images that form the famous Mandelbrot set

essentially constitute a depiction of a simple arithmetic process in the complex plane. If after repeating a specific arithmetic process at a given point of the complex plane the result leads to infinity, then the point is not in the set. If, however, starting from a given point, the result does not lead to infinity but is trapped in a perpetual loop, then the point does belong to the set. Depending on whether or not each point belongs to the plane and how quickly it can be established, these various points of the complex plane are coloured differently, thereby providing us with these images.

4. Here it should be noted that in fractal geometry there are three self-similar cases:

- The exact self-similarity, which is the strongest type of self-similarity, where the fractal is presented simply as it is at different scales. Such fractals are defined by the mathematical function of repetition (iterated function).
- The quasi self-similarity, which is a loose form of self-similarity, wherein the fractal appears almost the same (but not exactly) at various scales. The quasi self-similar fractals contain small copies of the set in distorted and degenerated forms and are defined by equations that determine sequences retroactively (recurrence relations).
- The statistical self-similarity, which is the weakest type of self-similarity, where the fractal retains some statistics or figures while running scales. Random fractals are examples of fractals which are statistically self-similar, without being exactly or nearly self-similar.

5. Immanuel Kant refers to the concept *ens realissimum* as a condition of possibility of all things (Kant 2005, 140).

6. Catastrophe theory, as formulated by the mathematician René Thom, also expresses the idea of the transition of a system into a different operating scheme in order to escape its destruction, when limit conditions are exceeded. Catastrophe theory essentially expresses the possibility of maintaining the movement of a function, even when its limits are exhausted. The process remains active even when points exceed the limits, even though logically, a system would be destroyed. This is achieved by a discontinuous and instantaneous leap from one branch to another, just at the point beyond which the system will be destroyed. There is a momentary rupture in the system's motion, which ensures the continuation of its mobility and its function in another branch, in a different scheme. For example, water that turns into ice or corn when heated above a certain point pops but is not destroyed after it becomes popcorn. The destruction here is more a concept of metamorphosis than of an end or a permanent loss (Thom, Lejeune and Duport 1978).

7. There are specific points of instability – dissipative points, in which there is a possibility of the system's metamorphosis to a new state. Bifurcation could be defined as any critical point beyond which new outcomes become possible. The unstable points, where an infinitesimal disruption can influence the mode of operation of a system, are such bifurcation points (Prigogine and Stengers 1984, 224).

8. This transition from one condition of functionality to another is analogous to the discontinuous leap of catastrophe theory, or to falling into chaos. It is also expressed by Deleuze and Guattari with the concept of scapegoating and flight lines/deterritorialisation, which it represents. Through a line of flight, a given complexity

can change its nature in a manner similar to the way a scapegoat, having been suspended from one social regime, escapes to another.

9. Object to Field. Field Conditions in Architecture snd Urbanism.' AD Magazine/After Geometry, no 127.

10. Mae=Wan Ho, "The New Age of the Organism," AD Magazine/New Science-New Architecture, no 129.

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Chapter 7

A Thousand Models of Realisation: Towards a Deleuzoguattarian Critical Urban Theory

Keith Harris

Abstract

This chapter argues that Deleuze and Guattari's political philosophy, as outlined in the 'Nomadology' and 'Apparatus of Capture' plateaus of *A Thousand Plateaus*, can serve as a powerful framework for understanding the complex relationships between the economic and extra-economic forces driving the contemporary process of urbanisation. The chapter begins by reviewing a recent debate between advocates of critical urban theory and what has come to be called assemblage urbanism, and then steps through Deleuze and Guattari's theorisation of the relationship between state and war machine. It then draws on a case study of an urban revitalisation project (South Lake Union, Seattle, Washington, United States) to sketch out a new paradigmatic relationship between the state and war machine that is specifically adapted for analysing the urbanisation process.

In a lecture entitled, 'The Urban Age in Question: Towards a New Epistemology of the Urban', Neil Brenner (2015) outlines some of the central aspects of what he calls Critical Urban Theory. The occasion was the opening of an exhibition at the University of Melbourne called *Operational Landscapes*, which featured visualisations created by Brenner's students and colleagues at Harvard's Graduate School of Design, and the lecture was intended to arm the audience with an understanding of the theoretical underpinnings of the work on display.

Brenner begins by noting the paradoxical position of urban theory today. On the one hand, the field is marked by strong disagreements about almost everything: What is the urban? What methods do we use to study it? What methods do we use to influence it? But on the other hand, he argues, the field still presupposes underlying spatial taxonomies that are a relic of the period

of late nineteenth/early twentieth-century capitalist development in which it emerged. Foremost among these presuppositions is the notion that the world is divided into urban and nonurban space, which suggests that the urban is a particular type of settlement that must be studied according to its own principles and categories. In fact, the *Operational Landscapes* exhibition and the cover of Brenner's (2014) most recently edited volume – which is a haunting photograph of the heavily industrialised Athabasca oil sands in Alberta, Canada – both highlight how spaces traditionally thought of as nonurban are becoming increasingly used as support for city life. Critical urban theory's role is therefore to engage with and clarify the concepts, categories and interpretations of contemporary urban processes and questions.

What about critical urban theory's own presuppositions? In an earlier article, Brenner (2009) explains the philosophical basis of his approach, which relies on reformulated elements from Frankfurt School critical theory to shape critiques of ideology, power, inequality, injustice and exploitation in the urban context. Specifically, he outlines four central propositions: first, critical theory is indeed *theory*, and is therefore 'unapologetically abstract':

> It is characterized by epistemological and philosophical reflections; the development of formal concepts, generalizations about historical trends; deductive and inductive modes of argumentation; and diverse forms of historical analysis. It may also build upon concrete research, that is upon an evidentiary basis, whether organized through traditional or critical methods. (Ibid., 201)

Second, he asserts that critical theory is reflexive in that it is a product of, and directed towards, specific sociohistorical circumstances. It therefore both rejects all claims of standing outside spatio-temporal context and focuses on how forms of consciousness, subjectivity and knowledge can emerge in and against specific social configurations. Most importantly, the internal contradictions of capitalist society enable critique; without them, Brenner argues, critical consciousness would be unnecessary. Third, and drawing on Max Weber, critical theory rejects instrumental reason, or the use of means-end rationality that streamlines activity without an explicit interrogation of the ends. This orientation thus rejects technocratic visions of efficiency or apolitical engagement with the world, and demands explicit normative orientations in both practical and political registers. Finally, critical theory focuses on the disjuncture between actual forms of oppression/domination and the underlying possibilities of emancipation.

Taken together, these principles delineate a powerful approach to thinking about contemporary urbanisation, but a problem arises when advocates of critical urban theory come into contact with another subset of urbanists who have been inspired by Deleuze and Guattari. Specifically, Brenner (2013,

92) highlights the 'particularly problematic tendency [of the] contextualist turn that has become fashionable among many urbanists who have been influenced by Latourian actor-network theory and associated, neo-Deleuzian concepts of assemblage'. He continues, 'Especially in their ontologically inflected variants, such approaches reject abstract or macrostructural forms of argumentation in favor of place-based narratives and thick descriptions' (ibid.). Above all, his concern is that this particular mobilisation of Deleusian thinking – often referred to as *assemblage urbanism* – fails to confront 'the broader geopolitical and geoeconomic dimensions of contemporary urbanisation processes and associated forms of worldwide capitalist restructuring, dispossession, and uneven spatial development' (ibid., 92–93). In the following I discuss a debate between proponents of these two forms of urban thought, but for now I simply want to point out that while I concur with his indictment of assemblage urbanism *as it has been formulated*, I nevertheless believe that this disagreement opens up a space to think about how Deleuze and Guattari's political philosophy can serve as the basis for critical urban thought. In fact, one does not have to look far to find philosophical, though not explicitly urban, support for such an endeavour:

> Deleuze and Guattari's political thought is largely neglected. . . . Mostly obviously, this elision allows the silencing of the crucial fact that Deleuze and Guattari's co-authored work results in a direct and explicit way, identifiable in perfectly circumscribed texts, in a working re-elaboration of a number of nodal problems of contemporary political thought: the State-form . . . relationships between economic processes and structures of social and state power . . . the intricacies between geo-economics and geopolitics, etc. (Sibertin-Blanc 2013, 7, my translation)

My argument proceeds from this tension and asserts that it is possible to ground a Deleuzoguattarian critical urban theory in the political and economic dynamics that Deleuze and Guattari (1987) lay out in the 'War Machine' and 'Apparatus of Capture' plateaus. From this perspective, it is possible to make a strong argument for how this relationship drives urbanisation in the traditional sense, as well as in the more contemporary mode of 'planetary urbanization' that Brenner (2013, 2014, 2015) has identified in his discussion of the breakdown of the dichotomy between the rural and the urban.

To complete such an undertaking is beyond the scope of this chapter. Instead, this is an attempt to bring together several important threads: the main points of the often-cited debate between proponents of critical urban theory and assemblage urbanism; the key aspects of Deleuze and Guattari's political philosophy, paying particularly close attention to its capacity to address both economic and extra-economic factors; a summary case study of a unique form of contemporary urban redevelopment; and finally, a slight

transformation of Deleuze and Guattari's political philosophy to orient it towards the complexities of contemporary urbanisation.

ASSEMBLAGE URBANISM AND CRITICAL URBAN THEORY

Colin McFarlane's essay 'Assemblage and Critical Urbanism' (2011a) initiated a debate in the journal *City* that lasted over all five issues published during 2011 and included a critique from Brenner and two of his associates (Brenner et al. 2011), a response by McFarlane, multiple articles supporting one side or the other and, finally, two responses from the original interlocutors. Here I will only focus on the initial essay and the critique thereof.

In his opening salvo, McFarlane (2011a) argues that assemblage thinking can contribute to critical urban theory in three ways. First, as a robust *descriptive orientation* that, via thick description, accounts for the historical dimensions of existing urban inequalities as well as potential directions that alternative urbanisms might take. He draws a connection to critical urban theory by arguing that this line of thinking is also 'concerned with analysing "the systemic, yet historically specific, intersections between capitalism and urbanization processes" [and strives to] "demarcate and to politicize the strategically essential possibilities for more progressive, socially just, emancipatory, and sustainable formations of urban life" ' (citing Brenner et al. 2009, 179).

Second, McFarlane sees assemblage thinking contributing to critical urban theory as a reconceptualisation of *agency*, specifically in relation to assemblage theory's focus on sociomaterial relationships. Borrowing Ignacio Farias's (2009, 15) definition of agency as 'an emergent capacity of assemblages', he focuses both on human agency and the alleged 'agency of the *materials themselves*' (McFarlane 2011a, 215). Citing Jane Bennett's (2010) concept of 'vital materialism', McFarlane claims that materiality must be seen as a process that modifies relations between the human and non-human elements of assemblages, as he has studied in depth in the informal settlements of Mumbai. For McFarlane, this reconceptualisation of materiality affects critical urban studies, because it illuminates how researchers must account for non-human processes when investigating causality.

Third, McFarlane argues that the notion of assemblage as a collection of heterogeneous elements has many affinities with the idea of cosmopolitanism, and therefore can serve as an *imaginary* for thinking about potential urban relationships where difference is central. Citing Henri Lefebvre's concept of the 'right to the city', McFarlane (2011a, 220) points out how the affirmation 'of both access to the city and active participation of a range of groups in the production of the city as a lived reality' that undergirds critical

urban thought resonates with assemblage thinking. Moreover, he links this inclusiveness back to the non-human elements of assemblages, thereby opening pathways for connecting critical urban thought with a wider range of political movements.

Brenner et al. (2011) are, however, not convinced that assemblage thinking has much to offer critical urban theory as ontology. They are, in fact, 'concerned that McFarlane's construction of an assemblage-theoretical urbanism remains too broadly framed, at times even indeterminate, to realize its proper analytical potential' (ibid., 229). Their first critique argues that although McFarlane seeks to highlight ways in which assemblage can augment critical urban theory, it displaces 'the key concepts and concerns of radical political economy' (ibid., 230). More broadly they argue that other assemblage urbanists are also confused with regard to how and if such concepts should even be used to explore the relationship between capitalism and urbanisation. Through an extensive literature review, Brenner et al. sort recent assemblage-based approaches into three articulations, highlighting the relation of each to urban political economy. At the *empirical* level, assemblages are used to describe particular research objects – such as technological networks or interconnected regimes of authority – which are then analysed using political-economic frameworks. The *methodological* level is also erected on a political-economic foundation, but extends and reformulates parts of it by investigating dimensions of capitalist urbanisation that are typically neglected, such as flows of energy or people. Finally, the *ontological* level consists of work that seeks to displace urban political economy as the fundamental analytical orientation towards how urbanisation occurs (ibid., 230). Brenner et al. endorse assemblage urbanism's contributions to critical urban theory from the first two categories, but remain sceptical of it at the ontological level. Their skepticism is indeed warranted, given the ways in which assemblage urbanism has been formulated, yet the fact remains that none of these conceptualisations expresses the underlying political dimensions of Deleuzoguattarian thought.

Their second critique argues that assemblage urbanism as ontology focuses too much on description and rejects structure to a point that it deprives itself of key tools for 'understanding the sociospatial, political-economic, and institutional contexts in which urban spaces and locally embedded social forces are positioned' (ibid., 233). Moreover, they claim it is unable to determine which actants are relevant, due to its ontological flattening, or what they term 'naïve objectivism'. Powerfully critiquing McFarlane's assertion that agency is demonstrated by the inhabitants of informal housing settlements in Mumbai who cobble together dwellings from materials at hand, Brenner et al. (2011, 234) assert that his approach 'leaves underspecified the question of what historical geographies of land ownership, deprivation and struggle

generated and entrenched the unequal distribution of resources and the precarious life-conditions in the areas under discussion'. They drive this point home by noting, 'without a sustained account of this *context of context*, the analysis remains radically incomplete' (ibid., 234).[1]

The tension here is therefore between a descriptive orientation that uses the concept of assemblages to think about relationships between people and matter, and an analytical perspective that is invested in understanding how these relationships are produced. Although McFarlane (2011b, 733) eventually claims that 'capitalist relations of power, oppression and exclusion occur through processes that enroll a long chain of actors and sites that do not just impact but *become realized in everyday urbanisms*' (my emphasis), he fails to note that it is precisely various models of realisation that channel flows of capital into different urban arrangements. By turning to Deleuze and Guattari's discussion on the global capitalist axiomatic, we can begin to see the complexity of the relationship between the economic and extraeconomic criteria that engineer the flows of capital.

THE GLOBAL CAPITALIST AXIOMATIC AND MODELS OF REALISATION

Two of Deleuze and Guattari's concepts that can be mobilised to respond to Brenner's critique are the global capitalist axiomatic and models of realisation. This relationship arises from the discussion of the state and war machine and is grounded in part in a reading of Georges Dumézil.[2] Regarding the state: to take the inseparable yet antithetical pair which serves as the title to Dumézil's book *Mitra-Varuna: An Essay on Two Indo-European Representations of Sovereignty* (1988) as an example, the Indic god Mitra is the 'organizer', the god of the daytime, rewards for proper sacrifice, the human world, cooking with steam, and milk: 'the sovereign under his reasoning aspect, luminous, ordered, calm, benevolent, priestly' (ibid., 72). Varuna, however, is the 'binder' and expresses the fearful aspects of political sovereignty: the terrible, tyrannical, magical and omnipresent deity with 'immediate prehension and action everywhere and over everything' (ibid., 67). He corresponds to night, poorly executed sacrifice, otherworldliness, roasting food over flames, intoxicating drink, in short, 'the sovereign under his attacking aspect' (ibid., 72). For Deleuze and Guattari, the state's interiority and sovereignty itself are defined by the oscillation between these two poles of organisation and binding (cf. Deleuze and Guattari 1987, 424).

On the other hand, there is a third figure that has a different nature altogether: it is a pure exteriority, directed against both heads of sovereignty, constantly untying bonds and betraying pacts. Dumézil (ibid., 61) cites the

myth of Manu, whose *śraddhā* – which should be understood not as faith or devotion but something more along the lines of 'correct training' in the Foucauldian sense – is so strong that he is prepared to hand over his wife to be sacrificed by two demonic priests. However, Indra the warrior god – who operates outside the bond-pact couple defined by Varuna and Mitra – appears and intervenes, saving the wife, beheading the priests, and guaranteeing that Manu receives the benefits of his discipline. This pure exteriority is precisely the essence of the war machine: whereas the state apparatus binds and organises, the war machine disaggregates and disperses.

Yet, despite the difference in nature between these two, the state can appropriate the war machine in the form of a military: 'the jurist-king' – the organising pole of the state – 'is a great organizer of war; but he gives it laws, lays out a field for it, makes it principled, imposes a discipline on it, [and] subordinates it to political ends' (Deleuze and Guattari 1987, 425). Such appropriation in no way implies a domestication once and for all, for like the state, the war machine also has two poles: at one pole, the war machine is appropriated by the state or uses the state to realise its own ends; at the other, the war machine's object is flight, escape and the creation of new alternatives. However, various acts of appropriation not only transform the war machine but also the state itself, and these interdependent transformations are key to understanding the emergence of the global capitalist axiomatic. Deleuze and Guattari highlight three categories of such appropriation, beginning with the classical situation in which states appropriate the war machine, make limited war its sole object and direct it towards particular political ends: 'striating, securing, and expanding territory' (Holland 2011, 26). The rise of fascism marks the point at which this type of appropriation gives way to an autonomous war machine, which still takes the state's ends as its aim, but now has a new object: unlimited war, which is characterised by a full mobilisation of resources as well as the annihilation of the 'entire [enemy] population and its economy' (Deleuze and Guattari 1987, 421). This development is crucial because it marks the first point at which the state has reconstituted a war machine that is out of its control, and in relation to which the state becomes secondary. Finally, following World War II, the war machine frees itself from state control and can pursue its own aim and object: the distribution of pure differences in a social field marked by a fragile equilibrium. Concretely, this takes the form of the global capitalist axiomatic and the attendant world order necessary for the accumulation of surplus value. In this new configuration, states do not vanish – neither do local differences, such as variations in population, natural resources, wealth, industrial capacity, education – but are instead transformed into constituent parts of the axiomatic, or what Deleuze and Guattari somewhat vaguely call 'models of realization'.

It is worth mentioning that these three functions at the two poles of the war machine are compatible with some of the foundational notions of critical urban theory. Throughout his expanding oeuvre, Brenner draws heavily on the work of Lefebvre, who defines two strategies driving urbanisation. The first strategy, which Lefebvre calls neo-dirigisme, corresponds to the state's appropriation of the war machine, and emphasises 'planning, which in the urban domain promotes the intervention of specialists and technocrats, and state capitalism' (Lefebvre 2003, 78). The second strategy is neo-liberalism, which corresponds to the war machine subordinating the state to its own ends and is marked by maximising 'the amount of initiative allowed to private enterprise, and with respect to urbanism, to developers and bankers' (ibid.). The pole of the war machine that strives to escape the state and create alternate forms of social life corresponds to Lefebvre's assertion that the two aforementioned strategies must be abandoned for an approach to urbanisation that is based on needs, desires and everyday life.

It is here at the intersection where flows of global capital are realised according to specific models that we can begin to construct another version of Critical Urban Theory that is loyal to Brenner's timely call for interrogating its underlying assumptions and Deleuze and Guattari's political philosophy. How, then, might we account for the specificity of these models, that determine the communications infrastructure or oil sands operations occuring in spaces traditionally understood as non-urban, or in the centres of cities? My proposition is that we can turn to Foucault's archaeological method for two reasons: first, because it provides a way of assessing the incredibly broad range of criteria directing, regulating, sorting, assessing, explaining and managing the flows of capital; and second, because it is methodologically compatible with Deleuze and Guattari's entire project.

Finally, I want to make it perfectly clear that although I am focusing on the models of realisation – that is, the set of axioms, statements, codes and values that shape the urban environment – I do not mean to downplay the worldwide violence of capitalism. Instead – and perhaps much of this is due to my own positionality, as a researcher from Seattle, Washington, where the de facto normativity is one celebrating liberalism, citizen participation, innovation, environmental and social sustainability, all in tandem with economic growth – I am trying to create a space for thinking critically about the belief systems that circulate in design schools, city council chambers and the press. Or, borrowing Deleuze and Guattari's (1983, 225) language, I agree that we live in an age of piety and cynicism, but feel that too much critical inquiry, especially in the realm of critical urban studies, only focuses on the latter (whereas design education, in my experience, almost exclusively focuses on the former). Above all else, though, I am trying to direct our attention to the ideas that are actually realised, with the hope that we might pay closer attention to what is

missing, but without falling into the trap of thinking that everything extra-economic is simply masking the underlying dynamics of capital accumulation.

DEFINING MODELS OF REALISATION

The empirical side of this approach is also compatible with Brenner's project but with one important difference. In an attempt to highlight the multiplicity of ways in which urbanisation unfolds, Brenner (2013, 104) argues that 'the urban-form under capitalism is an ideological effect of historically and geographically specific practices that create the structural appearance of territorial distinctiveness, coherence, and boundedness within a broader, worldwide maelstrom of rapid sociospatial transformation'. But rather than invoking a notion of ideology that 'expresses the way that the forms of appearance of social reality under capitalism are systematically distorted to the benefit of some and the detriment of others' (Wachsmuth 2014, 77), models of realisation are not an obfuscation of 'sociospatial transformation', but are essential to how such transformations are effected. My concern with models of realisation is none other than an inquiry into how the social machines producing various intensities of urbanisation function.

In order to go about this task, I am using a case study of South Lake Union (SLU), which is one of the largest urban redevelopment projects in the United States and is currently under construction immediately north of downtown Seattle. Over the past decade, SLU – which has been primarily developed by Vulcan Inc., the investment and philanthropic giving firm of Microsoft co-founder Paul Allen – has been transforming from a light industrial neighbourhood into something between an urban residential neighbourhood, a high-tech office park, a university campus and a contemporary urban core. It is typically considered to be Vulcan's brainchild, but if one submits it to a Foucauldian archaeology, it becomes clear that there is a vast set of operative systems that have emerged over the past half century at work. In *The Archaeology of Knowledge* (2010), Foucault outlines four approaches for describing a discursive formation, or what in this case amounts to the linguistic portion of a model of realisation: tracking discursive objects, enunciative modalities, concepts and strategies. Briefly, these can be defined as what the formation is talking about, who is empowered to speak, how seemingly different concepts are related and understanding the relationships between the field of possible enunciations and what actually gets said.

First, by turning to a wide variety of documents corresponding to regional growth plans, citywide strategies for accommodating growth, as well as design criteria, one can identify four social fields that SLU's discourse addresses: land use and character, institutional coordination, sustainable

economic development and quality of life. Land use and character involve governmental, private sector and citizen efforts to classify land in particular ways, from portions of the region targeted for intensive growth to visions of how neighbourhoods should grow. Institutional coordination refers to how these groups' activities are being drawn together, such that each group has a specific and interconnected role in influencing how urbanisation will occur. Sustainable economic development is an effort at coordination across multiple regional industries, including aerospace, high technology, biotechnology and global health, in such a way that the region's economy is highly diversified and tending towards sustainability in all senses of the word. Finally, quality of life refers to the region's amenities, including clean water and air, social diversity and tolerance, and appropriate density and urban vitality. Various aspects of these four fields are drawn into discourse and categorised through diverse systems of knowledge, including biology, civil engineering, political economy, architecture and psychology. The next question is one of determining who is using such knowledge to classify elements from the four social fields.

A second pass through the archive reveals at least three different groups of speakers empowered to perform this task: representative bodies, citizen-government-professional and private and semi-public enunciative modalities. The representative bodies include the state legislature, regional growth coalitions and the city council. Each of these bodies is constituted by a wide range of individuals whose power to speak is not directly grounded in professional training but instead comes through election. Their singular nature is especially clear when considering their function as relays in between the technical work of planners, designers, developers and consultants, and the subsequent production of the built environment. The citizen-government-professional assemblages are often established to address particular problems: water pollution in an early example of inter-jurisdictional cooperation (1950s), guidelines for a comprehensive plan to guide Seattle's growth (1970s) or the development of neighbourhood plans (1990s). Urban planners differ somewhat from representative bodies since they are empowered to speak based on their education. Finally, the private and semi-public enunciative modalities include private enterprises, such as Vulcan, or partnerships between elite groups and city officials, as in the case of the Committee for the Seattle Commons, which sought to build a large public park in what is now SLU (1990s). Vulcan's function is primarily one of developing and implementing plans, while semi-public ensembles organise information and create documents but with far less autonomy. The key point is that while each group produces discourse, it is only through the relationships between all of them that we can see the emergence of the SLU's models of realisation.

Third, an archaeologist tracks parallels between how concepts emerge in different fields by considering forms of succession, forms of coexistence and procedures of intervention. One form of succession that emerges in the history of SLU is how efforts to address water pollution fracture from centralised, integrated and reactive efforts to highly decentralised, coordinated and proactive efforts to obviate such problems without restricting growth or economic development. In terms of coexistence, a common thread can be drawn through the heterogeneous discourses that share a concomitant relationship to another concept altogether. In SLU, one such concept is innovation, which can be seen in a variety of discourses ranging from global health ('PATH: Driving Transformative Innovation to Save Lives'; PATH 2015), self-care service providers like exercise studios ('The Bar Method's Innovative Technique for Attaining a Neutral Spine'; Leonard 2012), to government programmes and rhetoric related to economic development and environmental sustainability ('A City of Innovation'; Nickels 2008). Finally, the primary intervention that ties these diverse discourses together is the explicit cross-referencing between documents through quotation and rewriting of, or excerpting sketches from, design or growth management criteria. The main point is that one can track common elements between altogether different undertakings, such as governance, high-technology and high-value industry, and service industry by looking at these relationships.

Finally, the archaeologist must consider the relationships between the field of possible enunciations and what actually gets said by tracking points of diffraction, the economy of the discursive constellation and the field of non-discursive practices.[3] One of several points of diffraction – points at which two incompatible sub-discourses emerge within one discourse – is found in the juxtaposition between discussions of affordable housing and the litany of advertising for luxury housing. The economy of the discursive constellation points to three broad categories of discourse that have circulated in the same temporal field: political economy, ethics and visions of the city. In each of these fields, one can identify the majoritarian discourses that have been taken up in SLU and the minoritarian discourses which serve as limits to SLU's form of urbanisation. In terms of the majoritarian political economy perspective, for example, the discourse of neo-liberal policy reducing barriers and providing incentives for private land development is attended by the normative positions of developers who consider their work to be in the service of 'progress' (cf. Allen n.d.), or reducing carbon dioxide emissions (Hurd and Hurd 2012) by invoking visions of compact and pedestrian-friendly cities. In terms of radical alternatives that have not been taken up in SLU, one could consider Holland's (2011) notion of 'free-market communism' as an alternative political economy, which could be accompanied by a robust vision of

justice (Fainstein 2010) and sustainability (Parr 2009), as well as a vision of the city drawn from the Situationist International's conceptions of urban spaces organised around variegated human impulses rather than economy or efficiency (cf. Pinder 2005).

This brief review of a much more involved archaeology of SLU's discourse provides a sense of what constitutes the models of realisation producing SLU. The question is now one of how these details relate back to the relationship between the historical war machine and state relationships.

NEW STRIATIONS

The three strategic dimensions of the models of realisation – political economy, ethics and visions of the city – constantly overlap and intersect to guide the production of SLU. The increasingly sophisticated, coordinated and integrated patterning of urban space that is emerging in SLU therefore cannot be reduced to either the state's impulse to striate nor to capital's predilection to smooth (Deleuze and Guattari 1987), but instead relies on a materially productive yet inherently contradictory relationship between these two trajectories. A Deleuzoguattarian ontology for critical urban studies therefore must pay particularly close attention to how the various models of realisation function or, put another way, how the striations of urban environments are produced.

In the case of SLU, we have only tracked statements that directly relate to how the material environment is being produced, from the perspective of who is making them, what they are talking about, how they are related to one another and how they are limited. Each of these approaches provides a different perspective on the models, but it is only the immanent totality of these parts – in their messy, fractured and splintered consistency – that we might call SLU's model of realisation. This approach emphasises who is speaking and according to what kind of logic. In the case of SLU, the question of who is controlling the production of these statements cannot be reduced to any specific group, be it municipal planners and officials, Vulcan, or the residents, professionals and scholars who have contributed to the multiple versions of the neighbourhood plan. Neither can it be neatly sorted into ethical categories such as utilitarianism or an emergent entrepreneurial virtue ethic. Instead, it is a much more textured composition that does not lend itself to simplification.

The effect of this sort of multiplicity on the basic Deleuzoguattarian framework is complex, but following Holland's (2011) typology of the six variants of the relationship between the war machine and the state apparatus, we can situate the problem of SLU's form of urbanisation somewhere between two

archetypes: in the first type, the state appropriates the war machine as an army and uses it to attain its own ends; in the second, capitalism has exceeded the state and the latter's function is to promote the accumulation of surplus value. But, again, in SLU, we are dealing with all of this and more: spatial striation, economic security and expanding economic territory, and profit accumulation in conjunction with ideas about protecting and restoring the environment and boosting the quality of urban life. The nature of the present project is to think about how a particular historical relationship has produced SLU and how this way of thinking can potentially grasp it.

Returning to the two variants of the war machine – state apparatus relationship that Holland sketches will help us define how a transformed relation is necessary to address our case of urbanisation. He notes that it is crucial to identify their four key components: 'aim, object, space (smooth vs. striated), and form of sociality (ultimately hinging on the distinction between denumerable and nondenumerable sets)' (ibid., 25). If we call the two aforementioned types of war machine-state apparatus relationship the state war machine and global capitalism, then we can summarise the key components thus:

Table 7.1. War Machine/State Relationships in *A Thousand Plateaus* (Adapted from Holland 2011)

	Aim	*Object*	*Space*	*Form of Sociality*
State war machine	Striating, securing and expanding territory	War	Increasingly striated on the state's own terms	Increasingly denumerable sets
Global capitalism	Capital accumulation	Capital accumulation	Oscillates: Increasingly striated to promote capital accumulation; smoothed by new capital growth and mobility	Increase in both nondenumerable and denumberable sets, with the growth in nondenumerable sets outpacing the denumerable

It is crucial to remember that Deleuze and Guattari historicise these archetypal forms of the relationship between the war machine and state. Holland extends this formulation to explain how the state, as a model of realisation for capitalism, oscillates between its aforementioned poles of binding and organising through a discussion of George W. Bush's administration as neoconservative and Bill Clinton's administration as neo-liberal (ibid., 58–59). However, the case of SLU, while it certainly fits best in this latter category,

also draws on the state war machine's aim of organising physical space, and thus takes a slightly different concrete form:

Table 7.2. War Machine/State Relationship in the SLU

	Aim	*Object*	*Space*	*Form of Sociality*
SLU	Managing growth and capital accumulation	Managed space and economic productivity	Increasingly striated on shared terms	Increasingly denumerable sets(?)

Much like the capitalist war machine that Holland defines, this particular form has a resonance between its aim and object, but the aim now combines elements of the capitalist war machine with a variation of its state war machine predecessor. Striating and securing the territory of SLU is certainly on the docket, but it is also expanding in at least two ways. First, the astronomical growth in SLU has fuelled aggressive development between SLU and the downtown core, in an area called the Denny Triangle, which has gone from a field of parking lots to a construction zone involving more office space for Amazon's corporate headquarters (replete with biodomes), as well as multiple luxury residential towers. Second, the global health and worldwide e-commerce work occurring in SLU takes the entirety of 'planetary space as the staging ground for Seattle's global ambition' (Sparke 2011, 65), thereby extending this territory in a virtual or abstract manner. In terms of space, this form of the war machine significantly differs in that both the state apparatus and the war machine itself are invested in striating on mutually beneficial terms. A decades-old vision of funnelling regional growth into particular areas is meeting a growing collection of private entities with sufficient capital and compatible ideas of what this growth should look like, not only in terms of potential profits but also according to a vision of what is right and how that can be accomplished physically. The resulting form of sociality is also an important element, but it cannot be sufficiently addressed due to my empirical focus on the production of the built environment that does not yet include a detailed examination of how it is being used. However, the general pattern of increasingly integrated, or perhaps even curated, development suggests that sociality is tending towards standardised relations between an increasingly homogeneous group, hence my suspicion that there is an increase of denumerable sets in SLU.

In sum, although Deleuze and Guattari's political philosophy was certainly not developed specifically in order to analyse urbanisation, it can serve as a basis to that end, but nevertheless requires a degree of transformation to address current conditions. We might also note that a direct application of Deleuze and Guattari's formulation of the global capitalist war machine,

wherein the state primarily aids capital accumulation by 'proving compensatory reterritorializations' (Deleuze and Guattari 1987, 455), resonates with critical urban theory's current preoccupation with the neo-liberal restructuring of planetary space. However, given the nature of the form of urbanisation that has captured my attention, I find that paying attention to the models of realisation that modulate capital flows enables a more nuanced understanding of what is occurring on the ground. This approach is robust enough to address the softer forms of urbanisation such as SLU but could also be used to investigate the more violent forms of natural resource extraction or the production of urban slums worldwide, and for that reason it is a powerful foundation for critical urban inquiry.

NOTES

1. This notion of context – meaning the political and economic situation in which urbanisation occurs – forms a central element of Brenner and his various co-authors' perspective on studying urbanisation (cf. Brenner et al. 2010).
2. Deleuze and Guattari's reading of Pierre Clastres's work (1989, 2010) is equally as important, but I am focusing on Dumézil here for the sake of brevity.
3. I have bracketed out the analysis of non-discursive practices for future work.

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Chapter 8

Non-Correlational Athens

Stavros Kousoulas

Abstract

This chapter wishes to examine the relationship between active morphogenetic processes and their perceptual classification as typological categories of any kind. Considering architectural and urban design as substantial drivers of material production, I will outline an understanding of matter as an active agent in morphogenesis: the birth of form. Shifting from an essentialist, correlationist approach, which focuses exclusively on actual products and conceives matter as an inert receptacle of forms from the outside, thus prioritising concepts over matter, central will be the gradual replacement of an obedient matter with one in which its own tendencies and capacities drive its reproductive processes. Using materialism as its ontological and epistemological base, and with the aid of assemblage theory, dynamical processes such as architectural and urban production will be coupled with the subsequent representation and interpretation of the urban environment by its inhabitants, as well as with the behavioural patterns that they formulate. The aim is to bridge a significant gap between ongoing material production and its a posteriori classificatory examination, proposing a non-hierarchical methodology of architectural research, which regards materiality neither fully empirically nor solely speculatively, while being able to remain abductive and pertain to real constraints. In that sense, the contemporary urban ecology of Athens will be used as a case study for highlighting the outlines of a new form of architectural research. By dramatising the emergence of the most prominent element of the Athenian urban ecologies, the multiple residential apartment unit referred in Greek as the *polykatoikia*, what hopefully becomes evident is the need for a paradigm shift in which the given is considered as incapable of explaining anything; on the contrary, it itself needs to be explained by

determining the conditions under which it is actualised. It is through this shift that the complexities of urban ecologies can be examined in their full potential, excluding interpretations which fail to avoid being attached to the correlationist circle.

INTRODUCTION

As any analysis of contemporary architectural discourse will attest, the discipline is presently revealing a series of unprecedented concerns in regard to its methods of inquiry and practice. At a time when the heavy reliance on critical theories as a means and a method to interrogate the practice of architecture has been loosened in favour of other more pragmatic and pluralistic approaches, contemporary discourse and theory formation are reacting accordingly. Arguably, the reflective character of architectural theory, its dependence on taxonomic classification of form and typology, its tendency to react to the presentation of flamboyant design concepts via a critique of a host of forces influencing the practice from outside are all but examples of how architectural theory and its discourse are grappling to deal with the rapid changes brought about by the increasingly complex reality of the present.

In this chapter I aim to depart from the problem of critique in contemporary architectural discourse, and rather argue that new or renewed attitudes and perspectives are necessary to further contribute to the body of knowledge of the discipline. Critique, as an analytical tool, bares the potential of giving little insight on the concepts it wishes to engage with, let alone advance them in a more productive state. Though arguably it has been the main methodological tool for entangling in a conversation within an epistemological apparatus, it is a tool which performs unilaterally, judging rather than enmeshing. As Bruno Latour points out, critique is a tool engrained in our habitual practices but nevertheless of little use in the situation we are currently facing.[1] The enhanced version of *episteme* as an ecology of practices that I support is one that literally leaves no space to critique; there is no room for a transcendent a priori or a posteriori which would function as the ground for articulating it, no room for *Doxa*, or even more, for *Urdoxa*, a rationalised opinion able to demolish all other in a never-ending competition.[2]

On the contrary, I follow Karen Barad's reading, who, influenced by Latour's take on Alan Turing's notion of the critical, proposes a method not of subtraction and negative judgement but one of affirmative diffraction.[3] For Turing the notion of critical instead that of critique refers to that of the critical mass; when a single neutron enters a critical sample of nuclear material, it produces a branching chain reaction that explodes with ideas.[4] Moving from ideological judgement to aesthetic encounter, then change can be understood

as the relation between intuition (aesthesis) and the processes that condition it. Change and its possibility are then to be problematised in plural terms, placing them within an environment of relationships.

If one is to provide an alternative account for the plurality of, intentional or other, morphogenetic processes that take place constantly within the urban field, then one has to examine the intricate relationship between the field itself and our access to it. In this dividing line between ontology and epistemology, any attempt to prioritise one of the two will lead to fallacies of all kinds, which would eventually exclude our access to a world from the world itself. Ontology, epistemology and the Athenian urban world are put forward in this chapter. When material forms are seen not as systems which operate on an input/output base – from the concept to the built, from the plan to space – but as the incorporation of functions as intricate ways of dealing with complexity, then it becomes clearer that an appropriate position from where to formulate architectural theory is on the immanent middle point, not on the edges. In other words, architectural discourse and material practices have to meet halfway. For architects, the 'carpenters of materiality', this implies to step out of a reflective loop and into a trajectory of emergence.

ON THE WHOLE

In order to develop further what I coin as *urban correlationism*, I first have to provide some clarifications on the concept of 'world'. The basic premise of this chapter is that there is not one world but rather a multiplicity of worlds, and their ecological, thus relational, coexistence. In Levi Bryant's words 'to be in a world is to be decentralized, to lack all mastery, and to be a participant in an assemblage, network, or composition that exceeds society, culture, and oneself'.[5] A world ought not to be diminished to our access to it and its particular peculiarities. On the contrary, a world – or rather an ecology of worlds – is what violently intrudes any structure of meaning and signification that we might have been based on so as to give account of it. In a sense, a world is a 'strange stranger'. Timothy Morton describes the strange stranger as

> something or someone whose existence we cannot anticipate. Even when strange strangers showed up, even if they lived with us for a thousand years, we might never know them fully – and we would never know whether we had exhausted our getting-to-know process.[6]

A world is not only a strange stranger itself, it is composed of strange strangers. However, as with any strange stranger, it would be a methodological mistake to confuse them within the binary of an 'other' than 'us', any binary

based on identity and variations of sameness. Therefore, a world itself and the component parts that constitute it shall not be reduced to our conscious experience of them nor exhausted by it, since strange strangers remain loyal to no other principle than that of their perpetual 'strangeness'; objects in mirror are not closer than they appear, on the contrary the closer one gets, the greater the distance.

Before I move on, I will make a rather provocative statement: throughout this chapter I will continuously assume that the world does not exist. I could rephrase, the whole does not exist. And this is so because the world – the Athenian urban world in our case – must not be treated as a container, one which is able to include other minor beings or objects in a mega milieu. In the flat ontology I wish to advance, there is no such being that has the capacity of encompassing all others in it. As it will be clear later on, when the basic premises of assemblage theory will be briefly highlighted, being is composed exclusively of individual populations of beings. Any greater population of beings can never be conceived as their container but rather as the emergent form of their aggregate, one which is not higher in a vertical hierarchy of existential value but exists parallel to the very objects that constitute it. As Deleuze puts it, it is a 'One or Whole so special that it results from the parts without altering the fragmentation or disparity of those parts, and, like the dragons of Balbec or Vinteuil's phrase, is itself valid as a part alongside other, adjacent to others'.[7] A world is just one among others. Further on, by examining Uexküll's concept of *Umwelt*, it will become evident why every object constitutes a world itself, based on its operational, practical and relational openness or closure.

URBAN BIFURCATIONS

Against the assumptions of any philosophy of access, which privileges the human being over any other entities, proposing therefore well-established or revived forms of anthropocentrism, I assume the independent existence of both human and non-human agents, regardless of whether or not the latter are product of deliberate human actions. In definition, correlationism stands for the idea that we only ever have access to the correlation between thinking and being, and never to each term apart from the other.[8] In its urban version, correlationism can never fully explore the independent existence of the very object it aims to examine, always posing the matter of access as privileged to that of material configurations themselves. In addition, representation, of any kind, is advanced as the main epistemological tool of architectural and urban research, as it is the one that best fits for artificially overcoming the limitations of an ontology that categorically denies existential pluralism.

As a philosophical term, correlationism was first coined by French philosopher Quentin Meillassoux. In his book *After Finitude*, Meillassoux departs from a long-lasting debate among philosophers, that of primary and secondary qualities.[9] As he mentions, both terms come from Locke, however, the basis for such a distinction is already developed by Descartes.[10] The distinction between them is one of significant effects on both an ontological and epistemological level; it is the very nature of reality that is at stake here. In brief, primary qualities can be considered the properties of objects which exist independently from the observer, while secondary ones are those that produce sensations to the observer. Motion, figure, number, solidity and extension, versus colour, sound, taste, smell, just to name a few. It is evident that we are dealing with issues of subject-object relations, of the knower and the known.

A world, or rather worlds, characterised by such a distinction are places that once you remove the observer 'the world becomes devoid of these sonorous, visual, olfactory, etc., qualities, just as the flame becomes devoid of pain once the finger is removed'.[11] Meillassoux himself admits, and he is quite keen in underlining it, that there is no reason to question the fact of emergent properties as a result of relational capacities between matter, be that of what is traditionally called as subject or object. The sensible is a relation and not a set of properties inherent and intrinsic to singular agents.[12] What he considers problematic is the very distinction itself: the emphasis given on the presence of a subject in order of for the secondary qualities to be able to manifest. To put it in simpler words, correlationism is the ontological stance according to which an object is unable to be conceived as in itself, in isolation from its relation to a subject observing it. But in order to do so, one must assume and adopt a version of nature which is radically separated from *sentience*; one must fully adopt what Whitehead refers to as the *bifurcation of nature*.

I will claim that the root of such a bifurcating assumption, one that categorically denies experience, in its passive and active forms, is what Roy Bhaskar named *epistemic fallacy*. In his words, the epistemic fallacy

> consists in the view that statements about being can be reduced to or analysed in terms of statements about knowledge; i.e. that ontological questions can always be transposed into epistemological terms. The idea that being can always be analysed in terms of our knowledge of being, that is sufficient for philosophy to 'treat only of the network, and not what the network describes', results in the systematic dissolution of the idea of a world (which I shall here metaphorically characterize as an ontological realm) independent of but investigated by science.[13]

The epistemic fallacy is not an attempt to accuse epistemology in general and describe it as inherently faulty or unable to stand as a specific domain

of knowledge. It rather deals with the systematic displacement of questions about how we know in the place of those that deal with what being is; it deals with the elimination of speculation and metaphysics from any attempt of human thought to deal with the exteriority of its relations with the worlds. Essentially, the epistemic fallacy consists in the need to formulate a *given*, capable of operating as the transcendent, magical missing element which would be able in its presence to explain – rationally and objectively – how everything else is given; a given that gives. A given that gives the foundations up to which all knowledge can be securely deposited. Once a God, later a Law, later a (royal) Science, at times a Language or Reason, transcendent concepts intrude the immanence of the worlds, denying their equally immanent origin as emergent outcomes of relational, differential scientific-conceptual intensities, only to name themselves as its undisputable jury committee. The passage from opinion to *Doxa* to *Urdoxa*.

A correlationist, therefore essentialist, approach to urban entities considers them as a seamless whole, a totality, whose parts formulate relations of interiority leading later to its emergent properties.[14] Most of essentialist approaches consist of a very specific process: starting with finished products, discovering through logical reductive analysis the enduring properties which characterise those products, and then elevating this set of properties into a defining essence. This process eventually generates the inability of conceiving architectural objects and urban entities as emergent products, enforcing a constant resemblance to archetypical taxonomies. On the contrary, by applying non-linear causality in urban research, the city is examined as a whole of multiple components and their interactions; material environment, inhabitants, their expressive behaviours, flows of energy, networks of people, which in their synthesis endow the city with properties that are not necessarily retraceable back to them. The linkages between the components of a city are not *logically necessary* but *contingently obligatory*.[15] And while logically necessary relations may be investigated by thought alone, contingently obligatory ones require an empirical research, one focusing on the history of the co-evolution of the related parts, inserting crucial parameters such as their heterogeneity, and thus conceiving them as historical products.[16]

FLAT ONTOLOGIES

The principal characteristic of an assemblage is that its components are determined by relations of exteriority. That stands for their independent existence, set apart from their interactions, in which they possess both material and expressive roles. In performing their agency, the constitutive elements

of an assemblage may stabilise or destabilise it – territorialisation and deterritorialisation – or alter its identity – coding and decoding. The result would be large-scale entities which have properties that are irreducible to the initial components.[17] The formulation of the diagram of the assemblage would be the equal of the development of a body plan: the tracing of the – universal and individual – singular as opposed to the ordinary. By gathering the variables that form the populations of singularities, focus is shifted from an imaginary end product which is to be accessed – the correlational circle – to the relational processes that maintain or destabilise it.

Focusing on relations instead of *relata* would require not only a dismissal of scalar, hierarchical and typological taxonomies but also, most importantly, their substitution for a methodology which makes uses of non-linear, dynamic and heuristic tools. What can be named as structural couplings is just but an example of a tool which exchanges a scale-based approach for a relational one. No more one-to-one spatial analogies but entities that enter into relations, regardless of shapes and sizes, functioning now as mediums which expand or diminish agential capacities.[18] It is the limit of action replacing that of figure, with bilateral relations measured by their effects and not by their analogical spatiality; no longer differences in kind, but only differences in degree. Through the investigation of structural couplings within the urban field, one can trace the transversal modifications of multiple becomings while simultaneously prioritising the agency of their relations – and not the terms themselves. It is a tool that is able to shift constantly between scales, constituting a mobile and immanent practice of urban analytics. When matched with a flat ontology, it bares the potential of establishing a paradigm shift within the correlationist saturated urban studies.

Flat ontologies consist of self-organising systems, assemblages, where the dynamic properties of matter produce a multiplicity of complex relations and singularities that sometimes lead to the creation of new, unique events and entities, while more often to relatively repetitive orders and practices.[19] For a coherent flat ontology to support a new urban research, three transdisciplinary conceptual fields have to be considered: the analytics of composition and decomposition, which resist the practice of representing urban ecologies as a mere jumble of flows; the focus on differential relations, which constitute the driving forces of material composition and problematise axiomatic tendencies to classify architectural objects and urban spatialities; and finally, the attention on localised and non-localised emergent events of differential relations actualised as temporary sites where the social unfolds.[20] A transdisciplinary flat ontology will eventually be able to account for sociospatial emergencies within the urban ecology without demanding any prior, static and monotonous conceptual category.

ATHENS BEYOND CORRELATION

The potential of Athens as a promising case study for applying a non-correlational urban research lies on the artificiality of its denomination as a capital and to the persistent existence of a multilayered environment. Athens, as a historically emergent urban assemblage possesses characteristics of a spatio-temporal intensity which complexifies even more the material environment and the patterns of inhabitance developed in it. It is through the consideration of the specific characteristics of the Athenian ecologies – urban, social and physical – that the singularities of the urban environment can be located. In the fundamental element of urban reproduction, the multiple residential apartment unit, the condominiums that spread throughout the city, what in Greek is simply called the *polykatoikia*, one can trace the rhythm of order and disorder which catalyses the Athenian asymmetries. Broadly, the *polykatoikia* model is the generic and prevalent scheme of privately owned but mixed and informally inhabited apartment blocks, which has served as the basic unit of modern metropolitan growth and urban reproduction. Studying the urban unit and its structural-relational couplings with a wide population of factors can expose the actualisation of policies and practices in space, and the ways that through whose replication, differentiated forms of sociospatial attitudes emerge.

By examining the differentiating character of practices of inhabitance through their spatio-temporal recurrence, one will be able to trace connections between multiple fields, which gave rise to various patterns of inhabitance. A situated flat ontology will be able to account for the intricate ways that the layout of the built environment operates as an ordering force in relation to the practices that humans arrange in it.[21] Therefore, in order to examine the Athenian ecologies from a non-correlational standpoint, one must follow a population of interactive urban practices through their localised interconnections. These practices are material actualisations of their virtual space of possibilities. That translates them as fully contingent, meaning that other arrangements of actual and virtual urban relations would result into different practices altering the arrangement and agency of the assemblage itself. It is this move that permits to place a non-essentialist architectural research in the in-between of the extensive mimesis of urban reproduction and its immanent intensive capacities for the emergence of the new.[22]

URBAN IMMANENCE

When King Otto laid foot on the port of Piraeus on 1 December 1834, the fate of Athens as the modern capital of the newly formed Greek State was sealed. Approximately 170 plans for new towns, including Athens, were executed

in Greece during the nineteenth century. The majority of the plans for these new towns were based on standard orthogonal grids, with the urban legislation of the period being the locomotive for the creation of a homogeneous urban environment. The new Athens followed a relatively symmetrical plan, originally devised by German and Greek architects Schaubert and Kleanthes (1831–1833), designed for a population of approximately 40,000 residents.[23] Its most important element was the isosceles triangle, nowadays referred to as the 'historical triangle', from where the city's most significant landmarks found their rearrangement. The new plans of the city forced its development towards the northeastern side of Acropolis with the Royal Palace placed at the centre of the project, positioned on the top of the isosceles triangle, facing the Parthenon, while the other monuments were aligned as fundamental parts of the urban tissue. The remaining Ottoman-occupied land was subdivided and distributed among small land owners, thus making it almost impossible to conceive an urban grid based on the city block as its key element.[24] This plan became the basis from where thousands of small- and large-scale buildings were to be constructed.

This is the precise moment of deviation in any analytical historical approach which would aim to account for the emergence of the Athenian contemporary urban unit, the *polykatoikia*. A correlational approach would advance the archetypical resemblance of the emergent urban unit with the dominant modernist principles of the time, considering therefore its formulation and articulation as a matter of merely applying a set of transcendental design rules. As a consequence, the matter at hand would be the agency of a privileged meta-subject, the architect, who being in control and detached from the complexities of the urban assemblage, performs as a solution-making machine. On the contrary, if one aims following a non-correlational approach, then the *polykatoikia* has to be seen not only as a solution but also as a problem-making machine, abstract yet concrete in its actualisation. The emergence of form is neither the input nor the output of a linear process, but the complexity of micro-macro couplings which occur in the mesoscale. For the study of the *in-betweenness* of the mesoscale, one has to shift the conceptualisation of the environment from transcendentality towards immanence.

German biologist Jakob von Uexküll's concept of *Umwelt* would significantly contribute to that. *Umwelt* stands for 'the biological foundations that lie at the very epicenter of the study of both communication and signification in the human [and non-human] world'.[25] In other words, it is the continuous surrounding practical world, in which an object – animate or inanimate – does not exist in a habitat, but it itself is the habitat. Through the concept of *Umwelt*, Uexküll set the very first principle of immanence, the foundation of ecological thinking; organisms, objects, environment, inside and outside do not differ in kind but in degree. It is via immanence that any linear account

for the emergence of form can be exchanged with a research methodology which handles form similarities as populations of contiguous ways of treating complexity, formulating and reshaping environments. In that sense, the *polykatoikia* is not an archetypical resemblance to a modernist fantasy but rather the complex solution to complex problems – and vice versa, the complex problematisation of given solutions.

A PROBLEMATISED FORM

For Deleuze, an organism poses both a problem and a solution. For a moment, let's enhance Deleuze's statement by replacing organism with form; form poses both a problem and a solution. Why is it then that the form is the enemy?[26] Claiming that the form is the enemy intends to radically raise the issue of going beyond form, past any taxonomical categorisation and straight to unravelling the pure ontological processes that give birth to it. It redefines form as the actualised solution to a problem.[27] However, it moves the understanding of form a step forward. It simultaneously problematises form, conceptualising a formal solution not as something fixed and reified. To do so would mean to conceal the immanent problems that made it emerge and which partially – or not – pertain. Because if form is a solution then it is a solution to its own problem.

The question therefore would be how one can locate the problem. What makes form emerge? What can it do? Out of which relations is a form composed? How are these relations articulated? This is the problematique that one must face in order to account for the double nature of form. It is the problematique of tracing becoming in being. In order to answer these questions, form has to be placed under the mode of a problem, forcing one to consider its implications on a population of heterogeneous fields. The 'question – problem complex' needs us to ask not only *what is* form but rather *who, which one, how many, how, where, when, in which case* and *from which point of view*.[28] In these minor questions lays the potential of tracing a formal continuous variation. It is due to the need of posing these minor questions that the *polykatoikia* can serve as a conceptual persona in tracing the singular out of the ordinary.

The *polykatoikia*, the unit of the Athenian urban reproduction, has to be placed in an environment of practices; a relational field of gravitational forces, attractors, which eventually give spatial and temporal shape – as spaces and rhythms of inhabitance – to the everyday life. The *polykatoikia*, seen as the set of practices it affords and can afford to its environment, is then an assemblage of active agencies formulating the Athenian milieu. A milieu which as Deleuze and Guattari remind us, is always fourfold: an exterior

milieu (the material 'outside' of the organism), an interior (its structure), an intermediary (the membrane) and an annexed or associated milieu (energy that transverses it).[29] Any form is in fact a composition of these environmental differential relations, and by recognising it as such we are able to account for its dual nature of a problematised solution.

The practices of the Athenian urban unit, as they unfold in the multiplicities of the milieu, can be distinguished to those it partakes and to those that take place in it. These are the relations that Sanford Kwinter names, respectively, as 'micro architectures' – the ones that saturate and compose the urban unit – and 'macro architectures' – relations that comprehend or envelop it.[30] For Kwinter, practices are clusters of action, affectivity and matter which correspond less to formed, distinguishable objects than to a specific 'regime' that for a specific time inhabits the urban field. A regime imposes a certain configuration on a field as it manages to organise, ally and distribute bodies, materials, movements and techniques in space while simultaneously controlling and developing the temporal relations between them.[31] If one aims to examine the emergence of the *polykatoikia*, its persistence, its transformations, as well as the differentiating patterns of inhabitance it produces, then one has to trace, interrelate and diagrammatically expose the – both virtual and actual – regimes that manifested it as a formal solution to the problems they posed.

It is then that historical facts seize to be part of a linear data insertion on a correlational and non-temporal account of urban form emergence, and rather set on to formulate populations of attractors which drive the morphogenetic processes. Influences of the modernist avant-garde, the demands of a state devastated after World War II, the influx of internal migration to the expanding capital, the appraisal and consequent fall of the *polykatoikia* as an ideal model of inhabitance, the use of a massively produced building material, closely connected to the industrialisation of the periphery of Athens, just to name few of them, are both singularities – moments of fundamental qualitative change – and intricate parts of a relational web which shapes the overlapping regimes of urban production.

OPEN-END

My aim so far has been twofold; on the one hand, to highlight aspects of a pluralist methodology which would encompass population, intensive and virtual modes of thinking so as to provide a non-correlational account for the emergence of complex urban phenomena. On the other, to dramatise the emergence of the most prominent element of the Athenian urban ecologies, the *polykatoikia*. Apparently, the specificities of an exact application of a

non-correlational approach to both the Athenian ecologies and its urban unit exceed the scope of the present contribution. However, and while I am fully aware of their importance, what hopefully becomes evident is the need for a paradigm shift in which the given is considered as incapable of explaining anything; on the contrary, it itself needs to be explained by determining the conditions under which it is actualised.[32] It is through this shift that the complexities of urban ecologies can be examined in their full potential, excluding interpretations which fail to avoid being attached to the correlationist circle, while maintaining the capability of synthetically incorporating future events. Thus, a heuristic and open-ended research process of urban and architectural emergence is structured, able of being simultaneously analytic and synthetic, aiming to affect not only the way we comprehend and consume space but also the way we record and produce it. A research process which problematises abstract design principles and choices via linking them with their actualisation: urban and architectural theory with material practices.

NOTES

This chapter includes reworked parts of the following online published text: Stavros Kousoulas, 'Becoming Form: Athenian Urban Ecologies', in *What's the Matter, Materiality and Materialism at the Age of Computation*, International Conference Proceedings, Barcelona, September 2014, ISBN 978-960-89320-6-7.

1. Bruno Latour, 'Why Has Critique Run Out of Steam? From Matters of Fact to Matters of Concern', in *Critical Inquiry* 30 (Winter 2004), pp. 225–48.

2. *Urdoxa* comes from the use of the German prefix *Ur-* and the Greek *doxa*, used by both Plato and Aristotle as opinion. Within contemporary philosophy, and especially in poststructuralist discourse, it adopted a negative connotation as it assumes a transcendental subjectivity which privileges a humanistic conception of Being. One can find references to it in the works of G. Deleuze, F. Guattari and M. Foucault among others.

3. Karen Barad, *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning* (Durham, NC: Duke University Press, 2007).

4. Alan Turing, 'Computing Machinery and Intelligence', in *Mind* 59 (1950), pp. 433–60.

5. Levi Bryant, *Ontocartography: An Ontology of Machines and Media* (Edinburgh: Edinburgh University Press, 2014), p. 114.

6. Timothy Morton, *The Ecological Thought* (Cambridge: Harvard University Press, 2010), p. 42.

7. Gilles Deleuze, *Proust and Signs: The Complete Text* (London: Continuum Books, 2000), pp. 164–65.

8. Quentin Meillassoux, *After Finitude* (London: Continuum Books, 2009), p. 5.

9. Quentin Meillassoux, *After Finitude: An Essay on the Necessity of Contingency* (London: Continuum, 2008).

10. Meillassoux, *After Finitude: An Essay on the Necessity of Contingency*, p. 1

11. Ibid., p. 1.

12. Ibid., p. 2.

13. Roy Bhaskar, *A Realist Theory of Science* (London: Routledge, 2008), p. 36.

14. Manuel DeLanda, *A New Philosophy of Society* (London: Continuum Books, 2006); and Manuel DeLanda, *Intensive Science and Virtual Philosophy* (London: Continuum Books, 2002).

15. 'A seamless whole is inconceivable except as a synthesis of these very parts, that is, the links between its components form logically necessary relations which make the whole what it is. But in an assemblage these relations may be only contingently obligatory' (DeLanda, *New Philosophy of Society*, p. 11).

16. 'What is an assemblage? It is a multiplicity which is made up of heterogeneous terms and which establishes liaisons, relations between them, across ages, sexes and reigns – different natures. Thus the assemblage's only unity is that of a co-functioning: it is a symbiosis, a sympathy. It is never filiations which are important, but alliances, alloys; these are not successions, lines of descent, but contagions, epidemics, the wind', see Gilles Deleuze and Claire Parnet, *Dialogues II* (London: Continuum Books, 2002), p. 69.

17. Gilles Deleuze and Félix Guattari, *A Thousand Plateaus* (London: Continuum Books, 2004); DeLanda, *New Philosophy of Society*.

18. Levi Bryant, *Ontocartography: An Ontology of Machines and Media* (Edinburgh University Press: Edinburgh, 2014), p. 31.

19. Sallie A Marston, John Paul Jones III and Keith Woodward, 'Human Geography without Scale', in *Transactions of the Institute of British Geographers* 30 (2005), p. 422.

20. Sallie A Marston, John Paul Jones III and Keith Woodward, Human Geography without Scale, pp. 416–32.

21. Ted Schatzki, *The Site of the Social: A Philosophical Account of the Constitution of Social Life and Change* (University Park: Pennsylvania State University Press, 2002); Sarah Whatmore, *Hybrid Geographies: Natures, Cultures, Spaces* (London: Sage, 2002).

22. Manuel DeLanda, *Intensive Science and Virtual Philosophy*.

23. Eleni Bastéa, *Neoclassic Urbanism and Greek National Identity* (Athens: Libro, 2008), p. 155.

24. It is important to mention that here I am dealing with the period that covers the initial planning and its implementation. The distribution of land was posterior to the planning, and was executed factually only on state-owned land in central areas. In the remaining areas, which were considered less important, the grid followed the guidelines of the master plan. In later stages, the plots were subdivided, something that gave access to all the small properties within the block.

25. Jakob von Uexküll, *A Stroll through the Worlds of Animals and Men: A Picture Book of Invisible Worlds* (International Universities Press: New York, 1957); Thomas Sebeok, *Contribution to the Doctrine of Signs* (University of Toronto Press: Toronto, 1976).

26. Paraphrasing the quote 'the organism is the enemy' from Deleuze and Guattari, *A Thousand Plateus*, p. 196.

27. Deleuze, *Difference and Repetition*, trans. Paul Patton (London: Continuum, 2001), p. 272.

28. Deleuze, *Difference and Repetition*, p. 66.

29. Deleuze and Guattari, *A Thousand Plateaus*, pp. 49, 52, 384; Brett Buchanan, *Onto-Ethologies: The Animal Environments of Uexküll, Heidegger, Merleau-Ponty and Deleuze* (New York: SUNY Press, 2008).

30. Sanford Kwinter, *Architectures of Time* (Cambridge: MIT Press, 2001), p. 14.

31. Ibid., p. 14.

32. Deleuze, *Difference and Repetition*.

Chapter 9

Architecture at the Age of Its Digital Production: The Force, Differentiation and Humanity of the Fold as an Architectural Principle

Constantinos V. Proïmos

Abstract

The architectural interest in the fold emerged soon after the publication of Gilles Deleuze's seminal book *The Fold: Leibniz and the baroque* during the end of the 1980s in Paris. The fold is a cultural force carrying the traits of many eras of cultural development but originating in the baroque. For in the baroque the fold leads to infinite inclusion, represents constant movement and continuously refers to other folds. Matthew Krissel argues that the fold is a differentiating principle in architecture and means a constantly evolving architecture like computer software which defies its classicist premise, *firmitas*, strength or longevity. Many examples may be given of such a constantly evolving architecture: Stephen Perella's pixel architecture which investigates formal plasticity with no regard to function or other determining factors of architecture; Marcos Novak's liquid architectures; and Greg Lynn's exact, anexact and inexact architectural making and formless design as it is occasioned by computational processes. This architecture that no longer seeks stable forms but continuous variation and movement through complex computational processes strikes as particularly inhuman. Nevertheless, the fold also gives the opportunity to ask the question of humanity while at the same time conceptualises and theoretically grounds developments in design, issuing from digital technology that is now vastly employed in architectural practice and thought. The stake is an immanent ethics of sustainable design issuing from the fold as proposed by Luke Feast.

During the past thirty years there has been a growing architectural interest in Gilles Deleuze's concept of the fold as this was launched in his seminal volume *The Fold: Leibniz and the baroque* (Deleuze 1993), originally published in the end of the 1980s in Paris. This interest in the fold is indicative

of the social pressures exerted on architectural design from an increasingly digitalised world on the one hand, and the social perplexity in contemporary multicultural societies, on the other. The interest in the fold is emblematic then of the dramatic changes taking place not only in architectural theory and thought, in the epistemology of the discipline, but also in architectural practice.

The three classical principles for architectural design, namely *venustas*, *firmitas* and *utilitas* (beauty, strength and utility, respectively), professed by the Roman architect Vitruvius during the first century BC and reiterated during the Italian Renaissance by Alberti, have been the long-lasting values of architectural composition which have ineffably marked the history of architecture. Classicism always accorded a primacy to beauty in comparison with strength and utility, and modernism favoured utility and function to a greater extent than the other two principles. At the beginning of the twenty-first century and especially during the past thirty years, it is *firmitas*, strength, which is being questioned on account of the fold. Strength of course means not only static perfection and stability but also endurance in time. The exemplary model of architectural strength is perhaps the Egyptian pyramid which aimed at safeguarding the rule and reign of the king, extending this rule to eternity after he crossed the border to the afterlife. Nowadays, buildings like the Tel Aviv Museum of Art, Israel, designed by Preston Scott-Cohen or the Klein Bottle House at Rye, Victoria, Australia, designed by Charles Ryan McBride seem to be on the antipodes of the pyramid. First, their intense curvilinearity or folding needs high maintenance in order to be preserved intact, in contrast to the minimal care that the pyramid requires for its longevity. Second, and more importantly perhaps, such contemporary buildings are generated as an architectural response to the increased desire and need for flexibility in design. Flexibility means the ability to correspond to changing circumstances. Such folded buildings, as they are commonly called, are one step before movement as they represent frozen movement, to recall the famous definition of architecture by Schelling. They represent the architects' struggle with time not in the sense of making a long-lasting product, like in the past, but in the sense of designing with a view to changing and conflicting requirements, stemming from complex and multicultural societies which may require upgrading buildings like upgrading software at all times to cope with new social needs and desires.

In what follows, I shall first furnish a reading of the fold in the way Deleuze develops it in his seminal book with the view of making it useful to architecture. Then I shall attempt to see how the fold applies in contemporary architectural discourse and practice and what its ramifications are, especially with regard to the epistemology of the discipline. Finally, I shall attempt to consider the consequences of tectonic intensification produced by the fold

and check the hypothesis of an ecologically sustainable design attributed to folded architecture by Luke Feast.

THE FOLD AS A CULTURAL FORCE OF THE SOUL

From the outset, Deleuze defines in his highly metaphorical language the fold as a cultural force carrying the traits of every era of cultural development (Deleuze 1993: 18, 3). There are folds from the Greek, Roman, Romanesque, Gothic and classical periods, but the one that preoccupies him most is the fold in the baroque. For in the baroque the fold leads to infinite inclusion or inherence (ibid., 22), represents constant movement (ibid., 12) and continuously refers to other folds from which it originates (ibid., 8, 10) and which constantly generates. In fact, the activity of folding and unfolding means enveloping and developing, involution and evolution (ibid., 8, 10). Furthermore, folding means diminishing and reducing, whereas unfolding means increasing and growing (ibid., 8, 10). The fold therefore suggests a cultural trait, manifesting itself as a force, representing infinite work in progress, namely, infinite receptivity and infinite spontaneity (ibid., 35). According to Deleuze, folding works like origami, the Japanese art of folding paper (ibid., 6).

As mentioned previously, the fold appears in every cultural era. However, it is also the operative concept of the baroque in the seventeenth century (ibid., 33). It appears most conspicuously in the baroque for there it 'knows an unlimited freedom' (ibid., 33). Baroque itself is a dynamic operative function for it entails fluidity and curvilinearity (ibid., 3, 4). In the baroque the world appears 'as an infinite series of curvatures or inflections' (ibid., 24). Whereas classicism, through its Cartesian coordinates, privileges firm points, angles and centres, in short essential forms, in some sort of abstract universalism, (ibid., 24), baroque favours the infinite variation of folding (ibid., 14, 19, 21, 16). Moreover, the contrast that exists in baroque between the interior and the exterior, the exacerbated exterior and the serene interior, in the churches, for example, (ibid., 28), is the contrast between folding and unfolding.

As Simon O'Sullivan contends, 'the concept of the fold allows Deleuze to think creatively about the production of subjectivity . . . one's relation to oneself' (O'Sullivan 2010: 107). This is an important dimension of the fold related to Leibniz from whose work the fold is directly inspired. It is Leibniz's monad, the name he ascribes to the soul or the subject (Deleuze 1993: 23), that Deleuze has in mind. The monad is a metaphysical point which is however not privileged or centred, for the world has an infinity of monads or souls (ibid., 25). The subject is a monad which is a point, a soul and, in fact, it is a point of view (ibid., 19). Accordingly, the object is an event (ibid., 19). The world is marked by perspectivism, relativism and pluralism (ibid., 20). The world is

in the monad which is like a cell, like a room with neither doors nor windows (ibid., 28). All activity in the soul which is full of folds takes place inside it (ibid., 28). The soul or the monad is marked by an absolute interiority or a dark background, in the sense that 'everything is drawn out of it and nothing goes out or comes in from the outside' (ibid., 27). The monad is characterised by absolute interiority and matter by absolute exteriority (ibid., 28). Between monads and matter there is force exerted (ibid., 6) although elsewhere Deleuze affirms that 'masses and organisms are coextensive as two kinds of force or two kinds of fold' (ibid., 9). Deleuze offers the famous allegory of the two-storey baroque house to explain the human condition: the ground floor has window openings and represents matter which affects the senses, while the upper floor is destined for the mind and interiority, for 'the vertiginous animality of the soul', a pure inside without outside (ibid., 12, 19).

THE FOLD AS A DIAGRAM

Graham Livesey admits that it is often difficult to translate Deleuze's concepts into concrete or material reality despite the immediate and global impact that Deleuze's concept of the fold has had in the world of architecture as soon as it was published (Livesey 2010: 109). Likewise, Matthew Krissel argues that when considering the fold in an architectural context, it encompasses continuous differentiation 'where new and unanticipated possibilities (between folded, enfolding and yet to be enfolded) occur without predetermined outcomes' (Krissel 2004). This entails 'a pliable topological space . . . no longer detached from program and event but where the folds become the events themselves' (Krissel 2004). Krissel continues wondering how this active space can occur in tangible reality beyond theory or computer space. His answer seems tentative: the ability to have a series of surfaces programmed to fold and unfold means an endless and unpredictable architecture, progressive and regressive at the same time and, in any case, continually evolving (Krissel 2004). Is it possible to imagine an architecture not solely in theory, diagram or static variation that materialises in building construction and is continually evolving or is constantly upgraded like computer software? This is the question that Krissel's readers immediately face. Despite the fact that famous architects, who are theorists at the same time, have already discussed such a possibility and it suffices to mention here Peter Eisenman, Greg Lynn and John Rajchman, this architecture is still a future projection and is nowhere to be found. It is noteworthy, nonetheless, that existing folded architecture, nowadays, already points to this direction of a constantly evolving architecture defying its static premise, *firmitas*, as an architecture in constant progress. Such an architecture which Krissel locates in the future, may,

according to him, actualise the fold 'within programmable matter' through the 'development of nanotechnology' (Krissel 2004).

When the fold is adopted on the level of formal manipulation alone (Jackson, 6), and even worse when it is justified with recourse to architectural imagination and its purported freedom and right to endless innovation, the radical core and consequences of the fold are lost, and folded architecture is simply delivered to advanced capitalist consumerism. Architects have always faced this danger and challenge of satisfying the desire of rich clientele for conspicuous consumption, a desire which threatens to reduce architecture to fashion. Mark Jackson argues that what appears to be in the cutting-edge architectural avant-garde, namely, cyberspace and virtual reality, as occasioned by folded architecture, sometimes tend 'to be construed on the most conservative of metaphysical principles' (ibid., 2). This being said, it must also be noted that since the 1980s there has been a growing concern over tectonic expression in architecture which, according to Kenneth Frampton, is 'the structural component *in se* but also the formal amplification of its presence in relation to the assembly of which it was part'. The term tectonic 'not only indicates a structural and material probity but also a poetics of construction, as this may be practiced in architecture and the related arts' (Frampton 1996: 520). Frampton's attention to the structural unit as the essence of architectural form that merits more attention than spatial invention and novelty is obviously polemical and is directed against postmodernists like Robert Venturi and their commodification of architecture. In his advocacy of the tectonic nature of architecture, Frampton refers to Marco Frascari and the crucial role the latter attributes to the architectural joint (ibid., 526). Marco Frascari as early as 1984, in his essay 'The Tell-The-Tale Detail' advocates Carlo Scarpa's work by valuing his sense of detail comprised by his expressive and perfect building joints. According to Frascari:

> Architecture is an art because it is interested not only in the original need for shelter but also in putting together spaces and materials in a meaningful manner. This occurs through formal and actual joints. The joint that is the fertile detail is the place where both the construction and the construing of architecture take place. (Furthermore, it is useful to complete our understanding of this essential role of the joint as the place of the process of signification to recall that the meaning of the original Indo-European root for the word *art* is 'joint'.) (Frascari 1996: 506, 511, 512)

Frascari's and Frampton's advocacy of architectural tectonic expression had a specific target, as mentioned previously, but may have contributed to the recent hypostatisation of intensive tectonic expression in folded architecture. The ill-fated idea that tectonics, spatial novelty and imaginary formal manipulation may be, after all, all that architecture is about, did not spring

out of nothingness but was gradually incubated in the context of the most authoritative theory and architectural practice during the 1980s and 1990s.

As much as the superficial applications of the fold in architectural theory and practice are genealogically understandable, a serious consideration of its immanent role will reveal its far-reaching consequences and ramifications. Mark Jackson pinpoints the dangers of commercialism in folded architecture (Jackson, nd: 4) but argues that folded or cyber architecture actually redefines 'our notions of spatiality, habitation and form making' with serious ethical consequences (ibid., 1). The problem is that folded architecture is still new, and this is precisely the reason why it is still quite inexpressible (ibid., 3), namely, far from being totally understood. Jackson critically examines the work of Stephen Perella whose hypersurface architecture indexes Deleuze's fold. Perella's pixel architecture delineated on the surface of a computer screen invents through digital technology a new plasticity in activated surfaces, but Perella's endeavour is restricted to formal manipulation alone with no regard to function or any other considerations, and is thus quite conventional (ibid., 3–6). Marcos Novak's liquid architecture identifies architecture with information space or cyberspace, a transcendent order which again proves to be a quite conventional interpretation of Deleuze as the latter emphasises the immanent place and role of the fold (ibid., 6, 7). Jackson favours more Greg Lynn's appropriation of the fold via the latter's notions of exact, anexact and inexact architectural making (ibid., 7). Architecture, according to Lynn, may not continue to be fascinated with static and closed forms but must venture into anexact or formless design as it is occasioned by computational processes (ibid., 7). Computational processes allow flexible surfaces, measurable constraints of parameters that enclose limits without defining them and evolution in time (ibid., 8). The stake is animate form but Lynn, according to Jackson, fails to understand Deleuze's fold as a diagram which mediates between virtuality and actuality and mistakes it for form. However, always according to Jackson, it is not a form, it is rather a force (ibid., 9). What is therefore meant by force and form is different in Deleuze and Lynn. Lynn, again according to Jackson, folds too soon, thereby limiting his engagement with the virtual as force. Virtual reality is not a threat to reality but is rather its surfeit. Virtual reality means the plenitude of reality. Deleuze's virtual abstraction is inherent in actuality: 'it is the potential for the new in what is actual', (ibid., 11), according to Brian Massumi. The virtual should not be confused with the complex space allowed by the technology of multimedia but is inherent in the actual world (ibid., 11).

> Hence the stakes for the discipline of architecture are not in the presentation of new forms responding to the technological innovation or imperatives of digital technologies. Rather, they are in the activating of virtualities in what is already

> actual, the horizons of which are revealed in the capacities of digitized technologies to confound the near and the far as the non-local. . . . We may begin to recognize architecture, abstractly, as a diagram of power, or technology of power, and it is at this level, of regimes of power productive of our forms of knowing that the imbrications of digitization and architecture may be recognized. (ibid., 13)

THE ETHICAL IMPLICATIONS OF THE FOLD

Although Jackson alludes to the ethical ramifications of his elaboration on folding, these are far from expressly stated. When architecture seeks no longer the stable forms but rather the 'architectonics of continuous variation and movement' (Loo 2013: 245), then the ethical questions become pressing and difficult to handle, in the context of a discipline that disavows one of its principal premises, namely, strength and stability. Of course theorists like Karen Burns and Cameron Duff have attempted, fairly recently, a demarcation of the ethical realm in Deleuze's reflections on the fold and architecture but I believe that the point of the ethics which issue from the fold is still quite pressing (Burns 2013: 15–39; Duff 2013: 215–29). Deleuze himself has alluded to ethics when he claimed that the permanence of the law is substituted by the fluctuation of the norm or even more explicitly when he argued that 'the series of curves imply constant parameters for each and every curve and the reduction of variables to a single and unique variability of the touching or tangent curve: the fold' (Deleuze 1993: 19). One might easily wonder here whether these parameters in question for each and every curve might be ethical or representational of forms of life from the point of view of course of a single and unique variability which is that of the fold. I cannot think of a reason to exclude such a conclusion or an ethical consequence of Deleuze's writings which would nonetheless require extended qualification.

The person however who systematically exploited the question of ethics as it issues from the fold is Luke Feast in his article 'The Discrete and the Continuous in Architecture and Design' (Feast 2006). There, Feast argues for an immanent ethics of sustainable design issuing from the fold rather than an external moral code of green design, associated with deconstructivist architecture (ibid., 1). His argument is based on the distinction between discrete and continuous multiplicities or on a difference between a metric space that can be reduced to an aggregate of points and a non-metric continuum that can only be qualitatively differentiated in time (ibid., 1). Obviously Feast favours the latter, namely, the continuous multiplicities which are inherently dynamic and temporal (ibid., 6) and are based on a non-metric continuum which supports folded architecture with its new logic of curvilinearity (ibid., 2). Folded architecture espouses a 'flexible complexity' which is considered as organic

or biomorphic and is thus characterised as ecological (ibid., 3). This flexible complexity concerns more practice than architectural form and means a concern with the qualitative change of duration (ibid., 8); it is a complexity which can never be completely materialised or explicated but always involves 'the potential for divergence with any given unity' (ibid., 2). Flexible complexity entails a kind of design that constantly intervenes with open proposals and welcomes change (ibid., 2) in order to correspond to the increasing complexities of today's societies. The design ethics is not external to the process of design but immanent to it, and this is precisely the reason why Feast views it as an ecologically sustainable design. The designer is a conceiver of scenarios (ibid., 3) devoted as she or he is to strategic planning and intervention which are morphogenetic (ibid., 9). The architect is thus not a producer of material forms but someone with an ethos of care (ibid., 4) privileging becoming rather than being (ibid., 9), with an engagement with duration and the virtual rather than with space and the actual (ibid., 10). Feast uses the well-known, standard notion of sustainability. In his own words,

> Sustainability in design is a temporal concept that is linked to the idea of sustainable development, in which human activity is conducted in a manner so as to maintain the environment and quality of life for future generations. The concept of sustainability developed as an alternative to throwaway culture and refers to the ability of ecosystems to maintain a form of dynamic stability which enables them to continue over long periods of time. (ibid., 10)

Feast's idea is to make 'a longer lasting product' rather than 'making a product last long'. Apparently, the point is not to make eternal products that resist time but products which are flexible and change according to the changing demands imposed on them by their users (ibid., 10). Such design products involve 'individualisation' and 'personalisation' for, in fact, they are less conceived as finished formal entities but rather as 'assemblages of elements' which evolve in time through upgradeability and repair, exactly like computer software, to meet the people's changing needs and desires. Folded architecture is marked by flexibility because of its adaptation to complexity, namely, to the unforeseeable array of events constantly emerging in contemporary societies and the uncertainty of future situations (ibid., 11).

Apparently, the distinction between a 'longer lasting product' and 'a product that lasts long' is a subtle one and perhaps a little difficult to demarcate clearly or qualify distinctly. That being said, the reader can clearly grasp the gist of Feast's proposal, namely, a flexible architecture which welcomes change and allows upgradeability to meet our highly complex contemporary world. The immediate consequence of Feast's claims is that architecture is viewed as some form of *bricolage*, the term Claude Levi-Strauss coined to describe one of the typical traits of mythical thought in his seminal book *La pensée sauvage* (Levi-Strauss 1962: 26, 27). According to Levi-Strauss,

mythical thought which is typical of the *bricoleur* contrasted to scientific method which is typical for the engineer. The *bricoleur* selects his means from his immediate environment and whatever is readily available, no matter their heterogeneity, and adapts to them in order to accomplish her or his task in contrast to the engineer who secures the appropriate materials and tools specifically geared for her or him in order to attain her or his purpose (ibid., 26, 27). Despite Le Corbusier's admiration for 'the aesthetic of engineers' professed in his early seminal book *Vers une architecture* (Le Corbusier 1995: xvii), his example is used by Colin Rowe and Fred Koetter to illustrate architectural bricolage rather than monumental, total planning and design (Rowe and Koetter 1978: 86–117) deemed more typical of Mies van der Rohe's architecture. It goes without saying that architectural bricolage is by far more ecological and sustainable than architectural engineering in accordance with Feast's thinking.

Thus, in my opinion, folded architecture does not reflect an aesthetic mannerism from the part of architects who have easy access to funds and digital high-end experimentation, despite the prevalence of this view in some architectural circles. Furthermore, folded architecture, according to Krissel, means 'a pliable topological space' ruled by the event of the fold (Krissel 2004). Thus it becomes clear, as Jackson argues, that the notion of the fold redefines our idea of spatiality which is neither formal nor formless, as perhaps Lynn might argue, but should be conceived in relation to the virtual and force (Jackson). The virtual means not only confounding the near and the far but also espousing the logic of the non-local. Feast qualifies the architecture of the fold as no longer a science of metric space which can be reduced to an aggregate of points but as a study of a non-metric continuum, allowing for flexible complexity and concerning practice rather than form (Feast 2006: 11). Folded architecture is therefore still interested in abstract space and local places but conceives both space and place in extremely complex ways that involve the non-space, time, care, events, virtual reality and similar considerations which cannot apparently be accommodated by geometry alone. Folded architecture therefore calls for an interdisciplinary approach to design and building.

Folded design is the functional response to the high demands imposed on architecture from our increasingly complex world to satisfy a number of social conditions like social minority concerns and concomitant problems pertaining to the human condition which require that design becomes an immanent developing force of constant adaptation in architecture. Thereof stems the questioning of *firmitas*, whether we understand strength or longevity by this very term and architectural principle. The fold itself as a cultural force of constant movement and development, infinite inherence or inclusion and infinite spontaneity finds its purity of apparition in the historical era of baroque but serves in the writings of Gilles Deleuze as a diachronic mark of

perspectivism, relativism and plurality, inherent in the logic of contemporary subjectivity. Luke Feast's ecological application of the fold and understanding of it in terms of sustainable design is in my view its most compelling interpretation and one which can be counted upon to take best advantage of the intensified tectonic signification and concomitant digitalisation of architecture.

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Chapter 10

Design of Earth Movement: Objects, Buildings and Environment Conceived as Landscape Formations

Konstantinos Moraïtis

Abstract

Though primarily philosophical, Deleuzean theory is in many ways relevant to cultural references of practical importance. We shall discuss in particular its relation to a contemporary conception of landscape, which approaches earth surface as a morphogenetic environment under constant transformation. Initially related to the baroque period, the Deleuzean proposal on 'fold', finally refers to the contemporary interest in topological geometry and moreover to the possibility of expressing this contemporary topological 'anxiety' in design and constructional practices. Originally initiated as a seventeenth-century Leibnizian effort, topology was not largely developed till the end of nineteenth century and the beginning of twentieth century, while its application in urban and landscape design occurred only recently, as a consequence of the advancement of contemporary computational simulation techniques. Thus the well-known Gilles Deleuze's book *Le Pli: Leibniz et le baroque* (*The Fold: Leibniz and the baroque*) seems to present in detail the first pole of the growth of topological thinking; that of the seventeenth century. The second pole, the contemporary one, could be presented by Deleuze's relation to Bernard Cache and through him to the maturation of the computational design practices. What is certainly missing from the Deleuzean project on fold is a detailed reference on geometrical 'anamorphosis' as applied in baroque garden design. Through anamorphic treatment, a hidden undulation of earth surface was created, clearly related to landscape folding formations, as conceived by contemporary architectural, urban and landscape designs.

CORRELATING EARTH MATERIALITY WITH CULTURAL PERCEPTION AND THEORY

Philosophy centres its effort, as Gilles Deleuze and Félix Guattari indicate, upon the formation and elaboration of concepts. However, concepts are not produced in a state of cultural void, in the absence of cultural perception. In this common frame of 'complicity' between theory and social expression, the these two intellectuals were not only proved sensible receivers of social stimuli but also seem to have offered, directly or indirectly, important suggestions, concerning contemporary design at many different scales of approach.

It is probable that the most important contribution of Deleuze and Guattari, in the above field of design practices, beyond the concept of geophilosophy (Deleuze and Guattari 1991) and the description of the *Three Ecologies* (Guattari 1989), concerns the inquiry on folding formations, as presented in Deleuze's book *Le pli: Leibniz et le baroque (The Fold: Leibniz and the baroque)* (Deleuze 1988). Though initially associated to the historic period of baroque, this theoretical approach may be correlated to a more extended research concerning 'normative patterns of variation', able to produce a 'regulatory law', a research that reaches its highest interest during the contemporary era. In this overall context, Gilles Deleuze, and his scientific companion of Bernard Cache, indicate topological mathematics as an epistemic paradigm which, initially corresponding to the intuition of earth bas-relief under transformation, may be applied in many different design perceptions, concerning objects, buildings as well as urban or natural landscape.

Thus the multiple modes for the design of human constructions may converge towards one need only, that for the definition of transformation principles; principles for which earth movement seems to present the fundamental perception of reference, as the homonym text of Bernard Cache's states (Cache 1995). 'Earth', Deleuze and Guattari insist on, 'is not just an element among others (of equal value); it reconnects all other elements in one and the same embracement' (Deleuze and Guattari 1991). However, such a monistic 'embracement' could only be realised in the interior of a uniform conception, able to include both subject and objects, and able to include, without diffidence, both territorial res extensa and cultural res cogitans. Thus we may comprehend the continuous effort for the 'deterritorialisation' of earth, the continuous theoretical effort to move from the critical issues concerning the immediate object materiality to critical issues concerning culture. It is under this scope that we conceive geophilosophy as a term conducing to an extended synergy of supplementary aspects concerning many different epistemic and epistemological features of contemporary societies.

Such a multivalent approach may refer to the materiality of earth, while simultaneously transcending tactual feeling, by proposing polymorphic inquiry in fields concerning topological geometry and computational

simulation, or psychoanalysis and neurophysiology, or even moves to domains closer to my own interests as those correlated to design practices. Moreover, such a multivalent approach could be easily associated to a landscape orientation, where material, natural or man-made substratum, coexists with cultural immanence; a cultural landscape approach, where both geohistory and geophilosophy occur, as offsprings of earthy materiality.

It is towards cultural landscape that Guattari's affirmation for environmental awareness is also directed, we believe, when insisting on the co-development of three directions of ecological awareness, with natural, social and mental partial orientation. Thus on the alleged uniform singularity of cultural landscape concept, a plurality of partial assumptions may be projected, a multiplicity of references, concerning nature and society simultaneously, describing cultural present or historic past, contemporary stimuli or the emblematic past forms, as those associated to the folding perception of the baroque period.

On this very last reference our chapter would like to insist, in order to indicate an absence, in Deleuze's exhaustive presentation of folding baroque formations, an absence concerning the lack of indications about the French seventeenth-century garden, the French garden and landscape 'formal' art of the baroque period. This garden and landscape art, using folded formations, seems to be paradigmatic for the comparison of the contemporary epistemological and epistemic landscape persistence, with the central, in its own period, landscape conception of the seventeenth-century European civilisation. However, if baroque cultural landscape is characterised by the hidden perspective of anamorphosis, the 'Anamorphosis Abscondita' (Weiss 1992), contemporary cultural landscape is conspicuously insisting on folding forms.

THE REFERENCE TO 'GEOPHILOSOPHY'

It appears rather imprudent, for somebody not belonging to the discipline of philosophy, to face philosophical paradigms standing on philosophical 'ground'. Let us insist on the term 'ground'. Let us insist on the affinity of Gilles Deleuze's proposals, as well as of Félix Guattari's, and in addition, of Bernard Cache's, with the domain of thought described as 'geophilosophy'.

'Earth', Deleuze and Guattari remark (1991; we have already quoted this citation previously), 'is not just an element among others'. 'It unifies all other elements in one and the same embracement'. Thus Deleuze and Guattari speak in favour of the land, correlating its material substratum to the total production of human civilisation and culture. They speak about land essence, comparable to the multiplicity of 'cultural landscape', and they allow us to refer to mental abstraction to this kind of mental 'schematism' described as philosophy, while using, at the same time, the consolation of material reference.

PHILOSOPHICAL CONCEPTS AND THEIR RELEVANT CULTURAL AND 'EPISTEMIC' DISCOURSE

Meanwhile, we shall step aside from our main subject, in order to refer to another lateral reasoning, important however for the 'external' perception of our central topic.

Every theoretical, philosophical approach may be correlated to a general context of civilisation or culture. It may be correlated to a global 'epistemic'[1] domain, produced by systematic scientific formulations or by rather indeterminate, generalised ideological attitudes, out of which, in every historic period, knowledge emerges. In our case of reference we may identify, in a clear way, the effort for response to central, actual problems of our contemporary civilisation and culture. We may identify a tendency for 'applied' theory; the determination for answers that refer not only to our intellectual past, as for example to Leibnizian philosophy, but, even more, to subjects concerning our immediate, contemporary interest. They may refer to the quality of the 'movement-image' (Deleuze 1983), or of the 'image-temps' (Deleuze 1985), to the three possible directions of the contemporary ecological awareness (Guattari 1989), or to the branch of the mathematical theory described as 'topology', a mathematical approach striving for the last three centuries to subvert Euclidian stability by replacing it with a geometrical approach of continuous transformation.

Insisting on the last part of our previous remark, we may comment that for Bernard Cache, a theoretician interested in design at many different scales, from furniture and industrial design to landscape design, such a turn towards topological geometry seems expected, as a result of his immediate relation to computational science. In comparison, for Deleuze, the interest for 'folding' forms reveals his effort for the meditative repetition of what, in Cache's case, appears to be an active immediate response. It clearly reveals Deleuze's effort, and we shall insist on the word 'effort', to place his theoretical investigation in the epicentre, in the focal point of the relevant epistemological inquiry. Evidently, contemporary topological theory was not first brooded in this French philosopher's pages. Deleuze's book *Le Pli: Leibniz et le baroque* (*The Fold: Leibniz and the baroque*), first appeared at its initial French edition in 1988, while René Thom's text *Stabilité structurelle et morphogenèse* (*Structural Stability and Morphogenesis*), a stirring topological contribution to contemporary thought, was already published in 1972. In Thom's writings, two centuries of previous mathematical efforts seem to attend a conclusive and moreover a culturally influential expression.

We have to remind in addition that René Thom was also awarded, in 1958, with the Fieldsen Medal, an award for mathematical science comparable to the Nobel Prize, and was appointed, in 1972, as one of the directors

of the seminar 'Philosophie et mathématiques – 'Philosophy and Mathematics' of the École Normale Supérieure. The object of the seminar was presented under the following explicit description as 'la confrontation des idées vivantes, sur les rapports entre la philosophie et les mathématiques – the confrontation of the living ideas between philosophy and mathematics'.

Thus we may insist on the fact that the interest of the contemporary thought for topological mathematic approach was already vivid since the middle of the previous century and, moreover, surpassing mathematic enclosure, was already extended to unexpected domains of theoretical inquiry.

In 1961, in his seminar 'L' identification – The identification', the famous psychoanalyst Jacques Lacan referred to topological formations as those concerning 'Möbius Strip' or 'Möbius Loop', the topological ring of 'Torus' and the 'Klein Bottle', first described by the German mathematician Felix Klein as 'Fläche – surface', 'Kleinsche Fläche – Klein surface' and then misinterpreted as 'Flasche – bottle' (Evans 1996). In 1971 Lacan added to his theoretical references 'the Borromean Rings' or 'Borromean Knot', a formation of three interlocking rings that can describe, according to him, the impossibility to dissolve, the inseparable correlation between the three aspects of mental life, the Real, the Symbolic and the Imaginary (R-S-I) (Evans 1996). We may express our skepticism about these topological references of Lacan's and comment that they seem to be limited to the level of iconic metaphor principally, without a deeper mathematical intuition. However, they prove that topology already acquired important publicity during the period of 1960s and 1970s. That it succeeded in attracting the wider scientific interest and produced a special 'topological Zeitgeist', a special spirit of the time, oriented towards topological references; a dominant school of thought, a 'topological epistemic' tendency, which typifies and influences our overall culture till now.

ONTOLOGICAL APPROACH AND LEADING SCIENTIFIC PARADIGMS

Let us firstly declare our absolute respect towards the theoretical contribution of Deleuze and Guattari. Then let us also declare our equal respect for the historic context enclosing their oeuvre. Then we may calmly validate their interest for the leading scientific paradigms of their era. Leading paradigms in the domain of social sciences, as well as in the domain of positive sciences, for example, in association with to mathematical theory, constitute the most important remark for the subject of our inquiry. If we agree with the crystal clear statement of Bernard Cache, according to which 'we ought to understand that nowadays (positive) sciences precede philosophy',[2] then we may estimate the crucial cultural 'trauma' that Western thought underwent, when

the leading proposals of philosophy ceased to guide scientific thought. Thus we refer to the ultimate contemporary maturation of an older epistemological tendency, already present during the eighteenth century, according to which ontological philosophy does not coincide anymore, as far as its representatives are concerned, with the central production of positive sciences.

TRYING TO REVERSE OUR PREVIOUS ARGUMENT

It is rather impossible, even if we try to move away from the academically constituted philosophical discipline, to detach the wider knowledge production from the need for abstract approaches and conclusions, concerning the constitutive principles of existence, knowledge and political or social praxis, or even aesthetic response. Thus we have to acknowledge to the representatives of the positive sciences, especially to the most important among them, an already-acquired philosophic tendency, certainly correlated with a fundamental intuition earned through their interest for concepts concerning the hermeneutics of human mental structure or the 'structure' of the external world. Thus, if Cartesian and Leibnizian contributions were not less in the scientific discipline of mathematics than in the realm of philosophy, we have also to assume that René Thom's mathematical propositions are not deprived of philosophical depth. On the contrary, they possess a central general theoretical interest, independently of their initiator's participation to organised activities of relevant academic orientation. If scientific discourse concerns in general, as well as in particular, an activity of 'symbolic' order, of symbolic schematic interpretation, we may also detect, in this schematic identity, a similarity to the abstract schematisation of philosophy. Then we may argue about the abstract schematisation in applied sciences or even representation practices and try to investigate the importance of Deleuze and Guattari's proposals, especially concerning such references as those on folding forms, for contemporary architects and landscape designers.

PHILOSOPHY AS ARCHITECTURE AND CONSTRUCTION AS PHILOSOPHY

It is not unexpected that contemporary architecture seems to be closely correlated to contemporary theory of social sciences and philosophy. Foucault's investigation on penitentiary architecture or even his 'heterotopia' concept, Jacques Derrida's deconstructivism and Deleuze's fold project offered central inspiring ideas for contemporary architects.

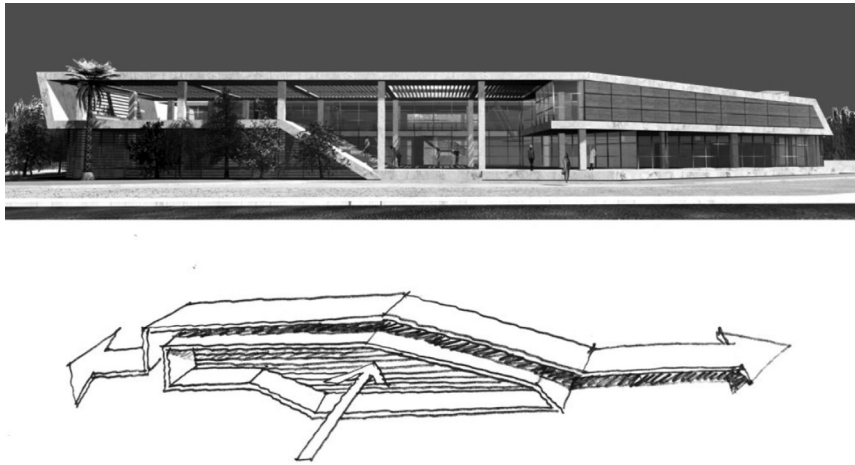


Figure 10.1. A building structure conceived as a landscape folding formation. Town hall building in Corinthia, Greece ('Arsis Architects' design team, 2003).

In a history-based explanation, architecture seems to present an emblematic perception of organised entities for mental, cultural, political as well as for expressive or constructional formations. What topology and the corresponding folding formations indicate is, at the same time, a change in the perception of external reality, not accepted as stable any more, as well as an alteration of the forms of representation and construction practices. A change that, though introduced in the baroque period by Leibnizian 'Analysis Situs',[3] was not vigorously correlated to constructional applications till recently. It was only after the development of computer-aided design that topological formations were largely applied in cultural expression. Thus, the Euclidean stabilised approach has been soon replaced by the interest for 'morphogeneses', for formal changes that are also described as 'animated' forms. If we describe earth surface as a 'plane of immanence', then we may detect in it a cultural identity, a spirit of culture correlated to central conceptions concerning place and society associations. A current designation of this association is the term the terms 'Cultural Landscape', already used at the introduction of our chapter. It is 'Cultural Landscape' that we correlated to Geophilosophy, and we may also use the same expression in order to explain Guattari's *Three Ecologies*, or the folding principles of contemporary design. We have to insist on the fact that folding formations of contemporary design are not 'natural'; they are not nature, or natural landscape in itself. What they offer is a schematised, culturally schematised, description of earth under transformation, of landscape schematised through cultural intervention.

It was René Thom who used the earth bas-relief metaphor, a landscape surface under transformation, in order to explain topology. If we accept this metaphor as a description able to present a new view of structures in a condition of formal changes, then we may also accept the perception of landscape surface as a structural paradigm unifying natural and man-made reality. This perception can even be used as a metaphor of social change, a 'field', a 'landscape' of continuous social transformation, which could probably be previsioned and prescribed.

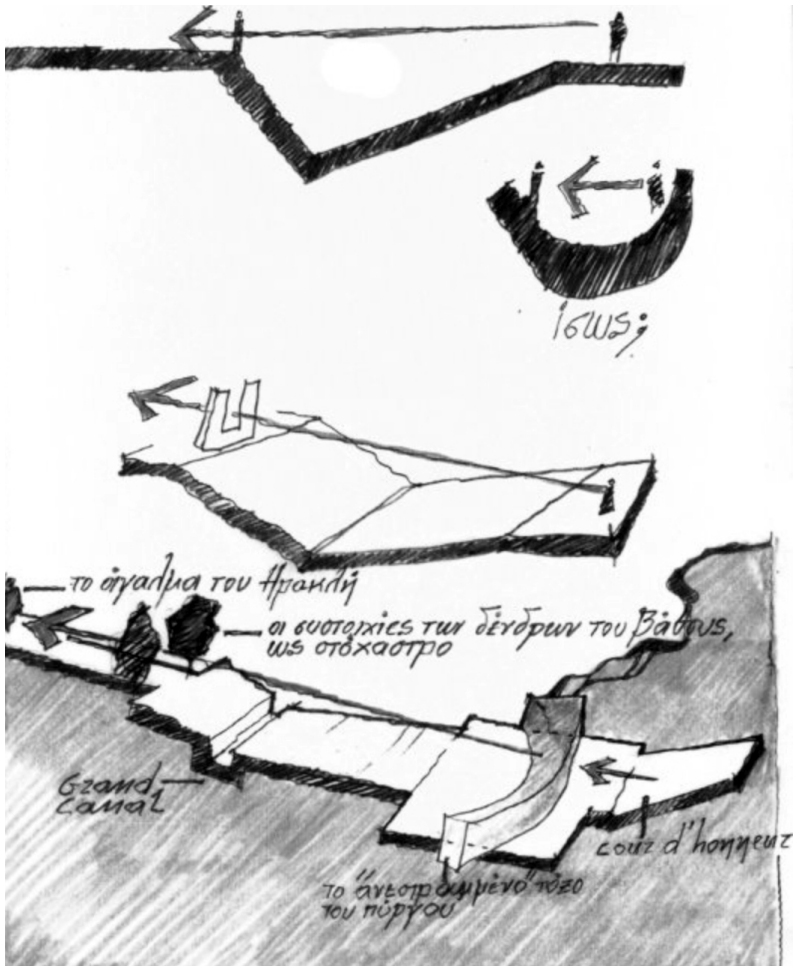


Figure 10.2. 'Une baliste optique – An optic ballista'. Author's sketches describing the illusionistic distortion of the space depth in Vaux-le-Vicomte gardens, through intentional use of the folding formation of the earth surface.

It is in that very way that contemporary architecture uses landscape formations; even human or social sciences use similar associations. Lacanian psychoanalysis introduced, as we already remarked, topological metaphors, in order to explain mental transformations, and George Heimonas, a Greek psychiatrist, neuroscientist and writer, illustrated mental activity using topological qualities, comparing it to a drop of ink, to a medusa or to the fish named ray. 'What may indicate the resemblance of the medusa to a drop of ink falling in the water? D' Arcy Thompson asked many years ago' (Heimonas, 1987). 'Which is the correlation of the thought, similar to the ray that suddenly shivers, which is the correlation of this inexplicable and sudden representation of mine, concerning intellect, to the widen forms of the topological transformations?' (ibid.)

In urban descriptions, we tend to speak in the past decade about urban 'sprawl', using the term 'informal', quasi-natural reference, in order to present the uncontrolled development of urban transformations, social and spatial at the same time. In this sense we may seriously discuss the term 'informal', as describing uncontrolled development, related concurrently with the negation of a stabilised Euclidean perception. Topology, we may think, could correspond to the negation of stabilised formality. However, it could also be associated with the search of principles, which could offer a 'prescription' of change, a normative prediction of transformative sequences.

THE LACK OF DELEUZEAN REFERENCE TO THE FRENCH BAROQUE GARDENS AND TO THE ILLUSIONARY FOLDS OF THE PERSPECTIVE 'ANAMORPHOSIS'

In Deleuze and Guattari's writings, as well as in those of Jacque Derrida and Michel Foucault, the tendency for theoretical approaches synthesising political, cultural and epistemological qualities in an overall epistemic atmosphere is immediately perceivable. Moreover the term 'épistémique – epistemic', as introduced by Foucault, is used in this very way, in order to describe the historic identity of historic periods, that of the Renaissance, 'l'âge classique – the classical age' and modernity. However, this epistemic periodisation may also be applied to the still unidentified physiognomy of the present cultural expression, of the contemporary historic period, out of which Foucault himself, as well as the other three intellectuals came from.

It is important to remark again that all four of them constantly prove an epistemic association to their cultural and scientific environment, insisting on their capability to assimilate in their discourse the perception of the arts, the space perception as well as more specific aspects, as for example, those

correlated to architecture and landscape design. The perception of architecture and landscape will not only offer to them the denotative images that will make their theoretical assumptions visible and accessible to the public vision, but it will also carry an important connotative meaning; it will declare that their theories could be culturally accomplished.

However, though the previous correlation proves the power of the architectural and landscape paradigms, we find neither in Foucault's *Les Mots et les Choses* nor in Deleuze's *Le Pli* any reference to the impressive cultural example of the baroque French Garden in general, or to its deliberately folded surface in particular. It is strange that none of them had commented on Anamorhosis, the illusionary perspective, extremely popular in the visual arts of the baroque period and extensively used in the design of the baroque gardens. Having as their final aim to impress the visitor, the baroque gardeners distorting on purpose the feeling of space depth, created to the visitor the expectation of a smaller distance to the end of his walk. The comparison of this false expectation to the real prolonged movement, produced the sensation of an endless itinerary, of a space extension much bigger than the one pre-estimated (Weiss 1992). The scenography of this perspective illusion, described by seventeenth-century mathematicians as 'optic thaumaturgy', or optic magic, was realised through distortion of geometrical schemes, in the reverse way that normal vision could expect. But it was also realised through the deliberate distortion of the earth surface, by well-organised folding land constructions, as described in our images (Moraitis 2011).

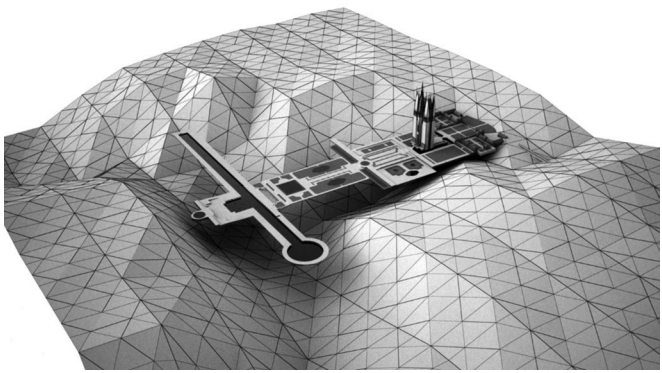


Figure 10.3. Illusionistic distortion of the space depth in Vaux-le-Vicomte gardens, through intentional use of the folding formation of the earth surface (vertical measurements are ten times exaggerated). Three-dimensional presentation designed by architect T. Petras as exercise in author's post-graduate seminar on 'History and Theory of Landscape' (NTUA – 2010).

It is not strange that the baroque spirit, accustomed to folding forms, to the deliberate distortion of matter, as well as to the perspective illusions of 'Anamorphosis', introduced both tendencies in landscape formations. Thus it proves a concrete correlation between 'Gea', earth in ancient Greek, with theoretical schematism, with mathematics and philosophy. It proves, namely, a 'geosophic' tendency, correlated to the historic idiosyncrasy of its era. Thus it is also not strange that, during this same period, Leibniz first introduced the theoretical predecessor of modern topology, his 'Analysis Situs'.

THE SEVENTEENTH-CENTURY PREDICTION OF THE CONTEMPORARY FOLDING LANDSCAPE VISION

Proposing a contribution to the Deleuzian analysis of the Fold would be extremely arrogant. However, our involvement in landscape design, and moreover our personal academic research on the subject of landscape and garden art (figure 10.3), may offer stimulating proposals.

As we have already stated, it is more than apparent that reference to Leibniz by Deleuze is not only correlated with the interest for the history of philosophy, but is rather largely triggered by the appreciation of topology, as a key theory of modern and contemporary epistemological avant-garde. Insisting on the fact that this 'restless', turbulent vision of the exterior world has been already prescribed by the seventeenth-century baroque period is for sure a brave historic realisation. Cultural sequence is not a matter of linear evolution. Thus Euclidean reasoning and neoclassical balance, as presented by the Enlightenment, could express, in some aspects, a cultural 'regression',[4] towards the consolidation of Cartesian stability, as applied in solid and well-proportioned building structures, in comparison to seventeenth-century epistemology of the 'Analysis Situs'.

However, landscape design had never returned to the formal attitude of the Renaissance garden art. It rather developed, during the baroque period, an interest for concealed folding formations, and it introduced, during the rococo period, design instability in the first curvilinear garden and park formations, which seem to be closer to folding iconography than the early baroque gardens. Little by little nature-like morphogeneses were introduced in landscape design by the eighteenth-century landscape architecture, in complete contradiction to the classical building immovable forms of the same historic period.

We could thus admit that a continuum of landscape design attitudes existed that might associate baroque gardens to the latter rococo curvilinear design principles and afterwards to the eighteenth-century nature-like landscape

architecture. We could then even correlate this continuum to the contemporary interest for 'animated' landscape formations, to the interest for earth movement, 'pour une terre-meuble' (for a moving earth)[5] as vividly described by Bernard Cache.

Yet this tendency already existed in baroque garden art, in embryonic state, as we previously remarked. It already existed in the illusionary treatment of the site that applied the trompe d'œil perspective of 'anamporhosis', which used gentle, not immediately perceivable undulations of earth surface, in order to distort the feeling of the real space depth. Nevertheless this gentle transformation of the earth bas-relief did not abolish, at the level of the horizontal projection, the orthogonal, symmetrical structure of the overall composition.



Figure 10.4. Bust of Louis XIV; marble portrait by Gian Lorenzo Bernini, proposed by Gilles Deleuze as characteristic of the seventeenth-century baroque agitated folding morphology.

It is probable that the lack of reference to baroque garden art, in Deleuze's presentation, has to do with the previously described controversy between concealed folding formations and apparent orthogonality. We have to mention in addition that detailed analyses on the application of 'anamorphosis abscondita'[6] appeared after the publication of Deleuze's *Fold*. Thus the French philosopher could not have known Allen S. Weiss's book *Mirrors of Infinity: French Formal Garden and 17th Century Metaphysics* (1992) nor the detailed terrain analysis of Vaux-le-Vicomte and Versailles baroque gardens produced by eminent landscape historians as Clemens Steenbergen and Wouter Reh (1996, 2003). It is under the contribution of similar recent scientific approaches that we may now assume that the baroque period does not only prevision modern and contemporary topology through Leibnizian 'Analysis Situs' but moreover prefigures modern and contemporary landscape formations.

ETHICAL ANALOGIES

Thus it would be interesting to ask about a deeper epistemological and epistemic correlation between the baroque seventeenth century and the contemporary era.

Is there, besides a superficial resemblance, a deeper ethical analogy between Leibnizian intuition of the world and contemporary computer-simulated sensitivity? Is there a similar 'plissage-folding' effort between Leibniz's short text *Principes de la nature et de la grâce fondé en raison*, also known as *Monadologie* (2002), and contemporary mathematical computer-aided modelling? 'Mind' or mental activity gradually descending towards external materiality, in order to 'abstract' it and pull it up towards a higher spiritual state – does this resemble to the perception and representation of earth surface motion, 'schematised' and thus mentally abstracted? Folding morphology, earth surface under continuous transformation, does it connote both mental and material animate reality as nowadays worked out and presented through computing simulation?

It seems that the previous two-sided metaphor of folding forms, correlated to earthly landscape surface and to human cognition as well, illustrates a central, ethical characteristic common in both the baroque seventeenth-century period and contemporary ingenuity. There is no isolated external extension, given in an immovable way, but rather a continuous confrontation of mind and of 'terre-meuble' (moving), of movable natural surface, agitated not only by natural disturbances but also by human mentality; the latter trying to participate to exterior forces and moreover to predict, being itself in motion, the animate, movable, active supposed this alleged exteriority. This 'Supposed' exteriority as the

contemporary figuration of the agitated, folded forms totally negates the Cartesian barriers between internal res cogitans and external res extensa.

To pose the previous statement in an additional supplementary way, continuous transformation of earth surface, of 'terre-meuble' (moving), may be presented as analogous to the continuous transformation of abstract and schematising reasoning, mentally exerted and technically introduced in our everyday culture, by computing simulation processes. Thus prefiguration of natural or constructed exteriority and description of human cognition as well, being both under continuous transformation, seem to converge in the domain of mathematical modelling, the latter being par excellence the vehicle of contemporary controlling processes, of representational and constructive contemporary ethos.

However, it was probably the same condition that underlaid the 'illusionary' geometry of baroque, in sculpture, architecture, dressing manners, even in culinary civilisation and in garden design expression principally. A folding feeling that was firstly a descending movement of human intellect towards earth basrelief and then a bottom-up movement of mute materiality towards the geometric eloquence of the French formal garden, in Vaux-le-Vicomte or Versailles. After the baroque design intervention, the surface of the landscape was no more a material surface only. It was also projected to the eyes of the observer as an immaterial optical illusion, created by the treatment of the 'anamorphic' perspective. What was seen was principally a representation, a 'simulation' of reality, a virtual condition that transformed earth perception in accordance with to cultural volition, 'à volonté culturelle', or, using a more exact expression, in relation to the volition of the schematised and structured civilisation.

It is this Leibnizian and landscape-oriented idiosyncrasy that Michel Foucault is missing in his illustrious panorama of the 'classical age', 'de l'âge classique'.[7] First of all, even the use of the word 'classical' is not the more appropriate to designate baroque-specific expressive style. Louis XIV's sculptural bust by Gian Lorenzo Bernini had nothing to do with the Euclidian classicist stability and balanced eurhythmy. Thus it would be better to accept Bernard Teyssèdre's[8] remark that we may observe, during the same period, differentiated cultural expressions. The classical rationalistic tendencies, as expressed in the seventeenth-century Dutch territory, appear to be rather antithetic to the baroque exuberance, as expressed during the same period, in the French dominion. In *The Order of Things and in The Fold: Leibniz and the baroque*, it is the exaggeration of the French baroque gardens, the desire of the books' authors to transgress common sensitivity, their 'thaumaturgic'[9] aspect, that is missing. What is a *thauma* in Greek, a wonder in English? It probably describes the introduction of a superior volition in conventional reality, in order to radically transform it. If we surpass theological

argumentation, then we may speak of a radical intervention to the 'logic of senses', as radical as the one introduced in our present sensitivity of place and time by computational simulation.

NOTES

1. Term introduced by M. Foucault, in his work *Les Mots et les Choses, The Order of Things* (Foucault 1966). It derives from the noun 'episteme' which, in its own turn, derives for the Greek work 'επιστήμη – science'. However, according to Foucault's usage of the term, 'épistème' does not describe the exact field of science anymore. It describes, in a more generalised way, the historical 'a priori' that grounds knowledge and its discourses; thus it represents the general cultural condition of their possible formation within a particular epoch.

2. An important Cache's remark, during his lecture on 'Albrecht Duerer and the Anticipation of Some Deleuzian Concepts in His Treatise of Geometry' ('Gilles Deleuze and Félix Guattari: Refrains of Freedom', International Conference, Athens, Greece, 24–26 April 2015).

3. Leibniz was the first to introduce the term 'Analysis Situs', later used in the nineteenth and twentieth centuries in order to refer to what is now known as topology. In 1985 Henri Poincaré published a seminal mathematics paper, in the *Journal de l'École Polytechnique*, using the same Latin title. This first publication of Poincaré was followed by five additional supplements between 1899 and 1904.

4. In psychoanalytic terminology, 'regression' refers to a backward movement towards a previous period of personal history, where present problems appear not to exist. In a certain way historic regression to classicism presents for modern European societies of the eighteenth century a reference of formal 'stability' in aesthetical and ethical terms, a 'solid' reference of 'ideal' cultural and political conditions.

5. Reference to the title of Cache's well-known book (1983).

6. Adjective derived from the Latin verb 'abscondo' that means to hide, conceal, cover, shroud; thus 'abscondita' means concealed or hidden.

7. It is in this description of the 'classical age' that the baroque period of the seventeenth century in France is presented, in Foucault's *Les mots et les choses: Une archéologie des sciences humaines* (Foucault 1966), translated in English under the title *The Order of Things: An Archaeology of the Human Sciences* (Foucault 1970).

8. In his introductory 'Presentation' in H. Wöllfin's *Renaissance und Barock*, French translation of the *Livre de Poche* editions (Wöllfin 1967, pp. 7–28).

9. 'Thaumaturgic'is an adjective etymologically related to 'thaumaturgy', to the working of wonders or miracles. The use of the term refers to the Latin title *Thaumaturgus Opticus* of a book concerning anamorphic perspective, written by the Minim friar Jean-François Nicéron in the early seventeenth century. The French edition of the same book *La perspective curieuse, ou magie artificielle des effets merveilleux* (*The Curious Perspective, or the Artificial Magic of Miraculous Effects*) is equally irritating in terms of non-ordinary, 'magic' references.

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Chapter 11

Spatial Transcriptions of the Concept of the Fold in Architecture as a Landscape-Sensitive Approach

Anthi Verykiou

Abstract

The parallel-in-time elaboration of the post-structural assemblies as differential and differentiating by R. Thom and G. Deleuze in the field of mathematics and in the field of philosophy, respectively, yields the departure from a static perception of space towards a dynamic performative engagement of it. This is described by the terms 'mathematical phenomenology' and 'transcendental empiricism', as considered in each disciplinary field. These terms, as we will see in the mathematical cusp catastrophe or Fold model and the conceptual approach to the Fold, are referring specifically to an extended topological approach. The topological mathematical processing relates to spatial assemblies of complex elementary objects. These mathematical and conceptual differentiating assemblies highlight landscape as the central model of contemporary cultural production in general. According to the extended topological view, landscape refers to the coherent condition of theoretical reflection as the deeper meaning of the notion of topos yet emphasising on the singularities and the intensities of a particular real place, and therefore provides a landscape and mental topological continuity.

The construction of architectural imagery which is characterised by the extensive use of computer-aided parametric and associative design approaches enhances the production of multiple outcomes according to generic geometrical definitions. Approaches that finally forming a procedural 'rubber sheet' topological surface.

Hence the abstract and multidimensional topological spaces relate to spatial articulation of theoretical approaches and through the architectural design procedures assisted by contemporary computer-aided constructions of imageries perform expressive approaches to various complex cultural objects

and their variations. These imageries can be described as multiple alternative realisations of the singular.

THE TOPOLOGICAL APPROACH AND THE PERCEPTION OF LANDSCAPE

Differential topological approaches summarise physical changes, mathematical models and contemporary design approaches. Dynamic formations, or dynamic morphologies, refer to a generalised conception of space and geometry and simulate the transformative conditions of physical phenomena and biological processes. The mathematical model of the fold in particular, as a mental representation of singularities defined as 'local events' and referred to as 'hills and valleys', does not yield the terms of gravitational potential of the landscape relief, but data flows eventually blurring the threshold between mental and real, tangible and intangible, introducing what the term mathematical phenomenology describes. 'Ecology is considered as another word for structure' (Sheppard 2012).

We will observe that the 'rigid' field of mathematics seems to be transformed according to the contemporary worldview that refers to a landscape model. The term *landscape* as perceived today concerns a complex cultural construction and bares meaning attributed to it through epistemological developments concerning an energetic or 'ecological' conception of place. Landscape is considered as a system that is in a continuous state of change due to the exchanges between the subsystems from which it is constituted. This view concerns a concept of 'resilience' which allows the overall system to maintain basic features of its structure through internal and external adjustments while exhibiting alternative formal appearances – the phenotype – through a dynamic time-sensitive systemic conception. The systemic conception examines the flexibility to adaptation of complex structures and not of simple objects. Those may also include theoretical constructions and design approaches. Systems theory is linked with the development of the mathematical field of topology and its current use as research and editing tool to different scientific disciplines. Moreover, systems theory underlies the 'ecological' aspect of the landscape.

Topology examines relations (functions) or networks of relations between complex objects considered as elementary. Emphasis on relations is establishing the condition of homeomorphism, and further it concerns the formation of transitional relations, as we will see, to be expressed through the Fold model. But topology acts as a 'spatial metaphor' that is the mental disposition of the perception of a system into a sensible place, topos or landscape (Kosona 2012).

The reference to landscape as 'a model or the fundamental metaphor' (Deleuze 1990) represents an abstract approach that enables the understanding of complex phenomena. When referring to the landscape, the degrees of freedom increasing complexity and diversity become keywords. The abstract mathematical representations or models are performing a variety of connections with different fields of knowledge. So the term 'metaphor' in the context of an ethos that is shaped within the boundaries of the contemporary so called scientific paradigm – marking the removal of the linguistic model – should be attributed to corresponding mathematical terms such as projection of a real phenomenon in a model. And if the term 'model' seems to be teleological or deterministic, it may be argued that it is not static but evolving. Due to this observation, we may eventually lift the teleology inherent to mathematical approaches, because they are constantly broadening their disciplinary boundaries in order to include new mathematical objects emerging through a creative intuitive production. Ultimately, it allows this rigorous discipline to share the terms sustainability, self-organisation processes, evolutionary development and so on, with the ecological approach – dominant in the scientific or naturalistic paradigm – where it seems to fit. Finally it shapes our current understanding of landscape.

Topology provides a framework to evaluate the properties of interest in composite elementary phenomena that are usually examined. Specifically, the catastrophic approach in terms of differential topology incorporates ruptures in this condition of continuity. The surface of the landscape through the notion of the 'fold' refers to gradient relations. Finally, it introduces the concept of weightlessness to the perception of matter considered as matter of expression. The landscape model brings the concept of continuity of differences and ruptures into the catastrophic differential approach and establishes a condition of theoretical reflection, as an intermediate, interstitial space.

The topological spatial metaphor consists mainly of the study of 'spatial adjustment' of a phenomenon, that is, the behaviour of its boundaries under the effect of external perturbations. This is the meaning of the main topological terms 'compactness', 'area or neighborhood' and 'continuity'. The perception of landscape relates to the establishment of complex and extensive interconnections between the natural environment and ecosystems on the one hand, and sociocultural structures and organisations, on the other.

The notion of an organic, fluid design meets ecology as a concept of dynamic correlation of living systems as forms that have temporal transient material nature, which are always in a state of change. And it refers to mathematical models. 'So what appears as inconsistent or complex conditions that may initially be considered as random or chaotic, refers to high-level structured entities that summarize a particular set of geometric and spatial organizations' (Corner 2006).

TOPOLOGICAL LANDSCAPE AND MORPHOGENESIS

Dynamical mathematical models through the geometric intuition of natural processes connect abstract dynamic formalities with spatial expressions. The ensuing morphology of this general approach refers to forms of data organisation. In contrast to the discrete presence of stationary figures in an Euclidean approach, a topological surface is characterised by the continuity of heterogeneous processes. The topological surfaces – as a result of the interaction of mathematics with physics and biology – are generic, regarding to families that describe the evolution of natural phenomena or effects and they serve as tools of description and design expression. They formulate relational 'lattices' or rules of formation and refer more to terms of growth and development corresponding to an ecosystem, rather than to static figural representations. These systems are characterised by their phase spaces, which are conceived as singular instantiations of objects, and the plastic or elastic transitions between their formations, due to the transformative procedural structures. We will call the surfaces formed as models of unfolding time-sensitive (temporal) behavioural processes (particularly biological and physical phenomena) mental or topological landscapes.

Catastrophe theory refers to events which are incidents unknown in nature, occurring in unknown time. We might say that in this way catastrophe theory includes parts of empirical reality as such that are considered irreducible to mathematical control. As René Thom writes, 'they form kinds of islands of determinism, separated by zones of instability or indeterminacy' (Thom 1969, p. 321). These 'islands' stand also as references to the intuition of real physical space, as indicated by the use of terms such as 'hills', 'valleys' and (attractor's) 'basins'. The topological a priori (Petitot 2004) is not metaphysical but a landscape a priori as it is interwoven with the observable morphologies of the real, that is, with landscape perception.

THE MODEL OF THE FOLD IN THE UNIVERSAL UNFOLDING SINGULARITY THEORY: CUSP CATASTROPHE MODEL

In the area of philosophy, the hyperstructural approach, introduced by Gilles Deleuze, suggests that the mathematical model of structural assemblies is the differential calculus (Deleuze 1990). In order to present the dynamic differential and differentiating structural approach developed in the 1960s that emphasises the transformation or the diachronic aspect of structures, we will refer to the mathematical catastrophe theory.

Catastrophe theory, introduced by René Thom in the 1960s, is part of the mathematical area of differential topology, while its main influences

are found in the field of theoretical biology and particularly the epigenetic landscape of Conrad Hal Waddington (Thom 1990). The surface, called the fold, is one of the seven elementary catastrophes in the homonymous theory of Thom (visual reference: Woodcockand Davis 1978, pp. 37, 46). This surface is formed by inclined – for technical reasons – projections of successive sections of a morphology, for example, of a natural terrain. It is formed as the parametric space of a morphology under consideration, which is based on the perception of the actual physical relief. This morphology determines the dynamic equilibrium of a body in this topography. Body and territory are understood as a system considered collectively as an ongoing dynamic process. This process is considered as dynamic due to the equilibrium conditions described for successive positions on the relief, that is, a body in motion. Thus, catastrophe theory examines the potential of the system locally, in positions which are defined as critical or singular points. The singular points, in the corresponding theory of the topologist Marston Morse, are points of local minima and maxima – in the example of the topography they are referred to as hills and valleys. The most singular of these, the points with the highest elevation, are considered as unstable equilibrium positions, on which the kinetic state of the body is more likely to be changed (Morse 2007).

The change of perception of the substrate space constitutes a departure from the Newtonian homogeneous spatial substrate for a relational, dynamic space-time substratum. Due to the relational formation of the body morphology, changes to the potential lead to the perception of a changing morphology of the substratum. This process is called morphogenesis in the dynamic relationship among substratum – potential is attributed to situations of structural instability, which cause the alteration of the morphology. The area of the fold represents the positions where a change of the morphological status is more likely to occur. So the structural instability, that is the fold, can support morphological discontinuity or change and is the carrier of generative procedures.

Notice also that, in the theory of critical points (an intuitive topological approach, as Marston Morse claims) local maxima, local minima and inflection points are considered to be singular points, that is, points listed in a state of unstable equilibrium but referring to different degrees of stability. It is interesting that Morse's theory is based on Maxwell's paper on topography (1870) and uses the landscape analogy, featuring extrema and inflection points as hills, valleys and passes. These points ultimately refer to different equilibrium conditions or the potential of a body on a land topography or morphology. The degenerate critical points of curves or surfaces represent the possibility to change the structure of a curve or surface.

The point of inflection as an intrinsic singularity unlike extrema, maxima (crests) and minima (valleys) is what allows us to think in terms of transformation. Gilles Deleuze in his book *The Fold, Leibniz and the baroque*

calls the inflection point 'the physical point', the plastic or elastic point-fold, which comprises the possibility of the production of new folds. The point-fold is then more fundamental than individual vectors that describe the gravitational or social potential. We are referring here to the example of the city of Lausanne which Bernard Cache includes in his book *Earth Moves* (1995), concluding that the earth is weightless, that it moves as a result of mental perception of change. The inflection point is ultimately synonymous with change. Deleuze reclaims this image for a hybrid approach to baroque in a hyper-structuralistic approach. The inflection point is the point-fold. 'In contrast to the geometric point, the end of the line, the point of inflection is anexact' (Deleuze 2006).

EXAMPLES OF ARCHITECTURAL ANALYSIS AND DESIGN

The abstract nature of the objects or phenomena being studied by the topological mathematics and especially differential topology may bare different conceptual contents. Therefore, the adoption and use of topological mathematics in the area of philosophy, especially in the writings of Michel Foucault and Gilles Deleuze, emphasise the spatial theoretical formulation and the synchronic or diachronic aspect of their structuralistic approaches. Additionally, the topological approach is linked to the design research in the field of the architectural practice, and particularly to the construction of architectural imagery.

In the area of architectural design, we consider the construction of images as the in-between, that is, the symbolic level that mediates and correlates the real and the imaginary. This practice depends upon the available means for producing these images, taking into account the technological possibilities of an era. The first example refers to Eugene Viollet-le-Duc who in the middle and late nineteenth century developed a theory for the structural formation of a landscape relief. So while drawing a series of naturalistic representations of Mont Blanc – static representations – he attempted to study the geodesic structural formation and the transformations of the relief due to glacier formations (1876). For this purpose he used a rhomboidal system in which he integrated and sought to define and control the formation and the transformations of the natural terrain. He attempted an abstract formulation of the constitutional conditions of the actual terrain using an 'Euclidean' geometrical method. In doing so, he imposed on the landscape regularities that refer to transcendental abstract geometric entities.

In opposition to this approach, design practice assisted by basic CAD tools such as NURBS surfaces – that we would characterise as topologically 'sensitive' – can simulate the natural relief, attributing to it the closest possible

approximation. Moreover, this representation is not static but refers to relevant parametric design software like Grasshopper plug-in for Rhino. This software is offering tools for controlling the transformations of this relief, thereby resulting to a representation nearest both to a realistic rendering of physical reality and its variations (visual reference: Issa 2010, p. 37).

I used examples of imagery construction considered to be the symbolic interspace of architectural practice, in order to illustrate the distinction and the correlation of the terms 'real/mental'. The first example is about the discrete construction of the picturesque representation and the geometric subsumption of it by Viollet-le-Duc as well as the comparison to the example of contemporary NURBS differential geometries.

The second example refers to the use of advanced point cloud surveying techniques by Christophe Girot (chair of ETH Zürich Landscape Architecture Department). These techniques permit the coexistence in the final result of both the 'natural' NURBS and the constructive mesh geometry, as well as the mapping of the texture and the gradation of the colour of the physical relief, which are no longer considered as secondary properties as René Descartes thought (visual reference: Girot 2009). I will suggest that this example might be transcribing, in terms of representation, René Thom's concept of 'mathematical phenomenology' or Gilles Deleuze's concept of 'transcendental empiricism'.

The concept of the fold and the concept of difference as well, unlike dialectic opposition, further attempt to overcome the dualism between terms as nature/culture, city/rural field, among others. It refers to a wide variety of architectural approaches. The term 'landscape formations', coined by Zaha Hadid, extends the term 'natural' to buildings, even if this relates to an inherent fluid plasticity of curvilinear forms controlled by the possibilities offered by differential geometry. Examples of the overcoming of this distinction from different architectural elaborations are many and varied, defining an infinite sequence that converges to landscape-sensitive approaches and tends to overcome the previously mentioned distinction. Consequently, we are witnessing the interest of the architectural practice turning to public space's design in general, to urban parks as well as the agricultural or extra-urban rural landscape, and regarding to the area of urban design, terms like parametric urbanism, landscape urbanism and the subsequent ecological urbanism, are appearing.

One more term emerged in architectural practice – the term 'soft', which refers neither to the engagement with the plant material or to the unstable soil and climatic conditions in landscape architecture, nor to the relatively recent trend of 'meteorological architecture' that summarises and transcribes atmospheric qualities. Rather, it highlights the expressive pluralism in design formulations attempting to escape from the academic schemas and constantly pushing the boundaries of the cognitive field of architecture. These

boundaries are expanding due to exchanges with other artistic and scientific areas, constantly producing new expressive formulations of the imaginary.

FINAL REMARKS AND FURTHER DISCUSSION

The scientific developments of the twentieth century seem to promote a change of perception regarding the nature and correlation of Kant's primary forms of perception, namely, space and time. The departure from the Newtonian homogeneous and isotropic space and the acceptance of relational space-time support evolutionary formal processes, rather than static patterns. This change is considered to be the central ontological progress of our times. It also defines and gives meaning to the contemporary worldview and highlights what Michel Foucault calls epistemic approach or discursive formations. The shift of the mathematical models which are supporting and determining the structural approaches towards the more dynamic poststructuralist approaches of the 1960s and the 1970s not only points to the departure from the Cartesian cogito but also signifies a complete transition to landscape-sensitive approaches considered as a model of the central epistemic conception of our era. Indeed, the contemporary conception of space and time points to the abandonment of the 'classicist' categorical associations and also of the mechanistic – algebraic model of the modernism for the procedural dynamic topologies.

We may conclude that topological thinking constitutes a framework which is not allowing for clearly designated boundaries between different disciplines and historical formations and that conclusion is further indicated by the correlation between the perception of the 'Fold' as a philosophical notion for Gilles Deleuze and René Thom's topological consideration of it.

The 'Fold' demonstrates a 'vivid' reality in constant transformation within which man is understood as an active agent. As a cognitive activity, the Fold 'curves' the surfaces of knowledge, producing thereby mental landscapes as meaningful places. Places of 'orgiastic meaning' that enclose all the knowledge past and current – this is the point where the 'Fold' refers to history as retrospective reflection being adapted to a present context of understanding.

Considering the folds of thought as concerning a real-time mapping, the famous software designer and architect David Rutten suggests that they describe our wandering in a foggy landscape. An example borrowed from the area of computational techniques will serve the landscape analogy of the generic problem-solving algorithms. Notice that Rutten calls the representation of mapping cognitive processes 'landscape geometries' (Rutten 2014).

We might suggest that this expressive possibility of thinking transforms every material into material of expression and determines as 'soft' or

'weightless' the perception of materiality, that is, it offers the possibility of opening a space in which we can think (Deleuze 1990), in other words, to problematise, as problematising means formulating both a problem and its conditions. The problematising space is multidimensional, plastic and adaptive. As the terms *interstitial* and *transitional* are being emphasised through the notion of the 'fold', we may further conceive the possibilities of interdisciplinary exchanges that are providing the construction of diverse expressive identities of cognitive objects. We might even suggest another term for the notion of landscape as an ecologically evolving process – the term *seeding*. The landscape as the cultural identification of place (Moraitis 2005) exemplifies both theoretical and mathematical processing. The effort to reach an ecologically coherent perception transverses all the areas of human knowledge and reflects the content of interdisciplinarity. *Interdisciplinarity* refers to the flexibility of methodological approaches to import geometrical translation of data being able to communicate among many different domains. Furthermore, this attitude intersects with architectural design and meets the contemporary construction of site imagery. It tends to imply conditions of consistency as the deepest content of the concept 'topos', emphasising on the specific features or singularities and intensities in terms of a topological geometrical approach. These geometrical approaches ultimately produce approximate representations of reality which nevertheless do not fail to be expressive approximations of the imaginary. 'Because the point is not how to finish a fold, but how to make it infinite' (Deleuze 2006).

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Part III

VITAL MATERIALITY

Chapter 12

Laocoon and the Snakes of History

Bernard Cache

EDITORIAL NOTE

Sophia Damianidou

Confronted by the publication in this volume of this chapter by Bernard Cache, one of the pioneer architects of 'non-standard architecture', indeed, the one who introduced the term, the unsuspecting reader may not, on his or her first reading, discern its relevance to a volume on Deleuze and architectural or urban theory. But there is relevance. As a student and collaborator of Gilles Deleuze, Cache translated Deleuze's philosophical discourse into architectural discourse. His ongoing inquiry may be summed up, in his own words, as an attempt to 'reestablish . . . a link between antiquity and the contemporary world'. Traversing ancient Greek philosophical thought and using examples from art history (Albrecht Dürer) and architecture (Vitruvius), his contribution to architectural theory focused on an intra-disciplinary exploration of geometry and the transformation of Euclidean geometry to topology. Focusing on the genesis of form, Cache opened up new prospects for the definition of the notion of curve, conceiving it as a flexible line in constant motion, relentlessly changing through time.

In cultural studies, the arts and the sciences, the concept of the formless or of the not yet specified form has become the centre of diverse theories and computational examples. In this chapter, through a marriage between concepts coming from the past and current practices, Bernard Cache, focusing on the intriguing serpentine curves of the marble sculpture of the Trojan seer Laocoon and the unique work of the German artist Albert Dürer, unfolds a dynamic logic for defining formlessness and offers an intriguing possibility for understanding the generation of forms. Aided by Dürer's work, particularly, the *Underweysung* and centred on the concepts of multiplicity, changeability and complexity, Cache analyses Dürer's numerical model in terms

of a creative computational machine. In doing so, he opts to work with the Deleuzian concept of becoming – a highly constructive procedure of natural systems where distinct entities coexist and closely communicate with each other. With this as a background, Cache intends to define form by means of sequential instances that derive from the evaluation and recombination of distinct parameters, and he goes on to propose a few new possibilities.

Laocoon's sculpture makes an admirable use of intensive, interwoven curves that seem to overcome the limits of its materiality. Deleuze is able to explain this accomplishment through his appeal to the law of curvilinearity, where the production of diverse curves is the result of external and internal conflicts. Forces from the outside transform the initial forms, whereas tendencies inside the body await for contextual triggers to be activated. It is precisely because 'the cause of movement is already in the body' (Deleuze 1993: 12) that the motions of desire and of the affects give rise to powers that form such a complicated mix. For Bernard Cache, flexible folds in variation represent the internal tensions and seem to be radically opposite to the general staging of the sculpture and its invisible yet unmistakable contours. Curved lines going from Laocoon's elbow to his knee and foot clearly contrast to the standardised frame of the serpentine cluster, permitting us to be transferred to the next scene, watching the new position of the bodies and mentally experiencing potential future transformations of the serpentine forms.

In the sequence, Cache concentrates on the generative mechanism of Dürer, who takes it upon himself to define complex curved lines geometrically. Unlike other artists, Dürer decides to use well-structured mathematical relations in order to produce accurate results. His work is oriented towards the systematisation of a generative process that could result in multiple transformations of curves. Similar to Klee who studied curved lines, Cache studies sequences of diagrammatic models that determine the attributes of each curve and map the final objects. The advantage of this attempt is in the choice of a logic that encloses with high-precision potential numerical elements able to describe a complex curve. The emerging diagrams do not function as a cul-de-sac or a predetermined situation, fixed and consolidated. Whereas the inflections of Klee's curves propose a 'weightless state of things'[1] and introduce active vectorial diagrams, Dürer's mechanism dynamically charts new curves as diagrams get progressively redefined. Different combinations produce different diagrams and, consequently, different curves, reminiscent of the Deleuzian view of vortical forms where the 'fluidity of matter, the elasticity of the bodies and the motivating spirit'[2] are being revealed.

Without fixed points or mathematical formulas that produce rigid or finite results, we experience the differenciation of the shape of the curves on the basis of an immanent transformational process. We cannot help but think of the evolutionary process of life where power of becoming and the tendency

of change nest in the very nature of living organisms as Deleuze highlights. In the same context, Cache's chapter explores Dürer's device that generates elements controlled by their internal or better their inherent characteristics and result in curves standing for instances of diagrammatic models, in a procedure of constant change and evolution. It is worth noticing that Cache does not attribute to this generative procedure any subjectivity or end. He emphasises the role of mathematical relations, and he succeeds in presenting Dürer's machine as a means to the free decomposition and recomposition of complex curves, in other words, as a machine of creative power.

In fact, Dürer's machine is rooted in a series of parameters, embedded in mathematical logic that defines the shape of his curves. They constitute factors susceptible to numerical values of a certain spectrum in a group of potential combinations. These factors constitute the genotype of the generated curves as they control the attributes of the complex lines. This way that Dürer derives his curves can be correlated with the genesis of life whereupon DNA biologically determines the existence of species and brings about a rich biodiversity through time. What Cache does is to show the combination of distinct elements in a symbiotic relationship that contributes to the generation of form so that 'multiplicity is continually transforming itself into a string of other multiplicities' (Deleuze and Guattari 2004, p. 249).

Whenever the inherent elements of the serpentine lines are modified however slightly, new and different serpentine lines rise up, and form is perceived in an incessant transformation and development. Moreover, entire groups of serpentine lines can be created as certain parameters are able to control lines with common attributes. Dürer's device produces sets of twisted objects that can be modified when a specific range of factors is changed in a way similar to the families of curves that Leibniz introduces in his description of the inverse tangents.

> Instead of seeking the unique straight tangent in a unique point for a given curve, we can go about seeking the tangent curve in an infinity of points with an infinity of curves: the curve is not touched, it is touching, the tangent no longer either straight, unique or touching, but now being curvilinear, an infinite, touched family. (Deleuze 1993, p. 18)

We should be clear that Dürer's aim is not to provide us with an absolute form but rather with a bunch of curves that can be modified only through their inherent attributes. Cache chooses to call such curves *objectiles* to the extent that they are flexible lines in constant motion. They relentlessly change through time and redefine their characteristics in terms of a 'continuous temporal mold'.[3]

All this can easily be recognised in the working of current technological applications and in the definition of form made possible by contemporary

computational methods. The idea of parameters is deeply rooted in such practices where initial factors are determined as capsules of information or more accurately as bits of memory. Their purpose is to store numerical values that can describe an object and determine its form. Mathematical logic undoubtedly functions as a fundamental tool for combining these factors, but in no case it predetermines the object or the generative process itself. An infinite range of possible solutions is plausible, and infinite variations can be obtained. Similar to Deleuze's 'teeming crowds'[4] the members of which are in constant motion and in close relation to each other, in this case too, and by means of sequential iterations, we are able to experiment with the genesis of differentiated forms and to observe a great diversity of sequences of flexible objects. In this process, we witness the constant renewal of an object's state.

Building a dynamic generative context for the definition of the machines that offer such a plurality of forms is definitely a remarkable achievement. Dürer's pioneering device introduces us to a vast field of forces and flows responsible for the production of infinite forms, the same way that a constant *zymotic* procedure does. According to Cache, it can be thought of as a machine that produces sequences of frames, describing the flux of the inflections found in serpentine lines.[5] The form becomes fluid and changeable through time. It can best be defined through the interaction of forces on a neutral diagrammatic system, always making room for new potential states. Cache, as much as Deleuze, unveils the tensions that preside over the creation of forms and the construction of worlds. In their analyses of natural or social systems or in their handling of cases taken from cinema or art works, they both concentrate on the constructive powers born by the idea of the possible or the virtual. In Deleuze's words, the process that both he and Cache have endeavoured to describe can be thought of as 'the abstract Machine of which each concrete assemblage is a multiplicity, a becoming, a segment, a vibration. And the abstract machine is the intersection of them all' (Deleuze and Guattari 2004, p. 254).

NOTES

1. Expression used by Bernard Cache for Klee's work. See Bernard Cache, *Earth Moves*, p. 50.

2. See Deleuze on Wolfflin's approach regarding baroque. Deleuze, 1993, p. 4.

3. This interesting phrase is used by Gilbert Simondon in *L'individu et sa genèse physico-biologique* and is cited in Deleuze's *The Fold: Leibniz and the baroque*.

4. Deleuze speaks of the behaviour of kinds that live in groups such as packs of wolves or swarms of bees. He presents the idea of multiple individuals that change their in-between relations in a dynamic redefinition of the group's behaviour. In Deleuze and Guattari, *A Thousand Plateaus: Capitalism and Schizophrenia*, p. 29.

5. With respect to the inflections of the curved lines in Lausanne's topography or Klee's active lines, see Bernard Cache, *Earth Moves*.

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—Sophia Damianidou

LAOCOON AND THE SNAKES OF HISTORY
BERNARD CACHE*

On the one hand, paintings and engravings, on the other, treatises and messy notes – the heritage of Albert Dürer is considerable and cannot be easily interpreted. In the graphic part of his work, *Melancolia§1* portrays a rather regular polyhedron in an engraving whose exceptional quality contrasts with the poverty and defects of the presentation of polyhedrons in the *Underweysung*. But on the other side of his work, the side of writing, the object that has the greatest priority, even in its most elementary procedures, is not the polyhedrons but the serpentine line. Far from reducing geometry to a composition of straight lines and arcs, Dürer intends to bring mathematical knowledge to the study of curves and complex surfaces which are the real objects of painting: folded textiles (*drapés*), textures and human physiognomies in all their diversity, even at the risk of having to face the deformity of the body or the formless witches riding their horses while facing backwards. But as he draws the serpentine line of his treatise on geometry, Dürer sticks to the instrument that permits him to generate this curve and to the diagrams that put it in variation, all the while leaving to the *Baldungs* and to the 'little godless masters' the task of exploring the postures which are characteristic of the witches of his time.

Even though his geometry provides Dürer with the means to generate the most tortuous curves, he sticks to an absence of figure that gives his serpentine line the status of a clandestine passer-by. In his entire graphic work, the formless waits to take us by surprise. We could start with the self-portrait of 1493, where the folds of the cushion suggest a whole series of deformations that Dürer pursues to the back of the image with six cushions. Dürer exposes his own face to the defiguration of cushions on the recto, which initiates human features on the verso. The painting *Hercules Wrestling with Stymphalian Birds* reveals the tension that animates the entire work of Dürer. We see a Hercules, whom Dürer clearly paints in his own resemblance, stretching an

arc towards one of the birds trying to escape from the marshy lake. It is more than a bird that Dürer offers to our sight, it is a kind of a dragon being reduced to a simple sign, an inflection resembling the letter S of the first of the figures to be truly extracted from the layout of the *Underweysung*, without losing its alphabetic character: IOS. We see here in plain light the tension that runs through Dürer in his effort to give form to the formless.

All sorts of reasons may have been behind this absence of representation of the serpentine line. For example, to the extent that Dürer knew the works of Cornelius Agrippa,[1] some saw in the snake the agent of the original sin, which for Agrippa was precisely the masculine member of Adam. Some commentators have claimed that in the denomination figuring on the banderole of *Melancolia§I* there is a kind of inscription '*§'* between the word '*Melancolia'* and the number *'I'* that permits the indication of different tails of serpents in the engraving, say, from the tail of the bat to the inflected configuration of the characters in the magic square. Certainly, negative connotations were not absent from the animal that gave its name to the main object of the *Underweysung der Messung*, leading Dürer to refrain from representing this line It is indeed worthwhile to examine the register of rivalry when the treatise of geometry was offered to the German painters as the way to avoid mere virtuosity exercises (*Brauch*) and to accede to the knowledge (*Kunst*) of their Italian colleagues, in order to compete with the ancients. Should we recall the monogram of Jacopo de Barbari, and the fact that Dürer asked for the notebook bequeathed to Marguerite of Austria – a caduceus that in German means 'serpents on stick [*Schlangenstabe*]'? At any rate, the written work of Dürer seems to be developed in response, if not in a spirit of rivalry, to the theoretical questioning initiated by the Italians and not just by Jacopo de Barbari.

LAOCOON: THE SNAKE IN THE ANCIENT WESTERN TRADITION

Let us return to this line which, without being drawn, bears the name of the serpent. That Dürer is ahead of his own language and that he, an artist, undertakes to write the first treatise of geometry in German[2] should not prevent us from paying attention to the becoming-animal that he introduces to the field of geometry, the very crucible of all rational discourses at least ever since Euclid. Not having an adequate vocabulary at his disposal, Dürer could easily have chosen other concrete terms in order to describe his lines. In fact, the citizen of Nuremberg had at least one illustrious predecessor regarding his reference to this animal in order to describe the movements of the body and the affections of the soul in painting. Indeed, in his *De Pictura*, Alberti

explains that the painter has no other means to express passions than to rely on seven movements: 'Anything that moves its place can do in seven ways: up, the first, down, the second; to the right, the third, to the left, the fourth; in depth moving closer and then away; and the seventh going around. I desire all these movements in painting'.[3] Thanks to a number of rules that will be exposed later, Alberti, unlike Hogarth, does not doubt that the most complex and subtle movements could be recreated with the help of the 'seven directions to move'. Thus, Dürer's instrument for generating serpentine lines is to the dynamic movement what the instrument, described by Alberti, in *De Statua*, is for the static representation of the human body. This is how Alberti describes the movements he has in mind: 'The seven movements are especially pleasing in hair where part of it turns in spirals as if wishing to knot itself, waves in the air like flames, turns around itself like a serpent'.[4] As a matter of fact, in the Latin version of his treatise, Alberti does not directly use the noun 'serpent', but the verb *serpere*, which means: to crawl, to creep, to spread insensibly or to reach something step by step.

In her comments on this passage, Aby Warburg sees, with Alberti's eyes, 'tangled[5] snakes'. It is true that in the iconography of the sculpted panels of his *Tempio Malatestiano*, Alberti found it necessary to ask specifically the sculptor Agostino di Duccio to amplify the characters' movements in accordance with the recommendations of the *De Pictura*[6] where the painter makes use of the wind to swell certain parts of the body and tightens the fabric[7] in order to reveal others. Warburg found convincing arguments for discovering in the ondulations of the wind, in hair and in clothing, an ancient artifice that gives an amplified external form to internal formless passions, in accordance with what he calls 'pathos formula [*Pathosformel*]'.[8] Movement is no longer a mere displacement of the body in space but rather the product of an impulse that expresses an emotion. A lot of extracts from Ovide,[9] Virgile[10] and Lucretius[11] are offered in support of this thesis, which applies both to Botticelli's *Venus* and to Leonardo da Vinci's *Bella Simonetta*. The latter is explicitly aligned with the ancients when he writes: 'And mimic as much as you can Greeks and Latins in their fashions of discovering body members when wind blows to their clothes'.[12]

Faced with the nude and impassive antiquity of Winckelmann, Warburg displays another, dressed and in motion. Let us therefore see how, under the cover of the verb *serpere*, which we find in Alberti's *De Pictura*, Warburg goes from the pattern of clothes and hair agitated by the wind to that of the tangled snakes. The figure that functions as an intermediary is that of Maenads: these young followers of Dionysus are often represented in their billowing dress, with their arm decorated with a snake coiled into a spiral. It is on the basis of the verb *serpere*, converted into the noun 'serpent' that Warburg will never cease to trace the serpentine lines of the Renaissance, as

for example, in the papier mâché the folded dragons that Buontalenti crafted for the third interlude[13] of a marriage in the Medici court in 1589.

WARBURG AND THE PATTERN OF THE POINTED KNEE

Warburg rediscovers, in works known to be the finest of the Renaissance, the remnants of a Dionysian[14] antiquity, at the mercy of the primitive fears still faced by the Hopi Indians that the historian was going to observe in the New Mexico of 1896, soon after the composition of his text on the *Interludes*. In the tableau 25 of the Mnemosyne image, Atlas is chiefly dedicated to the Malatestiano temple where Warburg saw the fruit of the collaboration between the Apollonian spirit of Alberti's architecture and the Dionysian spirit of the Agostino di Duccio[15] ornamentation. Warburg of course did not fail to detect the recurrence of such a pathic antiquity even in Dürer,[16] particularly in the drawing of *The Death of Orpheus*. The very idea of the *Mnemosyne Atlas* to which Warburg devoted the last years of his life was first expressed when the historian of images was working on 'Dürer and the Antiquity' – especially on the death of Orpheus – a work fit to evoke the primitive fears that art has as its mission to domesticate.[17]

Warburg had intended to show how Dürer, in the years 1494/1495, was pleased to imitate the Italian pathos of Mantegna and Pollaiuolo in its most expressive and exacerbated dimension. For example, he had no difficulty to study both Mantegna's *Bacchanales* and the *Battle of the Sea Gods* and to execute remarkable copies of them. However, for Warburg, the main evidence of the desire to intensify pathos is present in the posture of the knee pointed to the ground. This is Orpheus's posture that can be identified[18] as the resumption of a pattern characteristic of ancient violent scenes in painting and sculpture. In fact, it is before his first trip south of the Alps that Dürer takes up literally for his Orpheus the 'pathos formula [*Pathosformel*]' that Pollaiuolo used for the fallen character of his *Hercules and the Giants*. But, according to Warburg, Dürer was not of the same mind after his second trip to Venice, that is, after the discovery of the statue that would soon be recognised as Laocoon himself, 'the main figure as well as the children and the serpents (*draconum*) with their wonderful folds (*mirabiles nexus*)'.[19]

THE APPEARANCE OF A SCULPTURE

Which work can then be more serpentine than this singular *Laocoon*? We could even wonder whether the instrument used for drawing the serpentine lines could not have appeared in the eyes of his inventor, as a machine for the

production, if not of *Laocoons*, at least of the snakes that chocked the unfortunate Trojan priest and his children. Dürer would refrain from any figuration of this line because he had distanced himself from this 'southern and pagan liveliness'.[20] Warburg[21] invokes a sentence of Dürer's letter dated 7 February 1506:[22] 'And the thing that pleased me so well eleven years ago pleases me no longer. And if I had not seen it with my own eyes, I would never have taken another's word for it'. But if that were really the case, how can we explain the persistence and even the urgency with which Dürer promotes the serpentine line in his *Underweysung* of 1525? The distance he took from the Italian expressive style (*manière*) in order to create his own was not enough to turn the German artist's mind away from his interest in snakes and undulating lines. Even more, we could wonder whether the very absence of representation does reflect the way antiquity treats this animal. For no less than the Hopi Indians, the Greeks and later the Romans could not confront the so-called primitive fear of the snakes without constructing an artificial response that has to be listened as such, in defiance to any fascination with any 'originary instinctive lived experience (*vécu*)'.

Pliny mentions that the statue of *Laocoon*, which, despite the fact that it was the fruit of the joint labour of three sculptors, so that none of the three could assume a personal glory, was so much appreciated by Emperor Titus that it was placed in his personal dwelling. Names identical with those that Pliny gives to two of the three sculptors have been found in fifteen inscriptions in Rhodes and Italy, two of which can be given a date: the first is called Hagesander (around 55 bce), while the second bears the name Athenodoros (around 42 bce). Without being totally certain, it is very probable that these are two of the creators of *Laocoon*, the statue that the three associates sculpted between 40 bce and 20 bce, after they left Rhodes, as many of their colleagues had done after the destruction of the city by Cassius between 43 bce and 42 bce. Shortly after the period that we usually assign to *Laocoon*, the same partners, Hagesander, Athenodoros and Polydoros were able to visit the cave of Sperlonga, in the coast between Rome and Naples, in order to create this other famous group, *Scylla*, upon which the signatures of all three have been found.

Certainly *Scylla* and *Laocoon* show an exacerbation of pathos that contrasts to the reserve and serenity that will be imposed on the Augustan art of this period. This is why some other commentators, such as Bernard Andreae,[23] saw in these Roman marble sculptures copies of an unknown original bronze of a Hellenistic period, in other words, sculptures that can be rather associated with a 'baroque' style. More specifically, the alleged original bronze statue of *Laocoon* would have been melted by artists of the Pergamus School approximately in the vicinity of 140 bce, on the order of Attalus II, wanting to prove to Scipio Emilianus his desire for an alliance with Rome. Nevertheless, Salvatore Settis rightly prefers to rely on datable inscriptions[24] rather than on

stylistic similarities that have no chronological value, unless we imagine that styles mechanically succeed one another in time, without overlapping, that is, with no anticipation and no survival. Also, according to Settis, *Scylla* and *Laocoon* were created at a critical time when the pre-Augustan art had not yet eclipsed the post-Hellenic, and both tendencies coexisted:[25] one intended to exacerbate pathos such as the pain of Laocoon, while the other aimed at removing any trace of it, as in the *Athlete* of the Villa Albani. These tendencies therefore could be used according to circumstances, and the artists could select the style that seemed most appropriate to each request.

The core of Settis's argument consists in seeing in *Laocoon* a heterodox variant of a theme that was treated in different ways in ancient painting and sculpture. No ancient painting having survived, the most ancient graphic testimonies of Laocoon's history are those on Greek vases from South Italy which are very different from the sculpture that Pliny praised. In fact, soon we will have to listen to the story that these images tell us in order to understand what they have to say. Suffice it for now to point out that on these Greek vases the main character with the intertwining snakes exhibits a rather static posture, despite the fact that we see at his feet the scattered members of his children. It seems that it was at a later day that artists began to have access, for the description of the story of Laocoon, to the pattern of the pointed knee that we mentioned previously. In ancient art, reference to this pattern in order to evoke violent situations was common, no matter the point of view from which they were considered. Whether the fury of the aggressor was considered or the terror of the victim, both passions were represented with a bent knee, while the other leg would have remained stretched.

At any rate, Settis's words appear to be well founded when he writes of the sculpture of the Rhodians as a formal variation[26] of the formula (*Pathos-formel*) of the pointed knee. This variant occurs in the paradoxical context of a sculpture in the round (*en ronde bosse*) destined to be viewed from a unique point of view, namely, from the front, as if it were a painting or a bas-relief. This by the way justifies the two frontal stagings of this sculpture since its discovery in 1506: the permanent one in the Belvedere niche of Vatican and the temporary one, during its brief sojourn in France, in the Napoleon Museum from 1798 to 1815. In both cases, the statue was presented and viewed from the front like a bas-relief sculpted on a wall.

Salvatore Settis proposes to include *Laocoon* in a series of bas-reliefs that share the pattern of the pointed knee, whether this is the knee on the ground of the warrior on the aegis of Phidias's *Athena Parthenos* or whether it is the flexed knee that a woman raises in front of an altar on an urn of a Pergamon inspiration, found in Volterra. Definitely, in the variant of the marble statue discussed by Pliny, Laocoon's left knee is not completely flexed, nevertheless it is the body member which sticks out of the front plane of the Rhodian

sculpture and comes closer to the viewer. Between the knee bent on the ground of Phidias's warrior and the flexed but always protruding knee of the Rhodian Laocoon, Settis puts two figures of the Pergamon Altar: Alcyoneus of the Grand Frieze and Telephos, who drags the young Orestes in the Small Frieze. Rather than being a marble copy of an original bronze, the Rhodian *Laocoon* seems to be a very original variation of the common pattern of the pointed knee on the ground, conditioned by the frontal view of a sculpture in the round (*en ronde bosse*) being presented as a bas-relief. Consequently, this work of nearly archetypical value, ever since its being unearthed in 1506, has always stood for a singular or rather isolated moment, inside a series of variations that were spread over a very long period in Antiquity.

ROMAN STORIES: VIRGIL AND PETRONIUS

Virgil completes his *Aeneid*, just before dying in 19 bce, having inscribed the reign of Augustus, in a continuing tradition that dates back to the Trojan War. Unlike the silent figure of the statue carved in marble a few years ago by the three Rhodian sculptors, Virgil's Laocoon like the Philoctetes of Sophocles 'raises to the sky horrendous voice'. Is it because, in a written work, a poet would not care about the deformations a face suffers while uttering such cries? 'Enough that *clamores horrendos ad sidera tollit* (*Aeneid* I, 222) has a powerful appeal to the ear, no matter what its effect on the eye!',[27] writes Lessing, for whom the statue was sculpted after the poem was written by Virgil. Poetry can allow a pure verbal description of phenomena, the deformity of which would be unsustainable in an image. But then what about the snake? In this story, where their fellow citizens did not apprehend what beckoned Laocoon and Cassandra, wouldn't the snakes constitute the formless that Dürer had always feared to intrude in his work? It will be Petronius, even more than Virgil, who will find the words to show us the power of disfiguration that runs through this story. The serpent is nothing but the curve of the profile of the island of Tenedos, which, extending into the sea waves, will come to clasp to the members of Laocoon and bring down his body, dead, to the ground of the city of Troy, soon to become a mere heap of ruins.

This is an astonishing description that starts from the relief of the island of Tenedos plunging into the sea, as the hollows of the sea swell. These hollows soon will be mounted by Greek ships whose stern resembles the chests of snakes that will drag Laocoon to the ground – a prelude of the fate that awaits all Trojans. The undulation of the snake on the water amplifies the sonorous vibrations that Laocoon perceives after throwing his spear at the wooden horse, according to the definition of the voice given by Vitruvius: 'Voice is a flowing breath of air, perceptible to the hearing by contact. It moves in an

endless number of circular rounds, like the innumerably increasing circular waves which appear when a stone is thrown into smooth water'.[28] Couldn't this snake be the echo of the cries of pain uttered by Philoctetes on his island after the bite of another serpent? As in a continuous anamorphosis, all the elements of this sequence lose their proper form in order to blend into each other. In his *Satyricon*, Petronius does not really describe the events of Troy but rather their representation in a plastic work, even if it is in this case a painting and not a marble sculpture.

The serpent constitutes the power of the formless that always winds around the artwork. We have now to see, along with Warburg, how Dürer himself confronted the omens of the same order as those of Petronius. Without ascribing to the German artist a political reading of Laocoon, we could still maintain that Dürer saw clearly a power of deformation, analogous to that of the theme of a Laocoon, that he himself never represented despite the fact that he made use of the motif of the pointed knee. Isn't one of the major objectives of the *Underweysung* to conjure up the formless by giving it an acceptable form, with the help of mathematics? Without diminishing the quality of Salvatore Settis's analysis, shouldn't we notice that the historian concludes his book by inscribing the Rhodian sculpture in a strongly symmetrical composition, organised around the diagonal that runs from the elbow of the right arm of Laocoon to the extremity of the garment of his eldest son, falling to the ground? In order to account for the frame of this composition, the Italian commentator mentions a 'geometric cage shaken by lines of tension [gabbia geometrica . . . scossa da vive linee di tensione]',[29] in order to oppose the rigid geometry of the semicircle and the parallelogram to a lively and vibrant contour line. Of such a line, Settis has nothing to say, other than that it recalls the 'pictorial [*pittorici*]' experimentations of the end of Hellenism. It wasn't Dürer's priority to invent a geometry producing a variety of lines that in the end he himself would refrain from representing – a geometry of the serpent, which, at last, emerges from its polyhedral cage.

DÜRER'S *UNDERWEYSUNG*: BETWEEN THE LEGACY OF ANTIQUITY AND THE PREMONITION OF OUR PRESENT

If we return to these old stories, it is not just because the folds of the snakes that will choke Laocoon offer themselves as a possible figure of the serpentine line that Dürer refrained from drawing. It is not either, because we read his treatise on geometry as an aborted project of and we ask ourselves as a result about the stories, which according to Alberti, would have followed the lineaments of geometry. It is mainly because Dürer represents the middle term between our time and the antiquity. On the one hand, we tried to find out

the use he made of the three major sources – Plato, Euclid and Vitruvius – in a tradition that attributed significance to invariance only to the extent that it would allow the implementation and regulation of variations. But, on the other, we tried to pay attention to what preserves for us the value of a beginning,[30] despite the fact that the conditions of that time could not lead such primings to the developments that we are currently familiar with.

With the help of a few instruments designed in antiquity, Dürer will nurture the general project for designing with devices that are both numerical and mechanical. First among the numerical devices is the decreasing arithmetic series of the rays of the Ionic volute described by Vitruvius – a process the passage to the limit of which the German artist introduced together with the spiral of Archimedes. Before the invention of the instrument of Huygens, this curve was not traceable by a mechanical instrument. Dürer systematised a numerical method through the formation of numbered pairs of points, the way it was already done occasionally on Gothic sites. This numerical method, applied to a three-dimensional space, anticipates the descriptive geometry, the theory of which will be presented by Monge only at the end of the eighteenth century. Inside the framework of this descriptive method, Dürer will make use of a diagram that comes from the optics of the antiquity. Even though engravings such as that of Saint Jerome give him the opportunity to explore how the central ray of sight can be moved to the edge of the image, even though he is considered as one of the first to describe with precision the perspectival procedures before the vanishing point, Dürer, along with Leonardo da Vinci, demonstrates an attachment to an ancient conception of the image: the one that Plato problematised in his *Sophist*. The image, conceived as an apparent angle and not as a section of the visual cone will lead the Greek philosopher to accept that, in the artworks, the proportions of the objects are distorted, in order to appear in conformity with the model. Therein lies the entire ancient reflection on optical corrections that we will find again in Vitruvius, and that Dürer grasped in order to deform the spiral of Archimedes. That's how he invented a variety of twisted columns, the way we create today families of architectural components with the help of our computers.

The same numerical method will be used for the generation of a curve supplied with a retrograding loop by means of two uniform circular movements: deferent and epicyclical. Apollonius had demonstrated that, in the composition of these circular movements that could account for the apparent retrogradations of the planets seen from the Earth, it would suffice to invert the direction of one of the rotations in order to obtain an eccentric circular trajectory. This was just the change of the sign of one parameter that would allow the generation of such a variety of curves, the use of which in astronomy was the subject that Johan Werner was in the process of discussing with

Copernicus. This time, the numerical method can be incarnated in a mechanical instrument whose principle is described by Dürer. A little later, Carlo Urbino di Crema put in motion the *Vitruvian Man* of his master Leonardo da Vinci, figuring out the main articulations of the body: shoulder, elbow, wrist and hip, knee, ankle, as the different rotation centres of a mechanical instrument whose arms are multiplied. Undoubtedly, the heliocentric hypotheses of Aristarchus of Samos or the Pythagoreans are still remembered in the Renaissance, even though they were rejected for being heterodox. In fact, the neutralisation of the ideas of Copernicus by Osiander, who identifies the geocentric and the heliocentric hypotheses, was underpinned by the duplicity of the centres of the world that Theon of Smyrna invokes in his *Mathematics Useful for Understanding Plato*. And it is remarkable that this indifference regarding mathematical hypotheses had turned into one of the many strategies of contemporary science.

However, the montage that Leonardo da Vinci proposed for the two inscriptions of the human body, inside a circle and then in a square, was mainly rooted on the display on the same plan of two distinct paragraphs of Vitruvius concerning the postures of a reclined and then of a standing man. Also, in order to help the movements of the human body exit the plane, Dürer adjoined a vertical rod to his spider mechanism that composed the deferent and the epicyclical circles. It is precisely the rotation of this rod that was going to deploy in space the serpentine line that Dürer announced by the letter 'S' in his figure U.I, 2 – the first one to be extracted from the textual body of the *Underweysung*. But instead of the eagerly anticipated line, and the undulations of the snakes that wrap around Laocoon's body and of his children, Dürer offers only, side by side, one instrument and three diagrams. It is a question of inscribing the length of the successive arms of this instrument in the continuous proportion that can be established between the side of a square and its semi-diagonal. These diagrams are the tools that permit the implementation and regulation of the variations of this instrument, which are all too many to be represented explicitly. Today, we would have said that these tools permit the exploration of the virtual configurations of the highly variable serpentine line.

As always in Dürer, the diagram of continuous proportion is nothing but an initial proposal that will be followed by a dozen other diagrams. In fact, his objective is to produce an entire variety of serpentine lines by modifying the length of the different arms of his instrument, according to the rules that these new diagrams activate. This is exactly what Dürer wrote at a time when the divorce between mathematics and the arts was not yet consummated. But Hogarth, two centuries later, does no longer have the means to follow Newton and Leibniz on the pathways of the differential calculus, although these two had invented the means of describing the singularities of his own serpentine

line. The gap, open at this point, nips any effective encyclopaedic project in the bud. At the time that Diderot and Alembert wrote their works, sciences and technology were being so rapidly developed that art and literature would make reference only to subjective realities: romanticism, impressionism and symbolism belong here. But at the moment that Dürer was writing, the recovery of the corpus of the ancient science was still an ongoing task, with its concept of proportion, in its many forms, still being its core and its horizon.

Such a concept of proportion was still accessible to artists like Dürer who wished to ground their art on a knowledge process. The diagrams then are the tools that allow the regulation of the variations of a line as complex as the one of the snakes that choked Laocoon, by focusing on the proportions of the arms of the instrument, without any recourse to differential calculus able to account for the properties of the curve. Certainly, Dürer is not a mathematician, but he finds the means to use mathematics in his practice. Instead of piling up propositions for demonstration, he described processes that were destined to construct figures or instruments, driven by diagrams. It is this pragmatic relationship with science that can render the artist of Nuremberg our contemporary. With today's computer science, we rediscover the possibility to make use, in design, of the mathematical tools of our time. Of course, manipulation does not amount to knowledge. We bet, however, that a reasonable use of simulation tools opens up new paths for the exploration of mathematical concepts, without the necessity to master the techniques of calculus. We find again the possibility of building an encyclopaedia the project of which was aborted by the progress of science itself.

Indeed, this is only a mere possibility and it is not at all certain that the desirable results can be produced. To the extent that Dürer really makes geometry, it is because he does not object to exploring the great problems bequeathed by antiquity. Whether in the case of the trisection of an angle or whether, and especially, in the case of the duplication of the cube, our artist will design and experiment with figures of approximate solutions and the instruments that can provide us with exact results. He will not hesitate therefore to apply a tool such as *Vergleicher* (comparator) to such diverse fields as the study of human proportions or the design of fortresses. And precisely because he understands that the continuous proportions represent a crucial point, he will endeavour exceptionally to offer, step by step, a detailed demonstration of the methods of Heron and Pseudo-Plato. It is this effort to understand mathematics through experimentation that turns Dürer to a model of our time, as we are overwhelmed by the abundance of simulation tools. It is certain that a recourse to history constitutes one of the means for the understanding of the genesis of the mathematical tools that we can simulate and that, conversely, these simulation tools allow us to make a more effective use of historical archives, whether material or textual. We are in a position

today to explore the diversity of the serpentine lines that Dürer avoided. We are able to build and operate the instrument for solving the duplication of the cube that an ancient tradition attributed to Plato. And this promotes new readings of the Greek philosopher, very different from the idealistic readings to which he is often relegated.

Similarly, we are able to explore the variety of the curves that Eudoxus proposed, in the wake of Plato, in order to describe the dance of the planets. Today's simulations show that, far from being limited to the mere description of the positions and retrogradations of the planets, the composition of different rotations of homocentric spheres would permit the generation of interlaced trajectories, perhaps giving rise to the geometric model of weaving. It is this model that Plato qualifies as the paradigm of paradigms in his *Statesman*. In this way, simulation tools would make possible the renewal of the way we are doing philosophy in our reinterpretation of the most classic authors.

Let us observe how Dürer's effort to appropriate mathematical tools bequeathed by the antiquity is accompanied by the distanciation of the image, a thing remarkable given the fact that we have to do with a painter. Instead of the beautiful perspectival tableaus that Luca Pacioli expects from Leonardo, Dürer cares less about the presentation of regular polyhedra and provides us with the means to decompose and recompose them as we stretch them on a plane. When the artist undertakes to describe devices for the creation of perspective, he will go as far as to invent a totally mechanical device, without any trace of subjectivity.

NOTES

1. Cornelius Agrippa, *De Originali Peccato*, 1518.

2. Jeanne Peiffer noted that the terminology of Dürer continues to be used until Kepler, but it will subsequently be lost. See her introduction to Albert Dürer, *La Géometrie* (Paris: Seuil, 1995), p. 49.

3. Alberti, *De Pictura*, II, p. 43. Alberta Leon Battista, *On Painting* (first appeared 1435–36) Translation with 'Introduction' and 'Notes' by John R. Spencer (New Haven: Yale University Press, 1970, first printed 1956). www.noteaccess.com/Texts/Alberti, accessed 19 November 2016.

4. Alberti, *De Pictura*, II, p. 45, where 'sometimes they slip like snakes under other hair' is closer to the Italian grammatical form: '*parte quasi come serpe si tessano fra li altri*' and the Latin: '*modoque sub aliis crinibus serpant*'.

5. 'Alberti then sees tangled snakes, spouting flames or the branches of a "tree" ', in Aby Warburg, *Essais florentins. La Naissance de Vénus et le Printemps de Sandro Boticelli*. 1893. Presentation by Evelyne Pinto (Paris: Klincksieck,1990), p. 57.

6. Alberti, *On Painting*, II: 'Folds are in the same way emerging like the branches from the trunk of a tree. In this they adhere to the seven movements so that no part of the cloth is bare of movement'.

7. Alberti, *On Painting*, II, p. 45: 'It would be well to place in the picture the face of the wind Zephyrus or Austrus who blows from the clouds and makes the draperies move in the wind. Thus you will see with what grace the bodies, where they are struck by the wind, show the nude under the draperies in suitable parts. In the other parts, the draperies blown by the wind fly gracefully through the air. In this blowing in the wind the painter should take care not to display any drape against the wind'.

8. Marcus Andrew Hurttig: *Die entfesselte Antike. Aby Warburg und die Geburt der Pathosformel*, 2012.

9. Ovid, *Métamorphoses*, I, p. 528, about the pursuit of Io by Zeus: 'The winds bared her body, the opposing breezes in her way fluttered her clothes, and the light airs threw her streaming hair behind her'. Ovid, *Metamorphoses* (A. S. Kline's version), ovid.lib.virginia.edu/trans./Metamorph htm#488381109, accessed 19 November 2016.

10. Virgile, *Enéïde*, I, p. 317, about the appearance of Venus to Aeneas: 'She had loosened her hair to the winds. Her knees were bare and her flowing tunic was caught in a knot'. Publius Vergilius Maro, *The Aeneid*, trans. L. R. Lind (Bloomington: Indiana University Press, 1963), p. 11.

11. Lucretius, *On the Nature of Things*, I, p. 6, in the famous ode to Venus which opens the poem: 'Before thee, Goddess . . . flee stormy winds'. Lucretius, *On the Nature of Things*, Trans. William Elley Lorna. *The Internet Classics Ancient Archive/ On the Nature of Things*, classics.mit.edu/Carus/nature_things.html, accessed 19 November 2016.

12. Leonardo da Vinci, *Traité de la Peinture*, IV, n° 521, ed. Keller, 1977, p. 211, cited by Aby Warburg: *Essais florentins. La Naissance de Vénus et le Printemps de Sandro Boticelli*. 1893. Presentation by Evelyne Pinto (Paris: Klincksieck,1990), p. 88.

13. Aby Warburg: '*I costumi teatrali per gli Intermezzi del 1589 – I disegni di Bernardo Buontalenti e il libro di conti di Emilio de' Cavalieri*', Atti dell'Accademia del R. Istituto Musicale di Firenze, 1895.

14. In a much less tragic register, but equally significant, we notice the name of the weekdays in modern languages that refer to the seven heavenly bodies, known by Chaldean astrologers, whose order of succession throughout the seven days of the week resulted from the supposed influence of the planets on the successive hours of the day. The ecclesiastical authorities repeatedly tend to rename days in a more Christian terminology, but in Latin languages the only accepted change is that of Sunday, the Lord's Day, instead of the day of the denomination of the Apollonian Sun, which was preserved in the German *Sonntag* and the English *Sunday*. However, this change is the direct result due of the Jewish Sabbath which celebrated the god of the Bible on Saturday. Few additional everyday phenomena as the days of the week express the stratification of multiple legacies that come from Chaldean, Jewish, Greco-Roman and Christian traditions.

15. 'Thus, we see the influx of the ancient world in both Apollonian and Dionysian currents. . . . On one side, Agostino di Duccio and Nietzsche, on the other, Burckhardt and the architects: tectonic against line'. Aby Warburg, *Burckard-Übungen* (1927), p. 196, cited in Kurt W. Foster and Katia Mazzucco: *Introduzione ad Aby Warburg e all'Atlante della Memoria, a cura di Monica Centanni* (Mondadori, 2002).

16. Aby Warburg, 'Dürer and the Italian Antiquity (1905)', in *The Renewal of Pagan Antiquity: Contributions to the Cultural History of the European Renaissance*. Introduction by Kurt W. Foster. Translated by David Britt (Los Angeles: Getty Center for the History of Art and the Humanities, 1999).

17. Roland Recht, *Aby Warburg. L'Atlas Mnémosyne*, p. 26. On page 160, the board 57 can be found. There, Warburg has related numerous images so as to illustrate the pathos formulas in Dürer: the *Death of Orpheus*, *Hercules against the Stymphalian Birds* and *Hercules at a Crossroad*.

18. L. D. Ettlinger, *Exemplum doloris. Reflections on the Laocoön Group*. In *De Artibus Opuscula XL. Essays in Honour of Erwin Panofsky*. Edited by Millard Meiss. 1961, pp. 121–26.

19. Pliny the Elder, *The Natural History*, trans. John Bostock and H. T Riley, 36, 4 (Tufts University, New York: New York Press, 1855). Perseus Digital Library. Retrieved 19 August 2016.

20. Warburg, 'Dürer and the Italian Antiquity' (1905). Translation by Evelyne Pinto, *Aby Warburg. Essais Florentins* (Klincksieck,1990), p. 163.

21. See Dürer's letter cited by Warburg, 'Dürer and the Italian Antiquity (1905)'. In *The Renewal of Pagan Antiquity*, p. 556.

22. When reading again this sentence and its context in this letter to Pirckheimer, we could ask whether Dürer targets particular works of Bellini or rather the Italian Mannerism in general.

23. For a succinct summary of his thesis in French, see: Bernard Andreae, '*Problèmes d'histoire de l'art du Laocoon*', in *Le Laocoon Histoire et réception*. Edited by Elisabeth Décultot, Jacques Le Rider et François Queyrel (Paris: Presses Universitaires de France, 2003), pp. 33–57. For a more detailed argumentation in German, see: Bernard Andreae: *Laokoon und die Gründung Roms. (Kulturgeschichte der antiken Welt 37)* (Mainz am Rhein: Philipp von Zaben, 1988).

24. Despite the possible but unlikely existence of homonyms.

25. Settis particularly refers here to a text of K. Fittschen: '*Pathossteigerung und Pathosdämpfung. Bemerkungen zu grieschichen und römischen Porträts des 2 und 1. Jahrhunderts v. Chr*.' In *Archäologischer Anzeiger* (1991), pp. 253–70.

26. See Salvatore Settis, *Laocoonte. Fama e stile* (Donzelli editore, 1999), p. 70: '*Si tratta di una* variatio *pienamente consapevole e costruita con consumata abilità dello schema del "ginocchio puntato"*', where Settis refers on several occasions to the Latin word *variation* for reporting on the Rhodian statue.

27. Gotthold Ephraim Lessing, *Laocoön, An Essay on the Limits of Painting and Poetry*, translated with an Introduction and Notes by Edward Allen McCormick (Baltimore: The Johns Hopkins University Press, 1962), p. 23.

28. Vitruvius, *De Architectura*, V, pp. 3, 6. Marcus Vitruvius Pollio, *Vitruvius: The Ten Books of Architecture*, trans. Morris Hicky Morgan (Boston : Harvard University Press, 1914). www.guttenber.org/files/20239/20239-h/29239-h.htm#Page-137 accessed 19 August 2016.

29. Salvator Settis, *Laocoonte, Fama e stile*, (Milano: Donizelli, 1999), p. 74.

30. With respect to the return of the classic in Western history, see also: Salvatore Settis, *Futuro del classico* (Scampia: Vele, 2004).

*Translated from the French by Sophia Damianidou.

Chapter 13

Reterritorialising Concrete as an Actor of Comfort in Architecture

Athena Moustaka

Abstract

Idealist strands of philosophical thought and phenomenology implemented into architectural theory conspicuously neglect the role of matter. They treat it as a passive and inert receptacle of a creator's order, rather than an active presence. The active abilities of materials within a network of human and non-human actors have been limitedly explored with regards to the potential they offer to understand what shapes an architectural experience. My chapter investigates the listed concrete in a post-war Social Housing development. The way non-human actors interact with human ones is revealed through the lens of the material. The material exposes how properties emerge from a network of social, visual and aesthetic parameters. This chapter consists of three parts: In the first part I will start by drawing upon Deleuzian vocabulary to describe the built environment as an assemblage of material and immaterial elements that come together but are also continuously on the move. I then turn focus on the notion of comfort, which should not be confined to a tight environmentally defined concept, but should include parameters of the visual, social and the aesthetic in line with a Deleuzian framework. In the final part, I describe concrete as a deterritorialised material, in these assemblages of the built environment, forming relations with other parts. Comfort, itself a continuously changing notion, can be manifest through the relationships concrete in these assemblages.

Park Hill is a 1960s council estate that dominates the Sheffield skyline and imposes its presence on the city's landscape. Constructed during the late 1950s and early 1960s, the estate has stood in place for over half a century. But its presence has undergone continuous changes both physically and in the perceptions of the occupants and Sheffield residents. In the early years

following its creation, it was representing the vision of an optimistic Britain, looking towards a promising future, following the country's tumultuous past. As the estate was maturing, however, its popularity was waning and the rough Yorkshire weather was taking its toll. During the 1980s and 1990s its decline became representative of the harshness of Northern English cities, and in the early 2000s, following its listing in 1998 that prevented its demolition, it fell into a state of neglect and disrepair. Now a Grade II* listed by English Heritage, it is undergoing a redevelopment.

The way that the recent redevelopment has handled the concrete reveals a surprise: it has stripped the building of every other material leaving only the concrete in place, making its physical presence stand out. While a wide range of questions is raised by this observation (i.e., why only concrete and what is the purpose of this concrete preservation), this chapter will focus on only one of the plateaus of Park Hill where concrete is exhibiting agency, that of comfort. To describe the emergence of comfort through the interaction of concrete with other building components, I will firstly move away from Park Hill and reflect on the concepts that Deleuzian philosophy has to offer in describing these types of interactions.

BORROWING FROM DELEUZE

Assemblages

To describe comfort by acknowledging the physical presence of materials in the built environment alongside the constructed attribution of meaning in architecture, I first need to turn to Deleuzian thought and its implication in the description of architectural practices. The turn to Deleuze in the context of architecture has gained momentum in the past twenty years, predominantly in conceptualising ideas, practices and events (Ballantyne 2007). A great part of this focus has been on the potential of the digital, and on formalist interpretations of Deleuzian concepts (such as the fold). Yet, as Hélène Frichot points out, reaching out to Deleuze can bring down 'preconceptions about what architecture is, what architecture does and how architecture is implicated with the environment' (Frichot 2005). Moving away from idealist thought that has dominated architectural thinking, Deleuze can offer a way of appreciating material entities in the built environment alongside constructed ones. And since my intention is to highlight the active presence of concrete in the built environment, borrowing from this line of thought is crucial. Before I go any further in arguing how an immaterial object can have a significant presence in shaping the built environment, it is important to initially position the framework of this chapter ontologically in a manner that can acknowledge

matter as being also active, away from idealist strands of philosophy, and alongside an object-oriented framework. In order to read concrete as active and powerful rather than passive and inert, I consider two ways in which the redescription of architectural practices through the lens of Deleuzian thought has re-evaluated architecture.

The first is the attribution of meaning and symbolism in architecture, which viewed in the Deleuzian manner is found problematic (Nesbit 1996; Frichot 2005). Architecture is continuously shifting and changing, without a purpose or means to an end but constantly morphing and acquiring new lines of flight. Viewed this way, concrete cannot be a symbol of a style, or a practice or an intention, but rather the means of achieving it. This of course should have been anticipated by architects: Past attempts to attribute a teleological scope to architecture have failed (Frichot 2005); key texts by great masters like Vitruvius or more recent like Le Corbusier are rendered obsolete after a few years, leaving them with little value other than in themselves and none whatsoever for the current architectural discourses.

The second concept that will aid in describing the built environment away from transcendent notions is the 'assemblage'. Borrowing the description of an assemblage from Deleuze will aid in conceptualising comfort as an emergent property, through the interaction of an array of entities. In a similar manner that Deleuze describes to Claire Parnet the horse, warrior and weapon as an assemblage that gives rise to emergent properties, there is the potential to view the building as an assemblage of parts (Deleuze 2007). An equivalent description is presented by Deleuze and Félix Guattari in the opening chapter of *A Thousand Plateaus*, where a book assemblage has two axes: one of the paper, ink, typeface of the writing and another of the content description, ideas and lines, narrator and reader. They all exist simultaneously in a complex arrangement shaping and being shaped by one another. In a very similar manner, architecture can be conceptualised as such: the bricks, mortar and doorknobs coexisting alongside users, communities, architectural experience, appreciation, mice, cleaners and affectual triggers in the building. There is no way of considering architecture purely as the materials of a building, but neither can the physical matter in a building be viewed as disconnected from the experience.

THE PRESENCE OF MATTER IN ASSEMBLAGES OF THE BUILT ENVIRONMENT

The question that arises is how matter is implicated in these assemblages of the built environment. As Manuel DeLanda points out, adopting a materialist stance where the world exists independently of our mind assumes that

differences should be constructed by Aristotelian essences (DeLanda 2006: 26). DeLanda therefore suggests assemblages as a way of assuming a purely materialist position that relates to their open-endedness. Properties emerge for DeLanda (2005) in assemblages with open associations and through the continuously changing relationships of their entities. These emergent properties do not pre-exist in the individual entities but are only revealed through their interaction. As Graham Harman (2008) observes, the interactions become non-linear in a Deleuzian assemblage where there are no layers but a flat interaction, the properties emerge from the continuous interaction of these parts.

DeLanda borrows from chemistry the concept of 'catalysis' to describe the non-linear type of interactions of elements in assemblages. He presents causality in a manner that is rendered redundant when catalysis is brought in and breaks linearity by implying that 'different causes can lead to one effect . . . and that one and the same cause may produce very different effects' (DeLanda 2006: 20). Assemblages described in this mode can include heterogeneous entities with non-causal relationships among them. By describing the interaction of heterogenous entities in an assemblage that permits both human-constructed ideas (immaterial entities) and non-human (material) entities to be simultaneously accounted for, DeLanda enables us to think of an ontology that can be applied to the case of acquired connotations. This description allows matter in the assemblage to participate in the emergence of properties. DeLanda has previously highlighted material actions, mainly in the realm of morphogenetics: Matter in an assemblage can, in his view, be charged with the potential of form giving. It is not anymore an inert receptacle of commands, obeying to orders of a creator (DeLanda 2015), but has immanent morphogenetic properties with the potential of becoming. Matter in other words can have form-making powers being 'morphogenetically impregnated' in DeLanda's own words (DeLanda 2005), and these powers pre-exist any outside powers.

Through this lens of the immanent capacities of materials, concrete will be viewed as inherently active in assemblages of the built environment. This activeness of the material is revealed in the context of a network of other parameters participating in the perception of the built environment. DeLanda's breaking with the presence of Aristotelian essences allows the aesthetic values of concrete to be read free of any preconceptions for the material. The terms 'concrete jungle' and 'concrete monstrosity', commonly used in popular media as descriptions of concrete, can thus be understood not as a defining identity of concrete but as one of a myriad of its manifestations, brought about through diverse human experiences and interaction with the material. Catalysis, on the other hand, is important to describe how concrete cannot have a direct, a linear relationship to other components of the built

environment or the affects it triggers. The description of the built environment as an assemblage gives rise to properties through the interaction of its parts that are emergent.

By viewing the building in the Deleuzian frame, one can understand that any discrete parts can produce any number of effects in the architectural experience with the different modes of interaction of its parts. These elements form relationships of interiority, that is, the relationship and interaction of the components is crucial to the identity of the assemblage, or of exteriority, that is, the interaction of components is external to the identity of the assemblage and its emergent properties. To think about these relations in the context of concrete, with steel and formwork it forms relationships of interiority that are important to the mode of an assemblage, but with users' preconceptions about concrete and historical information about its uses, it forms exteriority interactions that do not alter the nature of the assemblage.

COMFORT REDEFINED THROUGH DELEUZE: AN OPEN-ENDED DESCRIPTION

Describing comfort in a fixed definition has been elusive: it has continuously been given changing descriptions over time and/or place, depending on sociocultural norms seeking to stabilise it. A brief overview of the ways it has been approached has revealed a continuous shift. In the 1960s, attempts to conceptualise the notion of comfort have considered it as a positivist concept, composed of a limited number of environmental factors: temperature (mean radiant), humidity (relative levels) and ventilation (air velocity) (Fanger 1970). Users of a building were expected to conform to the rigidly and narrowly described conditions. Ole Fanger, a researcher of comfort, described these conditions tightly by working in a 'climate chamber' at the Technical University of Copenhagen. Framing comfort, however, in an absolute way was soon deemed limited as it leaves out the individual experiencing it. The personal experiences, environmental influences and personal biology of the individuals operating in a constructed environment were ignored. This is what the view of later researchers came to address.

A sociocultural line of thought that redefined the definition of comfort emerged from the 1990s onwards. In this view, comfort is presented as a constructed notion. The more recent work of Michael Humphries, of Fergus Nicol and of Elizabeth Shove (Humphreys 1995; Nicol and Humphreys 2002; Shove 2003) defies the previous positivist tradition of comfort thought as a quantifiable concept. Instead, it considers it as a sociocultural product, a construct of human behaviours. By doing so, comfort is relocated ontologically in a different realm: It is a notion established by sociotemporal processes.

Comfort in this view can be informed by instances of habits consolidated into culturally and socially acceptable practices. Through a continuous transformation of habits and experiences, Shove (2003) presents comfort as stabilised in its descriptions. But this culturally constructed description cannot be permanently stabilised, as cultural norms cannot themselves be stabilised. Allowing the changing norms to interact with the myriad possibilities of comfortable experiences and individual human biology, further destabilises the descriptions of comfort. And although this line of work is moving towards a constructed view where both the biological characteristics and socially and culturally constructed accounts are considered, it leaves out what might be acceptable in terms of the aesthetic or the visual aspects of an internal environment.

When this description of comfort is further allowed to interact with the intricate built assemblage described in the earlier section, new possibilities arise for the description of comfort: the visual and aesthetic properties can now be included in its description. There is an abundance of literature to suggest that the visual and aesthetic properties give rise to comfort: the impact of the visual of an interior on levels of productivity, seasonal affective disorder (SAD) syndrome prevalence and percentages of absenteeism are well documented in the bibliography, thus implying the powers of the visual and the aesthetics of interior environments. To assume that the experience of a comfortable built environment is manifest solely through the attainment of steady environmental conditions is like claiming that the sole purpose of a building is to provide shelter. Buildings serve a variety of purposes and operate on multiple levels: social, cultural, visual, social, political and so on.

The architectural styles of the twentieth century can be viewed as attempts to set universal constants for high levels of human attainment. However, and as I described in the previous section, the creation of a common aesthetic whether that has been called modernism, or brutalism, or any other '-ism', has clearly failed. This constantly shifting aesthetic can create differing manifestations of visual comfort, able to be grasped only temporarily, not diachronically. Visual and aesthetic comfort can be described as the formation of a common aesthetic, that is, sharing the same visual landscape. This shared ground creates a closeness that Christopher Wren would describe as 'customary beauty' constructing an appreciation in things 'not in themselves lovely' (Bennett 1982: 120). This closeness is attained in two ways: by creating a common visual experience and by attaching this experience to participation in everyday practices.

What is perceived as a comfortable condition has been mutating continuously to include varying concepts and definitions. In this respect, the work of Gilles Deleuze and Félix Guattari offers the necessary terminology to describe comfort as the emergent property of an intricate assemblage of components,

some material and some immaterial, all interacting together. Some are purely non-human, and some exist in the minds of the users; they are all related to the assemblage, and through their interaction, properties can emerge. The constructed experiences of users are not far from the material aspects of the building, and not far from the biology affecting their perceptions; all interact in a continuous network. To assume that any one presence of the assemblage can linearly predict and dictate the outcome of comfort would be a narrow reductionist view. They all simultaneously interact: the visual, the biological and the aesthetic. To follow the description of assemblages by Deleuze and Guattari (2008), on the one axis of the assemblage are the strictly material and physical entities and on the other the purely constructed ones. All of these form part of the same assemblage and through their interaction, emergent properties can arise.

MOVING BACK TO PARK HILL: OBSERVATIONS ON CONCRETE AND COMFORT IN A DELEUZIAN FRAMEWORK

To understand the emergence of comfort from a built assemblage that involves the presence of concrete, I conducted a series of semi-structured interviews with Park Hill's residents and construction professionals. All interviewees were involved with the renovated building either in a professional capacity or as users of the building. Over a period of eight months (April to December 2014) a total of twenty-one interviews were conducted. All names used here have been altered to ensure anonymity. The interviews were conducted with the aim of following the presence of the concrete and describe the emergence of comfort.

With respect to the morphogenetic capabilities of matter discussed earlier, Park Hill's material arrangement is crucial to understanding how it interacts with the landscape, its users, the city's networks and flows. It is not a coincidence for example that it was a popular spot for drug dealing in the 1980s and 1990s: its intricate typology made it the perfect drug dealing and hiding spot. And these properties were partly afforded by concrete. Park Hill's materials, its arrangements, its concrete balustrades, the drug dealers and the streets in the sky cannot be viewed separately. They operate in an assemblaged network. Viewing Park Hill solely as a result of the human's projection of ideas upon it neglects the role of the material arrangement, like in the case of drug dealing. But viewing the material arrangement as the sole means of the building's capacities also neglects a constructed experience by the user that does affect how the building is understood. The acquired connotations and the formalist interpretation of the architects' intentions come together in the use and experience of Park Hill.

Subject-object–oriented views and object-object descriptions cannot be viewed individually. To claim that Park Hill is the projection of modernist ideals would be ignoring the human interactions on the 'Streets in the sky'[1]. And one way to view such an intricate interaction is by conceptualising them into assemblages with intricate interacting networks. Once the concrete, Park Hill, users and experiences are understood to operate in such a network, it will be easier to describe concrete as an actor of comfort.

INSTANCES OF DETERRITORIALISATION

Concrete in the estate has been deterritorrialised from a structural presence. This is far from a rare occurrence in architectural history: The ancient frieze is a reinterpretation of a timber frame and the triglyph is the ending of a timber beam. The structural properties of timber have been deterritorialised into a visual effect in stone and marble. Marc-Antoine Laugier's primitive hut is another such occurrence (Laugier 1753). In Park Hill, this reterritorialisation of concrete is evident in its omnipresence of the structural frame that has been preserved intact to create an aesthetic for the development. The structural abilities of concrete have been reterritorialised into a visual aesthetic. The continuous synergy of its material effects with the constructed notions it has been ascribed, and the formalist representations the original architects intended, concrete has acquired emergent visual properties.

Further to the visual language emerging with concrete, the effect of the material has also triggered further comfort affects in the case of the concrete soffits of Park Hill. To comply with their initial ideas of keeping the concrete intact, the developers intended to leave the exposed concrete of the soffits untouched. It was decided purely for aesthetic, visual and historic reasons, according to the developers. Yet, during an initial survey, thin cracks were noticed on the exposed surface of the concrete. According to the structural engineer they were innocuous, posing no threat to the stability of the building. This type of cracks is common and results from the curing process of the concrete. The developers, however, decided against keeping them exposed and plastered over them instead. Future buyers, according to the developers, could see them as threatening to the structure, and this possibility was a deterrent. To address their concern, they were plastered in their entirety and finished with a smooth white paint. Comfort, in this case was emergent through the views of the stakeholders and the future buyers' perception of structural stability of the soffit cracks. What the engineer found as a harmless effect of the curing process, the users perceived it as threatening. With the nature of the interaction of material and immaterial entities in this case, comfort emerged.

MYRIAD ACTUALISATIONS – OPEN-ENDED POSSIBILITIES

The interaction of all the components of a built assemblage that gives rise to emergent properties produces effects impossible to preconceive. There is therefore no preconceived reality on the drawing board that can lead to the production of one absolute building. Rather, there are endless possibilities, and the potential interactions of these components are infinite. Interacting with the concrete in Park Hill reveals infinite potential for the emergence of comfort for Park Hill's residents. The interviews with the occupants on their engagement with the exposed concrete walls in their flats can reveal an infinite number of possible effects.

Karen is a young lecturer at a local University. She bought the apartment because she found it 'quirky', and concrete played a part in this view. The exposed concrete has even influenced the choice of furniture in her apartment. This part of her interaction with the material is triggered by the presence of concrete in the flat. However, there is a constructed aspect to her experience also: she was familiar with the concrete grand ensembles of France in her teens and had positive memories of her time spent there; Park Hill was more European in her view. As a result of this dual interaction with concrete, Karen has made it a prominent feature in her flat and has chosen to maintain the concrete walls as intact as possible.

Alan, residing on a different floor of Park Hill than Karen, has engaged with Park Hill's concrete in an entirely different manner. Having been offered a flat by the council, he has moved to Sheffield as a refugee. Exposed concrete in his flat was not well received. Other residents with similar attitude as Alan have asked for special permission by the residents' association to cover their exposed walls either with wallpaper, plasterboard or suspended plywood sheets. They are clearly not engaging in a positive manner with the material. It is virtually impossible to accurately find out whether this is related to a previous negative connotation or it is an effect triggered directly by the material. But what is certain is that the interaction effects are very different to those of Karen. Each user and resident interacts with the material in a different way, and each time exposed concrete is perceived, it gives rise to a different experience of comfort. Contrary to expectations by the developers, who were enthusiastic about the prospects of exposed interiors, the outcome every time is rewritten, and since there are infinite factors, material and constructed that can come to the built assemblage, there are infinite emergent properties.

Although historically it has been attempted to define universal constants that would lead to environmental and social comfort, in reality it could never be shaped by abstract terms. Arbitrarily constructed ideas have been unable to impose themselves and to create a continuous, steady and unified perception

of comfort. The articulation of comfort in the case of Park Hill depends both on the decisions taken by professionals in the built environment and on the associations users have had with the materialised building and their past experiences, rather than arbitrarily constructed definitions.

DISCUSSION

By viewing the material as an active presence in an assemblage, matter and, in this case, concrete can be repositioned in the experience of the built environment. Concrete has achieved with its presence tremendous capabilities – as a means of social transformation, a visual aesthetic and the driving force behind engineering marvels. Through the interaction of its randomly put-together components, with its day-to-day interaction with architects, planners and engineers, concrete has given rise to emergent capabilities. What can be observed for concrete and the emergence of comfort is that the latter has emerged as a surprise; a detailed examination of its components could not reveal its emergence. The emergent properties are virtual in that they can arise at any point, but not actual: they do not always become articulated in the experiences of the users. Comfort was never confined to performing in a certain way; its performance has always shifted to fit the performances it was expected to display. In some cases these performances were associated with the experience the material provided, in and around its built manifestations, articulated in built practice in equally unconfined ways.

Comfort is only one manifestation of the emergent properties of concrete. The experience the material affords, in relation to a comfortable built environment, has attracted and rejected meaning, perceptions and ideas continuously, not always having them imposed upon itself but also sometimes from within it. There are therefore myriad possibilities in how the material can shape comfort, never operating in a clear direct way but endlessly intertwined. It is the ability of the material to serve multiple purposes and morph into different roles that influence the behaviour of the building towards comfort or non-comfort. Redescribing concrete through Deleuze alters the relationship between concrete and comfort; never seeking to be finalised but always temporal; never defined but always changing meaning; never linearly but always lacking causality; never in an absolute manner but always in-between.

NOTE

1. ‘Streets in the sky’ was the term used by Park Hill’s original architects to describe the long corridors leading to the apartments, which they envisioned as a

space of social interaction. The term 'streets in the air' has been used by the Smithsons in their Golden Lane estate competition, and it is uncertain who initially coined the term.

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Chapter 14

Radicalising Architecture by Redefining the Monument

Mike Hale

Abstract

Deleuze's rich conceptual matrix is underpinned by a combination of an affirmative creative experimental ethos and a challenging orientation to the material world, including the nature of spaces, materials and sensations that this entails. Deleuze's unique position can be used as a platform to explore how architecture can creatively experiment in ways that resist the forces of organisation and control that otherwise prevent it from inventing new architectural responses to life. If Deleuze is understood as committed to the ongoing search for thought without image by using philosophy as the art of creating concepts, then what might the implications of this approach be in an architectural context? Using the concept of metallurgy as a springboard, this chapter will commence with a critical and propositional analysis of the concept of materiality in *A Thousand Plateaus* and *What Is Philosophy?*' Subsequently, it will explore how Deleuze and Guattari's provocative concept of sensation is integral to an understanding of its material constitution. How materials and sensation are composed to generate their effects is simultaneously a question of spatial relations and their capacity to resonate with each other and with us. This allows me to advance the proposition of architecture as a sensation machine. By detailing their concepts of the monument and the house (among others), I will argue that Deleuze and Guattari's work might suggest the traces of an experimental method for architectural practice to experiment with in order to reinvent itself into futures yet unseen.

The monument redefined is a call to creatively consolidate new structures oriented towards an open future resonating with the vitality of sensation as its enduring material. Rather than preserve politicised and idealised past events, the primary focus is placed on the creative production of art's capacity to memorialise

aggregates of sensation which persist beyond the process and its constitutive material forms. *What Is Philosophy?* explores a number of artistic fields as different moments of art's processes of monumentalising sensation, namely painting, sculpture, literature and music, however, architecture is described as the first art.[1] Architecture is important as a precursory structural organisational process, a territorialising movement which consolidates its building materials from the flows of a vitalising substrate of 'material and organic indeterminacy'[2] in order to construct the supporting structures for life, including art.

The resulting problem of how to constitute these structures and how they are understood to operate requires a deeper investigation of what relationship an architect might have to the materials developed for this primary purpose. Does the particularity of an architect's understanding, constitution and configuration of the materials themselves help to suggest an alternate conception of an architect as a metallurgist of sorts, wielding materials to translate physical forces into spaces for life, tracing lines of material and developing paths for expressive traits in order to generate the structural machinery for new aggregates of sensation?[3] It remains an open question how such a radical suggestion might come to influence experimental architectural practices and what future implications this may have. The following work seeks to examine how Deleuze and Guattari see the conjunction between architecture and art through the reimagined concept of the monument, before examining the implications for architectural practices reoriented this way. How might we understand the manner in which architecture achieves this aim, and what are the associated implications both in terms of architecture's functional role and the concept of its materials?

Setting out in the following with an oppositional definition of the traditional and redefined concepts of the monument, I will move on to review the connection between the architecture and art nexus through the concept of the refrain in *A Thousand Plateaus*, and its later development in the redefinition of the artistic function of monumentality in *What Is Philosophy?*. Combining monumentality as sensation with a conception of architecture as a practice that structures and consolidates vital materiality, I will argue that Deleuze and Guattari's thinking starts to sketch a way to broaden the practice of architecture beyond its current limits and suggest an exploration of possibilities for a variable, undetermined and unexplored future. Propositionally then I will argue it is the material dimension of architecture that makes its territorialising function so unique and is best explored as a practically applied or working concept. It is this idea of creating consolidations and consistencies through experimental practice, illustrated with fragments and clues throughout the development of the redefining of the monument in *What Is Philosophy?* that proposes a provocation for architecture to recompose its understanding of its very materiality, processes and purpose.

A TRADITIONAL CONCEPT OF THE MONUMENT

Traditionally understood, monuments are designed to resist the forces of time in order to preserve and project a particular inflection of selected memories. As a starting point, this definition raises at least two particular lines of inquiry around the concept of preservation. First, what is the genesis of the drive to preserve and promote a particular memory, life or event, that is, what are its intentions in a formal, spatial and socio-political sense? Second, how might the materiality of the monument resist the inexorable entropic processes of decay in order to preserve its capacity to emit its instructive purpose, a question about the intersection of the material and temporal aspects of the will to monumentalise. The problem of preservation and commemoration is characterised by the architectural expression of its social and political context as Bataille observes:

> Architecture is the expression of every society's being. . . . [But] only the ideal being of society, the one that issues orders and interdictions with authority, is expressed in architectural compositions in the strict sense of the word. . . . Thus great monuments rise up like levees, opposing the logic of majesty and authority to any confusion: Church and State in the form of cathedrals and palaces to speak to the multitudes, or silence them. It is obvious that monuments inspire social good behaviour in societies and often even real fear. The storming of the Bastille is symbolic of this state of affairs: it is hard to explain this mass movement other than through the people's animosity (animus) against the monuments that are its real masters.[4]

Bataille's thoughts on the monument as a preserver of order and state-sanctioned memory underline and alert us to some of the problems the static, controlling and idealising function of architecture creates. The concept of the monument is a general category that traces its way through all architecture as a function of its desire to preserve itself and in doing so a particular version of us as its makers. This is closely coupled with the literal materiality of the monument, a symptom of the underlying belief that in order to preserve an aggregate of memories, the processes of its genesis and formation, the materials themselves must resist the forces of chaos, weathering and decay inflicted by a hostile exterior environment. Monuments combine a formal agenda of instruction with a material programme of enduring solidity, a hardness which flies in the face of decay. This twofold agenda helps to explain the fascination with decay over time and the love of ruins, that is, monuments of lost events and empires ravaged by time, relegated to history, yet on rediscovery still imbued with the power to speak of their origins and their longevity. This is a primary purpose of the monument, illustrating an underlying idea that the event can be preserved in the material as long as the object itself persists. The museum's halls overflow with stone and bronze messengers from the past,

seeking to preserve the events that presided over their creation, as does the museum building itself. This aspect of architecture preserves the spaces of shelter and itself in order to house and protect the past. The resultant problem is the underlying concept of the event as static in the sense that it is inexorably and invariably coupled to the material forms that its creators hope will continue to exist through the passage of time. This is why the victors destroy the monuments of their conquered cities so that they may clear the ground and recreate future memories according to their own version of history. Without the constitutive and sustaining function of its own material structure, the monument ceases to maintain its capacity to speak of its origins and fails to convey its intended message to the world.

REDEFINING THE MONUMENT: FROM CONCEPTUAL TO MATERIAL VITALISM

'It is true that every work of art is a *monument*, but here the monument is not something commemorating a past, it is a bloc of present sensations that owe their preservation only to themselves and that provide the event with the compound that celebrates it' (Deleuze and Guattari, 1994:167–68).

The redefinition of the monument is no longer thought as a simple combination of materials and forms, rather it is a process of consolidation of materials structured to capture sensations through experimental methods of artistic production. Once the sensations are expressed through these material compositions, they transcend them and coagulate as blocs of sensation that persist beyond them.

The monument now alters its traditional reliance on the materials themselves enduring and translates the idea of preservation to a more mobile process carried out by the aggregates of sensation. Rather than the willful projection of idealised social beings into the future as immobile, resistant and unalterable monumental markers, Deleuze and Guattari focus on the process of translation and constitutive separation of sensation from the material structures that are created to support, capture and express them. Starting with this simple definition of a complex and richly nuanced conjunction of ideas centred around the concept of the monument, it is the act of consolidation and structuring that allows the elaboration of an innovative approach to what architecture does and how it can be reoriented to the materials it makes.

In the first instance, the task of preservation itself is coupled to art and its plane of composition: 'Art preserves, and it is the only thing that is preserved. It preserves and is preserved in itself, although actually it lasts no longer than

its support or materials – stone, canvas, chemical colour and so on'(WP: 163).[5] Initially this statement is paradoxical: how is art preserved if it can't outlast its materials? The answer comes from an unexpected direction, which posits a two-stage proposition. First, percepts and affects (the elements of sensation) are different from perceptions and affections, they are understood as something that is detached from the body that experiences them, setting up a shift away from the materially dependent idealised beings of society towards pure beings of sensation:

> Percepts are no longer perceptions; they are independent of a state of those who experience them. Affects are no longer feelings of affections; they go beyond the strength of those who undergo them. Sensations, percepts, and affects are *beings* whose validity lies in themselves and exceeds any lived. They could be said to exist in the absence of man because man, as he is caught in stone, on the canvas, or by words, is himself a compound of percepts and affects. The work of art is a being of sensation and nothing else: it exists in itself (WP: 164).

Second, the composition of a monument is a constitutive process which captures flows of sensation that endure beyond their genesis. Creating affects and percepts is not enough, they must be preserved through their ability to combine with other sensations and subsequently exist independently, this is the initial redefinition of the monument: 'The artist creates blocs of percepts and affects, but the only law of creation is that the compound must stand up on its own'(WP: 164). Materials remain an important part of the creation of sensation, but we are no longer required to remember the act and implied significance of a creation itself in the traditional 'monumental' way. Compounds of materials give way to the compounds of sensation they create, which take on a trajectory of their own without further reliance on the structures that supported their emergence.

One problem the reliance on the literal preservation of the traditional monument presents for Deleuze and Guattari is the way in which it misunderstands the idea of duration. By conceptualising materiality as modulations of matter-energy flows with variable states of mixture and consistency, the very notion of fixing something according to a logic of resisting the flow it wants to remember is problematic in its objectifying attempt to cut off a material from its source of vitality. Conceptually, material is better understood as a moment of consolidation according to the singularities of consistency and organisation that define it. Accordingly, any material consolidation will be subject to ongoing deformations and changes in its configuration, even at a physical level, and as such resists objectification in any permanent sense. Even so, how consolidation in this sense is achieved and how it interacts with the problem of how a monument as blocs of sensation might operate beyond its materials remain to be worked through.

REFRAIN (ART-ARCHITECTURE) MONUMENT)

In *What Is Philosophy?* Deleuze and Guattari redefine the concept of the monument within the broader discussion of the function of artistic endeavour.[6] Against the background of philosophy as the creator of concepts on a plane of immanence, and science of functions on a plane of reference, art is seen as the field that is capable of capturing sensations as Grosz neatly summarises: 'What distinguishes the arts from other forms of cultural production are the ways in which artistic production merges with, intensifies and externalizes or monumentalizes, sensation'.[7] But in order to capture and express sensation, Deleuze and Guattari detail a movement of material consolidation that forms the scaffolding of support and capture needed to achieve this aim. We can see the idea of consolidation introduced in an artistic context in the opening scene in 'Plateau 11. 1837: Of the Refrain' in *A Thousand Plateaus*, where the child in the dark consolidates a space though the song he sings to himself. The territorialising movement which starts to create a territory, a home, begins with a jump 'from chaos to the beginnings of order . . . home does not preexist: it was necessary . . . to organize a limited space'. (Deleuze and Guattari, 1987: 311). In other words, the internal refrain of the child consolidated as space is an early example of the territory-house system.[8] In *What Is Philosophy?*, this concept is developed by the example of the bird turning over leaves on the rainforest floor to attract a mate.[9] The bird constructs a territory-house system which transforms the functional task of consolidating leaf matter into an expressive demarcation of its territory-house. It is a simultaneous act of spatio-material consolidation and expression. The key movement of the territory-house system is the translation of functions into expressions, as Deleuze and Guattari write, 'with the territory and the house it (expressiveness) becomes constructive and erects ritual monuments of an animal mass that celebrates qualities before extracting new causalities and finalities from them'(WP: 184).[10] Importantly, this expressive consolidation is not given, it is creatively constituted by the self-expression of a refrain, an ordering of the world that comforts a child lost in the void, or that constructs a system to define the bird's home as an invitation to its mate.

The processes of consolidation are many, varied and operate across different registers. The child's refrain introduces the general concept of consolidation, however, in the present discussion we are particularly concerned with the relationship between how materials are made and how this might inform our use of them to construct works that monumentalise sensation. For Deleuze and Guattari, this brings the art of architecture into focus because it is seen as the first art, the initial framing movement of consolidation that sets up sections and frames over the vitalised matter-force movements in order to territorialise zones of stability. In *A Thousand Plateaus*, architecture is posited 'as the art

of the abode and territory' (ATP: 329), which evolves into the claim in *What Is Philosophy?* that 'Art begins not with the flesh but with the house. That is why architecture is the first of the arts' (WP: 186). In both cases we can see architecture as the process that creatively structures the spaces of consolidation from which artistic endeavour might spring forth. The house combines frames of consolidation in order to support new ways to experiment beyond its initial construction. It creates a constitutive structure which facilitates the possibility for the design of new compositions that remain open to the underlying vitality of the matter and forces it uses to create its enclosures. Bernard Cache's work explores the suggestion Deleuze and Guattari make in great detail by taking as its starting point the conjunction of the frames or sections of architecture as the place where compounds of sensation are held.[11] The creative consolidation of architecture is however a reflexive movement that not only builds but also holds itself open to destruction. It also embraces the deframing movements of decay and collapse because this counter-movement keeps it open to the vitalising flows of undifferentiated matter and force. So when we are told 'Consolidation is not content to come after; it is creative' (ATP: 329), the creative force operates both to construct and destroy so that we might continue to experiment anew. This is one aspect of the conjunction of an underlying concept of matter as vital and architecture as a consolidating territorialisation that resolves in the monument redefined. We must create the materials to frame the capture of sensation, which endures beyond the materials before dissolving in deframing movements in order to return to the revitalising flow of force-matter.

ARCHITECTURE AND THE MATERIALITY/ SENSATION RELATION

Matter itself is nothing without being subjected to force. Physical forces are part of what animate and vitalise matter. This is why movement for Deleuze and Guattari is always implicated in the shifting configurations of matter according to the forces that traverse, structure, deform and recompose it. Again, inflected slightly differently, they describe the problem of consistency and matter in order to harness these forces.

When forces become necessarily cosmic, material becomes necessarily molecular, with enormous force operating in an infinitesimal space. The problem is no longer that of the beginning, any more than it is that of a foundation-ground. It is now a problem of consistency or consolidation: how to consolidate the material, make it consistent, so that it can harness unthinkable, invisible, nonsonorous forces (ATP: 343).[12]

On the one hand, architecture frames and territorialises matter, but it attempts to achieve this in order to harness forces yet to be expressed and

experienced. Conceptually, this substitutes the traditional binary opposition between matter and forces by understanding them instead as fundamentally structuring each other, as interrelated vital flows consolidated, slowed, trapped and manipulated through experimental practices. So if matter is vital, then architecture can attempt to render the forces of life itself visible through the consistent arrangements of matter and space it draws. Architecture's task is to make perceptible the imperceptible forces that populate the world. However, architecture is required to go one step further. For it is not enough to simply summon the forces of gravity, heaviness, rotation, the vortex, explosion and so forth, the physical forces that configure materials themselves. Architecture must strive to allow us to feel these forces, which is something different to rendering them visible, habitable, and in doing so capturing them as sensory aggregates. It is a combination of consistency, matter and the spaces of sensation that architecture creates that offers us a way to consider the architectural problematic field differently, to feel the forces not just of the physical world but of life itself. The challenge for architecture then unfolds to see if it can not only reveal the hidden forces of our vital materiality but also to see if it can create spaces where we might discover new sensations. In the end, the task here is to understand architecture as a way to create new paths for life itself to explore. Materiality is not something separate to the sensations it produces or the compositional refrains that structure it. Rather it is a collision of the concept of what might be designed, qualified by its ability to support the production and capture of sensation, regardless of how fleeting this might be. A two-way movement between the creation of the materials and our experience of the spaces and sensory environments, this frames and captures.

VOID AS SENSATION

How materials and sensation are composed to generate their effects is simultaneously a question of spatial relations and their capacity to resonate with each other and with us. If the creative concern of the architect and artist is to produce the structures of capture in which sensations that hold up well, there needs to be literally the space around and within them, allowing the emission of their sensate flows with a certain degree of focus. Deleuze and Guattari mention the sensations of the mad, which although generally do hold up well, suffer from a lack of empty space.[13] It is this empty space that facilitates specific and unique oscillations throughout the composed aggregate. Or we could simply say that 'the void is sensation' too. As Deleuze observes in a particularly architectural passage, 'blocs need pockets of air and emptiness, because even the void is sensation. All sensation is composed with the void in composing itself with itself, and everything holds together on earth and

in the air, and preserves in the void, is preserved in the void by preserving itself'(WP: 165). This statement clearly makes a link between the concept of the monument (of sensations in the virtual) and the concept of the void as the non-space of composition of sensation. This could be interpreted to suggest an architectural vision that schizophrenically invades space, leaving behind the task of prying out or perhaps inserting or growing pockets of emptiness within the madness in order to release the sensations trapped by their dense overcrowding. The creative design task would be to reveal sensation by growing voids in order to filter material flows, in order to create the space in which blocs of sensation can resonate along different paths to consistency. Such a proposition, if nothing else, serves as an example of a potentially useful design 'tactic' especially in the context of today's oppressive bombardment of seemingly inescapable information overload and digital disruption. If an architect's task is to lay a plane out over chaos, then perhaps we might experiment with doing this by the mediation of voids and emptiness as alternative ways to order chaos and escape the hylomorphic material schema at the same time.[14] Such an idea is exemplified by Deleuze and Guattari when discussing the Chinese painter Huan Pin-Hung's comments on the work of art needing to 'save enough empty space for horses to prance in (even if this is only through the variety of planes)'(WP: 165–66). The insertion or simple presence of space between intersecting or transversally connected Deleuzian 'planes' creates the necessary void(s) and as such is not a response to a two-dimensional reading of space or emptiness but a multidimensional conception. Voids in the multiplicity as sensation in themselves. Another illustration can be found in the work of the Italian architect Carlo Scarpa, who Deleuze and Guattari describe as an architect who uses the idea of inscribing volumes in thickness, hollowing out space, or according to the proposition previously discussed, we might call this the insertion of voids as sensation into a plane of composition. They refer to Hubert Damisch's interpretation of Scarpa as one who 'suppresses the movement of projection and the mechanisms of perspective so as to inscribe volumes in the thickness of the plane itself'(WP: 195). While there is clearly a critique of representational thinking in this statement, there is also an underlying suggestion that subtraction is an effective tool as projection. In Scarpa's case we are presented with a complex, three-dimensional network of intertwining stone-clad sculptural spaces, the effect of which appears to generate a minimalist sophistication between the interacting planar volumes and voids.[15] This illustrates Scarpa's consistent attempt to place his spatial design thinking upon the 'plane of composition' by substituting the need to put something into perspective with the processes of hollowing it out from the depths of the material.

Sensation then is a complex concept for Deleuze and Guattari, but a crucial one that uses a philosophical rendering of materiality to construct the

platform for new experience and new ways of living; 'sensation refers only to its material: it is the percept or affect of the material itself, the smile of oil, the gesture of fired clay, the crouch of Romanesque stone and the ascent of Gothic stone. The material is so varied in each case that it is difficult to say where in fact the material ends and the sensation begins' (WP: 166). Correspondingly, the synergy between Deleuze and Guattari's neo-constructivist materiality and one of architecture's underlying aims can be described in terms of the concept of sensation and the way it uses and transforms materiality. John Rajchman provides us with a useful identification of what is unique about Deleuze's concept of sensation when he writes that:

> Materials become expressive (or unfold a 'will-to-art' or 'becoming-art') . . . the aim of art is, through expressive materials, to extract sensations from habitual sensibilia – from habits of perception, memory, recognition, agreement – and cause us to see and feel in new or unforseen ways. The composite of sensations that is the work of art (or the work in art) is not to be confused with its material support, nor with techniques. It is a peculiar thing, which precedes and may survive the physical supports and technological means without which it would nevertheless not exist.[16]

The construction of the right structural conditions for sensation to emerge is one part of the problem, another would be the right balance of spatial relations in order for sensation to resonate, however, the question still remains as to how the material and sensation interact against this background. Deleuze's comments on this particular and crucial moment are circuitous and somewhat enigmatic. For example, he asserts that sensation and material are different things, yet also claims that there is a blurring of the boundaries between them, a zone of indiscernibility where certain slippery transmissions between the two take place. As he states, 'the plane of the material ascends irresistibly and invades the plane of composition of the sensations themselves to the point of being part of them or indiscernible from them. . . . And yet, in principle at least, sensation is not the same thing as material'(WP: 166). We can clearly see Deleuze's use of the Bergsonian conception of duration as the bridge to explain how sensation is preserved without an ongoing reliance on its material components: 'Even if the material lasts for only a few seconds it will give sensation the power to exist and be preserved in itself in the eternity that coexists with this short duration'(WP: 166). Sensation is understood to be preserved in duration, so once it is created through the architect's use of steel and stone, the painter's use of colour, brush stroke, and so on, in other words, the literal materiality, the unleashing of sensation is preserved in duration and filters into the virtual realm ready to eternally return and affirm its ongoing productive affectivity.

Architecture can subsequently be understood as 'self-preserving' in its creative and expressive possibility, in its affirmation of sensation as its

monument, not the literal material actuality but a creation of sensation from the opposite direction, doubled back on its virtual preservation by capturing and expressing new affects through the following of 'actual' material traits. Materiality is thus a creative following in and out of striated constraints searching for different material consolidations and consistencies and the way they are translated into different architectural forms and spaces. A great game of capture and release. Experiment to capture sensation through our material structures, so that it may escape as expressed sensation forming a monument. Compounds of sensation as monuments in the virtual that eternally return in order to affirm their ability to stand up on their own thus develop the ground for further and future possibilities.[17] Architecture is not limited to building its own objects but retasked as a structuring process that captures sensation which exceeds its creative material genesis. Deleuze and Guattari speak of architecture as constructing the faces of a dice of sensation, framing each side before being rolled to reveal an unknown result, holding back the limiting force of predetermination.[18] This posits the redefinition of the monument as open to undetermined possibility, a radicalisation of architecture's powerful desire to court time in order to trap it in stone. Instead, physical material will dissolve only to rejoin and be revitalised by a more fluid pulse of matter-energy movement before finding another consolidation and material configuration. Notably this position counters the traditional monument's idealised and ultimately unsustainable aim of striving to physically persist unchanged through the ages. It is also possible to see a conceptual overlap with notions of impermanence and non-attachment in Buddhism, overlaid with a Nietzschean-Bergsonian affirmative drive towards creative evolutions beyond the existent and operatively enduring in the virtual.[19] Deleuze and Guattari's concept levels a difficult question at its traditional foundations by altering the relation between memory and the persistence of the physical object, simultaneously resisting the egotistical drive to be remembered according to how big, how well built, how expensive a building is.

In summary, Deleuze and Guattari's redefinition of the monument can be understood as a doubling of its traditional conceptual basis. Where at once the material itself must endure and resist the passage of time, now it must also produce the material structure that can become productive of sensation that persists beyond the material. The material once resistant is now more fluid and temporary in order to generate effects which endure. Material is no longer engaged in a fight against the physical forces of degradation and time, rather this idea of duration translates into the sensate aggregates which move beyond it. One could argue therefore that sensation becomes the material that endures. Now the placeholding, marking and commemorative function of the traditional monument persists, but again undergoes a transformation. Where place, time and history were marked by the material and form, now they are

released beyond it as sensation, carrying traces of their generative origins. In other words, the locus of the monument and its originary context are taken up in the production of sensation, but not fixed to it. In fact the radical element of Deleuze and Guattari's redefinition of the monument lies in the proposition that sensation endures beyond the materials. It does not matter then if the building, sculpture, painting or melody decays beyond recognition or recovery as the effects it has produced are preserved in their interaction with other sensate blocs which modulate and interact to take on a life of their own. It is the vitalising movement of trace and transformation that resonate into the future as possibility itself. Monuments may thus be described as evolving beyond an intended narrative reification of their commemorative and historically instructive functions and substituting sensate aggregates energised by an underlying material vitalism as possibility for the future to come. Traditional monumentality insists on its particular and immobile vision of the future, as a voice from the past claiming the future as its own. A paranoid and desperate claim against any alternate future interpretation of its fading past. By contrast, the monument redefined posits a future of possibility that remains resistant to predetermination, the controlling forces of the hylomorphic paradigm and the underlying conceptual position that matter itself is inert rather than vibrant and vitalising.

Importantly, there is no rule for how to construct blocs of sensate matter, but there is a requirement that they function in a very particular fashion if the artist's goal is to construct a monument in the sense Deleuze and Guattari outline: 'Standing up alone . . . is only the act by which the compound of created sensations is preserved in itself – *a monument*, but one that may be contained in a few marks or a few lines'(WP: 164–65; my italics).[20] It might be realised in a vast urban landscape, or contained within a few marks as they say, the field of exploration is not specified whereas the idea of its outcome is as a performative standard. How then might we develop our own methodology for consolidating materials and capturing sensations?

FRAGMENTS OF AN EXPERIMENTAL METHOD IN *A THOUSAND PLATEAUS* AND *WHAT IS PHILOSOPHY?*

For architects there is a persistent question which asks how this rich conceptual stance can be integrated into the practice of architectural design and production itself, that is to say, how can Deleuze and Guattari's provocations about architecture as a machine that structures the spaces of perceptual and affective capture translate into themes for experimental design practice? Without being prescriptive or suggesting that Deleuze and Guattari's thinking on art and architecture throughout their work offers a design method

that might literally translate into practice, the present discussion does trace a number of experimental lines for us to examine further.

The example of Carlo Scarpa composing with voids, stone and water, outlined earlier, is one of few instances we are given in *What Is Philosophy?* which traces a working process observed in an exemplary architect. Adrian Parr's rich and provocative reading of the design process, construction and surrounding politics of Maya Lin's Vietnam Veterans Memorial in Washington, DC provides us with an excellent illustration of the role sensation can play in a more contemporary architectural project. Designed as an open wound in the earth, the memorial is described as aspiring to achieve an absence from the landscape (after Cézanne's call to get too close to the wheat field [ATP: 493]), and resonate with the sensation of wounding the landscape as a visceral and material connection to the suffering of war. Parr picks up the use of materials in new ways to produce this particular 'block of sensation' when writing that 'Working with everyday architectural elements in unconventional ways infuses an affective intensity into the memorial, one that turns the self-justifying aesthetic of minimalism on its head, releasing a self-referential aesthetic in its place'.[21] Working with existing materials in new ways is one approach. Creating new material consolidations is another that can combine to vitalise architecture and art as creative experimentation.

Deleuze and Guattari suggest that in order to extract blocs of sensation, 'a method is needed and this varies with every artist and forms part of the work' (WP: 167). The procedures must be invented in the search for the beings of sensation and in the process form part of it: 'in this respect the writer's position is no different from that of the painter, musician or architect' (WP: 167). This idea was presented earlier in *On the Refrain*, inflected slightly differently as the process of artistic consolidation being a creative one. Here Deleuze and Guattari focused on the link between materiality and the indeterminacy of its origins, writing that the process of consolidation is structured by mechanisms of 'intercalated elements, intervals & articulations of superposition'.[22] How this consolidation occurs and what are its implications for our future attempts to create new material combinations remain to be experimented with, however, the typology sketched out in the following helps elucidate the position that sensation itself is the material beyond the materials.

THE GREAT MONUMENTAL TYPES OR VARIETIES OF COMPOUNDS OF SENSATION

A salient moment presenting a strong spatial elucidation of sensation as monument occurs when Deleuze and Guattari turn to a discussion of 'monumental types' (WP: 168). I have tabulated these in the following to illustrate

the physical even visceral sense that sensation has, as it vibrates, conjoins and opens up:

Table 14.1

Great 'Monumental Types' or 'Varieties' of Compounds of Sensation

The vibration (vibrating sensation)	Simple sensation
The embrace or the clinch (coupling sensation)	Two sensations resonate with each other in a clinch of what is no more than 'energies'
Withdrawal, division, distension (opening, splitting or hollowing out sensation)	Two sensations draw apart, release themselves, so as to be brought together by the light, the air or the void that sinks between them or into them forming a bloc that no longer needs a support

The creation of new sensations is a matter of preservation, contemplation and contraction but understood in a typically unusual manner. This is a description of a creative process, a mode of engagement and operation, not an abstract theoretical observation or claim. Looking at table 14.1, we can see the way Deleuze and Guattari understand a generic interaction according to differing principles of combination, deformation and alteration. Crucially we see the concept of resonance and vibration as processes of consolidation, of contraction. The subsequent thoughts about preservation Deleuze and Guattari add to this help to suggest a direct physical link which shows how the all important indeterminable spaces between sensation and materials might work. The relation is between material and sensate vibrations that interact and interfere with each other: 'Sensation itself vibrates because it contracts vibrations. It preserves itself because it preserves vibrations: it is Monument. It resonates because it makes its harmonics resonate. Sensation is the contracted vibration that has become quality, variety' (WP: 211). It is not the transformation of one into another that matters but that something passes from one to the other. This 'something' can be specified only as 'sensation'(WP: 173). This serves to illustrate a subtle difference between physical material change and immaterial change as matter and sensation commingle and interact. It is yet another facet of the recurring theme we are seeing of vague essences, the indeterminate and the in-between shifting spaces of exchange tracing throughout the discussion at hand, highlighting co-creation, transmission and indeterminate exchanges as the subtle yet definitive operations at play between material consistency and the production of enduring sensation. Indetermination is a theme central for Deleuze and Guattari, as it creates a vitalised mixture where matter becomes sensate: 'Life alone creates such zones where living beings whirl around, and only art can reach and penetrate them in its enterprise of *co-creation*. This is because *from the moment that the*

material passes into sensation, as in a Rodin sculpture, art itself lives on these zones of indetermination' (WP: 173; my italics).[23] Therefore, what is required is a process that translates or transforms and strips back the noise from the lived perceptions and affections that emerge and seeks the pure being of sensation, the percepts and affects, and for Deleuze and Guattari this process is only possible in the artistic plane.

The clear mandate given to art in order to carry out these operations of sensation remains in question. Why they couldn't at least be contaminated by interdisciplinary influences from the scientific plane of reference and the philosophical plane of immanence seems open to further debate. However, I think Deleuze and Guattari are more concerned with the working methods of artistic production as a means to access something unique to art, without specifying the many and varied ways this can be influenced from without. They would no doubt support all manner of cross-fertilisations between art, science and philosophy, while insisting that they each have different working methods and different materials, whether they be conceptual, functional or affective. Architecture as the first art, which attempts to lay out its planes over chaos, however temporarily, could hint at an answer to the preference for art, as art itself 'lives on these zones of indetermination', that is, the shifting combinations of structure and material arrangement that form planes over chaos, and as such reconceptualises architecture as the fabulation of sensate rather than literal monuments.

A monument in this sense never marks something which has already been, something that preserves the past, rather the redefined monument marks the material dissolution of forms into indistinction, or at least the proximity to indefinable zones of indeterminate relations where the results of new or simply different combinations of the sensation, as material relations in each particular manifestation remain open and unknown. Deleuze and Guattari argue that:

> Painting . . . needs the power of the ground that can dissolve forms and impose the existence of a zone in which we no longer know which is animal and which human, because something like a triumph or *a monument* of their nondistinction rises up – as in Goya or even Daumier or Redon. The *artist must create the syntactical or plastic methods and materials* necessary for such a great undertaking, which re-creates everywhere the primitive swamps of life
>
> (WP: 173–74; my italics).[24]

The monument dissolves literal materials and returns them to Goya's swamp, which seethes with matter-flows yet to be determined, and where the usually defined boundaries of the human decay into strangely familiar grey zones that cannot be construed as human or animal, organic or inorganic – the soup of becoming. Falling back into the unstructured flows of vitalised chaotic matter therefore sketches an oscillating feedback movement between

unstructured matter and force that vitalises, and the consolidations of it that compose and express the structures productive of sensations themselves. How we design and construct the methods and materials to produce conditions which preserve vitalised indeterminacy while consolidating the ground to do so remains the paradoxical task for any artistic endeavour.

CREATIVE FABULATION

Another instructive trace of an experimental method is encapsulated in the idea of 'fabulation', a Bergsonian inspired call for artists to go 'beyond the perceptual states and affective transitions of the lived. The artist is a seer, a becomer'.[25] The concept of fabulation is deployed to underscore Deleuze and Guattari's claim that the monument is constitutive of becomings produced through the affective triggers of sensation. It is a living, vital and transformative view of memory that constructs the present, fabricates, actively makes or 'fabulates' it, rather than a simple application of non-vital or otherwise disconnected memories applied to the present that can only remember it without altering it somehow: 'The monument's action is not memory but fabulation'(WP: 167–68).

One of the most challenging aspects of this proposition is the blurring of the traditional boundaries between the material and sensate as Deleuze and Guattari underline: 'Sensation is not realized in the material without the material passing completely into the sensation, into the percept or affect. *All the material becomes expressive*. It is the affect that is metallic, crystalline, stony and so on; the sensation is not coloured . . . but colouring' (WP: 167; my italics). By showing the difference here between the material as affect and the sensation as affective, the idea of affective compounds (blocs of sensation) influencing our compositional experiments becomes slightly more tangible. The material is both expressive in itself and expressive through the effects it has on us. In an attempt to distill this idea, it is useful to consider the relationship between sensation and materials as fluid and interactive, but it is difficult to grasp given the strong hold the hylomorphic model exerts on our intuitive understanding about how we are affected by and through our interactions with other material bodies. Fabulation calls the artist to create beyond the constraints of the world, or more precisely, by experimentally recomposing the way one structures it in order to unleash the potential for new sensations. The architect as artist then illustrates a two-way movement of materialism reconceptualised this way. It is constituted by us as a consolidation and draws us into the material at the same time through its affective impact. In effect we make the material and become part of it: 'It should be said of all art that, in relation to percepts or visions they give us, artists are presenters of affects,

the inventors and creators of affects. They not only create them in their work, they give them to us and make us become with them, they draw us into the compound' (WP: 175).

CONCLUSION

By redefining the concept of the monument in *What Is Philosophy?* Deleuze and Guattari also develop an original theory of materiality in opposition to the dominant hylomorphic paradigm. In response to their question 'How can a moment of the world be rendered durable or made to exist in itself?' (WP: 172), they present a rich reconceptualisation of what artistic practice is and how it can be oriented towards an open future of experimental undetermined possibility, including the particular role architecture plays. Material vitalism combines with a series of constructively creative movements that consolidate and harness force-matter flows in order to structure new spaces and capture the creation of new sensory aggregates that can endure beyond them. The architect's role is also conceptualised differently, modulating away from the use of materials as purely static, towards design as a process that actively constitutes the materials and structures they are yet to explore in a more variable, flexible and unexplored way. While Deleuze and Guattari do not proscribe a method, there are traces of how our experiments might be thought winding through this area of their theory of art, operating more as a rough patchwork of provocative notions than literal instructions. The monument redefined combats the limiting nature of materiality as hylomorphism, breaking open the way to experiment beyond by understanding the relationship between material vitality and the forces that traverse it. I have listed only a handful here to illustrate my proposition, however there are many more provocations, clues, fragments and lines of flight to further inspire a constructive reworking of architectural practice in this direction. The practical challenge ahead lies in our ability to consolidate new territories no matter how fleetingly in order to support unknown structures built from as yet unseen material compositions, capturing cascades of sensation that might endure beyond their genesis to build new monuments of life itself.

NOTES

1. See a more detailed discussion of this concept in the 'Refrain (Art-Architecture) Monument' section of this chapter.
2. Grosz (2008: 3).

3. For a detailed discussion of this idea see Hale (2013), 111–30, and Hale (2011), Chapter 3. 'Materiality: Metallurgy and Lines', 117–78.

4. Bataille, Georges (1971–1988). 'Architecture', in *Oeuvres Complètes*. Volume 1. Paris: Gallimard, pp. 171–72 cited in Hollier (1989: ix–x).

5. A similar question is posed when Deleuze and Guattari state, 'The young girl maintains the pose she has had for five thousand years, *a gesture that no longer depends on whoever made it*. The air still has the turbulence, the gust of wind, and the light that it had that day last year, and it no longer depends on who was breathing it that morning' (WP: 163; my italics).

6. See Deleuze and Guattari (1994), Chapter 7. 'Percept, Affect and Concept'.

7. Grosz (2008: 4).

8. For a lucid explanation and exploration of the 'territory-house' system and its architectural context, see both Ballantyne (2007: 38–60), Chapter 3. 'House'; and Grosz (2008: 1–25), Chapter 1. 'Chaos, Cosmos, Territory, Architecture'.

9. WP: 184.

10. My insertion of 'expressiveness' in parentheses.

11. See Deleuze and Guattari (1994: 187), where Deleuze and Guattari mention Cache's work; and Cache (1995). See also Deleuze and Guattari (1994: 197) where Deleuze and Guattari describe how 'The frames and their joins hold the compounds of sensation'.

12. ATP: 343.

13. See Deleuze and Guattari (1994: 165), where Deleuze and Guattari are working through different modes of generating sensation to see if they can stand up on their own, for example, through drug-induced creation, or the drawings of children. The creation of sensation through madness is understood as more likely to actually hold up, but this is because they are 'crammed full' of sensation. The problem then becomes the manner in which this unchecked overflow of sensation overwhelms us and effectively erodes our ability to properly feel or identify specific sensations. So although sensations can stand up through proliferation, as is the case with the mad here, there also needs to be enough space around then so that we can properly experience them.

14. Hylomorphism is defined as the philosophical doctrine that physical objects result from the combination of matter (hylo-; *hulē* in Greek) and form (*morphē* in Greek). It assumes that 'order displayed by material systems is due to the form projected in advance of production by an external producer, a form which organizes what would otherwise be chaotic or passive matter'. Therefore matter is 'dumb' and secondary to the ideas which transcend and precede it. See John Protevi's entry in Protevi (2005: 296–97). 'In *ATP* Deleuze and Guattari pick up the critique of hylomorphism in the work of Gilbert Simondon and follow him in developing a non-hylomorphic or artisanal theory of production. In this theory, forms are developed by artisans out of suggested potentials of matter rather than being dreamed up by architects and then imposed on passive matter. In artisanal production, the artisan must therefore surrender to matter, that is follow its potentials by attending to its implicit forms and then devise operations that bring forth these potential to actualize the desired properties'. See also Hale (2013: Note 6 and page 129).

15. See, for example, the canal front sculptural voids and stone block planar design in Querini Stampalia Foundation (Venice) which are filled and emptied of canal water on the tide. See Murphy (1993).

16. See Rajchman (2000: 135). See also Rajchman (2000: 113–42).

17. Refer to (WP: 177) which qualifies the importance of the possible, not virtual/actual of the event for the concept of the monument.

18. See the architectural passage describing architecture as framing and structuring sensation, the frames of which forming the dice which retains its connectivity to chance and possibility. 'These are the faces of a dice of sensation' (WP: 187).

19. This idea of the impermanence of materiality and Buddhist thought is taken up and developed by Francisco Varela in a paper entitled 'The Re-enchantment of the Concrete.' in Crary and Kwinter (Zone 6: Incorporations. New York: Zone Books, 1992: 320–39). Refer also to See and Joff (2016).

20. The generative concept of the diagram might well be described in a few marks or lines in just this fashion. What matters is how it produces the structure for a block of sensations to stand up with as a function of its architectural composition and the particular or singular way it arranges its spatio-material assemblages in order to do this. The 'diagram' concept in Deleuze and Guattari's work has been an important theme in contemporary architectural discourse. See, for example, Hale (2011: 57–116), Chapter 2. 'Immanence and Diagrams: Control or Design?'; and Garcia (2010).

21. Parr (2008: 65). While the restrictions of space prevent their inclusion here, I have detailed other more contemporary architectural examples to illustrate the thrust of my argument elsewhere. See the examples of Peter Zumthor's Kolumba Gallery in Cologne and the Yokohama Ocean Terminal by Foreign Office Architects in Hale (2011: 131–34;155–78) and R&Sie's *Une Architecture des Humeurs* in Frichot and Loo (2013: 121–28).

22. ATP: 329. We can see an earlier attempt to sketch the different forces of consolidation in *Of the Refrain* which identifies the mechanisms that act to produce consolidated aggregates as 'intercalated elements, intervals and articulations of superposition'(ATP: 329). I have discussed the associated examples of self-supporting structures and the use of concrete in Frichot and Loo (2013: 117–28).

23. Note the influence of Husserl's vague essences here. See (ATP:407) and Hale (2011: 138–39).

24. See the correlation with the wider concept of becoming and sensation (WP: 171–73).

25. (WP:171). The usual definition of 'fabulation' as an experimental approach to producing innovative literary works is also in keeping with Deleuze's position. We can also recognise the influence of Bergson's thinking in this term. Bergson used the term the 'fabulation function' to describe a facet of the imagination that creates voluntary hallucinations. See Bergson (1977). Deleuze and Guattari use the term in *What Is Philsophy?* to describe the active function of the monument: 'The monument's action is not memory but fabulation' (WP: 168), and then in its creative capacity: 'Creative fabulation has nothing to do with memory, however exaggerated, or with fantasy' (WP: 171), rather it is a matter of becoming. Deleuze and Guattari directly reference Bergson (WP: 230, Note 8) where they write, 'Bergson analyses fabulation

as a visionary faculty very different from the imagination and that consists in creating gods and giants.' See also Hale (2011: 198) on Deleuze and Proust's 'fabulations' in *What Is Philosophy?*.

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Chapter 15

Diagrammatic Narratives: Graphic Fields of Rupture and Catastrophe

Anthia Kosma

Abstract

This chapter intends to clarify the Deleuzian concept of diagram and how this one although it is against an ocularcentric culture, clichés and canons of representation, often is being dismissed from traditional attitudes and perceptions in the field of architecture. Diagram in painting for Deleuze is described with the stage of the catastrophe of the frame. It is the gesture that goes against the clichés and the bias and also the opening of an abysm that gradually is organised. This diagrammatic 'writing' and catastrophic gesture, a field of forces and links, is described in the article also through an example, a mural in a historical building in the centre of Athens. The example reveals aspects of the diagram which is interpreted more as an action, a pictorial way of exploration, a process where the 'destruction' of a familiar image opens the possibilities of another image, which was not evident until then (or inconceivable until then).

This chapter attempts to analyse the Deleuzian concept of *diagram* which often – in the field of architecture – is ignored or manipulated by traditional attitudes and perceptions of an ocularcentric culture, of clichés, of the rules of representation and of technological devices.

The views presented here are based mainly on the books *Francis Bacon: The Logic of Sensation* (Deleuze 2003: 99–110) and the *Pintura. El concepto del diagrama* [*Paint. The Concept of the diagram*] (Deleuze 2007) which resulted from the course that Deleuze offered in 1981 at the University of Vincennes on the topics of diagram, painting and the chaos-core theory. In both, *diagram* is presented from an 'inside' point of view, that is to say, through the subjective experience of the artist-designer, as an action-process

of production and not as an object. Therefore, it is being approached not as some kind of drawing, as a form of representation or a specific figure, but 'from inside', as a physical gesture. It is this dimension of the *diagram* that cannot be found often in the field of architecture.

ACTION, EXPERIENCE VERSUS OBJECT

Deleuze, speaking of the *diagram* in painting, presents it, at the birth of the image, as an inscribed gesture. Perhaps the infinitive 'diagramming' is more suitable to describe this approach. This 'inside point of view' and the ongoing process are empirical, almost invisible and thus difficult to be represented. The concept of *diagram* is being displaced away from an explanatory tool-operation-drawing towards a dynamic image of action. It becomes *aesthesis* instead of a simple medium for the organisation of thought. From being a static image, it is now accessed as a gesture-energy, investigated through experience, and also as a code that carries messages that are not based on similarity. This displacement is also associated with the critical thinking of an anti-ocularcentric trend in Western thought that questions the 'dominance' of vision. The *diagram* is almost invisible, and drawing is treated more like a field for the birth of possibilities, upon which – in the case of architectural design – human actions are organised. It is a design process where the architect uses diagramming to explore and differentiate himself from the cliché of well-known typologies (Seguí 2003) and to cause or better facilitate the 'opening' of the form. As Stan Allen observes:

> [T]he diagram is not a thing in itself, but a description of potential relationships among elements; not an abstract model of the way things behave in the world, but a map of possible worlds. (Stan 1998)

However, even this description does not cease to refer to the *diagram* as an explanatory tool, as an organisational chart, without investigating how this dynamic of relationships is formed. This is where the significance of Deleuze lies. He understands the *diagram* as a gesture, as an activity, that is, as directly connected to the body and even to a non-rational situation. In other words, the *diagram* is neither the image of a representation nor the product of a rational creative process but, it is being approached in terms of an ascent/creation of the image from the chaos. This perception is, to a great extent, in opposition to the dominant beliefs in the architectural field, where the image and its organisations arise and appear as products of a rational process.

CHAOS, CATASTROPHE, GERM

Deleuze in his analysis splits time in segments of 'before' and 'after' the *diagram*. Chaos is included in the pre-iconic, preparatory states of figuration and conditions of creations, not as a disorder but rather as an abyss where nothing can be discrete or where at the same time everything can be lost. This risk of loss is a purely pictorial experience. For example, Paul Klee's 'gray point' is this precarious limit where new possibilities might be born for the image to 'surpass itself' through the opening of its sensory dimensions or alternatively to be lost. So a skeleton can be seen in this abyss or rather pieces that can be transformed. To talk about this borderline between chaos and moulding, Deleuze refers to Francis Bacon, who speaks of the *diagram* as a catastrophe:

> The diagram is indeed a chaos, a catastrophe, but it is also a seed of order or rhythm. It is a germ of rhythm in relation to the new order of the painting. As Bacon says, it 'unlocks areas of sensation'. The diagram ends the preparatory work and begins the act of painting. (Deleuze 2003: 102)

Before the *diagram* and before the traces, the movements of the gestures are full of intentions; they are descriptive and perfectly controlled; they follow a narrative and operate more like beginnings. For Deleuze there is no blank page. The white pages are full of thoughts and clichés, and these in turn are full of intentions and trends. In these first movements, including the first traces, the eye follows the familiar, it controls and predetermines the hand movements. In controlled conditions, obviously, it is impossible for something to arise, something beyond the expected. 'The will to lose the will' (Deleuze 2003: 92), says Bacon. The 'work' can begin only when someone is able to reject the clichés and the preconceived images. For Julio Carlo Argan, to project-design or, as we say in Greek 'synthesize' (compose) is possible only through negation.

More specifically, Deleuze describes the *diagram* in painting according to Francis Bacon in this way: 'scrub, sweep or wipe the canvas in order to clear out locales or zones (colour – patches)' and accidentally destroy the painted figure, 'as if a zone of Sahara was suddenly inserted into the canvas' (Deleuze 2003: 93), revealing and creating a world of new possibilities and relationships. While the first images are being erased, previous clichés are deleted along with them, causing an artificial chaos, a 'borders capacitor' where new, unforeseeable forms begin to emerge.

When the *diagram* becomes a gesture-destruction, the body, we could say, is placed out of rational control. The movements are not controlled, the eye can no longer follow and control the hand that is being 'blind' and activated by internal forces. In this way, it opens up to randomness. Rafael Moneo in

'About the Concept of Arbitrariness in Architecture' (Moneo 2005) comments on this aspect of the architectural work and the need to accept this parameter.

The development process of a project – which in architecture is finally presented as a building – is the outcome of one of the possibilities that stood out from this mess-germ at the time of its birth. The arbitrariness of the designing process, the rupture with the clichés and the involvement of the senses are fundamental characteristics, which the discourse in the architectural field systematically chooses to ignore. Ever so often, *diagrams* are presented as steps of a method that seeks to justify and to integrate into a logical explanation the inevitable arbitrariness of conception and the creative process. I speak of course of the arbitrariness that often remains unspoken among the architectural elite (Bregazzi 2008: 147). However, it could be argued that the *diagram* as a process of opening a field of potential relations and as a gesture is important for the acquisition of consciousness – a kind of self-awareness around the creative process in architectural work. Could this intensely subjective perception of Deleuze for the *diagram* against the instrumentalist and ocularcentric approach of the image be liberating for architects?

CATASTROPHE, AN EXAMPLE

Perhaps, one could understand the meaning of *diagram* a little better, as a catastrophic gesture in the work of Deleuze, through the example of a provocative contemporary mural, located at the corner of an important historic building in the centre of Athens. This mural was recently dismantled but had already generated strong critique. For many people this black-and-white mural with random marks and diffused zones was a graffito. It was considered to be an act of vandalism since this part of the building belongs to the complex of the historic edifice in which the struggles of students associated with the 1974 fall of the dictatorship in Greece took place. Behind this building, the school of architecture of the Technical University of Athens is rising, so that the interpretation of the *diagram* may easily acquire a symbolic interest.

The background of the mural consisted of the two main sides of the facade of the historic building. It seemed like a diptych which was 'unfolded' and belonged to an oblong wing of a building with neoclassical features which, as it is often the case, has a base, a main body and a coronation. The mural was at the base and upper level, leaving the ridge of the building intact. The main part of the facade is divided by embossed decorative columns and openings – doors and windows – that give a rhythmical movement to the appearance of the building. It looks as if the mural was developed without any special attention to the classical decoration of the facade, its surface therefore has been treated as if it were flat.

At a first glance, the mural looked like it was a black background with white random marks and spots all over it, but a more close attention to it revealed that the black-and-white brush strokes interfered with one another, diffusing their limits. As a result, it was hard to distinguish between figure and background. The brushes did not represent an object but were following many different directions, contributing to the creation of an expressive gesture, which made the mural an abstract painting.

For many, the meaning of the mural – the gesture that marked the building's facade – is purely symbolic. 'Symvolo' stems from the verb *syn-vallo*. It means therefore 'to put together', 'to unify', 'to construct a distinct entity' 'under' some symbol or other. On the basis of this definition, we could say that the tension surrounding this transgressive mural was the result of the clash of two symbols, two images or better of one symbol and its catastrophy. The gesture of its creators managed to reveal and to display the 'degradation' of a symbol which, until that moment, had been left 'undisturbed'. To erase, in Greek, means also 'to wipe out'; in this strange mural the 'artist' does not represent something, an object located outside of its 'canvas' but rather with his or her gestures and brushwoks he or she begins, piece after piece, to wipe out and to erase the characteristics of a well-known symbolic canvas.

THE CATASTROPHE OF AN IMAGE

Diagram as a catastrophe, on this very symbolically charged canvas, has less to do with the destruction of the surface of the building and more with how this artistic gesture in a public place triggered a series of relationships-forces-capacities. Ultimately, this artistic intervention gives rise to the recognition of its *diagram* as a painting, indeed as a painting over a canvas placed in the particular circumstances of the city and the country where it took place. That is, the known and familiar visual aspect of the *diagram*, as a field of possibilities, as a gesture and behaviour, is connected to the urban space and the city's image, by interpreting and trying to understand a complex spatial intervention. Considering the diagram as an invisible gesture is more of an interpretation than an objective observation. As a subjective interpretation, it tries to find a different meaning on a given image, in this case shifts the perception of the diagram from an object to that of an action and presents the hermeneutic of an artwork.

As Deleuze characteristically says, by observing only what is represented in the frame, in this case, the destruction of the building's facade, 'we wouldn't be able to achieve much, because we are still in the frame, in what the frame represents. I am going to speak about another catastrophe. . . . To learn about a catastrophe that affects the act of painting by itself. See, we are moving from the catastrophe represented on the frame – whether it is a local

catastrophe, or a total catastrophe – a secret catastrophe, a catastrophe that affects the act of painting by itself' (Deleuze 2007: 24).

But the questions still remain: What did the artists do? How did the image in the frame emerge? What is the diagram? Perhaps, just like in the works of Francis Bacon, this mural is a crack. It is structured by many successive layers: preparation, cleaning, engraving, engraving on top of engraving, 'erase, write, rewrite, rewrite and erase once again'. It is diagramming, ultimately, without searching for a specific figure, there where the etchings are erasures on something else that precedes them, an energetic formation, a tension that dissolves. In the case of graffiti, marks or 'etchings' are made either by writing in black, leaving the white space to shine or by writing in white on black. A canvas full of divisions and separations. A gesture that adds-writes, removes-erases, allowing different parts to be revealed in a virtually unending testing ground. In this abysmal field, the canvas-building is not just perceived as background or outline but 'partakes', gets 'incorporated' in this world of black-and-white figures that get composed and decomposed and could extend even beyond its limits. The image appears gradually, without any particular centre, repetition of a figure or form. It can be called 'open' image in a sense of a 'window into chaos', as a part of a bigger image frame, or maybe even an opening to chaos, as an image germ of new images. The *diagram* of the mural certainly lies in the possible gestures of its creators, it extends all over the 'canvas' while consisting also in itself, as a whole, a gesture-catastrophe of a known image that once existed.

It looks therefore as if the mural succeeded, through this destructive gesture, to reveal a new *diagram*. This new image, the 'abuse' of a known image, is not disconnected from its creators or rather from the conditions in which they operate. The *diagram* of the mural, by 'destroying' the familiar image, the memory-image, indicates the possibilities of another image, which was not evident until then. A new field of relationships got created on a physical and symbolic level. Through its physical presence, the mural 'energised' the public space and through the use or transformation of the image it revealed the gap between the image-cliché and the (probably) truer picture of the building in the collective unconscious.

A CATASTROPHIC EPILOGUE

This unorthodox gesture in this symbolically charged canvas, through the 'invisible', 'silent' yet intense 'battle' between the artist and the image data, revealed that the *diagram* of the hackneyed architectural discourse is neither purely optical nor pure gesture but a potential image that can be seen in the remains of the known, 'damaged' images. *diagram*, as an almost extreme

cathartic gesture, motivated more by internal forces, invites the 'destruction' of the familiar, stereotypical images and clichés, disconnecting the creative process from the instrumentalist logic that stifles it. Simultaneously, it exonerates, releases and also calls in a game of emergence of new images.

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Chapter 16

The Concept of Map in the Homeric *Odyssey*[1]

Aspassia Kouzoupi

THE WASP AND THE ORCHID: THE MAP AS A PROCESS; EMERGENCE OF THE INTRUDER AND THE RECEPTACLE

The relationship between orchid and wasp shall be the stimulus for establishing the point of view through which the notion of map will be approached in the Homeric *Odyssey*. The action of mapping is understood as the perpetually evolving process of a relationship: the cartographers themselves are involved in the relationship, being among its major contributors.

'The orchid does not reproduce the tracing of the wasp; it forms a map with the wasp, in a rhizome' (Deleuze and Guattari 1980, 20). As an initial step, we approach the species of sexually deceptive orchids, addressing their deceptive call towards specific species of wasps that they need as pollinators. They target the senses of the pollinating subspecies of wasps which are adapted to each orchid subspecies. The optical, olfactory and tactile perceptive modes of the targeted wasps are stimulated by the orchids in ways that are still under investigation.[2] It is not mimicry, rather it is *transcoding*. *Transcoding* stands for the fact that the genetic evolution of orchids is triggered by the choices made by their pollinating wasps. The existing orchid subspecies owe their survival to their successful pollination, and thus they map – in a complex way – the choices of the wasps. The pollinators during pollination by sexual deception function as *deterritorialised*, tending to be incorporated into the reproductive system of the orchids, widening thereby the chosen orchid's distribution. The orchids *transcode* the wasps and seduce them, dragging them into a state of *relative deterritorialisation*, while involved in the sexual deception. This *deterritorialisation* has significant side effects: the reproduction of sexually deceived wasps is often limited after being involved in a sexual deception. The results are side effects that affect the

future generations, bringing about the possibility of extinction of orchids that bear certain characteristics: this is a matter of *absolute deterritorialisation* – or *overcoding*: 'As a general rule, relative deterritorializations (transcoding) reterritorialize on a deterritorialization that is in certain respects absolute (overcoding)' (Deleuze and Guattari 1987, 175). *Transcoding* and *overcoding* are thus parts of the mutual mapping of orchid and wasp.

Both orchid and wasp form their map as a dialogue, rather than as a result; both orchid and wasp are subjects in it: the orchid is perceived as receptacle and the wasp as the intruder. The relationship 'becoming Wasp of the Orchid, becoming Orchid of the Wasp' (Deleuze and Guattari 1987, 10) presupposes the vector of time. The mutual becoming is obvious in a long-term relationship, for example, in a relationship between a dweller and the landscape he or she inhabits; their transformations are mutual along the vector of time, but they are neither directly comparable nor directly symmetrical.[3]

ODYSSEUS AND THE UNKNOWN WORLD: THE ENCLOSED MAP AS A TIMELESS STRUCTURE

The biggest part of Odysseus's *nostos*[4] – his return to Ithaca – is his map as the object of his narration[5] to the Pheacians. 'Ulysses penetrates into a mythical universe, after his stay at the Cicones, people of Thrace completely real, known as well by Herodotus, . . . after the ten days of tempest that he meets on his detour of cape Maleas, the last "real" site of his journey before his return to Ithaca' (Vidal Naquet 1981, 45). The most significant segmentation in the narrative map of Odysseus is the 'clean cut' between the 'real' world, and the 'mythical universe' – as Vidal Naquet suggests. I will prefer to call them 'known world' and 'unknown world'. These two worlds do not appear to be connected in any clear way – neither Odysseus's narration nor the epic plot traces a clear path between these two worlds. They remain two unconnected spatial masses, especially because the position of the unknown World stays clouded and protected. Odysseus's narration is mostly of a map of a separate world encompassed by the unknown. A map of an unknown world, without any link to the facts of 'real' geography seems to stand as a timeless structure: It is a map that does not need to be updated. If we take it to be a part of the epic corpus of the century-old oral tradition, we may conclude that it would not have required any change or updating on the basis of any new geographic, ethnographic, anthropological or sociological discovery, of the kind that were indeed made during the eighth to the sixth centuries BC. The structure of Odysseus's map bears a self-protecting mechanism against alterations, while also protecting the identity of Odysseus as the only cartographer having intruded an unknown world. The unknown world is the receptacle that

resists having to disclose its secret position to its intruder: the resulting map as well as the intruding cartographer having been the beneficiaries of this resistance. Lost in the unknown world, Odysseus finds himself completely far away from any familiar territory: he is definitely *deterritorialised*.

I shall now discuss two strands of the narrated map: the way the map of the unknown world is structured as a complex whole; and the way the intruding observer and empirical cartographer responds and interacts with the situation of his complete *deterritorialisation*. In Odysseus's narrative of the way he navigated the unknown seascape, we discern along the plot of his narrative map two different but successive modes of travelling. The first modality is developed without the help of any map. Here, the narrative deploys a sequence of surprising experiences along Odysseus's journey. The second mode of travelling often follows the directions given in narrative maps that various narrators,[6] mostly with their own narrations, lend him and he quotes. I refer to the latter as 'encapsulated maps'. His narrative map, therefore, in the second travelling mode, becomes an enclosing structure that encapsulates other maps.

ODYSSEUS AND THE UNKNOWN ARCHIPELAGO: SMOOTH SPACE, RHYTHMIC STRIATION

The first mode of travelling into the unknown archipelago is dominated by the spatial law of *smooth space*, as expressed by Deleuze and Guattari: 'In smooth space the stop follows from the trajectory' (Deleuze and Guattari 1987, 478). During this travelling in a previously unmapped territory, the direction of the itinerary, and therefore the stops, are completely hazardous: Odysseus is travelling along one among many possible itineraries. Different winds and weather conditions could have brought about different routes of navigation. Along the vector of their hazardous navigation, Odysseus and his comrades encounter islands and lands that could have been temporary stops only. And yet, their intrusions are characterised by a cultural interest in these unknown territories; the intruders, turn into experimental observers and investigators.[7] Most of these lands and islands turn out to be inhabited by culturally distinct creatures. The sea is indeed the smooth articulation between the culturally different territories of the unknown world; and to the extent that Odysseus does not give any clues about the directions that his boats were following, his map lacks any Euclidean geometrical coordinates: it consists of a cluster of territories, of culturally distinct and detached pieces of land, joined in a random way. The analogy with the structure of the patchwork mentioned by Deleuze and Guattari is striking: 'An amorphous collection of juxtaposed pieces that can be joined together in an infinite number of

ways; we see that patchwork is literally a Riemannian space, or vice versa' (Deleuze and Guattari 1987, 476). There is a strong affinity between the unknown world of the *Odyssey* and what Deleuze and Guattari call *smooth space*, and again with their discussion of the *manifold*. As Plotnitsky remarks, 'both [Deleuzian smooth space and Riemannian manifold] are primarily defined by their constitution as conglomerates of local spaces and multiple transitions between them' (Plotnitsky 2006, 188). Indeed, what Odysseus discovered and mapped up in his travel is a detached world, structured as a conglomerate of distinct cultural – and perhaps ontological – territorial regions that could be linked among themselves by means of an infinite number of different lines of flight.

In the first part of his narration, where the first modality of travelling predominates, Odysseus is lost to the point that he cannot find a line of flight from the unknown, he cannot *reterritorialise himself* on previously known places. In these circumstances and as an intruder, he explores the unknown world with its culturally distinct and detached territories, whereas as a navigator he measures the time intervals – days and nights – between these distinct territories. Furthermore, as a narrator, he creates a repetitive structure, a refrain that he superimposes on them. He forms thereby in a narrative way the journey's pulse. However, it is not obvious whether this pulse pertains solely to the narration or whether it has also to do with the experience of the journey that generated the map. The important fact is that in his narrated map the refrain begins to mark visibly and steadily the *lines of flight* [*lignes de fuite*] away from each one of the intruded territories. This refrain is composed of a stereotypical formulation and a variable part which adapts to each situation and announces the entrance to the next encountered culturally distinct territory (table 16.1). The overall structure of the refrain conforms very well to the way Deleuze spoke about territories: 'There is no territory without a vector of exit from the territory. There is no exit from the territory – namely deterritorialisation – without simultaneously an effort of reterritorialisation elsewhere, on another thing' (Deleuze and Parnet 1989).

The stereotypical phrase 'we sailed on, grieved at heart' forms the invariable core of the refrain. Between the formulation of the territorial exit and the various and varying arrivals, the conjunction 'and' is being used. This conjunction plays the role of an articulation: in addition to its syntactic properties it bears spatial properties; it marks the way the different territories are linked among themselves. This conjunction, 'and', is the articulation of apposition and/or juxtaposition, at least when it operates as a spatial marker, proper to patchworks and smooth spaces, according to Deleuze and Guattari (1987, 476–77). Notice that the basic topographic or cultural information of each of the stops is always embedded in the arrival refrains: land, island or citadel. This information follows the variants of the arrival refrain – 'we came to', 'we set foot' – and emphasises the importance of the discovered lands

Table 16.1. Refrains in Odysseus's Narrative Map during the First Travelling Mode

Homeric Odyssey Verses/Books	*'Territorial Refrain' in Ancient Greek*	*Translation in English by A.T. Murray*
Book 9, verses 82–84 (Entrance into unknown world, arrival at the land of the lotus eaters)	ἔνθεν δ᾽ ἐννῆμαρ *φερόμην* ὀλοοῖς ἀνέμοισιν πόντον ἐπ᾽ ἰχθυόεντα· ἀτὰρ δεκάτῃ ἐπέβημεν *γαίης Λωτοφάγων*, οἵ τ᾽ ἄνθινον εἶδαρ ἔδουσιν.	Thence *for nine days time I was borne by savage winds over the fish-filled sea*; but on the tenth *we set foot on the land of* the Lotus-eaters who eat a flowery food
Book 9, verses 105–106 (after the land of the lotus eaters/arrival at the land of the Cyclopes)	ἔνθεν δὲ *προτέρω πλέομεν* ἀκαχήμενοι ἦτορ Κυκλώπων δ᾽ ἐς *γαῖαν* ὑπερφιάλων ἀθεμίστων ἱκόμεθ᾽	Thence *we sailed on, grieved at heart, and we came* to the land of the Cyclopes, an insolent and lawless folk.
Book 9, last verses 564–566 (after the land of the Cyclopes) Book 10, verse 1 (arrival at Aeolia)	ἔνθεν δὲ *προτέρω πλέομεν* ἀκαχήμενοι ἦτορ, ἄσμενοι ἐκ θανάτοιο, *φίλους* ὀλέσαντες ἑταίρους. *[ΡΑΨΩΔΙΑ Κ]* *Αἰολίην δ᾽* ἐς νῆσον ἀφικόμεθ᾽· ἔνθα δ᾽ ἔναιεν *Αἴολος* Ἱπποτάδης, *φίλος* ἀθανάτοισι θεοῖσιν	From there *we sailed on, grieved at heart*, glad to have escaped death, though we had lost our staunch comrades Then *we came to* the island of Aeolia, where dwelt Aeolus son of Hippotas, dear to the immortal gods.
Book 10, verse 77 (after the island Aeolia) Book 10, verses 81–82	ἔνθεν δὲ *προτέρω πλέομεν* ἀκαχήμενοι ἦτορ. ἑξῆμαρ μὲν ὁμῶς πλέομεν νύκτας τε καὶ ἦμαρ, ἑβδομάτῃ *δ᾽* ἱκόμεσθα *Λάμου αἰπὺ πτολίεθρον, Τηλέπυλον Λαιστρυγονίην*	From there *we sailed on, grieved at heart*. So for six days we sailed, night and day alike, and on the seventh *we came to* the lofty citadel of Lamus, to Telepylus of the Laestrygonians.
Book 10, verses 133–135	ἔνθεν δὲ *προτέρω πλέομεν* ἀκαχήμενοι ἦτορ, ἄσμενοι ἐκ θανάτοιο, *φίλους* ὀλέσαντες ἑταίρους. *Αἰαίην δ᾽* ἐς νῆσον ἀφικόμεθ᾽· ἔνθα δ᾽ ἔναιε *Κίρκη* ἐυπλόκαμος, *δεινὴ θεὸς* αὐδήεσσα,. . .	From there *we sailed on, grieved at heart*, glad to have escaped death, though we had lost our staunch comrades. And *we came to* the island of Aeaea, where fairtressed Circe lived a dread goddess of human speech.

and the materiality of cultural constructions emerging out of the vastness of an unknown sea. The territories are clearly announced by Odysseus in his refrains of arrival, forming a sort of an introductory preamble. Each preamble is a veritable condensed map comprising the basic cultural characteristics

of each culturally distinct territory. There is of course an anachronism in all these preambles: as Odysseus sails towards any of these lands, he has no idea what they are like and who lives there. However, the importance of what Deleuze and Guattari have to say about refrains is not lost on us: 'The role of the refrain has often been emphasized: it is territorial, a territorial assemblage' (Deleuze and Guattari 1987, 312). 'The marking of a territory is dimensional, but it is not a meter, it is a rhythm' (Deleuze and Guattari 1987, 315).

The combination of refrain and preambles functions as a sort of backbone to Odysseus's narration, as far as the unknown unmapped archipelago is concerned: they emphasise the smooth spatial properties of the unknown world and summarise Odysseus's access to each culturally defined territory. Three elements are present in this combination:

- The state of grief is the constant condition of the crew members throughout their crossings of the sea.
- The spatial articulation of the lands of the unknown archipelago is constantly identified as apposition or juxtaposition.
- The lands they encounter are primarily characterised by the cultural characteristics of their inhabitants: they are distinct culturally defined territories.

The refrain gives the journey's pulse; besides being rhythm, it is somehow characterised by striation: as long as the oral epic tradition lasted, the refrains of the orally performed epos[8] could be noticed by the audience; the smoothness of the unknown seascape was repetitively cut and interrupted by the refrain, accounting therefore for the audible striation. Deleuze, speaking of Boulez and the way 'smooth' or 'striated' musical space emerges, states:

> He [Boulez] distinguishes for music a space he calls 'the striated space', and a space that he calls 'the smooth space'. The striated space, he says, is a space that one has to count in order to occupy. It is a space that is defined by size and measure. In music what is it? It is the pulse, which is a base unit; it is the tempo, which is a certain number of units. You see, the pulse, the tempo defines a striated space.

THE REFRAINS OF ODYSSEUS BIND SMOOTH SPACE AND STRIATION TOGETHER

In a sense, striation has its own spatial results in the way the unknown archipelago is perceived: as a cluster of discrete territories. Yet, the striation is not responsible for any material spatial transformation: in Odysseus' map, the unknown archipelago does not exhibit any direct and obvious spatial traces.[9]

To the extent that the rhythmic striation in Odysseus's map leaves no material spatial traces, we cannot be certain as to whether or not a cultural expression is influencing directly the mentioned landscapes. However, the way Odysseus refers in each refrain to the cultures he is acquainted with has the potential to reveal on the one hand the kind of traces formed by the interaction between intruder and receptacle[10] and on the other hand the intruding cartographers' cultural background. Nor can we exclude that some of the mapping strategies of Odysseus that concern the unknown archipelago were influencing and influenced by the historically documented ways the first navigators would structure their own maps. Odysseus offers through his narrative a possible way of navigating and discovering new territories: the way his map displays and transmits his cartographic signature throughout the epic oral tradition must have influenced[11] cartography and its relationship with travelling. When the Homeric epic poems were officially recorded (second half of sixth century BC), the first scientific attempts to widen the knowledge of the known world were also registered under the name of *periplous*[12] and *periodos gẽs*.[13] In these first geographical databases, namely the *periodoi gẽs* distances were calculated on the basis of the temporal length of journeys between focal points. Odysseus had tried something similar, as he counted the days of his journey between unknown spatial spots.

ODYSSEUS MAP AND THE ENCAPSULATED MAPS: IDENTIFYING A MAPPING NODE

It is in the island of Aeaea that the second mode of travelling emerges in the Odysseus's narrative. A series of encapsulated maps form a superstructure that shapes and orients the plot that follows. The first in situ encapsulated map in this narrative is the map he himself presents after his ship arrives to Aeaea. Deciding to investigate the enigmatic – at first inspection desert – Aeaea island, Odysseus climbed on the highest peak he could find, in order to obtain the best viewing conditions of the whole island. He saw from up high that the island was surrounded by a vast sea, and a plume of smoke rising somewhere between its dense bushes that he interpreted as a well-hidden sign of civilisation [Book 10, verses 145–50]. Coming down, he described what he saw to his comrades, adding a significant, structural change to his map: the smoke was rising from the centre of the island [Book 10, verses 194–97]. But a centre to the island implies a perimeter – the coastline – the topographical limit of the island. The island took an almost Euclidian[14] geometrical shape, ceased to be shapeless, and was abstractly objectified. This map obviously transcodes the landscape of Aeaea, as the intruder projects his perceptive scheme. It is followed by the narrative map of Eurylohus,[15] which conforms to this transcoding, as Odysseus's comrade describes how he approached the

very place Odysseus had referred to as the 'centre' of the island [Book 10, verses 251–26].

Subsequently, a long chain of maps is being deployed in the Homeric story: the mapping instructions given to Odysseus by the god Hermes, containing the clues he needs to overcome the dangers represented by Circe. Following them, Odysseus obtained Circe's favour, which includes her advice expressed by means of two maps. Circe's first map leads Odysseus to the ghost of the seer Teiresias at the threshold of Hades, where he is to receive another important map. Her second map reinforces Teiresias's mapping directions, adding details concerning the journey into the archipelago in the neighbourhood of Aeaea. Together, Circe's and Teiresias's maps structure to a great extent Odysseus's *line of flight* away from the unknown world, and lead him to the way of his *reterritorialisation* onto his homeland – the island of Ithaca. We notice that the modality of the journey has changed, in that the territorial refrain of exit and arrival, which kept adding the rhythmic pulse throughout the first mode of travelling, is silenced right after the Aeaea island episode: it is neither heard when the ship sails off the island nor when it reaches new territories. The absence of the refrain signals a change in the narration rhythm.

In view of the quantity of encapsulated maps, the chain structure that links them, and also the way the mapping instructions are followed by Odysseus ever since his arrival to Aeaea, I conclude that Aeaea – with Hades's threshold – have the interesting and complex topological property of forming a 'mapping node'.

Two Maps and a Tomb: The Map as a Bearer of Potential Material Traces; the *Sema* as a Common Denominator between Map and Landscape

At the threshold of Hades, Odysseus meets his dead comrade Elpenor who died accidentally in Aeaea. Elpenor asks persistently for a tomb [*sema*] on the shoreline of Aeaea, and Odysseus promises to Elpenor that his tomb shall be constructed according to his wishes.

> σῆμά τέ μοι χεῦαι πολιῆς ἐπὶ θινὶ θαλάσσης,
> ἀνδρὸς δυστήνοιο καὶ ἐσσομένοισι πυθέσθαι.
> (heap up a mound [σήμα/sema] for me on the shore of the grey sea, in memory of an unlucky man, that men yet to be may know of me)
> [Book 11, verses 75–76]

The signifier *sema* is used in the *Iliad* and the *Odyssey* not only to refer to tombs (i.e., Achille's tomb in the *Iliad*, Book XXIII verses 255–56), but also to refer to signs of recognition (i.e., the clothes given to Odysseus by

Penelope, Book 19, verse 250). I argue that the tomb of Elpenor is functioning in both ways, it is a 'hybrid sema', or a 'hybrid sign'. It is simultaneously an agent of *territorialisation*, *deterritorialisation* and *reterritorialisation*.

TERRITORIALISATION

The tomb is constructed – 'heaped up' – according to the directions given by Elpenor, constituting a significant example of a verbal construction map (verbal plan) in the Homeric *Odyssey*. We may assume that the tomb of Elpenor, being a culturally defined material construction, was a *tumulus*. The tomb/tumulus is a territorial sign that consecrates the coastal landscape of Aeaea to the memory of Elpenor. On the other hand, the tomb may be seen as a trace in the landscape that maps in a concrete, material way the passage of Odysseus and his crew as well as the death of one among them. 'During the Middle Bronze Age in the Aegean, earthen mounds were used to conceal a number of different tomb types. . . . Despite sharing some basic characteristics, tumuli in the Aegean are almost individually idiosyncratic monuments' (Galanakis 2011, 219–20). This tumulus/tomb is a significant agent of *territorialisation* 'in memory of an unlucky man'.

DETERRITORIALISATION (FROM TRANSCODING TO OVERCODING)

I have argued that Odysseus's in situ map initiates a *relative deterritorialisation*: it consists of a *transcoding* of the island's landscape by the culturally oriented perception of the intruder. The *transcoding* initiated by Odysseus's *in situ* map affected the way the landscape was perceived by his comrades, including Eplenor: I specifically refer to the positioning of the *tumulus*. Elpenor requested that his tomb should be located at Aeaea's shoreline: its outmost border, as opposed to the 'centre' that Odysseus and his comrades perceived as the emitter of Circe's magical powers. The culturally defined nucleus of Aeaea, occupied by Circe's residence, was hidden within the bushy island, hence the island appeared desert to the intruders; in contrast, the tumulus occupies one of the most conspicuously visible positions. As Galanakis explains, the visual impact of the Bronze Age tumuli in the Aegean depends a lot on the topography of the surrounding terrain. Within the context of a complex geomorphology 'the visual impact would have been limited at close quarters', yet he opposes this type of topography to landscapes characterised by horizontality 'for example the vast grasslands of the steppes where mounds may have actually mapped the landscape'

(Galanakis 2011, 219–20). The coastline topography next to the vast horizontal sea allows the detection of Elpenor's tumulus from afar. It therefore changes the way the island will be perceived from a distance: As long as the tomb tumulus is on the shore, the island cannot be perceived or imagined as desert anymore. Furthermore, the *tumulus* signals the cultural identity of the intruders, before the traveler gets to the interior of the island and comes to know the culture of its dwellers. The culture of the intruders overcodes the way Aeaea is perceived.

According to Carla Antonacio, tombs and mortuary rituals were expressions of the cult of the dead during the Bronze, the Iron and the early Archaic era, marking the limit of separation between live and dead mortals (Antonacio 1993, 46–70). The tomb is the symbol of the absence of the dead from the world of the living, their presence in Hades, the underworld (Antonacio 1993, 46–70). Notice that there is a continuous transition zone that links Aeaea to Hades: sailing from Aeaea, Odysseus's boat could 'directly', without rowing oars, float to Hades, and as easily return from it to Aeaea. Aeaea is considered as a topological node, a location of passage between the rest of the unknown world and Hades. The presence of a tomb tumulus, as a symbol of demarcation between the living and the dead, seems to be in opposition to Aeaea's transitional properties.

The tomb tumulus represents an agent of *deterritorialization* which is in certain aspects *absolute* concerning the cultural landscape that used to be Aeaea's.

RETERRITORIALISATION

This tomb – Elpenor's *sema* – being visible from the sea, constitutes a potential sign for future seafarers. It is a sign-flag, demonstrating the previous presence of Achaeans in Aeaea. It is a sign of extension of the territory invested by the cultural system whose bearer is Odysseus – a sign of territorial extension. It is, therefore, a factor of *reterritorialisation* of an area of the unknown world. Furthermore, if we suppose that, among the seafarers apprehending the sign, there are some who have heard the Homeric epos, or of the adventures of Odysseus, the sign would directly link Odysseus's map with the landscape. Elpenor's *'sema* becomes a culturally defined landmark. It becomes a starting point for further investigations, including remappings and other, relative or absolute, striations, of the unknown world – an anchoring and orientation point placed in the midst of the erstwhile great unknown that would last so long as the tomb stands as an identifiable material cultural construction at Aeaea's shoreline.

NOTES

1. This chapter is part of Aspassia Kouzoupi's PhD thesis research entitled 'Natural Landscapes of the Homeric Odyssey: Investigating the Structures and Limits of Cultural Sediments along the Nostos of Odysseus', supervised by Professor Vana Tentokali, Department of Architecture, Aristotle University of Thessaloniki.

2. As the orchids tend to attract their pollinators, they emit sensory signs. These signs are olfactory, visual and tactile, according to Anne C. Gaskett (2010, 33).

3. We have been using in this text a number of words, expressions and formulations from the writings of Gilles Deleuze and Félix Guattari such as – applying to territorial concepts: *deterritotrialisation*, *reterritorialisation*, *transcoding* and *overcoding* – applying to space and rhythm the terms: *smooth* and *striated*, *manifold* or *multiplicity*.

4. The meaning of *Nostos* is: the journey back to the homeland.

5. Through the point of view that Gregory Nagy introduces the narrations which have been embedded to the epic plot are considered as 'quoted verbal testimonies': 'In line with this pattern of thinking, a Homeric narration or a Homeric quotation of a god or hero speaking within a narration are not at all representations: they are the real thing' (Nagy 1996, 61). Therefore we consider Odysseus and all the other narrators into the Homeric plot as more than implied narrators: they are quoted narrators.

6. Among the encapsulated maps in Odysseus's narration are also included the in situ maps that refer to the landscape of the unknown world. As in situ we refer to the maps that were described while at the site where the action narrated in the enclosing map takes place. Odysseus's enclosing map is structured as a narration in the island of Scheria, in front of an audience of Phaecians. While Phaecia is considered part of the unknown world as well, it is not described at all into the narrative map of Odysseus.

7. Odysseus's Nostos is led by the deep desire for his return to his homeland, thus he will not let himself be absorbed for too long by any of the lands and islands of the unknown archipelago he finds himself lost in.

8. Gregory Nagy argues that 'From the standpoint of mimesis, the rhapsode is a recomposed performer: he becomes recomposed into Homer every time he performs Homer' (Nagy 1996, 60). We cannot be certain of the implementation of these refrains on every oral performance, but we refer to their audible properties when present.

9. At least up to the island of Aeaea. What happens – concerning material traces – on the island of Aeaea is presented in section 6 of the present text.

10. For example, the Cyclops are announced as 'an insolent and lawless folk', and this enunciation somehow anticipates the violence that takes place at their land; the enunciation of the 'lofty citadel of Lamus' also bears an indication on the catastrophe that occurred to Odysseus's fleet there.

11. We apparently assume that the Homeric epos primarily influenced those travels, and not vice versa, as the broad use of the practice of mapping itineraries by pioneer navigators was posterior, at least to the period of oral transmission of the Homeric epic poems, and also posterior or almost synchronic to the Peisistratids era (second half of sixth century BC] when, as Pierre Carlier mentions, the official written version

of the Homeric Odyssey was registered by the city of Athens (Carlier 2003, 72–74). For instance, Hanno of Carthage and Scylax of Caryanda mapped their itineraries during the sixth and fifth centuries BC, respectively, followed by Nearchos during the fourth century BC, and many others (Livieratos 1998, 34). We are not in the position to exclude as a possibility that the entities we refer to, belonging to the most widely spread written version of the Homeric Odyssey (*vulgata*), could have been partially influenced by the synchronic advances emerging from the mapping of itineraries. However, this reverse assumption, according to which the epic corpus, when officially registered as a text, may have incorporated tools and methods introduced by the synchronically developed navigation and mapping practice, would be equally exciting.

12. For example, Hanno of Carthage, the navigator who performed his *periplous* along the coastline of West Africa by the ends of the sixth century BC.

13. Christian Jacob states that *periodos* is a signifier for 'map' in ancient Greek 'the usual formula being "periodos gēs", the circuit of earth, or the "voyage around the world" '. The signifier *Periodos gēs* refers to these circuits both as a map and as a journey (Jacob 2006, 19).

14. Euclid (born 300 BC) and the way he structured Euclidian geometry, are of course posterior to the Peisistratids era (second half of sixth century BC] when, as Pierre Carlier mentions, the official written version of the Homeric Odyssey was registered by the city of Athens (Carlier 2003, 72–74). Yet it is significant to notice this change in Odysseus map: the introduction of a centre in relation to a perimeter. The centre as a signifier thus shall not be understood as emerging from the Euclidean context: otherwise the relevant verses would have been a posterior addition.

15. Among the comrades of Odysseus that were sent to explore the centre of Aeaea island, Eurylochus kept a safe distance therefore only saw what happened in Circe's residence, and could turn back and inform the rest of the crew on his investigating experience.

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Part IV

THE CLINICAL

Chapter 17

From the Exhaustion of the Dogmatic Image of Thought that Circumscribes Architecture to Feminist Practices of Joy

Hélène Frichot

Abstract

This chapter will progress from the concept of the superfold, which I argue operates as a key, if underdeveloped, concept for Gilles Deleuze's 'control societies' – though it is not to be found in his brief, enigmatic essay 'Post-script for Control Societies' – to the woman-form and her current debts, and from there to feminist practices of joy. My aim is to draw attention to ecologies of practice (Stengers 2005, 2010, 2011) in architecture that do not privilege the insistence of architectural form, but are instead orientated around a complex latticework of ethico-aesthetic encounters and material relations amid environment-worlds, the kinds of relations that tend to get lost in the background of a masculinist discipline such as architecture. In the process I propose to shift the indebted woman-form (in architecture) towards the dynamic passage of becoming-woman and her practices of joy, with all the risks this entails. At the outset, I admit my hesitation, as both concepts, the superfold and becoming-woman, are deeply ambivalent, even despicable, depending on their context of use and deployment. My hope, following the progressive steps of this argument, or rather the uncoordinated leaps I will now test out, is to make a contribution to the meeting place between feminist architectural ecologies of practice and the enduring legacy of Gilles Deleuze and Félix Guattari's conceptual creations.

The superfold can be figured as a development of the fold so popular in the discipline and practice of architecture at the turn of the millennium into the twenty-first Century. In the 1990s architects commenced by manually developing the fold, initially via mechanical and hylomorphic procedures inflicted on mute materials, planar and porous. With the integration of computation into design they subsequently came to recognise that the fold could also be

conceptually related to the infinitesimal calculus, a logic taken to undergird the development of computational codes or algorithms and their promise of novel, formalistic ends (Parisi 2013: 98–101). All of which is simply to say that the fold, whether or not associated with the proper name Deleuze, quickly came to be associated with the so-called digital turn in architecture (Carpo 2013).[1] Yet what is often overlooked in the incremental architectural engagement in processes of folding is the discovery that amid the supple dynamism of folding, virtual and actualised, a sensitivity towards 'following the material' of a fold might also be fostered. This, I propose, is where feminist practices of joy are alert to the sufferings and enjoyments of an ecological encounter, and where a 'neo-avant-garde' practice in architecture that focuses on the autonomy of exquisite architectural form falls short.

THE EXHAUSTION OF ARCHITECTURE

Following a progression of the fold along the curvilinear route of the digital turn in architecture towards the concept of the superfold inaugurates something different in kind, because the superfold no longer takes us all the way to infinity, as the fold purportedly did with Leibniz and the baroque, but merely enables the development, through repetition, of exhaustive combinatorials, folds upon delimited folds, with the suspicion that all we are left with is more of the same, practiced perpetually through serial permutations. That is to say, a certain material and conceptual exhaustion, as architecture discovers it, is unable to think itself beyond its deeply conservative biases, or unable to follow the potentiality of the fold, at least not all the way through the exhaustion of the superfold towards encounters productive of joys, however fleeting.

In Deleuze's appendix to *Foucault*, entitled, 'On the Death of Man and Superman', the concept of the superfold, as successor to processes of folding, is briefly presented in its relation to Michel Foucault's configurations of life, labour and language, or biology, political economy and linguistics (Deleuze 1988a: 127). This tripartite conceptual organisation frames how 'man' is a living being, a working individual, and a speaking subject (97), which surely begs the question: and what of the woman-form? Following Foucault in *The Order of Things*, Deleuze identifies the impending reorganisation of the 'quasi-transcendentals' (Foucault 1970: 250) of life, labour and language, as the three privileged domains in which the superfold, as the organisational diagram of societies of control, makes itself manifest, actualising its effects. The story of the fold discovers its denouement in the emergence of the superfold, one disciplinary implication of which is the wholesale integration of the computer and computation, for instance, into the architectural studio and design school, and into all of our lives. This is not a situation that can be

judged as either good or bad, but it does demand a critical understanding that is undertaken less as a judgement than a mapping of symptoms.

What permutations of life, labour and language are produced? The superfold, or rather *superfolding* is a concept designating a dynamic milieu of expression coloured by the oppressive threat of societies of control, characterised by increasingly sophisticated information technologies, what Donna Haraway in her cyborg manifesto (1991: 101) has called the 'informatics of domination' (see also Galloway 2012). At the same time, it is exactly this feminist philosopher of science Haraway who introduces a vibrant rethinking of technologies, the prescience of which has anticipated much of what has more recently come to be called new or neo-materialism, especially where it intersects with a feminist orientation in thought and practice (see Dolphijn and van der Tuin 2012; Coole and Frost 2010; Alaimo and Hekman 2009). Likewise, when Deleuze concludes his appendix to the book *Foucault* he suggests that the problem remains as to whether the advent of this new form, the superfold, will prove better or worse than what came before, that is, presumably, whether it will be better or worse than the 'man-form' (Deleuze 1988a: 132). The lingering question that we are left with, through the finite, yet unlimited recombinations of the superfold, is how will 'man' and his environments come to be reciprocally altered? And what role will the woman-form take? What human and non-human relations come to be organised via the noopolitics of cognitive capitalism amid networked societies?

The superfold anticipates the death of man, characterised as a line drawn in the sand that is washed away by an incoming tide. As Foucault himself points out, when it comes to the man-form he is less monolithic than confusedly composed, as so many 'scattered beings, stable for an instant, [which] are formed, halt, hold life immobile – and in a sense kill it – but are then in turn destroyed by that inexhaustible force' (Foucault 1970: 278). Only the event so depicted is even less dramatic than this, because what is witnessed by a non-human landscape is merely the obsolescence of the man-form; a former continuity of folded forms that held global sway for a while, subsequently entering into slow dissolve (279). As noted earlier, another implication of the concept of the superfold is that it functions to displace a former classical sense of the infinite as that which raises relations all the way to infinity, in its place introducing an unlimited finitude wherein 'a finite number of components yields a practically unlimited diversity of combinations' (Deleuze 1988a: 131). Caught up amid these finite delimitations, which are unlimited in their combinations, 'man's' life, labour and language perpetuate through the 'modulations' of what Deleuze has denominated societies of control (Deleuze 1995). It is exactly in this small, late essay 'Postscript to Control Societies', where I argue the superfold unfurls in relation to 'ultrarapid forms of apparently free floating control' (Deleuze 1995: 178; Galloway 2012),

mobilising the 'dividual' (Deleuze 1995: 180) as quasi-individual reduced to a noopolitical cluster of data and behavioural tendencies, in a state of perpetual incompletion, indefinitely forestalled, neither here nor there, monitored not individually, but at the scale of populations. It really doesn't matter because in any case the 'dividual', a concept Deleuze appropriates from Guattari, merely functions as the unwitting contributor to a larger compositional tendency.

So here, to abruptly shift the focal point from the place taken up by a man-form under duress to a woman-form would be to risk merely reproducing a reversal, and with such a gesture there is no guarantee that anything is in fact achieved. This is simply the well-rehearsed caution issued about foregrounding the repressed or overlooked half of an inferred binary couplet. Still, this is a journey I find I must partially take, to get where I want to be. Something I would like to test, to insert directly into a specific discipline, and that is architecture. The superfold in architecture alerts us to how new techniques and technologies transform 'man' *and* his environments, the places man takes up and transforms, beyond recognition. Following Deleuze, 'it's not a matter of hoping for the best, but of finding new weapons' (1995: 178) with which to critically and creatively engage with such reciprocal transformations. This becomes a challenge for architecture, and I will come to this challenge in the following. But first I need to take a detour through 'feminist futures' to find a way of moving through the 'woman-form' towards becoming-woman, and beyond.

FUTURE FEMINIST PRACTICES

I want to use the superfold to get past a mere alternative to 'man' in his engagements with life, labour and language, which suggests the tired old binary that switches like a chemical clock between a man-form and a woman-form, towards experiments in becoming-woman in relation to architecture. The misgivings that have plagued the formulation of becoming-woman must be acknowledged, but Claire Colebrook has offered a way of recuperating this ambivalent concept towards methodological opportunities, mobilising it less as a concept-tool than as a weapon (2008: 2). The challenge becomes how to work with a despicable, 'useless and paralyzing' concept (Colebrook 2008: 3), how to extract it from the usual quibbles about essentialism, and binary logics, and well-known sexed personae. Because a becoming-woman will not suffice to save us, and I would add, to enunciate however loudly, 'I am a feminist' probably never really helped anyone (save for offering some courage, like a song sung in the head as one finds one's way home through the dark), but certainly we can continue to ask what a feminist theory might do when it transforms material (architectural) practices.

Prior to her editorial work on *Deleuze and Gender*, in her introduction to *Deleuze and Feminist Theory*, Colebrook (2000) had already alerted readers to all the issues that could be associated with the concept of becoming-woman. And yet at the same time, depending on its deployment, it might continue to challenge the status quo, create concepts, thwart mere repetition and enable an experimentation with altered practices (Colebrook 2000: 7). It is worthwhile returning to this volume because it is too easy to forget these precursors, and to underestimate what might happen if we at least try to 'restore an incommunicable novelty to our predecessors' (Deleuze and Guattari 1994: 204), which sometimes requires just a little demolition too.

By the end of this early volume, *Deleuze and Feminist Theory*, Elizabeth Grosz promises the future, one open to 'contingency and transformation' (2000: 229), and an acknowledgement of the novelty posited through an ineluctable durational unfurling. Including a trust that all that the future offers is liberatory, emancipatory (necessarily so), as it is animated in contact with the virtual as self-generating different/cial force, as that which makes processes of *actualisation* possible: 'An openness of things, (including life) to what befalls them' (Grosz 2000: 230). 'The new, the future, what is yet to come, is the mutual horizon', the challenge then being to 'bring out latencies, the potentiality of the future to be otherwise than the present' (230). For Grosz, to imagine 'feminist futures', even an inkling of a post-feminist possibility (214), what is required is a new orientation towards futurity, rather than the predictable transformation of the already known, or the revision of exhausted vanguard gestures (Marxist, socialist, anarchist, feminist). Reflecting Colebrook's introduction, Grosz insists that feminist theory must take as its task an orientation not towards the past, and the repetition of already established 'political strategies and conceptual dilemmas' (230), instead it must construct a 'politics of the future' (231).

What happens when a 'politics of the future', or a politics projected towards a future collapses into the coils of the serpent that the superfold becomes when entangled with control societies? Deleuze's cautionary tale of the superfold is a story told not only of a past but a story that is also imagined for a future, a coming people, and new approaches to material and immaterial admixtures. Although he speculates upon hybrids of biological and machinic parts that are post-anthropomorphic in scope, and even though the superfold anticipates a biotechnogenesis of the human condition, we cannot yet know what this means, or what kind of peoples, and things, and non-human landscapes will emerge. As distinct from the man of discipline who came before, the 'control man' undulates, suggests Deleuze, he surfs and his business dealings have a gaseous, irrepressible quality (Deleuze 1995: 180). As Grosz rightly suggests, we must think and imagine the future, or else suffocate. And yet what does this future promise?

Fast-Forward Future Feminist

Fast-forward to a post-9/11 world, the fall of the World Trade Center in New York (already an old story, an oft-told architectural event), and the permanent exceptions of a perpetual war on terror as our indefinite promise to what risks becoming an evacuated future. Here, in the future, we have also born witness to the toxic sub-prime mortgage fallout, and the subsequent global financial crisis of 2007–2008. We encounter the closed-in horizon of a foreclosed future. All promise of the future has collapsed into the smallest possible margin. Despite all this, or because of these events, we are still playing the same old game of property, the one in which the best intentions of architects cannot help but become recuperated towards the repackaging of architecture as spatial commodity. The promise of the durational unfolding of the 'subjective' component of an existence, entangled as it is with the spatiality of 'objective' relations (Grosz 2000: 225) as promised by Grosz, here devolves into mere subsistence. A base unit of survival of human and/or other organism and environment is just barely sustained. This is what has become of a life shrunken by the closing gap between the virtual and processes of actualisation: a getting by, a paying off of debts, from one month to the next. The cerebral interval of 'zones of indetermination', as 'indices of life', the small delay between sensation and action that Grosz calls forth (2000: 221) has subsequently been filled with anxiety over meeting one's repayments, whether these be for a mortgage or for the privilege of education (Lazzarato 2015).

Enter the superfold of control societies, a 'mutation of capitalism' (Deleuze 1995: 180) where the material and immaterial labour of searching, groping (Bergson cited in Grosz 2000: 225) for 'indices of life' is near exhausted, reified, and in any case, only available for an inflated market price. Even a promising mutation of how a purportedly subjective durational flow is consumed amid new experience economies (Pine and Gilmore 1999) becomes a mutation properly belonging to neo-liberal capitalist flows and the gaseous effects of business. Where, as Deleuze explains, 'Control is short-term and rapidly shifting, but at the same time continuous and unbounded, whereas discipline was long-term, infinite, and discontinuous. A man is no longer a man confined but a man in debt. One thing, it's true, capitalism keeps three quarters of humanity in extreme poverty, too poor to have debts and too numerous to be confined: control will not only have to deal with vanishing frontiers but with mushrooming shantytowns and ghettos' (1995: 181). Specifically, and again, 'A man is no longer a man confined but a man in debt', and this too pertains to a woman-form and her new-found debts as she enters the fray, attempting to claim some self-sovereignty, a room of her own, freedom of movement and expression, and the ability to make decisions for

herself, including when and where she has the opportunity to play the role of architectural thinker and/or practitioner.

INDEBTED FORMS

This is the refrain, from Deleuze's 'Postscript on Control Societies', that Maurizio Lazzarato integrates into his recent works, *The Making of the Indebted Man* (2012) and *Governing by Debt* (2015: 61): 'A man is no longer a man confined but a man in debt'. Taking and expanding upon Deleuze's control societies, undertaking a merger between biopolitics and noopolitics – or the politics of how minds are managed at the scale of populations – Lazzaratto asserts that the virtual is composed of little but debt, and that the short-lived neo-liberal glimmer of the entrepreneurial self amid knowledge societies, or 'cognitive capitalism', has promptly given way to societies of debt, and with them the 'making of the indebted man', or what Deleuze has earlier called the 'control man' (1995: 180). The making of the indebted man is the composition of a subjected figure, another kind of 'durational formation', a man-form suffering under extreme duress in relation to his ever-curtailed, over-curated spaces and times, including his global movements: 'Control through debt, however, is exercised within an open space and an unlimited time, that is, the space and time of life itself' (Lazzarato 2015: 69). And our complaint (as architects concerned with a pragmatics of feminist practices, even feminist futures, after Grosz) is a peevish, querulous one that asks, what of the making of the indebted woman in this formula? It would seem that what the feminist future has delivered to those who have been quietly waiting is a bad return, that is, an indebted woman-form.

In a recent essay for *Avery Review: Critical Essays on Architecture*, Helen Runting and I have taken hold of Lazzarato's *The Making of the Indebted Man* in order to ask after the indebted woman and her contemporary situation and future status (Runting and Frichot 2015). Together we argued that a former cry such as the 'personal is political' has all too easily become the 'personal is economic', and a hard-won fight for individual rights has turned out to be an excuse for a shift in responsibility, for now the indebted woman must take upon herself the costs and risks expropriated by the state and corporations (Lazzarato 2012: 51). The indebted woman has succeeded, she has become a neo-liberal feminist, but only to rediscover her 'newly born self' (Cixous and Clément 1986) mired in debt, and with no future promising the open possibilities of which Grosz speaks. And it is exactly her future that is at stake, the time of her existence, which she has offered by way of a guarantee that she will pay back her loan (Lazzarato 2012: 60). The woman-form, admittedly a near empty abstraction, departs from one stratum of historical

bondage to man, only to discover herself beholden to another, and that is her mortgage, a debt most likely further burdened by her student loans (Lazarato 2015). The indebted woman has mortgaged her future, most often in pursuit of some security in housing, in the meagre (or more lavish) means to shelter life, but the deal also involves the over-determined performances of her subjected position.

Lazzarato argues that the contemporary subject is compelled to be free (2012: 31), but free only to pursue a style of life that best fits what he identifies as the ubiquitous global logic of finance amid a debt economy, where finance is 'a formidable instrument for controlling the temporality of action', neutralizing possibilities, the 'moving present', 'quivering uncertainty' and 'the line where past and future meet'. It locks up possibilities within an existing framework while at the same time 'projecting into the future' (2012: 71). The temporal mechanics of finance lie in the promise we make to pay back our debts, so that we effectively offer up our future and our capacity to participate in a workforce as a form of security. Lazzarato explains that 'Making a person capable of keeping a promise means constructing a memory for him, endowing him with interiority, a conscience, which provide a bulwark against forgetting. It is within the domain of debt obligations that memory, subjectivity, and conscience begin to be produced' (2012: 40). Here, amid a relation to debt, is also where the 'quasi-transcendentals' of language, life and labour are arranged in specific compositions. Furthermore, it is exactly by this logic of the construction of memory that all memorialising, all memorial architectures might be revisited as reified instances of a debt owed in perpetuity. Memory, subjectivity, conscience, we might argue, are all recent possessions that the 'indebted woman' has been able to secure for herself in terms of her modes of life and her expressions of subjective existence, including the environment-worlds she manages to carve out for herself only to instantaneously discover that her possessions threaten to exhaust her possible future, because she has been extended an elastic line of credit that she will now be obliged to pay back.

And yet debt can be taken in its positive and negative forms. There is the debt that we might choose to *renege*, refusing the moral demands it places on us, as Lazzarato urges, and there is also the debt to those who have come before, those to whom we hope to restore an incomparable novelty: Our precursors. This is why I have returned to *Deleuze and Feminist Theory* and to *Deleuze and Gender*, where Australian feminists in particular play an important role. The debt here is how the intellectual labour undertaken can be aptly attributed, where I would insist that there persists another kind of debt that is worth owing to trailblazing 'daughters of chaos', those who have sought to make thinking consistent or sufficiently durable in relation to the problems they were seeking to tackle, finding ways to enable the collective enunciation of the voices of minoritarian others.

It is a debt that I cannot amortise but will continue to repay. In this, though, I suggest a debt that opens up and does not foreclose on possible futures.

Forms of Architectural Debt

Let's return again to the contemporary challenges facing architecture, where housing stress and associated social concerns are the most pressing issues, aggravated through structures of debt that do not open up, but close down on a future. Housing, shelter, which inevitably, too quickly, becomes someone else's property rearticulated as the spatial commodities of real estate, is an urgent question for architects, because architecture is a conservative discipline that too often contributes to the inflation of real-estate value through its image-making practices and signature edifices and other architectural 'effects'. Its remit, its key task is to conserve life, to shelter (human) existence from external storms (both real and existential). This means architecture necessarily tends towards the status quo, towards the presumed structural quality of good (design) sense, and the happy consensus of common (design) sense, towards recognisable forms, towards analogy, similitude, identity, simple opposition – what is good, and what is bad design. Architecture focuses on propositions, and how a proposition already shelters a ready-made solution and a viable, saleable, end product, in the process ever attempting to ward off error, except perhaps for the happy accidents that emerge amid a felicitous design process. All of which is to say, architecture, much like all the disciplines – and philosophy is by no means privileged here – is constrained by its own contemporary, hegemonic 'Image of Thought' (Deleuze 1994). For architecture its own Image of Thought pertains to the way it feels obliged to aid and abet in the capture and manipulation of atmospheres of affect in urban contexts and curated urban interiors, supporting at the same time the architectonic of over-mediatised lives lived out intermittently amid datascapes, collected and collated as the noopolitics of 'neural capital' (Braidotti 2013: 2), a domain, which all the while continues to privilege certain kinds of transcendent subjects and objects.

Feminist Practices

There are other approaches, even 'attacks of the castle' (Cixous 1997), the edifice of architecture, and attempts to dismantle the master's house with his tools, or else with newly constructed feminist design power tools (Bloomer 1993; Lorde 1984; Frichot 2016). There are other stories to be told, and stories yet to come, and we might yet focus on what Grosz calls a feminist future. Karen Burns, for instance, alerts architects to the fact that before digital architects and theorists like Greg Lynn, before Peter Eisenman (both

well-known US architect/theorists who have made use of Deleuze and Guattari's oeuvre), thinkers and practitioners like Jennifer Bloomer were reading Deleuze, and not just Deleuze alone, but integrating his work into broader constellations of thinkers and doers, allowing for more chatter and noise to emerge (Burns 2013). With an ill-formed question I have also previously wondered what might be the implications of thinking a new woman-form: 'Is a woman-form, and altered forms of practice possible to imagine across the plane of composition that is architecture?' It was a thought left unfinished, 'a promise to future work' (Frichot 2013b: 88), saved for a feminist future. The question was no doubt ill-formed exactly because of a disciplinary habit of form-thinking in architectural design research. Too often, where the question is architecture, the answer is assumed to be 'designed form', or 'built project' procuring a dogmatic Image of Thought that uncritically celebrates contemporary architecture's icons and idols. And while this thought risks promptly privileging a 'woman-form' over a 'man-form', as though such a simple displacement or reversal would suffice, yet, in a discipline in which, whether we like it or not, whether we want to admit it or not, women (however this category is performed, and reformed) remain under-represented. The 'woman-form' in architecture still proves insufficiently durable, disappearing rapidly, even while they enter the academy in ever-greater numbers.[2]

Less a 'woman-form' then, to undertake battle with a 'man-form' under erasure, than something closer to a becoming-woman. The task then is to venture new practices and practical identities deployed as tools and weapons, and alternative approaches to architecture. So, neither woman-form, nor exactly a becoming-woman, but something closer to an experimentation via 'feminist practices', which have meanwhile become evident more urgently in the discipline of architecture with books such as Petrescu's *Altering Practices: Feminist Politics and Poetics of Space* (2007), and Lori Brown's more recent *Feminist Practices* (2011). Even though feminist practices risk sounding anachronistic, because, 'didn't we already do this in the 1960s and the 1970s?', didn't this project either succeed or fail? Here the logic of feminist waves does no one any good, they make history sound like a series of linear, progressive success stories (or failures), and they also place women practitioners in a stalemate: to be accused of being passé, even anachronistic always produces an anxiety of irrelevancy, a bad debt.

While a course that is set towards feminist practices of joy may still seem for some an anachronistic project, superceded by work undertaken in the post-humanities and in relation to the non-human turn, architecture moves more slowly than these swift theoretical trajectories, and needs more time to explore the potentialities of alternative modes of feminist expression, before entirely letting go. Despite the noopolitical threats that can be associated with superfolding and its modulations and subjectivations (specifically, that we

are still apt to think collectively in rather over-determined ways), I want to argue that there persists the tenuous promise of becoming-other-than-what-we-have-been in relation to radically reformulated existential territories and their relational ecologies, that is to say, new imaginaries for new combinatorials of labour, language and a life. If we (the diverse, often disagreeing and loose collective of architectural thinkers and doers) move rather more slowly than the conceptual adventurers of the post-humanities, it is because there are still questions and demands that must be made for the practices of minoritarian expressions of becoming in architecture. I aim to conclude with one such example, an example I find myself frequently returning to, and that is the landscape encounters of architect-artist Margit Brünner, whose work challenges the decorous delimitations of a conservative field like architecture.

Fleeting Joys in Composition

Towards this end, and to conclude, I present the work of Margit Brünner, an Austrian-Australian architect. I offer this work as something of an after-image because this is really to conclude with a parable of a future for architecture that may well break its limits, dismantle its stern edifice or confound its boundaries, at the risk of dismantling its disciplinary 'consistency' and its material 'durability'. It is not that I propose that Margit provides any (architectural) answers, instead she opens up ethico-aesthetic 'test-sites' for investigation (Brünner 2015). When we pause to think of architecture, we are apt *to think that* architecture is composed of sturdy, delightful and useful forms, well-measured spatialities carved out in relation to well-selected materials, producing well-tempered environments. To introduce the work of Margit Brünner addles architecture's core assumptions with *indiscipline*,[3] less as a lack of self-control than a creative, even systematic and transgressive challenge that unsettles its conservative inclinations. Margit's work does not in any way *look like* architecture, and you will not recognise it as such. It is composed of embodied performances captured through photographic and video documentation of Margit engaging with urban and wilderness landscapes, and it is further supported by sketches and watercolour markings that capture traces of these events, but which also render 'communication' near opaque, preferring instead to operate towards the procurement of tremulous affects. Her work also deploys an idiosyncratic glossary that addresses the central role of atmospheres in her practice (Brünner 2015: 219).[4]

I propose that Margit's experiments call forth a feminist ecology of practices, and here the philosopher of science Isabelle Stengers and her cosmopolitical project may be of assistance, asking us to be alert to local requirements and obligations, to understand the validity of maintaining a diversity of practices: 'We do not know what a practice is able to become; what we know

instead is that the very way we define, or address, a practice is part of the surroundings which produces its ethos' (Stengers 2005: 195). Through processes of altering practices, the very environments, including disciplinary milieux (such as architecture) might be positively transformed. Margit challenges the core assumption of architecture that every architect should be beholden to a project, with an appropriate programme and brief. Instead Margit's labour, including her labour in concepts, is something of a refusal of the 'architectural project' and the disciplinary call to order it entails. Her labour, her life and her language made provisionally consistent as a (hetero)glossary of atmospheres of affect, enfolds and unfolds other concerns in contact with the earth, whatever we hope to mean by that. For instance, she challenges what it means to survey a site, to get to know a context, taking to a limit ad absurdum how to address the conservative architectural-phenomenological concept of *genius loci* where site and subject are reified and no ethological opening is possible.

Site for Margit is something that is atmospheric and fleeting, like dust rising above the battle, or mist rising over the prairie (to draw on two well-known images from Deleuze). Site, or rather situation produces another kind of sense in contact with the earth, and a reciprocal transformation takes place, so that neither environment-world nor practitioner-in-process remains untouched or unchanged. Its tactful capture requires the development of strange, near nonsensical 'cosmethical' tools that pass through the weight of dense urban air, or scratch at red earth and Australian mallee bush. What is it that Margit begins to imagine with her sensitive devices, strange tools that test the atmospheres of her environmental surroundings? Documentary photographs exhibit a paddle at the end of a stick that she waves in the vicinity of oblivious traffic in central Melbourne (Brünner 2015: 24, 86); there is another documentary image of something like sparklers or a burst of electric pollination escaping from her tender hands (24). There are clusters of sticks and a canvas skirt made to catch the scratches of the scrub when she leaves the city behind and heads for the South Australian outback. What there is little evidence of, or what recedes into the background as facilitative support, is what is usually recognised as built architectural form, the built environment.

What relation to architecture do her embodied performances amid dynamic non-human landscapes express? How do we apprehend her repeated gestures, insistently stroking grasses, navigating longitudes and latitudes, is she attempting to receive some reciprocating world-environment response?[5] She is a quasi-human assemblage in crisis seeking some equilibrium. Her work (and it is an enduring labour on atmospheres at stake here) enfolds non-human landscapes, things and thinkables, and what she is doing is 'following the materials' of her material encounters as an ecological process of folding that does not presume that form is an inevitable end product. Her aim (a Spinozist one) is to increase the proportion of joyful encounters, and reduce

the passive ones, a near impossible proposition, but all the more worthwhile for this great difficulty. Through her, what I wonder, with her reading of Spinoza (and Deleuze), her (hetero)glossary of atmospheric terms, her embodied assemblings and disassemblings with landscape, her pursuit of joy in a life, is how far language, labour and a life might be recalibrated? It's hard to imagine a world without work, a world in which other possibilities of creative and critical expression might be experimented with: 'We've got to hijack speech. Creating has always been something different from communicating' (Deleuze 1995a: 175). Could this be, as Lazzarato puts it, an example of the potential of a lazy action to suspend identities and open up new becomings? (Lazzarato 2014: 251). Or better still, these are practices of slowness, after Stengers, to slow down, to hesitate, to pause. While Margit's test-sites make for small, even incidental gestures and minor realignments, to shift an overwhelmingly dogmatic Image of Thought in the discipline of architecture where 'core concerns' are still determined by majoritarian actors is a difficult thing indeed.

NOTES

1. It must be noted that this supposedly authoritative account of the digital turn in architecture, edited by Mario Carpo, is unfortunately indicative of the exclusion of women architects and researchers when it comes to the production of discourse in publications treating architecture and digital technologies. Only one woman, and only as a co-author (Hina Jamelle) is noted on the contents page of Carpo's edited 2013 publication, in addition to the now disbanded architectural studio Foreign Office Architects where Farshid Moussavi was a director.

2. For further discussions concerning women, equity and leadership in architecture, see the various essays on the Australian website archiparlour.org.

3. 'Indiscipline' is a concept that is currently being explored by Jasmin Dücker in relation to her comparative reading between Deleuze and the poet Emily Dickinson. See http://anglistik1.phil-fak.uni-koeln.de/22327.html. Accessed 2 August 2016.

4. For further discussion of Margit Brünner's work, especially in relation to her reading of Spinoza and Deleuze, see Frichot (2017, forthcoming).

5. See, for instance, Margit's video *Zwischen Büschel* [*Among Tufts*], where the artist can be witness, dressed in a red that stands out against the greys and greens of the Flinders Ranges of South Australia, slowly moving through the low scrub caressing the sturdy flora with repetitive gestures. See https://vimeo.com/36272677. Accessed 2 August 2016.

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Mike Hale (Archispace, Sydney, Australia) is a registered practicing architect currently running the domestic studio Archispace, based in Sydney. He has worked on a wide range of projects, including the Sydney Olympic Tennis Centre complex, Stadium Australia, the MLC tower North Sydney, as well as many smaller commercial and residential projects. Mike attended the Architectural Association in London after completing his BArch at the University of New South Wales (UNSW) in Sydney, where he first encountered Deleuze's work while reading for his MArch in architectural histories and theories. He later followed this with an MLitt (Phil) at Sydney University where Moira Gatens supervised his treatise on 'Deleuze and Painting'. Mike completed his PhD under the tutelage of Paul Patton in 2012 at the School of History and Philosophy UNSW, which focused on Deleuze's concepts of the new, materiality, design and sensation. He published a chapter entitled 'The Architect as Metallurgist: Using Concrete to Trace Bio-digital Lines' in *Deleuze and Architecture* (2013), co-edited by Hélène Frichot and Stephen Loo. Mike continues to develop an experimental architectural practice which posits that to innovatively experiment requires a radical rethinking of the nature of the material world, of how space, matter and sensation combine via the architectural act and how to remain open to the vitalising metamorphic vagaries of difference itself.

Keith Harris is a lecturer in the University of Washington, where he teaches in the College of Built Environment and the Comparative History of Ideas Program on the Seattle campus, and in the Urban Studies Program on the Tacoma campus. His research focuses on the dynamics of large-scale urban redevelopment projects, which he uses to contribute to a broadly Deleuzoguattarian conception of critical urban theory. He received his PhD from the Interdisciplinary Built Environment Program at the University of Washington, in 2016.

Marko Jobst is Architecture Undergraduate Theory Coordinator at the Department of Architecture and Landscape, University of Greenwich, UK. He holds a Diploma in Architecture from Belgrade University and MArch, MSc and PhD from The Bartlett School of Architecture, UCL. He has practiced in Belgrade and London, and taught at a number of schools of architecture in London. His interests include the philosophy of Gilles Deleuze and

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Anthia Kosma (Technical University of Madrid) received her diploma of architect-engineer from Democritus University of Thrace (Greece). She holds a PhD and DEA from EscuelaTécnica Superior Arquitectura (ETSA) of Madrid and Universidad Politecnica de Madrid (UPM) of Spain, respectively, with a scholarship from (State Institution for Scholarships) IKY and Triantafyllidis Foundation.

Dr. Stavros Kousoulas studied Architecture at NTUA, Athens and at TU Delft. Since 2012, as a lecturer and researcher, he has been part of the Theory Section of the Faculty of Architecture of TU Delft. In addition, he is a PhD candidate at IUAV Venice participating in the Villard d' Honnecourt International Research Doctorate. His most recent publication is 'Urban Correlationism: A Matter of Access' in *Critical & Clinical Cartographies*, eds. A. Radman and H. Sohn (2017). He is a member of the editorial board of *Footprint Delft Architecture Theory Journal* since 2014.

Aspassia Kouzoupi holds a diploma in architecture from the Aristotle University of Thessaloniki (AUTH), a masters of advanced studies in landscape architecture from ETH/Zurich, and a diploma in fine arts from the National School of Fine Arts of Athens. She obtained her PhD at the Architecture Department of AUTH: her thesis under the title 'Natural landscapes of the Homeric Odyssey: Investigating the Limits of Cultural Depositions along the Course of Odysseus Nostos' – supervised by Professor Emeritus Vana Tentokali – was presented in June 2017. She has taught courses related to landscape, as an adjunct lecturer, at the Universities of Thessaly and Patras, since 2007. In 2000 she co-founded the team 'Sculpted Architectural Landscapes', in collaboration with landscape sculptor Nella Golanda, and since realises mostly urban scale landscape architecture projects, at urban and peri-urban sites. Some of these projects were finalists for significant European Prizes (Rosa Barba Award/European Prize for Urban Public Space), or have received honorary mentions in Greece. Her work is featured in international publications of architecture, landscape and art. The focus of her research and creativity concerns both the ways landscape is being perceived and mapped, and the ways the landscape maps the processes that transform it. She is involved into designing landscape at scales from 1:1 up to territorial scales. Her artistic work forms a dialogue with landscape as well, through her concept of 'signs' which functions as visual signifiers: 'hybrid signs'/2007, and 'complex signs'/2013.

Konstantinos Moraïtis wrote his doctoral thesis under the subject: 'Landscape – Allocating Place through Civilisation. Exposition and Theoretical Correlation

of the Most Significant Modern Approaches Concerning Landscape (School of Architecture, NTUA). He attended Postgraduate Studies of Ethical and Political Philosophy, Seminar of Aesthetic Philosophy (Université I de Paris, Panthéon-Sorbonne, 1980–1981), and Postgraduate Program of Arabic and Islamic Studies (Pantios School of Political Sciences, Athens, 1981–1982). He has been teaching in NTUA since 1983. He is responsible for the post-graduate seminar of 'History and Theory of Landscape Design'. He has published architectural projects and scientific articles, participated in collective editions and authored tutorial books concerning landscape design. He has achieved numerous distinctions in architectural competitions in Greece and Cyprus including two first prize distinctions in International Architectural Competitions: Urban and Landscape Design for the city of Lviv – Ukrania (2008), Design for the Centre of Holistic Medicine in Allonisos – Greece (1998). He is member of the Greek Philosophical Society and the Hellenic Society for Aesthetics.

Athena Moustaka (Aristotle University of Thessaloniki, Welsh School of Architecture, University of Manchester) is an RIBA/ARB architect with seven-year on-site experience in concrete structures, and currently lecturer at the University of Salford. Based on her observations of the everyday construction practices, Athena researches the presence of concrete as a vital material agent in architecture. In her work, concrete is viewed through the perspective of object-oriented ontologies, and becomes active in shaping construction practices and the experiences of the built environment on a daily basis. She recently completed a secondment at Northeastern University, Boston, as a visiting scholar; she has previously taught at Manchester School of Architecture and has prepared her work in the Universities of Liverpool, Arizona State, Central Lancashire and Manchester Metropolitan.

Constantinos V. Proïmos (Hellenic Open University) received a PhD in philosophy from the New School for Social Research, New York, NY (2001) after studying sociology, art history and philosophy in Athens, New York and Paris, respectively. He has published widely on aesthetics, philosophy of art and art history, his fields of interest and expertise, in Greek and in foreign venues. He received a state scholarship (IKY) for his postgraduate studies and was a 1993–1994 Helena Rubinstein Fellow at the Whitney Museum Independent Study Program. His first book, *On the Limits of Aesthetics: The Role of Art in the Writings of Martin Heidegger and Jacques Derrida* was published in Athens in 2003 by Kritiki Editions. He has taught at the University of Crete, the University of Cyprus and the Technical University of Crete, and in 2003 was awarded a state scholarship (IKY) for his postdoctoral work. He currently teaches at the Hellenic Open University while being an art critic and curator.

Chris L. Smith (University of Sydney, Australia) is an associate professor in architectural design and technê in the Faculty of Architecture at the University of Sydney. Chris has lectured internationally. His research is concerned with the complex connection of bodies and buildings – a connection he locates at the interdisciplinary nexus of philosophy, biology and architectural theory. He has published on the philosophy of Gilles Deleuze and Félix Guattari, technologies of the body and the influence of evolutionary biology on contemporary architectural theory. Presently Chris is concentrating upon an Australian Research Council project focused on the architectural expression of scientific ideals in biomedical laboratories, and has a book currently in production with Bloomsbury titled, *Bare Architecture: A Schizoanalysis*.

Vana Tentokali (School of Architecture, Aristotelian University of Thessaloniki [A.U.Th]) is a Dr architect and professor emeritus in the Department of Architecture, Aristotle Technical University Thessaliniki. She holds a diploma in architecture (A.U.Th., 1972) and a PhD in architecture (A.U.Th., 1988). She was a member of the teaching faculty, Department of Architecture, Roger Williams College (Bristol, Rhode Island, 1986), research fellow at the 'Behavioral Science Research Group in Architecture' (Department of Architecture) and 'Program for Gender Studies' (Department of Humanities), (MIT, Cambridge, MA, 1982–1985), visiting scholar at the 'Program of History, Theory and Criticism' (Department of Architecture, MIT, spring semester 1992), visiting research fellow at the 'Program in Hellenic Studies' (Princeton University, Princeton, New Jersey, fall term 2008). She presented lectures at MIT (1984); Rhode Island School of Design (1985 and 1986), Tampere University of Technology (Tampere, Finland) (1999); Master Program of Semiotics of the New Bulgarian University (Sofia, Bulgaria) (1999); Sushant School of Art and Architecture and I.P.Estate (New Delhi, India) (2001); and Princeton University.

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Stathis-Alexandros Zoulias (National Technical University of Athens [NTUA]) received a master's degree in architecture from the University of Patras in 2006; took part in KAM workshop in Chania in 2004 (part of the Greek representation in the Venice Biennale); 2007–2009: attended the postgraduate programme 'Space-Planning-Culture' at NTUA between 2007 and 2009; graduated in 2009; 2008–2010: attended the seminar 'Architecture et Philosophie' at ENSAPLV in Paris (professor: J. Boulet) and the University

Paris 8 seminar ‘W. Benjamin and the Question of Space’ (professors: J.-L. Deotte and V. Fabbri); and from 2010 to till date he pursues a PhD course of studies in Architecture at NTUA (supervisor: G. Parmenidis). He participated in three group exhibitions: September 2010: ‘Locus Solus’, Benaki Museum (curator: S. Goudouna); March 2011: ‘Art/Architecture competitional exhibition’, 11th Panhellenic Architect Congress, Zappeion Megaron, Athens (third prize); and November 2012: ‘Counter-Culture: The Emergence of a New Social Subject’, CAMP (curators: P. Arapinis and T. Moutsopoulos). Since 2010 he has been a self-employed architect.